

A LOCAL PROOF OF LANGLANDS'S SECOND MAIN LEMMA FOR LOCAL EPSILON FACTORS OVER NONARCHIMEDEAN LOCAL FIELDS

FUKUHIRO UEDA

ABSTRACT. We give a local proof of Langlands's Second Main Lemma for normalized one-dimensional local epsilon factors over every nonarchimedean local field, in both mixed and equal characteristic. The theorem asserts independence of the inducing field in a $C_\ell \times C_\ell$ extension, with the complete lower-field correction factors included. The one-dimensional inputs are the author's First Main Lemma and the local ramification and stationary-phase results established in its companion paper. All specialized two-field calculations are given here.

The odd-prime argument constructs compatible character models and uses weighted norm-trace estimates to reduce the comparison to a root-of-unity ambiguity removed by the First Main Lemma. The dyadic argument instead compares the complete residual functions, including their affine terms. Common norm representatives coordinate the stationary values, and the third quadratic field supplies the cancellation needed at a critical norm image of index two. Separate local constructions treat equal characteristic and the maximal and nonmaximal mixed-characteristic quadratic breaks. A characteristic-uniform tame calculation and an exhaustive ramification and conductor analysis complete the proof.

The methods retain the explicit local character of the work of Dwork, Langlands, and Lakkis. Their two-field comparison results are not used as premises. The remaining quoted inputs are stated classical results in local field theory. The leading finite-field Gauss congruence and the formal local trace-residue identity are proved in appendices; no global functional equation is used.

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1. INTRODUCTION

The First Main Lemma relates local constants along a cyclic extension of prime degree. The Second Main Lemma compares two different such routes through the same field diagram. A character on the top field can be obtained by norm pullback from different intermediate fields; the corresponding local constants, with their lower-field correction factors, must give the same answer. The purpose of this paper is to prove that compatibility locally, with every residue characteristic and both mixed and equal characteristic included.

Let F be a nonarchimedean local field with finite residue field of characteristic p . Let K/F be Galois with group $C_\ell \times C_\ell$, where ℓ is prime. Let $\Theta : K^\times \rightarrow \mathbf{C}^\times$ be a continuous quasi-character invariant under $\text{Gal}(K/F)$ and not of the form $\theta_F \circ N_{K/F}$. For each degree- ℓ intermediate field L choose θ_L with $\theta_L \circ N_{K/L} = \Theta$. Such extensions exist by cyclic Hilbert 90 and extension of characters from the norm image. Fix a nontrivial continuous additive character ψ_F , and put $\psi_L = \psi_F \circ \text{Tr}_{L/F}$. Write

$$S(L/F) = \{\nu \in \text{Hom}_{\text{cont}}(F^\times, \mathbf{C}^\times) : \nu \circ N_{L/F} = 1\}.$$

Here Hom_{cont} denotes continuous group homomorphisms.

Theorem 1.1 (Second Main Lemma). *The complete normalized induction expression*

$$\mathcal{A}_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F)$$

is independent of L . The assertion holds in mixed and equal characteristic, for every residue characteristic and every prime ℓ .

The proof is completed in Section 14. An actual family of fields and characters is the datum of the theorem; the existence of auxiliary stationary models is proved within the relevant branches, not assumed in its statement.

Here $\Delta_E(\theta, \psi)$ is the normalized local Gauss factor used in the companion First Main Lemma paper [8]; it is recalled in Section 2. The trivial character is included in $S(L/F)$ and contributes the factor 1. In a biquadratic extension the remaining lower-field factor is the local constant of the quadratic norm character. That factor is retained throughout the dyadic proof.

Historical background. The statement is Langlands’s Second Main Lemma [3]. Dwork’s and Lakkis’s explicit norm, trace, and Artin–Schreier calculations provide important methodological background for its local study [1, 2]. The present treatment retains that local framework while giving the specialized two-field constructions and comparisons within a single argument. These historical sources are acknowledged for the problem and methods; the reader need not consult them to supply an unstated step in the proof. No claim of historical priority is made for the individual constructions or alternative arguments. The leading Gauss congruence and the local residue formalism have classical antecedents in [6, 7]; the precise results used here are proved in Appendices A and B, so these two references are historical, not proof inputs.

Relation to the First Main Lemma. The one-dimensional foundation is the author’s companion paper [8]. We use its First Main Lemma, complete Lamprecht formula, stable-twist identity, stationary duality, norm filtration, conductor transport, and trace and different formulas. Their precise forms are recalled in Sections 2–4. References to that paper use its theorem and lemma numbers. The additional classical foundations are stated in Section 3. The diamond-level norm and ramification facts are proved in Appendix D from the cyclic-prime local results and elementary algebra; neither the general local reciprocity theorem nor the general Herbrand quotient theorem is required. Hilbert 90, character extension, different transitivity, and the order-discriminant bound are proved in Appendix C. Basic algebra already available in Mathlib is used without a redundant general development.

The First Main Lemma has a Lean formalization. The present paper is the written mathematical proof of the Second Main Lemma; no Lean verification of the new two-field argument is asserted. Its statements and proofs are presented independently of a formalization workflow.

Contributions of the present paper. The comparison must concern the given characters, rather than auxiliary models whose conductors happen to agree. We therefore construct the models, extend them to the full multiplicative groups, and prove that the prescribed characters are obtained from them by a common descending twist. Stationary representatives are then chosen with their complete character formulas in force. Every use of an exact norm is supported by an explicit construction or by a proved freedom of normalization; a subcritical norm approximation is not promoted to an exact norm.

In wild odd degree, truncated logarithms, actual-conjugate trace calculations, and two weighted terminal estimates control the full four-factor comparison. The cubic boundary is treated with both of its affine stationary corrections. In residue characteristic two, equality of squares does not settle the comparison. We instead match the complete residual functions, including the linear terms of their quadratic refinements. When a critical norm map has image of index two, the third quadratic field supplies a translation that cancels the complementary coset in the actual sum.

Logical organization of the proof. We first fix the local constant and conductor conventions, state the one-dimensional inputs, and isolate several common comparison principles. The tame case is treated uniformly when the extension degree differs from the residue characteristic. The wild cases are then separated according to inertia and, in the dyadic totally ramified case, according to the characteristic and quadratic breaks.

For orientation, the principal branches are as follows. In the two mixed-characteristic dyadic rows, $e = v_F(2)$ and the three entries are the lower breaks of the quadratic intermediate fields.

configuration	calculation used
$\ell \neq p$	tame comparison
$\ell = p > 2, I = p$	odd unramified–ramified comparison
$\ell = p > 2, I = p^2$	odd totally ramified comparison
$p = \ell = 2, I = 2$	dyadic unramified–ramified comparison
$p = \ell = 2, I = 4, \text{char } F = 2$	Artin–Schreier construction
$(2a - 1, 2e, 2e), \quad 1 \leq a \leq e$	maximal quadratic-break construction
$(2a - 1, 2r - 1, 2r - 1), \quad 1 \leq a \leq r \leq e$	nonmaximal quadratic-break construction.

Here I denotes the inertia subgroup. The final section proves that these possibilities exhaust the theorem. Within each branch, a common-twist realization reduces the given characters to consecutive minimal, intermediate, and high conductor ranges, or to a direct comparison at all higher conductors.

Mixed and equal characteristic. The odd-prime calculations are expressed by integral identities whose mixed-characteristic error terms are kept until their valuations have been checked. The corresponding terms vanish in equal characteristic. The dyadic totally ramified arguments require different constructions: local residues and Artin–Schreier equations are used in characteristic two, while quadratic norm symbols and aligned Kummer generators are used in mixed characteristic. Neither argument is obtained by simply substituting into the other. In each case the elementary stationary values and residual sums are compared together.

Organization. Sections 2–4 give the common local and stationary-phase preliminaries. Section 5 proves the tame comparison. Sections 6–9 give the wild odd-prime proof, beginning with an unramified intermediate field and then treating total ramification. Section 10 handles dyadic extensions with an unramified intermediate field. Sections 11–13 treat the totally ramified dyadic cases. Section 14 assembles the complete theorem. Appendix A proves the leading Gauss congruence by Jacobi sums and digit induction; Appendix B proves formal residue substitution, trace compatibility, and the logarithmic norm identity; Appendix C supplies the elementary descent and ramification arguments named above.

2. LOCAL CONSTANTS AND THE ELEMENTARY REDUCTION

2.1. Valuations, conductors, and local constants. For a local field E , write $\mathcal{O}_E$ for its integers, $\mathfrak{p}_E$ for its maximal ideal, k_E for its residue field, and $v_E(\mathfrak{p}_E) = 1$. Put $U_E^0 = \mathcal{O}_E^\times$ and $U_E^j = 1 + \mathfrak{p}_E^j$ for $j \geq 1$. The multiplicative conductor $m_E(\theta)$, also denoted $a_E(\theta)$ in the later case calculations, is the least $m \geq 0$ for which $\theta|_{U_E^m} = 1$. The additive conductor $n_E(\psi)$ is specified by $\mathfrak{p}_E^{-n_E(\psi)}$, the largest *ideal* on which ψ is trivial. It is not generally the full kernel of ψ .

We use exactly the normalized local constant of [8]. For a nonzero complex number z put $\text{ph}(z) = z/|z|$. If $m = m_E(\theta) > 0$, choose an admissible $\Gamma \in E^\times$ with $v_E \Gamma = m + n_E(\psi)$ and set

$$(1) \quad \Delta_E(\theta, \psi) = \theta(\Gamma) \text{ph} \left(\int_{\mathcal{O}_E^\times} \psi(u/\Gamma) \theta(u)^{-1} du \right).$$

The nonzero Gauss integral, or its positive-scalar finite-sum realization, and independence of Γ are proved in [8]. For $m = 0$ this convention gives $\Delta_E(\theta, \psi) = \theta(\pi_E)^{n_E(\psi)}$. The choice of positive Haar normalization does not change the phase. The characters in the theorem need not be unitary; their values on nonunits remain in the displayed multiplicative factors.

All valuations are normalized on the field named. Thus for a totally ramified degree- ℓ edge E/F ,

$$v_E|_F = \ell v_F, \quad v_F(N_{E/F} x) = v_E(x).$$

The different exponent $D_{E/F}$ is measured in E . Conjugation of a character means $\theta^\sigma(x) = \theta(\sigma^{-1}x)$. In $A \subset B_i \subset B$, symbols N_i, n_i denote the upper and lower norms $N_{B/B_i}, N_{B_i/A}$, respectively; the named domains are part of every formula. Analogous upper and lower traces are distinguished in each case calculation.

2.2. The First Main Lemma and elementary identities. For a cyclic extension E/F of prime degree and a quasi-character χ of $F^\times$, the First Main Lemma is

$$(2) \quad \Delta_E(\chi \circ N_{E/F}, \psi_F \circ \text{Tr}_{E/F}) \prod_{\nu \in S(E/F)} \Delta_F(\nu, \psi_F) = \prod_{\nu \in S(E/F)} \Delta_F(\chi \nu, \psi_F).$$

Also

$$\Delta_E(1, \psi) = 1, \quad \Delta_E(\theta, \psi(c \cdot)) = \theta(c) \Delta_E(\theta, \psi),$$

and for a finite-order character μ , $\Delta_E(\mu, \psi) \Delta_E(\mu^{-1}, \psi) = \mu(-1)$. These are [8, Theorem 1.1 and Lemmas 2.8, 2.10–2.11]. A change of variables in (1) also preserves the local constant under a field automorphism when the additive character is trace-invariant.

3. RAMIFICATION AND CLASSICAL LOCAL INPUTS

3.1. Trace ideals, norm filtration, and conductors. For a finite separable extension of ramification index $e_{E/F}$,

$$(3) \quad \text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{\lfloor (j+D_{E/F})/e_{E/F} \rfloor}, \quad n_E(\psi_F \text{Tr}) = e_{E/F} n_F(\psi_F) + D_{E/F}.$$

Here j may be negative. To see the asserted generality, write $\text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^c$. The inverse different is the trace-dual of $\mathcal{O}_E$. For $y \in F$, the condition $y \text{Tr}(\mathfrak{p}_E^j) \subset \mathcal{O}_F$ is therefore equivalent to $y \mathfrak{p}_E^j \subset \mathfrak{p}_E^{-D_{E/F}}$, or $e_{E/F} v_F(y) + j \geq -D_{E/F}$. Comparing the two principal dual ideals gives the displayed floor formula. The conductor formula follows by applying it to the largest trivial ideal of ψ_F . The cyclic-prime instances used most often below are [8, Theorem 3.7 and Corollary 3.8]. For a totally ramified cyclic prime-degree edge of lower break t , $D_{E/F} = (\ell - 1)(t + 1)$, and the inverse Herbrand function on nonnegative real arguments is s up to t , and $t + \ell(s - t)$ above t . The norm filtration [8, Theorem 3.7] gives the whole unit images above the break, isomorphisms of the subcritical graded norm quotients, and an image of index ℓ at the critical quotient. Every nontrivial norm character has conductor $t + 1$.

For a base character of conductor m , the pullback conductor m' obeys $m' - 1 = \psi_{E/F}(m - 1)$ when $m > t + 1$, and $m' = m$ when $m \leq t$. At $m = t + 1$ it is at most m , with a drop precisely when twisting by a norm character lowers the base conductor. These are [8, Corollary 3.9

and Proposition 3.10]. Unramified norm pullback preserves conductor, and its unit norms are surjective at every positive depth.

Subcritical approximation is used in the following precise form: for $A \in F^\times$ and $0 \leq j \leq t$ there is $x \in E^\times$ with $v_E x = v_F A$ and $N_{E/F} x/A \in U_F^j$. This is [8, Lemma 6.6]; it makes no assertion at the critical relative precision $t + 1$. For a wild cyclic degree- p edge and $1 \leq i < p$, the strong symmetric-coefficient bound is

$$(4) \quad v_F(E_i(x)) \geq \left\lfloor \frac{iv_E(x) + (p-1)(t+1)}{p} \right\rfloor.$$

It includes fractional ideals ([8, Lemma 3.4]). Stationary duality always identifies coefficients from an identity on a *whole ideal*, never from the value of a character at a single point. The high stationary tables used in Sections 6–9 are the specified rows of [8, Proposition 6.8]; their actual depth hypotheses are checked at each application. Ordinary Hasse–Davenport is stated in Section 5.

3.2. Additional classical local results. The following classical local results are the additional prerequisites. They contain no assertion about the induction of local constants.

Hilbert 90 and extension of characters. For a finite cyclic extension with generator σ , the multiplicative norm kernel consists of $y/\sigma(y)$ (equivalently $\sigma(y)/y$), and the additive trace kernel consists of $\sigma(y) - y$. Both assertions are proved in Lemma C.1. The continuous character-extension statement used in realizing the models is proved in Lemma C.2. In particular, agreement on overlaps must be proved before the extension lemma is applied; existence of a desired character is never an extra assumption.

Norm characters in the diamond. The order and conductor of a cyclic-prime norm quotient are supplied by [8, Corollary 3.9]. Proposition D.4 proves that the lower norm kernels are distinct for every distinguished comparison used in this paper. Lemma D.5 then proves the crossed norm-character pullback by norm transitivity and the prime order of the two quotients. In the biquadratic case, Lemma D.6 gives every pair and the product relation between the three lower characters. These are the precise consequences usually obtained from [5, Chapters XIII–XIV]; here their proofs are included, without the general local reciprocity theorem.

Ramification and differentials. The single-edge trace, different, norm, and conductor formulas are quoted from FML. Lemma D.2 proves the cyclic unramified quotient, the tame inertia and Frobenius action, and the subgroup rule for lower numbering. Lemma D.3 derives all required prime-square break relations directly from the different formula. Thus no general quotient ramification theorem is an additional input. These local facts have their classical setting in [5, Chapters III–V], retained for attribution and context. Transitivity of the different, the order-discriminant bound, and the leading wild ramification term (including negative valuations) are proved in Lemmas C.3–C.5. The usual monogenic different formula, elementary finite Galois and finite-field theory, and Hensel lifting are the standard algebraic foundations; their existing Mathlib or FML formulations and the short specialization steps are listed in the separate dependency record. Elementary cyclic coordinates are recalled at the start of Appendix D; the equal-characteristic coefficient field is constructed in Theorem B.4.

Local residues in equal characteristic. For a finite separable extension of Laurent-series fields, we use

$$(5) \quad \mathrm{Tr}_{k_E/k_F} \mathrm{Res}_E(\eta) = \mathrm{Res}_F(\mathrm{Tr}_{E/F} \eta).$$

Here the trace of $x d\varpi_F$ is $\mathrm{Tr}_{E/F}(x) d\varpi_F$. This is proved in Theorem B.4, including wild ramification. Lemma B.5 proves $d \log N(u) = \mathrm{Tr}(d \log u)$ separately. Neither identity is quoted from Tate, and neither proof uses a global curve or a global residue theorem. The Artin–Schreier symbol needed in Section 11 is proved there from these internal results and the explicit Cartier operator.

Finite sums. Ordinary finite-field Hasse–Davenport is taken from [8]. The leading Gauss congruence used in Section 5, including its leading unit and its sign at every prime, is proved in Theorem A.3. Its proof establishes the needed Jacobi-sum relation and residue calculation directly; no further Gauss-sum theorem is quoted. Elementary finite character orthogonality

is used throughout. All special finite identities needed for the two-field comparison are proved below.

4. STATIONARY PARAMETERS AND COMMON COMPARISON PRINCIPLES

4.1. Stationary numerator, covector, and rescaled coefficient. A stationary numerator $b \in \mathcal{O}_E^\times$ relative to Γ satisfies $\theta(1+z) = \psi_E(bz/\Gamma)$ on the specified ideal. Its additive covector is b/Γ . If a calculation instead uses $\Psi_E = \psi_E(\alpha \cdot)$ and writes $\theta(1-z) = \Psi_E(Cz)$, the covector in the plus convention is $-\alpha C$. In particular its elementary Lamprecht factor is

$$\theta((-\alpha C)^{-1})\Psi_E(-C).$$

One must not interchange these three descriptions without their scalar and sign. Letters for local breaks, depths, and rescaled coefficients are reintroduced within each case calculation. This local use of notation does not change the definitions of conductors or normalized valuations.

4.2. The full stationary formula and exact stable twist. Let $m = 2d + \varepsilon > 1$, $\varepsilon \in \{0, 1\}$. The stationary numerator b is determined modulo $\mathfrak{p}_E^d$ by

$$\theta(1+z) = \psi_E(bz/\Gamma) \quad (z \in \mathfrak{p}_E^{d+\varepsilon}).$$

It is a unit. If $\varepsilon = 0$, Lamprecht's formula is

$$\Delta_E(\theta, \psi_E) = \theta(\Gamma)\theta(b)^{-1}\psi_E(b/\Gamma).$$

If $\varepsilon = 1$, choose $\delta\mathcal{O}_E = \mathfrak{p}_E^d$ and put

$$H_\theta(\bar{z}) = \psi_E(b\delta z/\Gamma)\theta(1+\delta z)^{-1}.$$

This is a well-defined function on k_E whose sum has magnitude $|k_E|^{1/2}$. The even formula is then multiplied by $|k_E|^{-1/2} \sum_{k_E} H_\theta$. Its polar pairing is

$$H_\theta(x+y)/(H_\theta(x)H_\theta(y)) = \psi_E(b\delta^2 \widetilde{xy}/\Gamma).$$

Changing b changes both its elementary value and this full residual function; their compensating change preserves the local constant. These are [8, Theorem 4.5 and Lemma 4.6]. If $m_E(\nu) \leq d$, the exact stable twist is

$$(6) \quad \Delta_E(\theta\nu, \psi_E) = \nu(\Gamma/b)\Delta_E(\theta, \psi_E)$$

([8, Lemma 4.7]). Neither a homogeneous replacement of H_θ nor an equality of squared sums is substituted for these formulas.

Lemma 4.1 (Normalized stationary factors and terminal invariance). *Let E be a local field, $\alpha \in E^\times$, and $\Psi_E(x) = \psi_E(\alpha x)$ with largest trivial ideal $\mathfrak{p}_E^J$. Write the conductor of θ as $m = 2d + \varepsilon > 1$, with $\varepsilon \in \{0, 1\}$. Suppose $\theta(1-z) = \Psi_E(Zz)$ for $z \in \mathfrak{p}_E^{d+\varepsilon}$, with $v_E Z = J - m$. Then*

$$(7) \quad \Delta_E(\theta, \psi_E) = \theta((-\alpha Z)^{-1})\Psi_E(-Z)g_E.$$

For even m , $g_E = 1$. For odd m , choose $v_E \delta = d$ and set

$$(8) \quad H_E(\bar{x}) = \Psi_E(-Z\delta x)\theta(1+\delta x)^{-1}, \quad g_E = |k_E|^{-1/2} \sum_{x \in k_E} H_E(x).$$

More generally, $\Psi_E(-Zv)\theta(1+v)^{-1}$ is constant as v varies modulo $\mathfrak{p}_E^{d+1}$ in $\mathfrak{p}_E^d$ when m is odd; it is identically one on $\mathfrak{p}_E^d$ when m is even.

Proof. Apply the full Lamprecht formula to the covector $-\alpha Z$. For terminal invariance, if $v' - v \in \mathfrak{p}_E^{d+1}$, write $1+v' = (1+v)(1+w)$ with $w \in \mathfrak{p}_E^{d+1}$. The quotient of the two expressions is $\Psi_E(-Zvw) = 1$, since $v_E(Zvw) \geq J$; the linear part cancels by stationarity. In even conductor stationarity holds already on $\mathfrak{p}_E^d$. $\square$

4.3. Common comparison principles. The following elementary lemmas isolate the shared character-theoretic and stationary comparison arguments. They do not assume the existence of a simultaneous stationary origin. That existence, and the full precision of each stationary formula, are obligations discharged within the individual branch arguments.

Lemma 4.2 (Conjugate twists and the corrected common restriction). *Let ℓ be prime, let K/F be finite Galois with group C_ℓ^2 , and let L_1, L_2 be distinct degree- ℓ intermediate fields with distinct lower norm subgroups. Put $N_i = N_{K/L_i}$, $n_i = N_{L_i/F}$ and $\epsilon_i = \prod_{\omega \in S(L_i/F)} \omega$. Let $\theta_i N_i = \Theta$ be primitive compatible characters. For a generator σ_i of $\text{Gal}(L_i/F)$, the quotient $\mu_i = \theta_i^{\sigma_i} / \theta_i$ generates $S(K/L_i)$. Every character in that group is fixed by $\text{Gal}(L_i/F)$, and the conjugates of θ_i are precisely its twists by $S(K/L_i)$. Moreover*

$$(9) \quad D = D_\theta = \theta_i|_{F^\times} \epsilon_i$$

is independent of i . For odd ℓ , $\epsilon_i = 1$; for $\ell = 2$, $\epsilon_i = \omega_i$ is the nontrivial lower norm character. The distinct-kernel hypothesis holds for the distinguished pairs in Proposition D.4, and for every biquadratic pair by Lemma D.6.

Proof. The common pullback is invariant under both upper subgroups and hence under $\text{Gal}(K/F)$. Thus μ_i kills $N_i K^\times$. If it were trivial, Hilbert 90 and character extension would make θ_i a lower norm pullback, contradicting primitivity. The upper norm quotient has prime order, so μ_i generates its character group. Lemma D.5 identifies that group with $S(L_j/F) \circ n_i$; it is therefore invariant under σ_i . Iterated conjugation gives all ℓ twists.

For $x \in L_j^\times$, compatibility gives $\theta_i(n_j x) = \theta_j(x)^\ell$, whereas multiplication over the conjugates gives

$$\theta_j(n_j x) = \theta_j(x)^\ell \mu_j(x)^{-\ell(\ell-1)/2}.$$

For odd ℓ the last factor is one, and the characters in each $S(L_i/F)$ pair with their inverses, so $\epsilon_i = 1$. For $\ell = 2$, that factor is $\mu_j(x) = \omega_i(n_j x)$ and $\omega_j(n_j x) = 1$. In either case the two corrected restrictions agree on $n_j L_j^\times$, and by symmetry on $n_i L_i^\times$. These distinct index- ℓ subgroups generate $F^\times$, proving the result. $\square$

Corollary 4.3 (The odd-degree common power). *In Lemma 4.2, suppose ℓ is odd and use trace-compatible additive characters. Then*

$$\Delta_{L_i}(\theta_i, \psi_{L_i})^\ell = \Delta_K(\Theta, \psi_K), \quad \prod_{\omega \in S(L_i/F)} \Delta_F(\omega, \psi_F) = 1.$$

In particular, the ratio of the two complete induction expressions has ℓ th power one.

Proof. On any odd-prime cyclic edge, inverse pairing from FML gives $\Delta(\omega)\Delta(\omega^{-1}) = \omega(-1) = 1$; the trivial factor is one. Apply the First Main Lemma on K/L_i . By Lemma 4.2, its twists are conjugate, and their local constants agree by an automorphism change of variables in the defining integral. This proves the power relation as well as both product claims. $\square$

Proposition 4.4 (The biquadratic common square). *For a primitive compatible biquadratic family with trace-compatible additive characters, let $\mathcal{A}_i = \Delta_{L_i}(\theta_i)\Delta_F(\omega_i)$. Then*

$$\mathcal{A}_i^2 = \Delta_K(\Theta) \prod_{j=1}^3 \Delta_F(\omega_j).$$

Thus the ratio of two such expressions belongs to $\{1, -1\}$.

Proof. By Lemma 4.2 and FML on K/L_i , $\Delta_{L_i}(\theta_i)^2 = \Delta_K(\Theta)\Delta_{L_i}(\omega_j n_i)$ for $j \neq i$. FML on L_i/F , applied to ω_j , gives $\Delta_{L_i}(\omega_j n_i)\Delta_F(\omega_i) = \Delta_F(\omega_j)\Delta_F(\omega_k)$, where $\{i, j, k\} = \{1, 2, 3\}$. Use $\omega_i \omega_j = \omega_k$ from Lemma D.6 and multiply the two equalities. $\square$

Lemma 4.5 (A characteristic-free quadratic identity). *For a biquadratic extension K/F , $C \in K$, and any quadratic intermediate field L_i ,*

$$E_2(C) = \text{Tr}_{L_i/F}(N_i C) + n_i(S_i C), \quad S_i = \text{Tr}_{K/L_i},$$

where $E_2(C)$ is the second elementary symmetric function of the four conjugates of C .

Proof. Write the four conjugates, paired by the involution fixing L_i , as c_1, c_2 and c_3, c_4 . The two terms on the right are $c_1c_2 + c_3c_4$ and $(c_1 + c_2)(c_3 + c_4)$. Their sum is exactly the sum of all six pair-products. This identity is integral, so it holds in mixed and equal characteristic without a sign convention or division by two. $\square$

Lemma 4.6 (The complete factor at a genuine common origin). *Use the biquadratic setting of Lemma 4.2. Fix $\alpha \in F^\times$ and put $\Psi_E = \psi_E(\alpha \cdot)$. Suppose all the inducing and lower norm characters under consideration have conductor greater than one. Let $C \in K^\times$ satisfy $S_iC \neq 0$, and suppose that*

$$Z_i = N_iC, \quad W_i = n_i(S_iC)$$

are actual stationary coefficients for θ_i and ω_i , respectively, in the convention $\theta_i(1-z) = \Psi_{L_i}(Z_iz)$ and $\omega_i(1-z) = \Psi_F(W_iz)$ on their full ordinary stationary ideals. Let g_i and h_i denote the respective complete normalized residual sums in Lamprecht's formula, with value one for even conductor. Then

$$\mathcal{A}_i(\theta_i) = D(-\alpha^{-1})\Theta(C)^{-1}\Psi_F(-E_2(C))g_ih_i.$$

Thus equality of the complete residual products proves the desired comparison for these characters.

Proof. The actual covectors are $-\alpha Z_i$ and $-\alpha W_i$. The two complete Lamprecht factors are therefore

$$\theta_i(-\alpha^{-1})\theta_i(Z_i)^{-1}\Psi_{L_i}(-Z_i)g_i, \quad \omega_i(-\alpha^{-1})\omega_i(W_i)^{-1}\Psi_F(-W_i)h_i.$$

Compatibility gives $\theta_i(Z_i) = \Theta(C)$. The value $\omega_i(W_i)$ is one because W_i is a literal lower norm, not merely an element with the valuation of a norm. Multiply, apply Lemma 4.2, and use Lemma 4.5. No residual sign has been discarded. In characteristic two the field minus signs disappear, but the complex residual values need not be signs. $\square$

Lemma 4.7 (An identity before taking residual sums). *Use a common origin C as in Lemma 4.6. For another $C' \in K^\times$ with all $S_iC' \neq 0$, put*

$$1 + z_i = N_iC'/N_iC, \quad 1 + w_i = n_i(S_iC')/n_i(S_iC).$$

Then the products

$$\Psi_{L_i}(-Z_iz_i)\theta_i(1+z_i)^{-1}\Psi_F(-W_iw_i)\omega_i(1+w_i)^{-1}$$

are independent of i . Their common value is

$$\Theta(C'/C)^{-1}\Psi_F(-E_2(C') + E_2(C)).$$

When z_i, w_i belong to the respective terminal stationary ideals, these are identities of the actual residual functions, including their affine terms.

Proof. The upper character value is $\Theta(C'/C)$ by compatibility. The argument of the lower character is a quotient of two nonzero norms, so that character value is one. The additive exponent is

$$-\mathrm{Tr}_{L_i/F}(N_iC' - N_iC) - n_i(S_iC') + n_i(S_iC),$$

which is $-E_2(C') + E_2(C)$ by Lemma 4.5. The terminal-ideal interpretation follows from the full Lamprecht formula; it is used only after the relevant ideal membership has been proved. $\square$

Lemma 4.8 (Cancellation of the missing norm coset). *For each index i , let V_i be a finite abelian group of the same order, and let $H_i : V_i \rightarrow \mathbf{C}^\times$ have $H_i(0) = 1$ and bicharacter $B_i(x, y) = H_i(x+y)/(H_i(x)H_i(y))$. Let Z be a finite abelian group and let $p_i : Z \rightarrow V_i$ be group homomorphisms with kernel of order two and image I_i of index two. Suppose $H_i \circ p_i = Q$ for the same function on Z . Suppose further that there exists $w_i \in I_i$ such that $H_i(w_i) = 1$ and $B_i(w_i, \cdot)$ is the nontrivial character of V_i/I_i . Then*

$$\sum_{x \in V_i} H_i(x) = \frac{1}{2} \sum_{z \in Z} Q(z).$$

In particular the full normalized sums are equal.

Proof. Translation by w_i preserves the complementary coset $V_i \setminus I_i$. On that coset it multiplies the summand by minus one. Thus its sum equals its own negative and is zero in $\mathbf{C}$. On I_i , every value has exactly two preimages under p_i , so summing $H_i p_i = Q$ gives the asserted identity. The factor $1/2$ is a counting factor in $\mathbf{C}$, not division in the local field. The hypothesis $H_i(w_i) = 1$ concerns the actual quadratic refinement and cannot be deduced from its polar form alone. $\square$

Lemma 4.9 (Norm choice within a proven normalization freedom). *Let L/F be cyclic of prime degree and let ω be a nontrivial character with kernel $N_{L/F}L^\times$. Let $H \subset F^\times$ be a subgroup. Assume an auxiliary normalization α may be changed to αu , for every $u \in H$, without changing any of the already fixed character formulas in an argument. If $\omega(H) = \omega(F^\times)$, then for each $A_* \in F^\times$ one may make $A = A_*/u$ a norm while preserving the raw coefficient αA_* .*

Proof. Choose $u \in H$ with $\omega(u) = \omega(A_*)$. Then $\omega(A_*/u) = 1$, so A_*/u is a norm by the specified kernel. Moreover $(\alpha u)(A_*/u) = \alpha A_*$ exactly. Preservation of all other formulas is the explicit hypothesis; the branch proofs verify it on their whole ideals before this lemma is applied. This lemma never asserts that a previously fixed coefficient was already a norm. $\square$

5. FINITE-FIELD IDENTITIES AND THE TAME COMPARISON

5.1. Finite sums and their local leading term. For a finite field k , a nontrivial additive character ψ_k , and a multiplicative character χ , put

$$\tau_k(\chi) = - \sum_{x \in k^\times} \chi(x)^{-1} \psi_k(x).$$

Thus $\tau_k(1) = 1$. For a finite extension κ/k all additive characters are composed with trace. The two ordinary Hasse–Davenport identities in Ueda’s finite-field preliminaries are

$$(10) \quad \begin{aligned} \tau_\kappa(\chi \circ N) &= \tau_k(\chi)^{[\kappa:k]}, \\ \chi(s^s) \tau_k(\chi^s) \prod_{j=1}^{s-1} \tau_k(\eta^j) &= \prod_{j=0}^{s-1} \tau_k(\chi \eta^j), \end{aligned}$$

where η has order s . The second formula is used only when s is prime to the characteristic, as required for such a character.

We use the following classical leading Gauss congruence, proved in Appendix A by an elementary Jacobi-sum argument. We state its local version here to fix the precise normalization. Fix an embedding of the cyclotomic values into $\mathbf{Q}_p(\mu_{p^f-1}, \zeta_p)$, put $\varpi = \zeta_p - 1$, and let ω be the Teichmüller character of $\mathbf{F}_{p^f}^\times$. For the additive character $x \mapsto \zeta_p^{\text{Tr}(x)}$ and $0 < a < p^f - 1$, write $a = \sum a_i p^i$. Then

$$(11) \quad \tau_{\mathbf{F}_{p^f}}(\omega^a) \equiv_{\times} \frac{\varpi^{\sum a_i}}{\prod a_i!}.$$

Here $X \equiv_{\times} Y$ means $X/Y \in 1 + \mathfrak{m}$ in this local cyclotomic field. This is a leading congruence, not merely a valuation formula. The sign in (11) is positive because our τ is minus the Gauss sum. Equivalently, in the usual plus-sum convention $G(\omega^{-a}) = \sum_{x \neq 0} \omega^{-a}(x) \zeta_p^{\text{Tr}(x)}$, the leading term is $-\varpi^{\sum a_i} / \prod a_i!$; our definition is $\tau(\omega^a) = -G(\omega^{-a})$. This specifies the sign without a normalization imported from any two-field proof. Theorem A.3 proves the congruence with this exact sign, not only its valuation. It applies also at $p = 2$, with $\varpi = \zeta_2 - 1 = -2$. No complex absolute-value argument at other places is involved.

For later use, Wilson’s theorem gives, for every nonnegative integer n with base- p digits n_i ,

$$n! \equiv_{\times} (-p)^{(n - \sum n_i)/(p-1)} \prod n_i!.$$

Indeed separate the multiples of p , reduce each full nonzero residue block to $(p-1)! \equiv -1$, and repeat for $\lfloor n/p \rfloor!$. Since $\varpi^{p-1} \equiv_{\times} -p$, we may group (11) into base- q digits for any power q of p :

$$(12) \quad a_* = \sum_{i=0}^{r-1} \gamma_i q^i \implies \tau_{\mathbf{F}_{q^r}}(\omega^{a_*}) \equiv_{\times} \frac{\varpi^{\sum \gamma_i}}{\prod \gamma_i!}.$$

Thus no assertion involving complex absolute values or a global cyclotomic ideal-class theorem is needed below.

5.2. The primitive finite-field identity. Let ℓ be prime, $\ell \mid q - 1$, and let κ/k have degree ℓ , where $|k| = q$. Suppose

$$(13) \quad \chi_\kappa^\ell = \chi_0 \circ N_{\kappa/k}, \quad \chi_0|_{\mu_\ell} \neq 1.$$

Let μ be a character of $k^\times$ of order ℓ .

Theorem 5.1. *With the trace-compatible additive characters,*

$$(14) \quad \tau_\kappa(\chi_\kappa) = \chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j).$$

Proof. First take the canonical additive characters. Put $S = (q^\ell - 1)/(q - 1)$ and $h = (q - 1)/\ell$. Write $\chi_0 = \omega_k^a$, $1 \leq a \leq q - 2$, $\ell \nmid a$. The exponent of χ_κ is aS/ℓ modulo $(q^\ell - 1)/\ell$. Multiplication by q changes it by $a(q^\ell - 1)/\ell$; hence all ℓ possibilities are Frobenius conjugates. Their Gauss sums are equal, so we may take the exponent exactly $a_* = aS/\ell$.

Set

$$L = \frac{\tau_\kappa(\chi_\kappa)}{\chi_0(\ell) \tau_k(\chi_0) \prod_{j=1}^{\ell-1} \tau_k(\mu^j)}.$$

The order- ℓ characters over κ are the norm lifts of μ^j . Their twists of χ_κ are the Frobenius conjugates just considered. The product and lifting formulas (10), and $\chi_\kappa(\ell^\ell) = \chi_0(\ell)^\ell$, give

$$(15) \quad L^\ell = 1.$$

We now determine this root of unity locally. Write $a = \ell m + k_0$, $1 \leq k_0 < \ell$. The unordered base- q digits of aS/ℓ are

$$(16) \quad \gamma_j = m + jh + \mathbf{1}_{j \geq \ell - k_0}, \quad 0 \leq j \leq \ell - 1.$$

Here “unordered” is sufficient for their sum and factorial product. For completeness, multiplication of $k_0 S/\ell$ by ℓ , using $q = \ell h + 1$, gives successive residues $i_j = ((\ell - 1 - j)k_0 \bmod \ell)$ and carries $c_j = \lfloor ((\ell - 1 - j)k_0/\ell) \rfloor$. The zeroth digit has the additional $k_0 - c_0 = 1$; at a later digit the additional $c_{j-1} - c_j$ is one exactly when $i_j \geq \ell - k_0$. The i_j permute $0, \dots, \ell - 1$, proving (16). All digits lie between zero and $q - 1$, since $m \leq h - 1$. In particular

$$(17) \quad \sum \gamma_j = a + \sum_{j=1}^{\ell-1} jh.$$

We claim the multiplicative factorial congruence

$$(18) \quad \ell^a \prod_{j=0}^{\ell-1} \gamma_j! \equiv_\times a! \prod_{j=1}^{\ell-1} (jh)!.$$

In fact

$$\frac{\prod \gamma_j!}{\prod (jh)!} = \prod_{j=0}^{\ell-1} \prod_{n=1}^m (n + jh) \prod_{j=\ell-k_0}^{\ell-1} (m + 1 + jh).$$

For the first product, $\ell(n + jh) = \ell n - j + jq \equiv_\times \ell n - j$: the positive integer $\ell n - j$ is less than $q = p^f$, so its p -valuation is less than f . The numbers $\ell n - j$ exhaust $1, \dots, \ell m$. For the second product the same argument applies to $\ell(m + 1) - j$, which exhausts $\ell m + 1, \dots, a$, again all less than q . Multiplication proves (18), even when some of its factorials are divisible by p .

Apply (12) to the numerator and each factor of the denominator of L . The Teichmüller value $\chi_0(\ell)$ reduces to ℓ^a . Equations (17) and (18) give $L \equiv_\times 1$. Reduction is injective on the group of ℓ th roots of unity because $\ell \neq p$. Together with (15), this proves $L = 1$.

For an arbitrary additive character of k , change the canonical one by multiplication by $c \in k^\times$. The two sides acquire factors $\chi_\kappa(c)$ and $\chi_0(c) \prod_{j=1}^{\ell-1} \mu^j(c)$. These agree. Indeed the restriction exponent of χ_κ is aS/ℓ modulo $q - 1$: it is a for odd ℓ , and $a + (q - 1)/2$ for $\ell = 2$ because a is odd. This is exactly the exponent of $\chi_0 \prod \mu^j$. Thus the identity has the asserted generality. $\square$

5.3. The tame local diamond and its restriction identity. Let F have residue characteristic p , let K/F have Galois group C_ℓ^2 with $\ell \neq p$, and let U/F be its unique unramified degree- ℓ subextension. Let E/F be any other lower field. Its tame cyclic inertia gives $\ell \mid |k_F| - 1$; K/E is unramified and K/U is totally ramified. Suppose $\theta_U N_{K/U} = \theta_E N_{K/E}$ is invariant and does not descend from F . With $S(L/F)$ as defined in Section 1, put

$$\epsilon_L = \prod_{\nu \in S(L/F)} \nu, \quad \Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu, \psi_F).$$

For odd ℓ , $\epsilon_L = 1$; for $\ell = 2$ it is the nontrivial quadratic norm character. All additive characters are trace compatible.

5.4. Conductors greater than one. Put $m = m_U(\theta_U)$. Its conjugate quotient has conductor one, so $m \geq 1$. If $m > 1$, unramified Hilbert 90 on U_U^1 , followed by norm surjectivity, gives a character λ of $F^\times$ such that

$$\theta_U = \chi_U(\lambda n_U), \quad \theta_E = \chi_E(\lambda n_E), \quad m(\chi_U) = m(\chi_E) = 1, \quad m(\lambda) = m.$$

Here $n_L = N_{L/F}$. To justify the conductor-one assertion, remove the descended restriction on U_U^1 . The nontrivial conjugate quotient forces the remaining U character to have conductor exactly one; all its upper-norm twists are conjugates. Ueda's minimal-orbit conductor formula gives conductor one for its pullback to K ; unramified pullback from E preserves conductor.

Let $g \in F^\times$ be an ordinary stationary covector of λ , so $g = \beta/\Gamma$ and $v_F(\Gamma) = m + n_F(\psi_F)$. The following local calculation shows that this same g is a stationary covector of both pullbacks. Their conductors are m over U and $M = 1 + \ell(m - 1)$ over E , and $n_E(\psi_E) = \ell n_F(\psi_F) + \ell - 1$. Hence the same $\Gamma \in F$ is admissible in both.

Lemma 5.2 (The full tame stationary covector). *For $m > 1$, a covector g satisfying $\lambda(1 + z) = \psi_F(gz)$ on $\mathfrak{p}_F^{[m/2]}$ satisfies the corresponding ordinary stationary formula for $\lambda \circ n_U$ and $\lambda \circ n_E$ on their full stationary ideals. This holds in every residue characteristic different from ℓ .*

Proof. On the ramified side put $s = \lceil M/2 \rceil$ and take $x \in \mathfrak{p}_E^s$. For each elementary symmetric function of its ℓ conjugates, restriction of valuations gives $v_F(E_j(x)) \geq \lceil js/\ell \rceil$. For $j \geq 2$ this is at least m , since $2s \geq M > \ell(m - 1)$. Thus

$$n_E(1 + x) \equiv 1 + \text{Tr}_{E/F} x \pmod{\mathfrak{p}_F^m}.$$

The trace has valuation at least $\lceil s/\ell \rceil$, which is at least $\lceil m/2 \rceil$: for $m = 2b$ use $s > \ell(b - 1)$, and for $m = 2b + 1$ use $s > \ell b$. Both units in this congruence are principal, so their quotient belongs to U_F^m . Evaluating λ proves $(\lambda n_E)(1 + x) = \psi_E(gx)$ on the entire required ideal. On the unramified side put $s = \lceil m/2 \rceil$. All symmetric terms of degree at least two have valuation at least $2s \geq m$, and the trace has depth at least s . The same norm expansion proves the assertion. No division by the residue characteristic, nor a half-depth formula on a shallower group, is used. $\square$

The conductor-one characters are in the stable-twist range on both sides, including $\ell = 2$ and $m = 2$. Thus

$$(19) \quad \Delta_L(\theta_L) = \chi_L(g^{-1}) \Delta_L(\lambda n_L).$$

The First Main Lemma and its exact stable-twist formula on F give

$$(20) \quad \Delta_L(\lambda n_L) \Lambda_L = \prod_{\nu \in S(L/F)} \Delta_F(\nu \lambda) = \epsilon_L(g^{-1}) \Delta_F(\lambda)^\ell.$$

All lower norm characters have conductor zero or one, hence at most $\lfloor m/2 \rfloor$. Combining (19)–(20) with Lemma 4.2 for the compatible pair (χ_U, χ_E) proves

$$(21) \quad \Delta_U(\theta_U) \Lambda_U = \Delta_E(\theta_E) \Lambda_E$$

for every $m > 1$. No comparison of incomplete quadratic factors was made; the stable-twist formula retains the complete Lamprecht factor.

5.5. Conductor one and the sign. Now $m_U = m_E = 1$. Let $k = k_F = k_E$, $\kappa = k_U$, and denote the two residue characters by χ_κ, χ_0 . Norm compatibility gives $\chi_\kappa^\ell = \chi_0 N_{\kappa/k}$. The Frobenius conjugate quotient is a nontrivial character of order ℓ . Writing the exponents as in the proof of Theorem 5.1 shows that this is exactly $\ell \nmid a$, or $\chi_0|_{\mu_\ell} \neq 1$. In particular both residue Gauss sums are nontrivial-character sums.

Write $n = n_F(\psi_F)$ and choose $\Gamma = \pi_F^{n+1} \in F$. It is admissible in both fields: $n_E = \ell n + \ell - 1$, while $n_U = n$. The conductor-one local formulas in Ueda's normalization give

$$\begin{aligned}\Delta_U(\theta_U) &= -\theta_U(\Gamma) \text{ph } \tau_\kappa(\chi_\kappa), \\ \Delta_E(\theta_E) &= -\theta_E(\Gamma) \chi_0(\ell) \text{ph } \tau_k(\chi_0).\end{aligned}$$

The factor $\chi_0(\ell)$ is present because the residual additive character over E is $x \mapsto \psi_k(\ell x)$, not $\psi_k(x)$. For the lower norm-character products,

$$\Lambda_U = (-1)^{(\ell-1)n}, \quad \Lambda_E = (-1)^{\ell-1} \epsilon_E(\Gamma) \prod_{j=1}^{\ell-1} \text{ph } \tau_k(\mu^j).$$

Indeed $\epsilon_U(\pi_F) = (-1)^{\ell-1}$ by the product of the ℓ unramified characters, and every nontrivial member of $S(E/F)$ has conductor one. Lemma 4.2 gives $\theta_U(\Gamma)/\theta_E(\Gamma) = \epsilon_E(\Gamma)/\epsilon_U(\Gamma)$. Since $\epsilon_U(\Gamma) = (-1)^{(\ell-1)(n+1)}$, all the displayed signs cancel in the quotient of (21). What remains is precisely the phase of the quotient in Theorem 5.1, namely one. This proves (21) at conductor one as well.

Theorem 5.3 (the complete tame branch). *For a tame C_ℓ^2 diamond over a local field of arbitrary residue characteristic $p \neq \ell$, every compatible non-descending inducing pair has the same normalized induction expression. This holds in both characteristics, in every conductor, and includes $\ell = 2$ with the exact sign.*

Proof. The unramified lower field is unique. The arguments above compare it with every ramified lower field, so transitivity compares any two. The higher-conductor case is (19)–(20); the conductor-one case is the final calculation. $\square$

6. WILD ODD DEGREE: THE UNRAMIFIED–RAMIFIED CASE

6.1. Statement and notation. Let F be a nonarchimedean local field of residue characteristic $p > 2$. Let K/F be Galois with group $C_p \times C_p$, and suppose its inertia subgroup has order p . Write U/F for its unramified degree- p field and E/F for any ramified degree- p field. Then $K = EU$, K/E is unramified, and E/F and K/U are totally ramified cyclic of degree p , with the same break $t \geq 1$. Set

$$T = t + 1, \quad D_0 = (p-1)T, \quad q_0 = \lceil T/p \rceil, \quad e_0 = \lfloor D_0/p \rfloor = T - q_0.$$

No hypothesis $p \nmid t$ is imposed. All valuations are normalized on the field displayed. In particular $v_K|_E = v_E$, $v_K|_U = pv_U$, and $v_E|_F = pv_F$. Use distinct norm symbols

$$n_E = N_{E/F}, \quad n_U = N_{U/F}, \quad N_E = N_{K/E}, \quad N_U = N_{K/U}.$$

Fix a nontrivial additive character e_F and set $e_L = e_F \circ \text{Tr}_{L/F}$ for all four fields. Let $\Delta_L(\theta, e_L)$ mean Ueda's normalized one-dimensional local constant, with the same admissible-denominator convention.

Theorem 6.1. *Suppose θ_U and θ_E have the same norm pullback*

$$\theta_K = \theta_U \circ N_U = \theta_E \circ N_E,$$

and this character does not descend through $N_{K/F}$. Then

$$\Delta_U(\theta_U, e_U) = \Delta_E(\theta_E, e_E).$$

The cyclic lower-field products in the Second Main Lemma are both 1, so this is also equality of its complete normalized induction expressions. The assertion holds in mixed and equal characteristic.

We use the one-dimensional results recalled in Sections 2–4, together with the subcritical norm representatives of [8, Lemma 6.6] and the stable part of its high-conductor stationary table [8, Proposition 6.8]. The following consequences specify the precise normalizations used here. For a totally ramified degree- p edge L/M with break t ,

$$(22) \quad \mathrm{Tr}_{L/M}(\mathfrak{p}_L^a) = \mathfrak{p}_M^{\lfloor (a+D_0)/p \rfloor},$$

$$(23) \quad v_M(e_i(y)) \geq \lfloor (iv_L(y) + D_0)/p \rfloor \quad (1 \leq i < p),$$

$$(24) \quad v_M(Ny) = v_L(y).$$

Here $e_i(y)$ is the i th elementary symmetric function of the p conjugates, even when y lies in a proper subfield. The inequalities apply to negative as well as positive valuations. For unramified edges, the norm is onto every positive unit group and norm pullback preserves conductors. If $c \in M^\times$ has valuation a and $0 \leq r \leq t$, there is $x \in L^\times$ with

$$(25) \quad v_L(x) = a, \quad c - N(x) \in \mathfrak{p}_M^{a+r}.$$

We use Hilbert 90 and extension of characters of finite abelian groups. Every additional two-field argument is proved below.

6.2. Truncated logarithms and the actual norm character. Put

$$P(Z) = \sum_{j=1}^{p-1} \frac{Z^j}{j}.$$

Its coefficients lie in $\mathbf{Z}_{(p)}$. Our sign convention is that $P(z)$ is the truncation of $-\log(1-z)$.

Lemma 6.2 (logarithm charts). *The polynomial $P(z+w-zw) - P(z) - P(w)$ has no monomial of total degree less than p . Consequently, for integers $1 \leq r \leq M$ with $pr \geq M$, the map*

$$1 - z \mapsto P(z)$$

is a group isomorphism $U_L^r/U_L^M \rightarrow \mathfrak{p}_L^r/\mathfrak{p}_L^M$. The map $P : \mathfrak{p}_L^r \rightarrow \mathfrak{p}_L^r$ is bijective and preserves valuations.

Proof. Compare degrees less than p in the formal logarithm identity over $\mathbf{Q}$. The truncated expression has coefficients in $\mathbf{Z}_{(p)}$, so the assertion is valid in both characteristics. The congruence is additive modulo $\mathfrak{p}_L^M$ under the stated inequality. Finally $P(z) = z + z^2 R(z)$ with R integral and $P'(z) \equiv 1 \pmod{\mathfrak{p}_L}$. Hensel's lemma solves $P(z) = w$ uniquely in $\mathfrak{p}_L^r$, or equivalently one can invert successively on its successive quotients. This also proves the valuation assertion and the group isomorphism. $\square$

Lemma 6.3 (full trace–norm logarithm identity). *Let L/M be a totally ramified degree- p cyclic edge of break t . For $h \geq 0$ and $z \in \mathfrak{p}_L^{h+q_0}$,*

$$(26) \quad P(1 - N(1 - z)) \equiv \mathrm{Tr} P(z) + N(P(z)) \pmod{\mathfrak{p}_M^{T+h}}.$$

For an unramified degree- p edge and $z \in \mathfrak{p}_L^{q_0}$, one instead has

$$(27) \quad P(1 - N(1 - z)) \equiv \mathrm{Tr} P(z) \pmod{\mathfrak{p}_M^T}.$$

Proof. Here is a universal proof, so no earlier assertion about a norm representative is needed. Let $z_1, \dots, z_p$ be indeterminates and e_i their elementary symmetric functions. Give e_i weight i and put

$$u = 1 - \prod_{a=1}^p (1 - z_a) = e_1 - e_2 + \dots + e_p, \quad \mathscr{D} = P(u) - \sum_a P(z_a) - \prod_a P(z_a).$$

The polynomial $\mathscr{D}$ belongs to $\mathbf{Z}_{(p)}[e_1, \dots, e_p]$. Every monomial has weight at least p and at least two elementary-symmetric factors. Indeed formal logarithms kill all weights below p , and the weight- p term before the final product is $(\sum_a z_a^p - e_1^p)/p$. Newton's identity gives coefficient 1 for e_p in this polynomial; the final product cancels it. There is no possible one-factor monomial of larger weight. Moreover every coefficient of a pure power e_p^j is divisible by p : modulo p ,

specialize $e_1 = \cdots = e_{p-1} = 0$, adjoin w with $w^p = e_p$, and note that $\sum_a P(z_a) = 0$ and $\prod_a P(z_a) = P(w)^p = P(e_p)$. The injectivity of $\mathbf{F}_p[e_p] \rightarrow \mathbf{F}_p[w]$ justifies this coefficient test.

Write $q = h + q_0$ and specialize to the conjugates of z . Then $v_M(e_p) \geq q$ and, for $i < p$,

$$v_M(e_i) \geq \lfloor (iq + D_0)/p \rfloor \geq \frac{iq + (p-1)t}{p}, \quad v_M(e_i) \geq e_0.$$

A monomial containing both e_p and a smaller-index factor has valuation at least $q + e_0 = T + h$. A monomial consisting of $r \geq 2$ smaller-index factors and having total weight at least p has valuation at least

$$q + \frac{2(p-1)t}{p} \geq T + h,$$

because $q_0 + (p-2)t/p - 1 \geq 0$. In mixed characteristic $e = v_M(p) \geq e_0$, by $p = \text{Tr}_{L/M}(1)$ and (22). A pure norm-power error has valuation at least $e + 2q \geq T + h$. In equal characteristic it is zero. This proves (26). For an unramified edge, every monomial of weight at least p has valuation at least $pq_0 \geq T$, and the norm of $P(z)$ has that valuation also. This gives (27). $\square$

Choose a nontrivial actual norm character τ_F for E/F . It has conductor T . Lemma 6.2 and the additive trace pairing give $\alpha \in F^\times$ such that

$$(28) \quad \tau_F(1 - z) = e_F(\alpha P(z)) \quad (z \in \mathfrak{p}_F^{q_0}), \quad v_F(\alpha) = -n_F(e_F) - T.$$

Set

$$\Psi_L(z) = e_L(\alpha z).$$

The largest ideal on which Ψ_L is trivial is $\mathfrak{p}_L^T$ for each of F, U, E, K . For E/F , this is the different calculation $-pT + D_0 = -T$; the other edges follow by unramified base change. Thus all ideal exponents below refer to normalized valuations, not rescaled ones. Set $\tau_U = \tau_F \circ n_U$. It is the actual norm character for K/U , and (27) gives its P -formula on $U_U^{q_0}$ with scalar 1 relative to Ψ_U .

Corollary 6.4 (whole-subgroup conversion). *For either ramified edge $L/M = E/F$ or K/U and every $w \in \mathfrak{p}_L^{q_0}$,*

$$(29) \quad \Psi_M(\text{Tr}_{L/M} w + N_{L/M} w) = 1, \quad \Psi_L(w) = \Psi_M(-N_{L/M} w).$$

Proof. Write $w = P(z)$ with $z \in \mathfrak{p}_L^{q_0}$, using Lemma 6.2. The norm of $1 - z$ belongs to $U_M^{q_0}$: the intermediate symmetric terms have valuation at least q_0 by (23), and so does the endpoint. The actual norm character is trivial on this norm. Apply its P -formula and Lemma 6.3 with $h = 0$. This proves both assertions. In particular this is a character identity on a whole additive subgroup, not an inference from one test value. $\square$

6.3. A quadratic refinement of the stationary representative.

Lemma 6.5 (enhanced coefficient and its exact affine correction). *Let θ have conductor $m \geq 2$ on a local field L of odd residue characteristic. Let $\Psi_L = e_L(\alpha \cdot)$ have largest trivial ideal $\mathfrak{p}_L^J$, where $J \in \mathbf{Z}$. Put $d = \lfloor m/2 \rfloor$ and $s = \lceil m/2 \rceil$. There is $B \in L$ of valuation $J - m$, determined modulo $\mathfrak{p}_L^{J-d}$, such that*

$$(30) \quad \theta(1 - z) = \Psi_L(BP(z)) \quad (z \in \mathfrak{p}_L^d).$$

If $C = B + \eta$ with $v_L(\eta) \geq J - s$, it is an ordinary stationary coefficient in the same sign convention, and

$$(31) \quad \Delta_L(\theta, e_L) = \theta((- \alpha C)^{-1}) \Psi_L(-C) \Psi_L\left(\frac{\eta^2}{2B}\right) g_L, \quad g_L \in \mu_4.$$

For even m , $g_L = 1$ and the correction is 1. If $v_L(\eta) \geq J - d$, the correction is 1 in either parity.

Proof. Since $pd \geq m$, Lemma 6.2 identifies the character on U_L^d with an additive character on $\mathfrak{p}_L^d/\mathfrak{p}_L^m$. The trace pairing gives B and its ambiguity, and the exact conductor gives $v_L(B) = J - m$. On this depth $P(z) \equiv z + z^2/2$ at the character precision, since $3d \geq m$. Thus

$$\theta(1 + z) = \Psi_L(-Bz + Bz^2/2).$$

The conventional Lamprecht linear coefficient is $-\alpha C$; the minus sign must not be dropped. Choose any admissible denominator Γ and use the unit numerator $-\alpha C\Gamma$. Its elementary factor is $\theta((-\alpha C)^{-1})\Psi_L(-C)$. For odd $m = 2d + 1$, choose $\delta\mathcal{O}_L = \mathfrak{p}_L^d$. The exact residual function formed with C is

$$\bar{z} \mapsto \Psi_L(-\eta\delta z - B\delta^2 z^2/2).$$

The shift $\eta/(B\delta)$ is integral, by the assumed ordinary precision. Completing the square produces the factor $\Psi_L(\eta^2/(2B))$. The remaining homogeneous, nondegenerate quadratic Gauss phase is a fourth root of unity. Indeed, writing $G = \sum_{x \neq 0} \chi_2(x)\psi_0(x)$ for a nontrivial residue additive character, orthogonality gives $|G|^2 = |k_L|$, and the substitution $x \mapsto -x$ gives $\bar{G} = \chi_2(-1)G$. Thus $G^2 = \chi_2(-1)|k_L|$; scaling the quadratic coefficient only multiplies G by χ_2 of that coefficient. In even conductor the ordinary linearization already holds on $\mathfrak{p}_L^d$ and Lamprecht has no residual sum. Finally $v_L(\eta^2/B) \geq J$ in the even case, and also whenever the stronger depth $J - d$ holds. This proves every assertion. $\square$

6.4. Realizing an actual pair with an unramified Artin–Schreier element. Let k be the residue field of F, E and κ that of U, K . Write $Q = |k|$ and let ϕ be arithmetic Frobenius in $\text{Gal}(U/F)$. Choose $\bar{d} \in \kappa$ with $\bar{d}^Q - \bar{d} = 1$. Then $\bar{c} = \bar{d}^p - \bar{d} \in k$ has absolute trace 1. Choose any lift $c \in \mathcal{O}_F$ of $\bar{c}$. The polynomial $X^p - X - c$ is irreducible and unramified over F , so we can choose

$$(32) \quad d \in U, \quad d^p - d = c, \quad d \bmod \mathfrak{p}_U = \bar{d}.$$

In mixed characteristic, Hensel's lemma at $d+1$ gives $\phi(d) - (d+1) \in p\mathcal{O}_U$. In equal characteristic this difference is zero. Also

$$(33) \quad e_0 + q_0 = T, \quad \lceil T/2 \rceil \geq q_0, \quad 2e_0 \geq T, \quad e_0 \geq \lceil T/2 \rceil.$$

The elementary ceiling inequalities hold for every $T \geq 2, p \geq 3$.

Proposition 6.6 (global compatible models, not merely leading classes). *With τ_F, τ_U, Ψ_L as above, there are characters χ_U^0 and χ_E^0 of conductor T , with a common invariant primitive norm pullback, such that*

$$(34) \quad (\chi_U^0)^\phi / \chi_U^0 = \tau_U,$$

$$(35) \quad \chi_U^0(1 - z) = \Psi_U(d^p P(z)) \quad (z \in \mathfrak{p}_U^{q_0}),$$

$$(36) \quad \chi_E^0(1 - z) = \Psi_E(cP(z)) \quad (z \in \mathfrak{p}_E^{q_0}).$$

Here $\chi^\phi(y) = \chi(\phi^{-1}y)$.

Proof. First define the right side of (35) on $H = U_U^{q_0}$. It is a character by Lemma 6.2, with exact conductor T . Its conjugate quotient is τ_U on H because

$$\phi(d^p) - d^p - 1 \in p\mathcal{O}_U, \quad v_U(p) + q_0 \geq e_0 + q_0 = T.$$

This uses the ideal conductor of Ψ_U , so it is valid for every coefficient in $p\mathcal{O}_U$, not only for rational multiples.

We now extend while preserving the *exact* conjugate quotient. Put $V = U_U^1$ and $I = \{\phi^{-1}(v)/v : v \in V\}$. Prescribe on I

$$\kappa(\phi^{-1}(v)/v) = \tau_U(v).$$

This is well defined: the kernel of $v \mapsto \phi^{-1}(v)/v$ is U_F^1 , and $\tau_U(f) = \tau_F(f^p) = 1$ there. If $\phi^{-1}(v)/v \in H$, unramified Teichmüller expansions imply $v = fh$ for $f \in U_F^1$ and $h \in H$; indeed every coefficient of a Frobenius-fixed truncated expansion lies in k . Thus the prescribed character on $I \cap H$ agrees with (35). The two characters define a character on IH , trivial on U_U^T . Extend

it across the finite abelian quotient V/U_U^T to a character of V . Take its value to be 1 on the prime-to- p Teichmüller subgroup and on a fixed uniformizer of F . The character τ_U is trivial on these two factors, so (34) now holds on all of $U^\times$. This gives χ_U^0 with conductor T and the required full P -formula.

Put $\chi_K^0 = \chi_U^0 \circ N_U$. It is invariant under $\text{Gal}(K/U)$ by definition and under $\text{Gal}(K/E)$ by (34), since τ_U is a norm character. Hilbert 90 therefore makes it trivial on $\ker N_E$. It descends to $N_E K^\times$ and extends to a character χ_E^0 on $E^\times$; this extension only involves the valuation factor, since K/E is unramified.

For $z \in \mathfrak{p}_K^{q_0}$ the ramified norm of $1 - z$ is in $U_U^{q_0}$. Equations (35), (26), and (29), using $N_U(d) = d^p$ exactly, give

$$\chi_K^0(1 - z) = \Psi_K(d^p P(z)) \Psi_U(N_U(dP(z))) = \Psi_K((d^p - d)P(z)) = \Psi_K(cP(z)).$$

The unramified congruence (27) shows that this is the pullback of the right side of (36). The norm $U_K^{q_0} \rightarrow U_E^{q_0}$ is onto, so (36) follows on its entire stated subgroup. Since c is a unit and the ideal conductor of Ψ_E is T , χ_E^0 has exact conductor T . If the common pullback descended from F , then χ_U^0 would be a base norm pullback times an upper norm character. Both are fixed by ϕ , the latter by Lemma D.5. This contradicts its nontrivial conjugate quotient τ_U , proving primitivity. $\square$

Proposition 6.7 (actual common twist). *Choose τ_F using Lemma 4.2, and construct the models of Proposition 6.6. There is a character λ of $F^\times$ such that, after changing only the unramified value of χ_E^0 if needed,*

$$(37) \quad \theta_U = \chi_U^0(\lambda \circ n_U), \quad \theta_E = \chi_E^0(\lambda \circ n_E).$$

Write $m_U = T + h$. Then $h \geq 0$, $m_E = T + ph$, and for $h > 0$ the character λ has conductor $T + h$. For $h = 0$ its conductor is at most T .

Proof. The quotient θ_U/χ_U^0 is ϕ -invariant, so Hilbert 90 makes it a norm pullback of a character of $F^\times$ after extension from $n_U U^\times$. Compatibility implies that $\theta_E/(\chi_E^0(\lambda n_E))$ is an unramified norm character for K/E . Absorb it into χ_E^0 ; the full unit formula (36) is unchanged. The conductor of θ_U is at least that of its commutator, namely T . If it exceeds T , its common twist has the same conductor because χ_U^0 has conductor T and unramified norm pullback preserves conductors. The remaining assertion about λ follows similarly. All twists of θ_U by $S(K/U)$ are conjugate and have the same conductor. Therefore the minimal-orbit conductor formula in Ueda applies, both at $m_U = T$ and above it; it gives $m_K = T + ph$. The unramified edge K/E preserves conductors, so $m_E = m_K$. $\square$

6.5. Exact norm choices at the precision actually available. First suppose $0 \leq h \leq t$. Set

$$m_U = T + h, \quad m_E = T + ph, \quad d_U = \lfloor m_U/2 \rfloor, \quad d_E = \lfloor m_E/2 \rfloor.$$

In particular $d_U, d_E \geq q_0$. By the logarithm chart and the additive pairing choose $A_* \in F$ so that

$$(38) \quad \lambda(1 - z) = \Psi_F(A_* P(z)) \quad (z \in \mathfrak{p}_F^{d_U}).$$

For $h > 0$, $v_F(A_*) = -h$ and its ambiguity is $\mathfrak{p}_F^{T-d_U}$. For $h = 0$, its class lies in $\mathcal{O}_F/\mathfrak{p}_F^{T-d_U}$; a zero class may be represented by 0.

Lemma 6.8. *There is $x \in E$ with $A = n_E(x)$ having the following properties. If $0 \leq h < t$, one may take $A_* = A$ in (38). If $h = t$, one has instead*

$$(39) \quad A_* = A + \epsilon, \quad \epsilon \in \mathcal{O}_F.$$

For $h > 0$, $v_E(x) = v_F(A) = -h$. For $h = 0$, x is integral; the zero coefficient is treated by $x = A = 0$.

Proof. For $0 < h < t$, the relative precision of the enhanced class is $T + h - d_U = \lceil m_U/2 \rceil \leq t$. Thus (25) supplies an exact norm in that class. For $h = 0$, a nonzero class has valuation $a \geq 0$; its relative precision is $T - d_U - a \leq \lceil T/2 \rceil \leq t$. Again (25) applies. If the class is zero, use

$x = 0$. At $h = t$, $m_U = 2t + 1$ and $d_U = t$. Use (25) at relative precision t , not $t + 1$. Its absolute error is integral, giving (39). No exact norm representative at the critical successor has been assumed. The valuation assertions follow from (24). $\square$

Set $\epsilon = 0$ in the nonboundary cases, and put

$$(40) \quad B_U = A + d^p + \epsilon \in U, \quad B_E = A - x + c \in E.$$

Proposition 6.9 (enhanced coefficients of the actual pair). *For the actual characters in (37),*

$$(41) \quad \theta_U(1 - z) = \Psi_U(B_U P(z)) \quad (z \in \mathfrak{p}_U^{d_U}), \quad \theta_E(1 - z) = \Psi_E(B_E P(z)) \quad (z \in \mathfrak{p}_E^{d_E}).$$

Thus these are full quadratic stationary representatives, not just coefficients on the deeper linearization layer.

Proof. For the unramified pullback of λ , the logarithm identity through an unramified norm is valid modulo $\mathfrak{p}_F^{m_U}$ at depth d_U , because $pd_U \geq m_U$. It gives coefficient A_* upstairs. Combine it with (35) to obtain the first formula.

For $z \in \mathfrak{p}_E^{d_E}$ put $u = 1 - n_E(1 - z)$. The symmetric estimates give

$$v_F(u) \geq \min\{d_E, \lfloor (d_E + D_0)/p \rfloor\} \geq d_U.$$

The last inequality follows from $d_E \geq d_U$ and $d_E + D_0 - pd_U = (p - 1)(T + \varepsilon)/2 \geq 0$, where $\varepsilon \in \{0, 1\}$ is the common conductor parity. Also $d_E - h \geq q_0$, since $d_E - h = \lfloor (T + (p - 2)h)/2 \rfloor \geq \lfloor T/2 \rfloor \geq q_0$. For $h = 0$ the same inequalities hold with x integral. Lemma 6.3 therefore gives

$$P(u) \equiv \text{Tr } P(z) + n_E(P(z)) \pmod{\mathfrak{p}_F^{T+h}}.$$

Multiply by A , whose valuation is $-h$ for $h > 0$ and nonnegative for $h = 0$. Conversion (29) applies to $xP(z)$, and the *exact* equality $n_E(x) = A$ gives

$$\Psi_F(AP(u)) = \Psi_E(AP(z))\Psi_F(n_E(xP(z))) = \Psi_E((A - x)P(z)).$$

At $h = t$, $d_E = (p + 1)t/2 \geq T$. It follows from the same norm estimate that $u \in \mathfrak{p}_F^T$. Hence the extra coefficient $\epsilon \in \mathcal{O}_F$ in (39) contributes $\Psi_F(\epsilon P(u)) = 1$. Thus the displayed formula is the actual pullback of λ at this boundary too. Multiply by (36) to obtain the second formula. $\square$

For $h > 0$, $v_U(B_U) = -h$ and $v_E(B_E) = -ph$, with unique dominant term A . For $h = 0$, if $\xi = \bar{x} \in k$, their residues are $\xi^p + \bar{d}^p$ and $\xi^p - \xi + \bar{c}$. The first is nonzero because $\bar{d} \notin k$; the second has absolute trace 1, so is nonzero also. Thus the coefficients have exactly the valuations required by their stated conductors in every case.

6.6. Two actual norms of a single common element. Put

$$(42) \quad C = x + d \in K^\times, \quad Z_U = N_U(C), \quad Z_E = N_E(C).$$

These are literal elements in their respective fields. Because d lies in U , it is fixed under $\text{Gal}(K/U)$. Because x lies in E , it is fixed under $\text{Gal}(K/E)$. Hence, writing $a_i = e_i^{E/F}(x)$,

$$(43) \quad Z_U = d^p + \sum_{i=1}^{p-1} a_i d^{p-i} + A,$$

$$(44) \quad Z_E = x^p - x + c.$$

The second identity is the exact polynomial $X^p - X - c$, evaluated at $-x$; it uses actual conjugates in mixed characteristic as well. Define

$$\eta_U = Z_U - B_U = \sum_{i=1}^{p-1} a_i d^{p-i} - \epsilon, \quad \eta_E = Z_E - B_E = x^p - A.$$

Lemma 6.10 (ordinary and enhanced depths). *For $0 \leq h < t$, Z_U and Z_E are enhanced representatives in Lemma 6.5; both affine corrections in (31) are 1. For $h = t$, they are ordinary stationary representatives. The E one is enhanced if $p > 3$. In this boundary case, if $b = a_{p-1}$, then*

$$(45) \quad \eta_U \equiv bd - \epsilon \pmod{\mathfrak{p}_U}, \quad b \in \mathcal{O}_F.$$

Proof. For integral x , (23) puts all the a_i in $\mathfrak{p}_F^{e_0}$. The characteristic-polynomial identity gives

$$(46) \quad x^p - A = \sum_{i=1}^{p-1} (-1)^{i+1} a_i x^{p-i}, \quad v_E(x^p - A) \geq p v_E(x) + (p-1)t.$$

The bound follows from $p \lfloor (ia + D_0)/p \rfloor \geq ia + (p-1)t$ with $a = v_E(x)$. Thus for $h = 0$ the errors have depths at least e_0 and $(p-1)t$, respectively, both at least $T - \lfloor T/2 \rfloor$. If $x = 0$ the assertions are immediate.

For $h > 0$, (23) and (46) give

$$(47) \quad v_U \left(\sum_{i=1}^{p-1} a_i d^{p-i} \right) \geq \lfloor (p-1)(T-h)/p \rfloor, \quad v_E(\eta_E) \geq (p-1)t - ph.$$

If $h < t$, put $r = T - h \geq 2$. The first bound is $r - \lceil r/p \rceil \geq \lceil r/2 \rceil = T - d_U$. Here $\lceil r/p \rceil \leq \lfloor r/2 \rfloor$ for $r \geq 2$, $p \geq 3$. For the second, $h \leq t - 1$ gives

$$2((p-1)t - ph) - (T - ph) = (2p-3)t - ph - 1 \geq (p-3)t + p - 1 \geq 0;$$

hence $(p-1)t - ph \geq T - d_E$. These are precisely enhanced depths.

At $h = t$, $m_U = 2t + 1$, $m_E = (p+1)t + 1$, and the ordinary ambiguity depths are 0 over U and $-(p-1)t/2$ over E . The two bounds $v_U(\eta_U) \geq 0$ and $v_E(\eta_E) \geq -t$ suffice. If $p > 3$, also $-t \geq 1 - (p-1)t/2 = T - d_E$, so the E representative is enhanced. Finally, at this boundary a_{p-1} is integral and all a_i with $i \leq p-2$ have valuation at least $\lfloor (t+p-1)/p \rfloor = \lceil t/p \rceil \geq 1$. This proves (45). $\square$

Lemma 6.11 (exact reduction to a scalar phase). *Let*

$$(48) \quad S = \mathrm{Tr}_{U/F} Z_U - \mathrm{Tr}_{E/F} Z_E.$$

The quotient of actual local constants satisfies

$$(49) \quad \frac{\Delta_U(\theta_U)}{\Delta_E(\theta_E)} \in \mu_4 \Psi_F(-S) \frac{\Psi_U(\eta_U^2/(2B_U))}{\Psi_E(\eta_E^2/(2B_E))}.$$

Furthermore one has the exact identity

$$(50) \quad S = pA - \mathrm{Tr}_{E/F}(x^p) + p \mathrm{Tr}_{E/F}(x).$$

Proof. Apply Lemma 6.5 with the ordinary representatives Z_U, Z_E . The factors at $-\alpha \in F^\times$ cancel by (9), and

$$\theta_U(Z_U) = \theta_K(C) = \theta_E(Z_E).$$

Thus all multiplicative stationary values cancel exactly, leaving (49), with the displayed signs forced by (31). Newton identities for $X^p - X - c$ give

$$\mathrm{Tr}_{U/F}(d^j) = 0 \quad (1 \leq j < p-1), \quad \mathrm{Tr}_{U/F}(d^{p-1}) = p-1, \quad \mathrm{Tr}_{U/F}(d^p) = pc.$$

Take traces in (43) and (44) and subtract. This gives (50). No approximate translated roots or discarded norm errors enter either assertion. $\square$

6.7. Vanishing away from the last cubic row.

Lemma 6.12. *One has $S \in \mathfrak{p}_F^T$ if $h = 0$, if $1 \leq h < t$, or if $p > 3$ and $h = t$.*

Proof. For integral x , all smaller elementary coefficients and all power traces $\mathrm{Tr}(x^j)$ have valuation at least e_0 . Newton's identity at degree p puts $\mathrm{Tr}(x^p) - pA$ in $\mathfrak{p}_F^{2e_0}$, and $p \mathrm{Tr} x$ has valuation at least $v_F(p) + e_0 \geq 2e_0$. Use $2e_0 \geq T$.

For $h > 0$ use the following integral universal identity:

$$\mathrm{Tr}(x^p) - pA = a_1^p + pQ(a_1, \dots, a_{p-1}),$$

where every monomial of Q has weight p and at least two factors. Indeed the power sum is congruent to a_1^p modulo p as an integral symmetric polynomial, its a_p coefficient is p , and there is no one-factor monomial of weight p among $a_1, \dots, a_{p-1}$. The estimates

$$v_F(a_i) \geq \frac{-ih + (p-1)t}{p}, \quad v_F(p) \geq \frac{(p-1)t}{p}$$

give the respective lower bounds

$$v_F(pa_1) \geq \frac{2(p-1)t - h}{p}, \quad v_F(a_1^p) \geq (p-1)t - h,$$

and, for every monomial of pQ , a lower bound $3(p-1)t/p - h$. All three bounds are strictly greater than t when $p > 3$ and $h \leq t$, or when $p = 3$ and $h < t$. Since valuations are integral, they are at least T . In equal characteristic the terms divisible by p are zero and the same conclusion follows. This proves the lemma. $\square$

For $h < t$, both corrections in (49) are 1 by Lemma 6.10, so Lemma 6.12 already puts the quotient in μ_4 . At $h = t$ one has $B_U = A + d^p + \epsilon$ with $v_F(A) = -t$ and $\eta_U \equiv bd - \epsilon \pmod{\mathfrak{p}_U}$. Therefore

$$(51) \quad \Psi_U\left(\frac{\eta_U^2}{2B_U}\right) = \Psi_U\left(\frac{(bd - \epsilon)^2}{2A}\right).$$

The numerator replacement costs depth $t+1$; the inverse-denominator replacement costs depth $2t \geq t+1$. These are exact character-kernel estimates. If $p > 3$, both $\text{Tr}(d)$ and $\text{Tr}(d^2)$ vanish, while the trace of 1 is p . The right side of (51) is $\Psi_F(p\epsilon^2/(2A)) = 1$, since its argument has valuation at least $v_F(p) + t \geq T$ (and is zero in equal characteristic). The E correction is 1 by its enhanced depth. Lemma 6.12 therefore also settles this last row for $p > 3$.

6.8. The cubic boundary with both affine terms retained. It remains to treat $p = 3$ and $h = t$, without any restriction on divisibility of t . Write

$$a = \text{Tr}_{E/F} x, \quad b = e_2^{E/F}(x), \quad A = n_E(x).$$

The symmetric estimates give

$$(52) \quad v_F(a) \geq \lceil t/3 \rceil \geq 1, \quad v_F(b) \geq 0, \quad v_F(A) = -t, \quad v_E(x) = -t.$$

The full coefficients remain $B_U = A + d^3 + \epsilon$ and $B_E = A - x + c$. Because $\text{Tr}_{U/F}(d^2) = 2$ and $\text{Tr}_{U/F}(d) = 0$, (51) gives

$$(53) \quad \Psi_U\left(\frac{\eta_U^2}{2B_U}\right) = \Psi_F(b^2/A).$$

The $3\epsilon^2/(2A)$ term is in $\mathfrak{p}_F^T$. Thus the critical ambiguity of the approximate norm choice has disappeared by the unramified trace, not by an assumed exact norm at depth $t+1$.

The characteristic polynomial of x gives the exact identity

$$(54) \quad \eta_E = x^3 - A = ax^2 - bx.$$

One has $v_E(\eta_E) \geq -t$, and $v_E(1/B_E - 1/A) \geq 5t$. Consequently

$$(55) \quad \Psi_E\left(\frac{\eta_E^2}{2B_E}\right) = \Psi_F\left(\frac{\text{Tr}_{E/F}(\eta_E^2)}{2A}\right),$$

since the omitted E -argument has valuation at least $3t \geq T$. The trace maps $\mathfrak{p}_E^T$ into $\mathfrak{p}_F^T$ by (22).

Lemma 6.13 (exact cubic completion identity). *Put*

$$\Lambda = S - \frac{b^2}{A} + \frac{\text{Tr}_{E/F}(\eta_E^2)}{2A}.$$

Then

$$(56) \quad \Lambda = 3a + a^3 - \frac{b^2 + b^3}{A} + \frac{a^2(a^2 - 3b)^2}{2A}.$$

Moreover $\Psi_F(\Lambda) = 1$.

Proof. Newton's identities give

$$\mathrm{Tr} x^2 = a^2 - 2b, \quad \mathrm{Tr} x^3 = a^3 - 3ab + 3A, \quad \mathrm{Tr} x^4 = a^4 - 4a^2b + 2b^2 + 4aA.$$

Using (54), expand

$$\mathrm{Tr} \eta_E^2 = a^2 \mathrm{Tr} x^4 - 2ab \mathrm{Tr} x^3 + b^2 \mathrm{Tr} x^2 = a^6 - 6a^4b + 9a^2b^2 - 2b^3 + 4a^3A - 6abA.$$

Equation (50) reads $S = 3a - a^3 + 3ab$. Substitution proves (56) as an exact identity in both characteristics.

The last summand of (56) belongs to $\mathfrak{p}_F^T$: (52) gives $v_F(a) \geq 1$, $v_F(a^2 - 3b) \geq 1$, and $v_F(A^{-1}) = t$. For the first remaining pair, apply (29) to the element $a \in F \subset E$. It lies in $\mathfrak{p}_E^{q_0}$, since $v_E(a) \geq 3 \lceil t/3 \rceil \geq t \geq q_0$. It gives

$$\Psi_F(3a + a^3) = 1.$$

For the other pair apply (29) to b/x . This element also has E -valuation at least $t \geq q_0$, and the exact reciprocal-trace identity is

$$\mathrm{Tr}_{E/F}(b/x) = b \frac{e_2(x)}{n_E(x)} = b^2/A, \quad n_E(b/x) = b^3/A.$$

Hence $\Psi_F((b^2 + b^3)/A) = 1$. Combining these two genuine norm-character identities with the ideal estimate proves $\Psi_F(\Lambda) = 1$. This is a phase assertion; no unjustified claim that Λ itself belongs to $\mathfrak{p}_F^T$ is made. $\square$

Equations (49), (53), and (55) now give

$$\frac{\Delta_U(\theta_U)}{\Delta_E(\theta_E)} \in \mu_4 \Psi_F(-\Lambda) = \mu_4.$$

This finishes every conductor with $0 \leq h \leq t$, including the affine odd-conductor endpoint in residual characteristic 3.

6.9. Stable conductors and completion of the branch. Suppose $h \geq T$, so $m = m_F(\lambda) = T + h \geq 2T$. Choose a stationary numerator $\beta \in F^\times$ for λ with admissible denominator $\Gamma \in F^\times$. Ueda's stable high-conductor table permits the same Γ, β for its norm pullbacks to U and E . The denominator assertion includes the different:

$$m_E(\lambda n_E) + n_E(e_E) = T + p(m - T) + pn_F(e_F) + (p - 1)T = p(m + n_F(e_F)) = v_E(\Gamma).$$

The minimal models have conductor T , at most half the conductor of each pullback. Exact stable twist gives

$$\Delta_L(\theta_L) = \chi_L^0(\Gamma/\beta) \Delta_L(\lambda \circ n_L), \quad L = U, E.$$

The multipliers agree by the common restriction to $F^\times$. By Ueda's First Main Lemma on each lower edge and exact stable twist on F , the product over its norm characters is

$$\prod_{\nu \in S(L/F)} \Delta_F(\nu \lambda) = \Delta_F(\lambda)^p \prod_{\nu \in S(L/F)} \nu(\Gamma/\beta) = \Delta_F(\lambda)^p.$$

The character product is trivial in an odd cyclic group, and the product of its individual local constants is 1 by inverse pairing. Therefore $\Delta_U(\lambda n_U) = \Delta_E(\lambda n_E)$ and the stable quotient is exactly 1.

Proof of Theorem 6.1. The stable argument covers $h \geq T$. For $0 \leq h \leq t$, the exact common norm representatives, the scalar phase evaluation, and the retained cubic affine correction prove that the actual quotient belongs to μ_4 . Its p th power is 1 by Corollary 4.3. Since p is odd, $\mu_p \cap \mu_4 = \{1\}$, and the quotient is 1. The lower cyclic products are 1 by the same corollary. These cases exhaust all possible conductors, and every step was valid in either characteristic. $\square$

7. ARTIN–SCHREIER COORDINATES AND TRACE–NORM ESTIMATES

7.1. The ramification data and valuation conventions. Let $A \subset B_1, B_2 \subset B$ be a totally ramified Galois diamond with $\text{Gal}(B/A) = C_p^2$, $p > 2$. Choose the two lines so that their breaks are $t = t_1 \leq t_2$, and the breaks of B/B_2 and B/B_1 are t and $t' = t + p\delta$, respectively, where $\delta = t_2 - t \geq 0$. Write

$$N_i = N_{B/B_i}, \quad S_i = \text{Tr}_{B/B_i}, \quad n_i = N_{B_i/A}, \quad T_i = \text{Tr}_{B_i/A}.$$

All valuations v_B, v_i, v_A are normalized on their own fields. In particular $v_B|_{B_i} = pv_i$, $v_i|_A = pv_A$, and $v_A(n_i y) = v_i(y)$. The different exponents of the lower edges are $D_i = (p-1)(t_i + 1)$; that of B/B_1 is $D' = (p-1)(t' + 1)$.

We use Ueda's proved cyclic trace ideals, norm filtration, and strong symmetric-function estimate. For a totally ramified degree- p cyclic extension L/M with break u they read

$$(57) \quad \text{Tr}_{L/M}(\mathfrak{p}_L^a) = \mathfrak{p}_M^{\lfloor (a+(p-1)(u+1))/p \rfloor},$$

$$(58) \quad v_M(E_i^{L/M}(y)) \geq \lfloor (iv_L(y) + (p-1)(u+1))/p \rfloor, \quad 1 \leq i < p.$$

They hold for fractional ideals as well. Additive Hilbert 90 and the first nonzero ramification term are standard local algebraic inputs; no local constant comparison is used. In mixed characteristic put $V = v_B(p)$. Applying (57) to 1 on B_2/A gives

$$(59) \quad V \geq p(p-1)t_2.$$

In equal characteristic all expressions divisible by p below are zero.

Lemma 7.1 (a consequence of the characteristic polynomial). *For every $y \in L$ in the preceding cyclic extension,*

$$v_L(y^p - N_{L/M} y) \geq pv_L(y) + (p-1)u.$$

Proof. For $1 \leq i < p$, (58) implies $pv_M(E_i(y)) \geq iv_L(y) + (p-1)u$. Every intermediate term $E_i(y)y^{p-i}$ of the characteristic polynomial therefore has the displayed valuation. Its constant term is $-Ny$ because p is odd. $\square$

7.2. Construction of an exact Artin–Schreier coordinate.

Proposition 7.2. *There exists $\Delta \in B$ such that*

$$\Delta^p - \Delta = a \in B_2, \quad v_B(\Delta) = -t, \quad v_2(a) = -t.$$

In this situation $p \nmid t$ and $B = B_2(\Delta)$.

Proof. Let σ generate $\text{Gal}(B/B_2)$. First suppose the characteristic is zero and put $\kappa = V - (p-1)t > 0$. The trace-ideal formula for B/B_2 gives

$$S_2(\mathfrak{P}_B^\kappa) = p\mathcal{O}_{B_2}.$$

Choose $\eta \in \mathfrak{P}_B^\kappa$ with $S_2\eta = -p$. Since $S_2(1 + \eta) = 0$, additive Hilbert 90 supplies $d \in B$ with $\sigma d - d = 1 + \eta$.

Subtract an element of B_2 from d so that its valuation is maximal in its additive B_2 -coset. A maximum exists because B_2 is closed in B . This maximal valuation is not divisible by p : otherwise one could cancel its leading term by an element of B_2 and increase it. The first ramification term therefore gives

$$v_B(\sigma d - d) = v_B(d) + t.$$

The left side is zero, so $v_B(d) = -t$ and $p \nmid t$.

Put $a_0 = d^p - d$. The equation $\sigma d - d = 1 + \eta$, the binomial theorem, and $v_B(d) = -t$ give $v_B(\sigma a_0 - a_0) \geq \kappa$. Choose a best additive approximation $a \in B_2$ to a_0 . If $a_0 \notin B_2$, the same first-ramification-term argument applied to the remainder gives

$$v_B(a_0 - a) \geq \kappa - t = V - pt > 0.$$

For $f(X) = X^p - X - a$, the shifted polynomial $f(d + Z)$ has integral coefficients, unit linear coefficient, and constant coefficient in $\mathfrak{P}_B^{V-pt}$. Indeed $v_B(pd^{p-1}) = \kappa > 0$, and the other

intermediate coefficients are integral. Hensel's lemma gives a root $\Delta \equiv d \pmod{\mathfrak{P}_B^{V-pt}}$. It has valuation $-t$. The valuation of $\Delta^p - \Delta$ is $-pt$, proving $v_2(a) = -t$. Since $p \nmid t$, this root is not in B_2 .

In characteristic p , additive Hilbert 90 directly solves $\sigma d - d = 1$. The same best-approximation argument gives valuation $-t$, and $d^p - d$ is already fixed by σ . $\square$

Set

$$e_i = E_i^{B/B_1}(\Delta), \quad b = N_1\Delta = e_p, \quad s = -e_{p-1}, \quad K_0 = 1 + pt_2.$$

Then $N_2\Delta = a$ and $v_1(b) = -t$. Because the polynomial over B_2 is exactly $X^p - X - a$,

$$(60) \quad S_2(\Delta^j) = 0 \quad (1 \leq j < p-1), \quad S_2(\Delta^{p-1}) = p-1, \quad S_2(\Delta^p) = pa.$$

7.3. Coefficient extraction and twisted estimates.

Lemma 7.3 (coefficient extraction). *If $\theta \in B_1$ and $\theta = \sum_{i=0}^{p-1} Y_i \Delta^i$ with $Y_i \in B_2$, then*

$$v_2(Y_i) \geq v_1(\theta) + it \quad (0 \leq i < p).$$

Proof. Since $p \nmid t$, the valuations $pv_2(Y_i) - it$ are distinct modulo p . Each is at least $pv_1(\theta)$, proving the assertion for $i = 0$ and the preliminary bound for every i . For each nonzero $\xi \in \mu_{p-1} \subset A$, a conjugate of Δ is

$$\Delta_\xi = \Delta + \xi + \eta_\xi, \quad v_B(\eta_\xi) \geq V - (p-1)t.$$

This follows by applying Hensel's lemma to the shifted polynomial at $\Delta + \xi$; its constant coefficient has that valuation and its derivative is a unit. In equal characteristic $\eta_\xi = 0$.

Induct simultaneously for all $\theta \in B_1$ on j . The restriction of this conjugating automorphism to B_1 has break t , so $v_1(\theta_\xi - \theta) \geq v_1(\theta) + t$. The error obtained by replacing Δ_ξ by $\Delta + \xi$ in the expansion of θ_ξ has valuation at least

$$pv_1(\theta) + V - (p-2)t.$$

Its Δ^{j-1} coefficient has v_2 at least $v_1(\theta) + jt$, because $V \geq p(p-1)t \geq (p-1)(j+1)t$. The induction hypothesis applied to $\theta_\xi - \theta$ thus yields

$$\sum_{i=j}^{p-1} \binom{i}{j-1} \xi^{i-j+1} Y_i \in \mathfrak{P}_{B_2}^{v_1(\theta) + jt}.$$

Multiply by ξ^{-1} and sum over μ_{p-1} . The exact power sums kill every term except $(p-1)jY_j$; this multiplier is a unit. $\square$

Lemma 7.4 (twisted trace and symmetric functions). *For $Y \in B_2$ and $1 \leq i < p$,*

$$(61) \quad v_1(S_1(Y\Delta^i)) \geq (p-1)t_2 - it + v_2(Y),$$

$$(62) \quad v_1(E_i^{B/B_1}(Y\Delta)) \geq (p-1)t_2 + i(v_2(Y) - t).$$

Also $v_1(S_1Y) \geq (p-1)t_2 + v_2(Y)$. In particular

$$(63) \quad v_1(e_i) \geq (p-1)t_2 - it \quad (i < p), \quad v_1(s) \geq (p-1)\delta.$$

Proof. The assertion for $S_1Y = T_2Y \in A$ follows from (57). For $i \geq 1$ put $c = (p-1)t_2 - it + v_2Y$. By trace duality it suffices to show $T_1(\theta S_1(Y\Delta^i)) \in \mathcal{O}_A$ for every $v_1(\theta) \geq -D_1 - c$. Write $\theta = \sum Y_j \Delta^j$ and use trace transitivity. For $1 \leq k \leq 2p-2$, the only nonzero power traces over B_2 are those of degrees $p-1, p, 2p-2$, with values $p-1, pa, p-1$. Thus

$$S_2(\theta \Delta^i) = (p-1)Y_{p-1-i} + paY_{p-i} + \mathbf{1}_{i=p-1}(p-1)Y_{p-1}.$$

Lemma 7.3 shows that, after multiplication by Y , all three terms lie in $\mathfrak{P}_{B_2}^{-D_2}$. For the first this is exactly

$$v_2Y + (p-1-i)t - D_1 - c = -D_2;$$

for the second use $v_2(pa) \geq -t$, and for the last there is an additional nonnegative $(p-1)t$. This proves the trace estimate. Newton's identity for $E_i(Y\Delta)$ involves the power traces $S_1(Y^j \Delta^j)$, $1 \leq j \leq i < p$. Each product with total weight i has valuation at least $(p-1)t_2 + i(v_2Y - t)$, and division by i is legal. $\square$

7.4. The trace–norm congruence.

Theorem 7.5. *One has*

$$T_1(b) - T_2(a) \in \mathfrak{p}_A^{1+t_2}.$$

Proof. Compare the coefficient of Z^p in the exact identity

$$n_1(N_1(1 - Z\Delta)) = n_2(N_2(1 - Z\Delta)).$$

The upper polynomials are

$$N_1(1 - Z\Delta) = 1 + \sum_{i=1}^{p-1} (-1)^i e_i Z^i - bZ^p, \quad N_2(1 - Z\Delta) = 1 - Z^{p-1} - aZ^p.$$

The right coefficient is $-T_2a$. Besides $-T_1b$, the left coefficient contains $-n_1(e_1)$ and traces of monomials in conjugates of the e_i with $i < p$, at least two factors, and total weight p . Indeed the cyclic group permutes the selections from the p factors; every orbit has size p , except the constant selection e_1 from every factor, which gives $n_1(e_1)$. By (63), each trace of a nonconstant selection has valuation at least

$$\left\lfloor \frac{2(p-1)t_2 - pt + D_1}{p} \right\rfloor.$$

This is at least $1 + t_2$ whenever $(p-2)t_2 - t - 1 \geq 0$. The same condition makes $v_A(n_1e_1) = v_1(e_1) \geq (p-1)t_2 - t \geq 1 + t_2$. This covers $p \geq 5$ and $p = 3, t_2 > t$.

For $p = 3, t_2 = t$, first note that $T_1e_1 = T_1S_1\Delta = 0$ by (60). The trace map on $\mathfrak{P}_{B_1}^t/\mathfrak{P}_{B_1}^{t+1} \rightarrow \mathfrak{p}_A^t/\mathfrak{p}_A^{t+1}$ is an isomorphism: (57) gives its surjectivity, and both spaces have dimension one over the common residue field. Since $v_1(e_1) \geq t$, this proves $v_1(e_1) \geq t + 1$. The norm of e_1 is consequently deep enough. Every remaining weight-three selection has one e_1 factor and one e_2 factor. Its trace has valuation at least $\lfloor (t+1+2(t+1))/3 \rfloor = t+1$. $\square$

7.5. The weighted translated-root estimate. Put $H = pt_2 + t$. For $x \in B_2$ and $1 \leq k \leq p$, write

$$X = n_2x, \quad R = v_B(x\Delta^{k-1}) = pv_2x - (k-1)t.$$

Lemma 7.6. *If $R \geq 1$, then*

$$XT_1(b^k) \equiv X \sum_{\xi \in \mathcal{T}} N_1(\Delta + \xi)^k \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}, \quad \mathcal{T} = \{0\} \cup \mu_{p-1}.$$

Proof. It is exact in characteristic p . In mixed characteristic fix $\xi \neq 0$, put $D = \Delta + \xi$ and write the actual conjugate as $D(1+u)$. Polynomial division gives the exact polynomial

$$Q_\xi(\Delta) = \frac{(\Delta + \xi)^p - (\Delta + \xi) - (\Delta^p - \Delta)}{\Delta + \xi}.$$

It has degree $p-2$, leading coefficient $p\xi$, and every coefficient is in $p\mathcal{O}_A$. The equation for u is

$$u = Q_\xi + pD^{p-1}u + D^{p-1} \sum_{j=2}^p \binom{p}{j} u^j.$$

Hence

$$v_Bu \geq V - (p-2)t, \quad v_B(u - Q_\xi) \geq 2V - (2p-3)t.$$

For $p \geq 5$, the trace of the latter error has valuation at least H : subtracting H from its lower bound gives

$$(p-3)t - \lfloor (p-2)t/p \rfloor + (2p-3)\delta \geq 0.$$

The symmetric functions $E_i(u)$ with $2 \leq i < p$ have at least this precision, and $v_1(N_1u) = v_Bu \geq V - (p-2)t \geq H$. Terms of degree at most $p-3$ in Q_ξ have traces of valuation at least $2(p-1)t_2 - (p-3)t \geq H$ by Lemma 7.4.

For $p = 3$ retain one further iterate:

$$Q_\xi = 3\xi\Delta, \quad u = Q_\xi + 3D^2Q_\xi + E, \quad v_BE \geq 3V - 5t.$$

The trace of E has valuation at least $V - t + 2\delta \geq H$. Moreover

$$3D^2Q_\xi = 9\xi\Delta^3 + 18\xi^2\Delta^2 + 9\xi^3\Delta.$$

Using $\Delta^3 = \Delta + a$, each trace on the right has valuation at least $6t_2 - 2t \geq H$. The other norm coefficients satisfy $v_1(E_2(u)) \geq 2V/3 + 2\delta \geq H$ and $v_1(N_1u) \geq V - t \geq H$. Thus in all cases

$$(64) \quad N_1(1+u) \equiv 1 + \xi L \pmod{\mathfrak{P}_{B_1}^H}, \quad L = pS_1(\Delta^{p-2}), \quad v_1L \geq pt + 2(p-1)\delta.$$

Since $2v_1L \geq H$, the same formula raised to the k th power has linear term $k\xi L$.

Now $N_1(\Delta + \xi) - b$ is integral by (63); hence $v_1(N_1(\Delta + \xi)^k - b^k) \geq -kt + t$. After multiplying by X , the error term in (64) is in $\mathfrak{P}_{B_1}^{pt_2+R}$ because $v_1(Xb^k) = R - t$. In the linear term, replacing $N_1(\Delta + \xi)^k$ by b^k costs at least $R + pt + 2(p-1)\delta \geq R + pt_2$. The remaining expression vanishes after summation because $\sum_{\xi \in \mu_{p-1}} \xi = 0$ exactly. Finally, abelianness identifies the sum over actual conjugates with $T_1(b^k)$. $\square$

7.6. A filtered Newton-polynomial calculation. Define

$$P_k = b^k + \sum_{\xi \in \mu_{p-1}} N_1(\Delta + \xi)^k, \quad Q_k = S_1((\Delta^p - \Delta)^k) = T_2(a^k).$$

Lemma 7.7. *With X, R as above and $R \geq 1$,*

$$X(P_k - Q_k) \equiv (p-1)\mathbf{1}_{k=p-1}X(1 + e_{p-1}^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Proof. For a monomial $e_\lambda = \prod e_i^{\lambda_i}$ put $r = \sum \lambda_i$, $n = \sum i\lambda_i$, and $\nu = \sum_{i < p} \lambda_i$. The exact Teichmüller power sums show that P_k contains it only if $n \equiv k \pmod{p-1}$, with coefficient

$$(p-1) \binom{k}{r} \frac{r!}{\prod \lambda_i!},$$

plus the separate term b^k . Write $n = (p-1)v + k$. Newton's generating-function formula shows that its coefficient in Q_k is, for $0 \leq v \leq k$,

$$(-1)^{r+v} \binom{k}{v} \frac{n(r-1)!}{\prod \lambda_i!}.$$

These are integral coefficients, even when written as factorial quotients. The valuation estimate (63) gives

$$(65) \quad v_1(Xe_\lambda) \geq R - t + (p-1)((r-v)t + \nu\delta).$$

For $v \geq 0$, if $r \geq v+2$, the deficiency $pr - n = v + p(r-v) - k \geq p$ forces $\nu \geq 2$. Then (65) is at least $R - t + 2(p-1)t_2 \geq R + pt_2$. If $r = v$, only $v = r = k$, $e_\lambda = b^k$ is possible; its coefficients are $1 + (p-1) = p$ and p , so they cancel.

Suppose $r = v+1 \leq k$. Their coefficient difference is exactly

$$\frac{(r-1)!}{\prod \lambda_i!} \left((p-1)r \binom{k}{r} + n \binom{k}{r-1} \right) = \frac{pk \binom{k}{r-1} (r-1)!}{\prod \lambda_i!}.$$

Here $r \binom{k}{r} = (k-r+1) \binom{k}{r-1}$ and $n = (p-1)(r-1) + k$. The factorial denominators are units at p : if $r = p$, then $k = p$ and $n \equiv 1 \pmod{p}$, so no λ_i can equal p . Except for the pure norm monomial, one has $\nu \geq 1$. The added valuation $v_1p \geq (p-1)t_2$ then makes (65) at least $R + pt_2$. The pure norm possibility is $k = p, r = 1, v = 0$, $e_\lambda = b$; its coefficient difference is p^2 , again more than sufficient.

If $v = k$ and $r = k+1$, the first coefficient is absent and the difference is $pk k! / \prod \lambda_i!$. It is divisible by p except when $k = p-1$ and $e_\lambda = e_{p-1}^p$, in which case it is exactly $p-1$. Indeed for $k \leq p-2$ all factorial denominators are units; for $k = p-1$ the only nonunit denominator comes from that constant selection; for $k = p$ the numerator contains at least three factors of p and the factorial denominator at most one. The remaining monomials with $r \geq v+2$ were already dealt with.

Finally $v = -1$ contributes only the constant monomial when $k = p - 1$, giving $p - 1$, or e_1 when $k = p$, giving $p(p - 1)e_1$. The latter is deep enough by (63) and (59). These cases exhaust the possible monomials and leave exactly the two terms displayed in the lemma. $\square$

Theorem 7.8 (weighted norm–trace defect). *Define*

$$h(z) = T_1 N_1(\Delta z) - T_2 N_2(\Delta z).$$

For $x \in B_2$, $0 \leq j \leq p - 1$, and $R = v_B(x\Delta^j) \geq 1$,

$$(66) \quad h(x\Delta^j) \equiv (p - 1)\mathbf{1}_{j=p-2}n_2(x)(1 - s^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Proof. Put $k = j + 1$. Exact multiplicativity shows that replacing $h(x\Delta^{k-1})$ by $n_2(x)(T_1 b^k - T_2 a^k)$ costs $T_2((n_2 x - x^p)a^k)$. By Lemma 7.1, the argument has v_2 at least $R - t + (p - 1)t_2$. Its trace, viewed in B_1 , has valuation at least $R - t + 2(p - 1)t_2 \geq R + pt_2$. Apply Lemmas 7.6 and 7.7. Since p is odd and $s = -e_{p-1}$, $1 + e_{p-1}^p = 1 - s^p$. $\square$

7.7. The linear defect without translated-root expansions. Put

$$g(z) = T_1(bS_1 z) - T_2(aS_2 z).$$

Expand $b = \sum_{l=0}^{p-1} Y_l \Delta^l$ over B_2 and $s = \sum s_l \Delta^l$ over B_2 .

Lemma 7.9. *The coefficients satisfy*

$$\begin{aligned} v_2(Y_l) &\geq (p - 1)t_2 - (p - l)t & (2 \leq l < p), \\ v_2(Y_0 - a) &\geq (p - 1)t_2 - t, & Y_1 = 1 - s_0 + E, \quad v_2 E \geq (p - 1)t_2. \end{aligned}$$

Proof. Use $b = a + \Delta + \sum_{i=1}^{p-1} (-1)^i e_i \Delta^{p-i}$. Expand each e_i over B_2 by Lemma 7.3; its Δ^k coefficient has valuation at least $(p - 1)t_2 - it + kt$. If $k + p - i \geq p$, reduce once using $\Delta^p = \Delta + a$. The unreduced contribution to Y_l has valuation $(p - 1)t_2 - (p - l)t$; each reduced contribution has valuation at least $(p - 1)t_2 + (l - 1)t$. For $l = 0$ only the terms with $k = i$ contribute, with a factor a . For $l = 1$ the sole low term is the constant coefficient of e_{p-1} , namely $-s_0$; all others have valuation at least $(p - 1)t_2$. $\square$

Proposition 7.10. *Suppose $R = v_B(x\Delta^j) \geq 1 + \delta$. If $j \neq p - 2$, then $g(x\Delta^j) \in \mathfrak{p}_A^{1+t_2}$. For $j = p - 2$ there is a fixed $\omega \in B_2$ such that*

$$g(x\Delta^{p-2}) = T_2(x(\omega - 1)), \quad x\omega - xs \in \mathfrak{P}_B^{K_0}.$$

Proof. For $1 \leq j < p$, the exact moment identities give

$$g(x\Delta^j) = T_2(x((p - 1)Y_{p-1-j} + paY_{p-j} + (p - 1)\mathbf{1}_{j=p-1}(Y_{p-1} - a))).$$

For $j \leq p - 3$, the first term has v_2 at least $v_2 x + (p - 1)t_2 - (j + 1)t \geq 1 + t_2$; here $v_2 x \geq 1$ and $(p - 2)t_2 - (j + 1)t \geq 0$. The pa term is at least as deep by (59) and Lemma 7.9. For $j = p - 1$ combine $Y_0 - a$ with the other displayed terms and use $v_2 x + (p - 1)t_2 - t \geq 1 + t_2$. These elements are in the ideal whose trace is $\mathfrak{p}_A^{1+t_2}$.

For $j = 0$, trace transitivity gives

$$g(x) = T_2(x)T_1(b) - pT_2(ax).$$

Replace $T_1 b$ by $T_2 a$ using Theorem 7.5. The two resulting terms have valuations bounded below by $(v_2 x - t + 2(p - 1)t_2)/p > t_2$, using (57) and (59).

For $j = p - 2$ define $\omega = 1 + (p - 1)Y_1 + paY_2$. The formula for g is exact. Lemma 7.9 gives

$$\omega - s = p(1 - s_0) + (s_0 - s) + (p - 1)E + paY_2.$$

Every term except $s_0 - s$ has sufficient valuation immediately after multiplication by x . For the latter, Lemma 7.3 gives

$$v_B(s - s_0) \geq p(p - 1)\delta + (p - 1)t.$$

Since $v_B x = R + (p - 2)t$, the resulting valuation exceeds or equals K_0 ; the difference of the two lower bounds is $(p - 3)t + (p - 1)^2\delta \geq 0$. $\square$

7.8. A floor-corrected projection-to-norm lemma.

Lemma 7.11. *Let $x \in B_2$, $w \in B_1$, and $y \in B_2$ satisfy*

$$xw - y \in \mathfrak{P}_B^{K_0}, \quad h_0 = v_B(xw) \geq 1 + t_2 + (p-3)t.$$

Then

$$N_1(xw) - n_2(y) \in \mathfrak{P}_{B_1}^{K_0}.$$

Proof. The zero case is immediate, so write $z = xw \neq 0$ and expand $z = \sum_{i=0}^{p-1} y_i \Delta^i$, with $y_i = xc_i \in B_2$. Lemma 7.3 gives

$$v_B(y_i \Delta^i) \geq h_0 + (p-1)it.$$

Thus $v_B(y_0) = h_0$, and every nonconstant term is strictly deeper. The valuations of the nonzero terms in

$$z - y = (y_0 - y) + \sum_{i=1}^{p-1} y_i \Delta^i$$

have distinct residues modulo p , because $p \nmid t$. Since their sum has valuation at least K_0 , each term does. In particular $v_B(y_i \Delta^i) \geq K_0$ for $i \geq 1$ and $v_2(y_0 - y) \geq t_2 + 1$.

First keep the linear term and put $z_\ell = y_0 + y_1 \Delta$. Norm multiplicativity, not scalar homogeneity, gives

$$N_1(z_\ell) = n_2(y_0)N_1(1 + (y_1/y_0)\Delta).$$

We have $v_1(n_2 y_0) = h_0$ and $v_2(y_1/y_0) \geq t$. The twisted symmetric estimate therefore bounds the intermediate terms in $N_1(z_\ell) - n_2 y_0$ by

$$h_0 + (p-1)t_2 + j(v_2(y_1/y_0) - t) \geq h_0 + (p-1)t_2 \geq K_0.$$

The endpoint is $N_1(y_1 \Delta)$ and has v_1 equal to $v_B(y_1 \Delta) \geq K_0$. Hence $N_1(z_\ell) - n_2 y_0 \in \mathfrak{P}_{B_1}^{K_0}$.

Now put $U = \sum_{i=2}^{p-1} y_i \Delta^i$. Then $v_B(U/z) \geq 2(p-1)t$ and $v_B U \geq K_0$. The exact identity

$$N_1 z - N_1 z_\ell = N_1 z \{1 - N_1(1 - U/z)\}$$

is expanded on the cyclic edge B/B_1 , whose different exponent is D' . For each intermediate coefficient the strong symmetric bound gives

$$v_1(N_1 z E_j(U/z)) \geq h_0 + \lfloor [2j(p-1)t + D']/p \rfloor \geq K_0 + \lfloor [(p-3)t + p-1]/p \rfloor \geq K_0.$$

The minimum occurs at $j = 1$, and the second inequality follows by inserting $h_0 \geq 1 + t_2 + (p-3)t$ and $t' = t + p\delta$. The endpoint is $N_1 U$, which has valuation $v_B U \geq K_0$. Thus $N_1 z - N_1 z_\ell \in \mathfrak{P}_{B_1}^{K_0}$.

Finally y_0, y are integral and $v_2(y_0 - y) \geq t_2 + 1$. To check the last norm comparison explicitly, set $a = v_2 y$. If $a \geq t_2 + 1$ (or $y = 0$), both norms have valuation at least $t_2 + 1$. Otherwise factor out y , apply the strong symmetric estimate to $(y_0 - y)/y$, and note that its shallowest intermediate term has valuation at least

$$a + \lfloor [t_2 + 1 - a + (p-1)(t_2 + 1)]/p \rfloor = t_2 + 1 + \lfloor (p-1)a/p \rfloor \geq t_2 + 1.$$

The endpoint has valuation $v_2(y_0 - y) \geq t_2 + 1$. Consequently $n_2 y_0 - n_2 y \in \mathfrak{p}_A^{t_2+1}$, which lies in $\mathfrak{P}_{B_1}^{p(t_2+1)} \subseteq \mathfrak{P}_{B_1}^{K_0}$. Combining the three comparisons proves the assertion. $\square$

8. STATIONARY MODELS FOR TOTALLY RAMIFIED ODD-DEGREE EXTENSIONS

8.1. Construction and realization of minimal character models.

The local data. Let $A \subset B_1, B_2 \subset B$ be totally ramified, $\text{Gal}(B/A) = C_p^2$, p odd. Write

$$t_1 = t, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t, \quad \kappa_2 = t_2 + 1, \quad T' = t' + 1.$$

The upper breaks are t' over B_1 and t over B_2 . Put $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$, $S_i = \text{Tr}_{B/B_i}$, $T_i = \text{Tr}_{B_i/A}$. Normalize each valuation on its own field. Section 7 constructs $\Delta \in B$ with

$$\Delta^p - \Delta = a \in B_2, \quad v_B \Delta = -t, \quad N_2 \Delta = a, \quad b = N_1 \Delta, \quad s = -E_{p-1}^{B/B_1}(\Delta).$$

It proves, independently of the SML, the following exact inputs:

$$(67) \quad T_1 b - T_2 a \in \mathfrak{p}_A^{\kappa_2}, \quad v_B(b - a) \geq -t,$$

$$(68) \quad v_1 s \geq (p-1)\delta, \quad v_B(p) \geq p(p-1)t_2.$$

For $w = \sum x_j \Delta^j$, $x_j \in B_2$, the term valuations are distinct modulo p . For a coefficient $x \in B_2$ set

$$g(z) = T_1(bS_1 z) - T_2(aS_2 z), \quad h(z) = T_1 N_1(\Delta z) - T_2 N_2(\Delta z), \quad R = v_B(x \Delta^j).$$

If $R \geq 1 + \delta$, $g(x \Delta^j)$ is in $\mathfrak{p}_A^{\kappa_2}$ except at $j = p-2$. There is $\omega \in B_2$, independent of x , such that in that exception

$$(69) \quad g(x \Delta^{p-2}) = T_2(x(\omega - 1)), \quad x\omega - xs \in \mathfrak{P}_B^{1+pt_2}.$$

For $R \geq 1$, the weighted norm defect is

$$(70) \quad h(x \Delta^j) \equiv (p-1)\mathbf{1}_{j=p-2} n_2 x(1 - s^p) \pmod{\mathfrak{P}_{B_1}^{pt_2+R}}.$$

Finally, if $x \in B_2, w \in B_1, y \in B_2$, $v_B(xw) \geq 1 + t_2 + (p-3)t$, and $xw - y \in \mathfrak{P}_B^{1+pt_2}$, then

$$(71) \quad n_2 x w^p \equiv n_2 y \pmod{\mathfrak{P}_{B_1}^{1+pt_2}}.$$

These are the individual lemmas proved in Section 7. All their proofs persist after an unramified extension of A .

Choose an actual nontrivial norm character τ for B_2/A . Let $e_L = e_A \text{Tr}_{L/A}$, and choose $\alpha \in A^\times$ so

$$\tau(1-z) = e_A(\alpha P(z)), \quad P(X) = \sum_{j=1}^{p-1} X^j/j, \quad z \in \mathfrak{p}_A^c, \quad c = \lceil \kappa_2/p \rceil.$$

Put $\Psi_L(x) = e_L(\alpha x)$. Their largest trivial ideals are

$$\mathfrak{p}_A^{\kappa_2}, \quad \mathfrak{P}_{B_1}^{T'}, \quad \mathfrak{P}_{B_2}^{\kappa_2}, \quad \mathfrak{P}_B^{T'}.$$

The full additive subgroup conversion proved in Corollary 6.4 gives

$$(72) \quad \Psi_A(T_2 v + n_2 v) = 1 \quad (v \in \mathfrak{P}_{B_2}^c).$$

Only Ueda's proved cyclic norm filtration, trace/different estimates, conductor formula and local duality, plus the explicit proofs in Sections 6 and 7, are used below. In equal characteristic expressions containing p are zero; their valuation is interpreted as $+\infty$.

The exact unit domains. Set

$$m_1 = 1 + t + t', \quad m_2 = 1 + t + t_2, \quad q_i = \lceil m_i/p \rceil, \quad q = q_1 = \delta + \lceil (2t+1)/p \rceil, \quad H_i = U_{B_i}^{q_i}.$$

Define characters on these groups by

$$(73) \quad R_1(1-z) = \Psi_1(bPz), \quad R_2(1-z) = \Psi_2(aPz).$$

They are characters because $pq_i \geq m_i$ and the truncated logarithm is additive for the multiplicative group modulo $\mathfrak{P}_{B_i}^{m_i}$. They have exact conductors m_i since $v_i a = v_i b = -t$ on the indicated side.

Lemma 8.1. *One has $q_i \geq 1$, $q \leq t'$, $q_2 \leq t_2$ and*

$$q \geq q_2, \quad \lfloor [q + (p-1)(t+1)]/p \rfloor \geq q_2.$$

Consequently

$$(74) \quad N_1^{-1}(H_1) = U_B^q, \quad N_2(U_B^q) \subset H_2.$$

For any unramified base extension the same assertions hold.

Proof. The inequalities follow by writing $q = \delta + \lceil (2t+1)/p \rceil$ and $q_2 = \lceil (2t+\delta+1)/p \rceil$. For the floor inequality it is enough that $q + (p-1)(t+1) \geq pq_2$; use $pq_2 \leq 2t + \delta + p$ and $(p-3)t + \lceil (2t+1)/p \rceil - 1 \geq 0$. The inequalities $q \leq t'$ and $q_2 \leq t_2$ follow from $2t+1 \leq pt$ for $p \geq 3, t \geq 1$ and $\delta \geq 0$. Below the upper break t' , the norm preserves the first nonzero unit depth; at the critical depth its image is still in that depth. Thus a norm in H_1 has a preimage of depth at least q , including the case $q = t'$. The converse follows from the norm filtration. The second assertion follows coefficientwise from the strong symmetric bound, using the two displayed inequalities. Norm valuations first exclude elements outside U_B^1 from the first preimage assertion. $\square$

The monomial trace–norm phase.

Lemma 8.2. *For $x \in B_2$, $0 \leq j < p$, $v_B(x\Delta^j) \geq q$, one has*

$$\Psi_A(g(x\Delta^j) + h(x\Delta^j)) = 1.$$

Proof. We have $q \geq 1 + \delta$. Outside $j = p-2$ this follows at once from (69)–(70): the norm defect belongs to A and to $\mathfrak{P}_{B_1}^{1+pt_2}$, hence to $\mathfrak{p}_A^{\kappa_2}$. For $j = p-2$, $pv_2x \geq q + (p-2)t \geq \kappa_2$, so $v_2x \geq c$. Put $K_0 = 1 + pt_2$. The difference between $p-1$ and -1 in (70) is in $\mathfrak{P}_{B_1}^{K_0}$ because

$$v_1p + pv_2x \geq (p-1)t_2 + q + (p-2)t \geq K_0,$$

and s is integral. Also

$$v_B(xs) \geq q + (p-2)t + p(p-1)\delta \geq 1 + t_2 + (p-3)t.$$

Using (69) and (71) gives $n_2x s^p \equiv n_2(x\omega) \pmod{\mathfrak{P}_{B_1}^{K_0}}$. The approximation $x\omega \equiv xs \pmod{\mathfrak{P}_B^{K_0}}$ also implies $v_2(x\omega) \geq c$. Thus the sum in question is congruent, now between elements of A , to

$$-T_2x - n_2x + T_2(x\omega) + n_2(x\omega) \pmod{\mathfrak{p}_A^{\kappa_2}}.$$

Both pairs have trivial phase by (72). $\square$

Finite-field rigidity under unramified extension.

Lemma 8.3. *Let M be a local field of residue characteristic p , and let Ψ_M be a continuous additive character. Write $f(Z) = \sum_{n=1}^D a_n Z^n \in M[Z]$, where $D < p^2$, and suppose $x \mapsto \Psi_M(a_n x)$ has order dividing p on $\mathcal{O}_M$ for each n . Let M'/M be unramified of degree prime to p , with absolute residue degree at least 3, and set $\Psi_{M'} = \Psi_M \circ \text{Tr}_{M'/M}$. For $\lambda \in k_{M'}$, let $[\lambda]$ be its Teichmüller lift (the coefficient-field lift in equal characteristic). Suppose*

$$\lambda \mapsto \Psi_{M'}(f([\lambda]))$$

is additive on $k_{M'}$. Then

$$\Psi_M(f(1)) = \Psi_M(a_1 + a_p),$$

where missing coefficients are zero.

Proof. A Teichmüller carry belongs to $p\mathcal{O}_{M'}$, and hence is killed by each coefficient character. Therefore these characters on Teichmüller lifts are residue additive characters, of the form $\psi_{\mathbf{F}_p} \text{Tr}_{k_{M'}/\mathbf{F}_p}(\beta_n \lambda)$; unramified trace shows that each β_n is the scalar extension of its base coefficient. The displayed function is consequently

$$\psi_{\mathbf{F}_p} \text{Tr}_{k_{M'}/\mathbf{F}_p} \left(\sum_n \beta_n \lambda^n \right).$$

Under the absolute trace, exponents in the same Frobenius orbit are identified. Among positive exponents less than p^2 , with at least three base- p digits available in the residue field, the

only identifications are n with pn when both occur. This follows by rotating the base- p digits modulo $|k_{M'}| - 1$: a two-nonzero-digit exponent cannot return below p^2 except at itself. Distinct polynomial monomials of degrees less than $|k_{M'}|$ are linearly independent as functions. An additive function can therefore retain only the orbit of the exponents $1, p$; every other orbit has zero trace contribution. Evaluating at 1 proves the assertion over M' . Both sides are p th roots of unity and the unramified degree is prime to p , so the equality descends to M . $\square$

Compatibility on the full unit subgroup. Define on U_B^q

$$\Xi(u) = R_1(N_1 u) R_2(N_2 u)^{-1}.$$

It is an actual multiplicative character. For $z \in \mathfrak{P}_B^q$ put

$$(75) \quad F_z(Z) = T_1(bP(1 - N_1(1 - Zz))) - T_2(aP(1 - N_2(1 - Zz))).$$

This is an exact polynomial of degree at most $p(p-1) < p^2$ and $\Xi(1 - Zz) = \Psi_A(F_z(Z))$ for integral base scalars Z .

Lemma 8.4. *Every coefficient of F_z defines an additive character of order dividing p on $\mathcal{O}_A$. Its degree-one coefficient is $g(z)$; its degree- p coefficient is $h(z)$ modulo $\mathfrak{p}_A^{\kappa_2}$.*

Proof. The coefficients of $1 - N_i(1 - Zz)$ belong to $\mathfrak{P}_{B_i}^{q_i}$, by the coefficientwise proof of Lemma 8.1. Products of these coefficients have at least that valuation. Multiplication by b or a loses t . In mixed characteristic $v_i p \geq (p-1)t_2$, and

$$(p-1)t_2 - t + q \geq T', \quad (p-1)t_2 - t + q_2 \geq \kappa_2.$$

For the first inequality the difference is $(p-3)t + \lceil(2t+1)/p\rceil - 1$; the second follows from $(p-3)t + (p-2)\delta + q_2 - 1 \geq 0$. Thus p kills the phase of every coefficient, including after any integral base scalar. In equal characteristic this assertion is automatic.

The linear coefficient follows by differentiation at $Z = 0$. At degree p , the norm endpoint gives $h(z)$. All other terms contain at least two intermediate elementary coefficients of total weight p . If the upper break is u , their valuation after multiplication by b or a is at least

$$q - t + 2(p-1)u/p.$$

For $u = t$ this is strictly greater than $\kappa_2 - 1$; for $u = t'$ it is strictly greater than $T' - 1$, using $q = \delta + \lceil(2t+1)/p\rceil$. Integral valuations therefore put each term in the corresponding trivial ideal. Trace to A puts it in $\mathfrak{p}_A^{\kappa_2}$. This argument concerns *degree p only*; no assertion that the higher coefficients are individually deep is made. $\square$

Theorem 8.5 (full compatibility). *The characters (73) satisfy $R_1 N_1 = R_2 N_2$ on U_B^q . This holds for all odd p , in mixed and equal characteristic.*

Proof. First Ξ is trivial on $U_B^{m_1}$. At that depth the norm-logarithm congruence of Lemma 6.3 applies on both upper edges, with target moduli $\mathfrak{P}_{B_i}^{m_i}$. Indeed its sufficient input depths are $t + \lceil(t'+1)/p\rceil$ for the first edge and $t_2 + \lceil(t+1)/p\rceil$ for the second, both at most m_1 . The two weighted norm endpoints are killed after losing t . The remaining difference is $\Psi_B((b-a)Pz) = 1$ by (67) and $m_1 - t = T'$.

Proceed by descending induction, uniformly over all unramified base extensions. If triviality on U_B^M , $M \leq m_1$, is known, put $s_0 = \max(q, \lceil M/2 \rceil)$. On $\mathfrak{P}_B^{s_0}$ the map $z \mapsto \Xi(1 - z)$ is additive, since every multiplicative discrepancy has depth at least $2s_0 \geq M$. In mixed characteristic a Teichmüller carry multiplied by z has depth at least $v_B p + s_0 \geq M$; the carry is zero in equal characteristic. For each fixed z in this ideal, choose an unramified extension of degree prime to p and absolute residue degree at least 3. Apply Lemmas 8.3 and 8.4 to obtain

$$\Xi(1 - z) = \Psi_A(g(z) + h(z)).$$

Expand $z = \sum x_j \Delta^j$. Valuation separation puts every term in $\mathfrak{P}_B^{s_0}$. Additivity of the *actual character* on this ideal, and Lemma 8.2 for each term, give $\Xi(1 - z) = 1$. This proves triviality on $U_B^{s_0}$. Repeated halving reaches q . All conductors and calculations are stable under unramified extension, so the simultaneous induction and prime-to- p descent are legitimate. $\square$

Extension to the full multiplicative groups.

Theorem 8.6. *Every extension θ_2 of R_2 to $B_2^\times$ has a compatible extension θ_1 of R_1 to $B_1^\times$. Both characters have their exact minimal conductors m_i , and their common pullback is primitive.*

Proof. Such θ_2 exists by extension of a character of an open subgroup of an abelian group. Put $\Theta = \theta_2 N_2$. Let ρ generate $\text{Gal}(B/B_1)$, whose break is t' . For every $z \in B^\times$, not just a deep unit, the ramification inequality gives $\rho(z)/z \in U_B^{t'} \subset U_B^q$. Its N_1 -norm is 1. By Theorem 8.5 and (74),

$$\Theta(\rho(z)/z) = R_2(N_2(\rho(z)/z)) = R_1(N_1(\rho(z)/z)) = 1.$$

Hence Θ is invariant under $\text{Gal}(B/B_1)$ on all of $B^\times$. Hilbert 90 makes it trivial on $\ker N_1$, so it descends to the open norm image and extends to a character $\tilde{\theta}_1$ of $B_1^\times$. On $H_1 \cap N_1 B^\times$ it agrees with R_1 : any norm preimage belongs to U_B^q by (74), where full compatibility holds. Thus $\tilde{\theta}_1/R_1$ on H_1 is a character of the image of H_1 in $B_1^\times/N_1 B^\times$. Extend it across this finite cyclic quotient and divide $\tilde{\theta}_1$ by the resulting norm character. This gives $\theta_1|_{H_1} = R_1$, without changing the common pullback.

The exact conductors are already visible in R_i at depth $m_i - 1$. For a generator σ of $\text{Gal}(B_2/A)$ and $z \in \mathfrak{P}_{B_2}^t$, $\sigma^{-1}(1-z)/(1-z)$ has leading depth $t + t_2 = m_2 - 1$ and nonzero ramification coefficient t . Because $p \nmid t$, its leading map on the residue line is surjective. Therefore θ_2^σ/θ_2 is nontrivial, of conductor $t + 1$. Global invariance of Θ makes this a norm character for B/B_2 . If Θ descended from A , this commutator would be trivial, a contradiction. Here the action of the lower group of order p on the upper norm quotient of order p is trivial, since $|\text{Aut}(C_p)| = p - 1$. $\square$

Realizing a prescribed minimal pair. Let (φ_1, φ_2) be an actual primitive compatible pair with conductors m_i . Choose any initial pair from Theorem 8.6. Their second commutators are generators of $S(B/B_2)$. Choose $j \in \{1, \dots, p-1\}$ so that the commutator of φ_2 equals that of θ_2^j . Then $\varphi_2 \theta_2^{-j}$ is invariant under $\text{Gal}(B_2/A)$. Hilbert 90 and character extension give a character λ of $A^\times$ with the *global identity*

$$(76) \quad \varphi_2 = \theta_2^j(\lambda n_2).$$

Put

$$h_0 = \lfloor t/p \rfloor, \quad c = \lceil (t_2 + 1)/p \rceil, \quad q_0 = h_0 + c.$$

The cyclic conductor formula and $a(\lambda n_2) \leq m_2$ imply $a(\lambda) \leq \kappa_2 + h_0$. The following inequalities include the case $p \mid t_2$:

$$(77) \quad q_0 \leq q_2, \quad pq_0 \geq \kappa_2 + h_0, \quad n_2(H_2) \subset U_A^{q_0}, \quad (p-2)t_2 - ph_0 + q_0 - 1 \geq 0.$$

To verify them, write $t = ph_0 + b_0$, $1 \leq b_0 < p$. Then $q_2 = h_0 + \lceil (b_0 + t_2 + 1)/p \rceil \geq q_0$. For the norm inclusion use the strong trace bound and $q_2 + (p-1)\kappa_2 \geq pq_0$; the latter follows from $\kappa_2 \geq ph_0 + b_0 + 1$ and the preceding ceiling formula. The last inequality follows by replacing t_2 by t and q_0 by $h_0 + \lceil (t+1)/p \rceil$; the resulting lower bound is nonnegative.

Put $\Psi'_L = \Psi_L(j \cdot)$. The truncated-log chart gives a coefficient $\gamma \in A$, $v_A \gamma \geq -h_0$, such that

$$\lambda(1-z) = \Psi'_A(\gamma Pz) \quad (z \in \mathfrak{p}_A^{q_0}).$$

If the restriction is trivial take $\gamma = 0$ and $w = 0$. Otherwise choose an *approximate* norm representative by Ueda's ordinary subcritical norm filtration:

$$(78) \quad w \in B_2, \quad k = v_2 w = v_A \gamma \geq -h_0, \quad \gamma - n_2 w \in \mathfrak{p}_A^{k+t_2}.$$

Lemma 8.7. *For $x \in \mathfrak{P}_{B_2}^{q_2}$,*

$$(\lambda n_2)(1-x) = \Psi'_2((w^p - w)P(x)).$$

Proof. Set $u = 1 - n_2(1-x)$. It has depth at least q_0 . The error in replacing γ by $n_2 w$ contributes valuation at least $k + t_2 + q_0 \geq \kappa_2$, so (78) suffices. The norm-logarithm congruence from Lemma 6.3, now with target $\kappa_2 + h_0$ and input depth $h_0 + c = q_0 \leq q_2$, yields

$$(\lambda n_2)(1-x) = \Psi'_2((n_2 w)P(x)) \Psi'_A(n_2(wP(x))).$$

Since $k + q_2 \geq c$, the norm-character conversion changes the second factor to $\Psi'_2(-wP(x))$. The strong symmetric estimate applied to the characteristic polynomial of w gives $v_2(w^p - n_2w) \geq pk + (p-1)t_2$. Multiplication by $P(x)$ puts this error in $\mathfrak{P}_{B_2}^{\kappa_2}$ by (77). This proves the formula. $\square$

Theorem 8.8 (actual realization, with the allowed conjugate explicit). *There exist an actual norm-character coefficient $\alpha' = j\alpha$ and $\Delta_\lambda \in B$ such that*

$$\Delta_\lambda^p - \Delta_\lambda = a_\lambda \in B_2, \quad v_B \Delta_\lambda = -t,$$

for which the second full model is literally $\varphi_2|_{H_2}$ and the first full model is the restriction of a Galois conjugate of φ_1 . Consequently both local constants are unchanged by passing to these full models. No norm equation beyond (78) is assumed.

Proof. Set $a_\lambda = a + w^p - w$. Its valuation is $-t$ because $pv_2w \geq -ph_0 = -t + b_0 > -t$. In mixed characteristic the error of $\Delta + w$ in $X^p - X - a_\lambda$ has valuation at least

$$v_B(p) - (p-1)t + pv_2w \geq v_B(p) - pt + b_0 > 0.$$

Its derivative is a unit. Hensel's lemma gives the required exact root near $\Delta + w$, still of valuation $-t$. In equal characteristic $\Delta + w$ is already that root. It generates B/B_2 because $p \nmid t$. By (76) and Lemma 8.7, the new second model agrees with the actual φ_2 on *all* of H_2 . Apply Theorem 8.6 to the new generator and norm character, choosing φ_2 itself as the second extension. It supplies a first character $\tilde{\varphi}_1$ with the new full model and the same actual norm pullback. Thus $\tilde{\varphi}_1/\varphi_1 \in S(B/B_1)$. For a primitive compatible pair its nontrivial conjugate quotient generates this order- p group, on which the lower Galois group acts trivially. Hence every such twist is a Galois conjugate. Changing variables in the one-dimensional local integral proves equality of their local constants. This is a character-theoretic conjugacy argument, not an SML assertion. $\square$

8.2. The second terminal congruence.

Normalization. Use $A \subset B_1, B_2 \subset B$, $t_1 = t$, $t_2 = t + \delta$, $t' = t + p\delta$, $p \nmid t$, and N_i, n_i, S_i, T_i as in §7 and §8.1. Put

$$T' = t' + 1, \quad \kappa_2 = t_2 + 1, \quad m_1 = T' + t, \quad q_1 = \lceil m_1/p \rceil, \quad c_t = \lceil (t+1)/p \rceil = \lceil t/p \rceil, \quad q = \delta + c_t.$$

Let $\Delta^p - \Delta = a \in B_2$, $v_B \Delta = -t$, and

$$b = N_1 \Delta, \quad a = N_2 \Delta, \quad s = -E_{p-1}^{B/B_1}(\Delta), \quad H = n_1 s.$$

Choose actual norm characters τ_2 for B_2/A and τ_1 for B_1/A , with full truncated-logarithm coefficients α, β relative to the same additive character e_A . Write

$$\Psi_L(z) = e_L(\alpha z), \quad \varrho = \beta/\alpha, \quad v_A \varrho = \delta,$$

so $\Psi_A, \Psi_1, \Psi_2, \Psi_B$ have largest trivial ideals $\mathfrak{p}_A^{\kappa_2}, \mathfrak{P}_{B_1}^{T'}, \mathfrak{P}_{B_2}^{\kappa_2}, \mathfrak{P}_B^{T'}$. The actual first minimal character χ is taken in the conjugacy class supplied by Subsection 8.1, so that

$$(79) \quad \chi(1-y) = \Psi_1(bP(y)) \quad (y \in \mathfrak{P}_{B_1}^{q_1}), \quad P(X) = \sum_{j=1}^{p-1} X^j/j.$$

Its exact conductor is m_1 . Let σ be the restriction to B_1 of the upper automorphism of B/B_2 carrying Δ to the root near $\Delta + 1$. Our character convention is $\chi^\sigma(u) = \chi(\sigma^{-1}u)$.

A shallow commutator evaluated by a deeper character.

Lemma 8.9. *For $x \in \mathfrak{P}_{B_1}^q$,*

$$\frac{\chi^\sigma(1-x)}{\chi(1-x)} = \Psi_1((\sigma b - b)P(x)).$$

Proof. Put $d = \sigma^{-1}x - x$, so $v_1d \geq q + t$. The quotient $(1 - \sigma^{-1}x)/(1 - x)$ is $1 - d/(1 - x)$. We have $q + t \geq q_1$, hence (79) applies to this quotient even though it need not apply separately to $1 - x$. As an integral formal power series in x, d ,

$$P(d/(1 - x)) - P(x + d) + P(x)$$

is divisible by d and has total degree at least p . Its denominators are products of integers $1, \dots, p - 1$ and powers of $1 - x$. Its valuation is therefore at least $(p - 1)q + (q + t) = pq + t \geq T' + t = m_1$. Multiplication by b puts it in the kernel of Ψ_1 . Finally trace invariance gives $\Psi_1(bP(\sigma^{-1}x)) = \Psi_1((\sigma b)P(x))$. $\square$

Lemma 8.10. *Set*

$$R = T' - q = t + 1 - c_t + (p - 1)\delta.$$

Then $\sigma b - b \equiv 1 - s \pmod{\mathfrak{P}_{B_1}^R}$.

Proof. Norm expansion gives

$$N_1(\Delta + 1) - b = 1 - s + \sum_{i=1}^{p-2} E_i^{B/B_1}(\Delta).$$

Each summand has valuation at least $(p - 1)t_2 - (p - 2)t = t + (p - 1)\delta \geq R$ by Section 7. For the actual translated root write $\sigma\Delta = (\Delta + 1)(1 + u)$. The elementary Hensel estimate gives $v_B u \geq V - (p - 2)t$, where $V = v_B p$. For intermediate coefficients in $N_1(1 + u) - 1$ the strong symmetric bound is smallest at the trace coefficient, and after multiplying by $N_1(\Delta + 1)$ it gives at least

$$-t + \lfloor [V - (p - 2)t + (p - 1)(t' + 1)]/p \rfloor = V/p - t + (p - 1)\delta + c_t \geq (p - 2)t + 2(p - 1)\delta + c_t \geq R.$$

The norm endpoint has still larger valuation. In equal characteristic $u = 0$. This proves the assertion without dropping any mixed-characteristic $p = 3$ term. $\square$

Thus the actual commutator $\mu = \chi^\sigma/\chi$ satisfies

$$(80) \quad \mu(1 - x) = \Psi_1((1 - s)P(x)) \quad (x \in \mathfrak{P}_{B_1}^q).$$

It is a nontrivial norm character for B/B_1 , of conductor T' . The same is true of $\tau_2 n_1$. We next identify the generator, rather than assuming that these two nontrivial characters coincide.

The second expression uses only an approximate norm. By the ordinary subcritical norm filtration, choose $c \in B_1^\times$ with

$$(81) \quad n_1 c = \varrho \varepsilon, \quad \varepsilon \in U_A^t, \quad v_1 c = \delta.$$

Lemma 8.11. *On $U_{B_1}^q$ one has*

$$(\tau_2 n_1)(1 - x) = \Psi_1((1 - \varrho/c)P(x)).$$

Proof. For $u = 1 - n_1(1 - x)$, the norm filtration gives $v_A u \geq \lceil \kappa_2/p \rceil$, so the defining full formula for τ_2 applies. The norm-logarithm congruence of Lemma 6.3 on B_1/A , with target κ_2 and input depth $\delta + c_t = q$, gives

$$(\tau_2 n_1)(1 - x) = \Psi_1(P(x))\Psi_A(n_1 P(x)).$$

The error in replacing the second factor by $e_A(\beta n_1(P(x)/c))$ has valuation, before multiplication by α , at least $q + t \geq \kappa_2$. This is exactly the error $n_1 P(x)(\varepsilon^{-1} - 1)$, not an exact norm equation. Since $v_1(P(x)/c) \geq q - \delta = c_t$, the norm-character conversion for τ_1 makes this factor $e_{B_1}(-\beta P(x)/c)$. $\square$

Exact generator normalization in the one-break case.

Lemma 8.12 (reciprocal trace identity). *If $\delta = 0$, then for every $Y \in \mathcal{O}_A$,*

$$\Psi_1(sY/b) = \Psi_2(Y/a) = \Psi_A(-n_1(Y/b)).$$

Proof. Elementary symmetric functions give the *exact* identities

$$s/b = -S_1(1/\Delta), \quad S_2(1/\Delta) = -1/a.$$

The second uses the coefficient -1 of X in $X^p - X - a$. Trace transitivity therefore gives the first equality. As $v_2(Y/a) \geq t \geq \lceil (t+1)/p \rceil$, the actual norm character τ_2 gives $\Psi_2(Y/a) = \Psi_A(-n_2(Y/a))$. Finally $n_1b = n_2a$ by norm transitivity and $Y \in A$, so $n_2(Y/a) = Y^p/n_2a = n_1(Y/b)$. $\square$

Proposition 8.13. *The actual commutator is exactly $\mu = \tau_2 n_1$.*

Proof. Both are nontrivial characters of the same cyclic norm quotient of order p . Their restrictions to its last nonzero unit layer distinguish the $p-1$ possible generators. If $\delta > 0$, $v_1s \geq (p-1)\delta$ and $v_1(\varrho/c) = (p-1)\delta$ are positive. Equations (80) and Lemma 8.11 agree on $U_{B_1}^{t'}$; here $P(x) \equiv x$ at the character modulus. Hence the characters are equal.

If $\delta = 0$, use $x = Y/b$, whose classes as Y varies span $\mathfrak{P}_{B_1}^t/\mathfrak{P}_{B_1}^{t+1}$. On this layer $P(x) \equiv x$. The intermediate coefficients of $n_1(1-x)$ other than trace and norm are in $\mathfrak{p}_A^{t+1}$ by the strong symmetric bound. Thus

$$(\tau_2 n_1)(1-x) = \Psi_A(T_1x + n_1x) = \Psi_1(x)\Psi_A(n_1x).$$

Lemma 8.12 identifies this with $\Psi_1((1-s)x)$, which is the actual commutator by (80). Their critical restrictions agree, and therefore the full norm characters agree. No R_2 congruence was used. $\square$

Whole-subgroup comparison and norm descent.

Theorem 8.14. *For every odd p , in both characteristics and both break configurations,*

$$(82) \quad H \equiv \varrho^{p-1} \pmod{\mathfrak{p}_A^R}, \quad R = t+1 - c_t + (p-1)\delta.$$

Consequently if m is the transition parameter with $pm > t$, then

$$H \equiv (\beta/\alpha)^{p-1} \pmod{\mathfrak{p}_A^{1+t_2-(p-2)m}}.$$

Proof. Proposition 8.13, (80) and Lemma 8.11 imply

$$\Psi_1((s - \varrho/c)P(x)) = 1 \quad \text{for every } x \in \mathfrak{P}_{B_1}^q.$$

The polynomial P is a bijection of this entire ideal, by Hensel's lemma and $P'(0) = 1$. Perfect additive duality therefore gives the absolute coefficient congruence

$$s - \varrho/c \in \mathfrak{P}_{B_1}^{T'-q} = \mathfrak{P}_{B_1}^R.$$

Put $d_s = (p-1)\delta$. Both ϱ/c and s have valuation d_s , since $R - d_s = R_0 := t+1 - c_t \geq 1$. Their ratio is in $U_{B_1}^{R_0}$. We have $1 \leq R_0 \leq t$, so the cyclic norm filtration gives a ratio of norms in $U_A^{R_0}$. Multiplication by the norm's valuation d_s yields

$$n_1s \equiv n_1(\varrho/c) = \varrho^{p-1}\varepsilon^{-1} \pmod{\mathfrak{p}_A^{d_s+R_0}} = \pmod{\mathfrak{p}_A^R}.$$

The remaining error has valuation at least $d_s + t \geq d_s + t + 1 - c_t = R$. This proves (82), including $\delta = 0$. Finally $pm > t$ and $p \nmid t$ imply $m \geq c_t$, and

$$R - [1 + t_2 - (p-2)m] = (p-2)(\delta + m) - c_t \geq 0.$$

This proves the required terminal modulus uniformly, with no exceptional $p = 3$ strengthening and no additional critical norm representative. $\square$

8.3. Simultaneous norm parameters and stationary factors.

Truncated-logarithm coordinates. Throughout p is odd and $P(X) = \sum_{j=1}^{p-1} X^j/j$. For a field E let e_E have largest trivial ideal $\mathfrak{P}_E^{-d_E}$. A multiplicative character θ of conductor $n \geq 2$ admits a coefficient C with

$$(83) \quad \theta(1-x) = e_E(CP(x)), \quad x \in \mathfrak{P}_E^{\lceil n/p \rceil}, \quad v_E(C) = -d_E - n.$$

This follows directly from perfect additive duality: the polynomial $P(X+Y-XY) - P(X) - P(Y)$ has total degree at least p , while P is a bijection of every positive ideal, with derivative a unit. It therefore identifies the multiplicative quotient U_E^q/U_E^n with $\mathfrak{P}_E^q/\mathfrak{P}_E^n$ when $pq \geq n$. Exact conductor gives the stated valuation. The full coefficient is determined modulo $\mathfrak{P}_E^{-d_E-q}$, not just by a half-conductor class.

Lemma 8.15 (choosing a norm ratio by moving a norm-character coefficient). *Let E/F be ramified cyclic of odd prime degree with break $t \geq 1$. Let ν be an actual nontrivial norm character and write*

$$\nu(1-z) = e_F(AP(z)), \quad z \in \mathfrak{p}_F^c, \quad c = \lceil (t+1)/p \rceil, \quad v_F(A) = -d_F - t - 1.$$

For any $\gamma \in F^\times$ there is A' with

$$A'/A \in U_F^t, \quad \nu(1-z) = e_F(A'P(z)) \quad (z \in \mathfrak{p}_F^c), \quad \gamma/A' \in N_{E/F}(E^\times).$$

Consequently there is $w \in E^\times$ with $Nw = \gamma/A'$ and $v_E(w) = v_F(\gamma/A')$.

Proof. The character ν has exact conductor $t+1$, so its restriction to U_F^t is a surjection onto μ_p . Choose $h \in U_F^t$ with $\nu(h) = \nu(\gamma/A)$ and put $A' = Ah$. The change in the additive coefficient on the entire prescribed subgroup has valuation at least

$$v_F(A(h-1)P(z)) \geq -d_F - t - 1 + t + c \geq -d_F.$$

It is therefore invisible to e_F on that whole subgroup. Meanwhile $\nu(\gamma/A') = 1$. The kernel of this nontrivial character of the order- p norm quotient is precisely the norm subgroup. This proves exact norm membership. Total ramification gives $v_F(Nw) = v_E(w)$. $\square$

Lemma 8.16 (Preservation of the full minimal models). *Let the totally ramified notation be as in Subsection 8.1, and suppose the actual minimal pair has full coefficients $\alpha b, \alpha a$. If $h \in U_A^{t_2}$, then replacing α by αh leaves both full model formulas unchanged. It also leaves the full coefficient formula of the same norm character of B_2/A unchanged.*

Proof. Write $q_1 = \delta + \lceil (2t+1)/p \rceil$, $q_2 = \lceil (2t+\delta+1)/p \rceil$. The old additive characters $\Psi_i = e_{B_i}(\alpha \cdot)$ have largest trivial ideals with exponents $1+t'$ and $1+t_2$. For $z \in \mathfrak{P}_{B_i}^{q_i}$ the change in the normalized exponent has valuation at least $pt_2 - t + q_i$. On B_1 its excess over $1+t'$ is

$$(p-2)t + \delta + \lceil (2t+1)/p \rceil - 1 \geq 0,$$

and on B_2 its excess over $1+t_2$ is

$$(p-2)t + (p-1)\delta + q_2 - 1 \geq 0.$$

Thus neither full character formula changes. On the lower field A , the change at depth $c_2 = \lceil (t_2+1)/p \rceil$ has normalized valuation at least $t_2 + c_2 \geq t_2 + 1$, proving the last assertion. $\square$

Remark 8.17. This lemma permits realization of the actual minimal pair first, including the scalar $j \in \{1, \dots, p-1\}$ in Theorem 8.8, and the exact norm choices afterwards. Thus an exact norm equation cannot be invalidated by a later, unrecorded change of generator of the norm-character group. It does not assert that a previously fixed ratio is already a norm.

Full coefficients of a descending pullback.

Proposition 8.18. *Let E/F be as in Lemma 8.15. Let λ have conductor $1+r$, where $r > t$, and let γ be its full coefficient relative to e_F . Choose A, w by that lemma so that $Nw = \gamma/A$. Put $h = r - t$, $n = 1 + t + ph$, and $q = h + \lceil (t+1)/p \rceil = \lceil n/p \rceil$. Then the actual pullback has exact conductor n and*

$$(84) \quad (\lambda N)(1-x) = e_E((\gamma - Aw)P(x)), \quad x \in \mathfrak{P}_E^q, \quad e_E = e_F \text{Tr}_{E/F}.$$

Its coefficient has valuation $pv_F(\gamma)$.

Proof. The conductor is the high cyclic norm-transport formula, with different exponent $(p-1)(t+1)$. One has $v_E(w) = -h$ and

$$v_E(Aw/\gamma) = (p-1)h > 0.$$

This proves the claimed coefficient valuation and matches conductor n , since $d_E = pd_F + (p-1)(t+1)$.

For $x \in \mathfrak{P}_E^q$, the norm $N(1-x)$ lies in the full λ -coordinate domain $U_F^{[(r+1)/p]}$. Indeed the norm endpoint has valuation at least q , and the intermediate terms at least $\lfloor (q + (p-1)(t+1))/p \rfloor$; both are at least the required depth. For the second inequality compare the numerator with $r+p$; its excess is at least $\lceil (t+1)/p \rceil + (p-2)t - 1 \geq 0$. Lemma 6.3, with $h = r-t$ and target modulus $r+1$, gives

$$(\lambda N)(1-x) = e_E(\gamma P(x))e_F(\gamma N(P(x))).$$

Here its error has depth $r+1$ before multiplication by γ , so is in $\mathfrak{p}_F^{-d_F}$ afterward. Now $wP(x)$ belongs to $\mathfrak{P}_E^{\lceil (t+1)/p \rceil}$. The actual norm-character phase, obtained by applying ν to a norm and using Lemma 6.3 at modulus $t+1$, says

$$e_E(AwP(x)) = e_F(-AN(wP(x))) = e_F(-\gamma N(P(x))).$$

Combine the last two displays to obtain (84). In the first expression A stays outside the norm. The negative sign is forced by moving the norm contribution across a product equal to 1. $\square$

The full Lamprecht formula with its odd term retained.

Lemma 8.19 (complete formula from a full P coefficient). *Under (83), there is an explicitly determined $g_E(\theta, C) \in \mu_4$ with*

$$(85) \quad \Delta_E(\theta, e_E) = \theta(-C^{-1})e_E(-C)g_E(\theta, C).$$

For even n , $g_E = 1$. For odd $n = 2d+1$, choose any δ_E of valuation d . Then

$$(86) \quad g_E = |k_E|^{-1/2} \sum_{x \in k_E} e_E(-C\delta_E^2 \tilde{x}^2/2).$$

The residue phase in this display is exactly the full odd-conductor Lamprecht term, not only its polarization.

Proof. Apply Lemma 6.5 with $\Psi_E = e_E$, ideal exponent $J = -d_E$, and enhanced coefficient $B = C$. Take the ordinary coefficient to be the same C , so the displacement η is zero. The full P -formula is valid on $\mathfrak{p}_E^{\lfloor n/2 \rfloor}$ because $\lfloor n/2 \rfloor \geq \lceil n/p \rceil$ for odd p and $n \geq 2$. Its residual function is therefore exactly $e_E(-C\delta_E^2 \tilde{x}^2/2)$, and the elementary factor is $\theta(-C^{-1})e_E(-C)$. The homogeneous quadratic sum has phase in μ_4 , as in the proof of that lemma. No affine displacement remains. $\square$

Application to the totally ramified transition. Let the breaks be $t = t_1 \leq t_2 = t + \delta$, $t' = t + p\delta$, $p \nmid t$. Let the actual pair have the form

$$\theta_1 = \chi(\lambda n_1), \quad \theta_2 = \varphi(\lambda n_2), \quad \chi N_1 = \varphi N_2,$$

where $a(\chi) = m_1 = 1 + t + t'$, $a(\varphi) = m_2 = 1 + t + t_2$. Suppose $a(\lambda) = 1 + r$, $m = r - t_2$, and $pm > t$. First realize the actual minimal pair by Theorem 8.8. Conjugation on B_1 also conjugates θ_1 , since λn_1 is invariant, and hence does not change its local constant. Denote the resulting full coefficients by $\alpha b, \alpha a$, where $b = N_1 \Delta$ and $a = N_2 \Delta$. The scalar α is the full coefficient of an actual nontrivial norm character for B_2/A . Now choose the full coefficient γ for λ , and choose any actual nontrivial norm character for B_1/A , with coefficient β . Apply Lemma 8.15 separately to these two norm characters. Lemma 8.16 shows that its replacement of α does not change the already realized full minimal models; replacing β affects neither minimal model. Renaming the adjusted coefficients, the exact choices give

$$(87) \quad \begin{aligned} n_2 w_2 &= \gamma/\alpha, & v_2 w_2 &= -m, & n_1 w_1 &= \gamma/\beta, & v_1 w_1 &= -(m + \delta), \\ v_A \alpha &= -d - 1 - t_2, & v_A \beta &= -d - 1 - t, & v_A \gamma &= -d - 1 - r. \end{aligned}$$

Both adjusted coefficients still represent the same actual lower norm characters on their full logarithmic subgroups. The full minimal formulas remain

$$\chi(1-x) = e_{B_1}(\alpha b P(x)), \quad \varphi(1-x) = e_{B_2}(\alpha a P(x)), \quad x \in \mathfrak{P}_{B_i}^{[m_i/p]}$$

on the appropriate side. In particular Theorem 8.14 applies to these very coefficients, with no extra norm condition on β/α . Proposition 8.18 now gives the full coefficients

$$(88) \quad C_1 = \gamma - \beta w_1 + \alpha N_1 \Delta, \quad C_2 = \gamma - \alpha w_2 + \alpha N_2 \Delta.$$

The actual conductors are

$$n_1^{\text{char}} = 1 + t' + pm, \quad n_2^{\text{char}} = 1 + t_2 + pm.$$

Write $\alpha_1 = \beta$, $\alpha_2 = \alpha$. The full formulas for (88) hold at depths $\lceil n_i^{\text{char}}/p \rceil$. Indeed $n_i^{\text{char}} > m_i$, $v_i(\alpha N_i \Delta/\gamma) = pm - t > 0$, and $v_i(\alpha_i w_i/\gamma) = (p-1)(r-t_i) > 0$. Thus no leading cancellation is possible, and the conductors and full-coordinate domains are exactly as stated.

Proposition 8.20 (Ueda-normalized orientation of the four factors). *Define $A_i = \gamma + \alpha N_i \Delta$ and use the corrected factors*

$$Q_1 = \frac{\chi(A_1)}{\varphi(C_2)}, \quad Q_2 = \lambda\left(\frac{n_1(A_1)}{n_2(C_2)}\right),$$

$$Q_3 = e_{B_1}(-\beta w_1) \theta_1(C_1/A_1), \quad Q_4 = e_{B_2}(\alpha w_2).$$

Then, with the explicit g_i from Lemma 8.19,

$$(89) \quad \frac{\Delta_{B_2}(\theta_2, e_{B_2})}{\Delta_{B_1}(\theta_1, e_{B_1})} = \frac{g_2}{g_1} Q_1 Q_2 Q_3 Q_4 e_A(\alpha[T_1 N_1 \Delta - T_2 N_2 \Delta]).$$

In particular, after the trace-norm estimate of Theorem 7.5 removes the last factor, proving $(Q_1 Q_2 Q_3 Q_4)^{-1} = 1$ implies that the actual ratio belongs to μ_4 .

Proof. Compatibility and odd degree give $\theta_1(-1) = \theta_2(-1)$: apply the equal norm pullbacks to $-1 \in B^\times$. Formula (85) therefore gives

$$\frac{\Delta_{B_2}}{\Delta_{B_1}} = \frac{g_2}{g_1} \frac{\theta_1(C_1)}{\theta_2(C_2)} e_{B_1}(C_1) e_{B_2}(-C_2).$$

The two γ traces both equal $p\gamma$ and cancel. Substitute (88); the remaining additive factors are the three shown in (89). Direct multiplication gives

$$Q_1 Q_2 = \frac{\theta_1(A_1)}{\theta_2(C_2)}, \quad Q_1 Q_2 \theta_1(C_1/A_1) = \frac{\theta_1(C_1)}{\theta_2(C_2)}.$$

The remaining additive factors are exactly those of $Q_3 Q_4$. The same trace-norm estimate is $T_1 N_1 \Delta - T_2 N_2 \Delta \in \mathfrak{p}_A^{1+t_2}$; its weight α puts it in $\mathfrak{p}_A^{-d}$, as required. $\square$

9. THE TOTALLY RAMIFIED ODD-PRIME CALCULATION

9.1. The linear trace calculation.

Notation and local estimates. Use the totally ramified diamond $A \subset B_1, B_2 \subset B$, $\text{Gal}(B/A) = C_p^2$, p odd, with

$$t_1 = t, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t.$$

Write $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$, $S_i = \text{Tr}_{B/B_i}$, and normalize $v = v_B$. The independently proved construction and estimates of Section 7 give

$$(90) \quad \Delta^p - \Delta = a \in B_2, \quad v(\Delta) = -t, \quad b = N_1 \Delta, \quad e_i = E_i^{B/B_1}(\Delta), \quad s = -e_{p-1},$$

$$v(e_i) \geq p((p-1)t_2 - it), \quad v(S_1(Y\Delta^i)) \geq p(p-1)t_2 - pit + v(Y) \quad (Y \in B_2).$$

Put $V = v_B(p)$ in mixed characteristic; in equal characteristic every term carrying p is zero. The local trace-ideal bound is $V \geq p(p-1)t_2$. Let $w \in B_2$ have $n_2 w = W$, put

$$u = W^{-1}, \quad v_A(u) = m, \quad M = pm > t, \quad K = 1 + pt_2, \quad S = p(p-1)\delta.$$

Thus $M \geq t + 1$, $v(w) = -M$, $v(W) = -pM$, $v(a) = v(b) = -pt$, and $v(s) \geq S$. All congruences below have modulus $\mathfrak{P}_B^K$ unless a different modulus is displayed. Set

$$Y = W + b, \quad D = Y^p - Y, \quad D_0 = W^p - W + a^p, \quad C = 1 + a/W.$$

Their denominators are nonzero: $v(Y) = -pM$ and $v(D) = v(D_0) = -p^2M$.

The actual upper norms of the conjugates. Let $\mathcal{T} = \{0\} \cup \mu_{p-1} \subset A$ and let Δ_ξ be the root congruent to $\Delta + \xi$ at positive relative depth. The construction of Section 7 gives

$$\Delta_\xi = \Delta + \xi + \eta_\xi, \quad v(\eta_\xi) \geq V - (p-1)t.$$

These are all B/B_2 conjugates; in equal characteristic $\eta_\xi = 0$.

Lemma 9.1. *For each $\xi \in \mathcal{T}$,*

$$v_{B_1}(N_1\Delta_\xi - N_1(\Delta + \xi)) \geq t_2 + 1.$$

Proof. In mixed characteristic put $z = \eta_\xi/(\Delta + \xi)$, so $v(z) \geq V - (p-2)t$. The different exponent for B/B_1 is $D' = (p-1)(t' + 1)$. The strong symmetric estimate gives

$$v_1(N_1(1+z) - 1) \geq \min \left\{ v(z), \left\lfloor \frac{v(z) + D'}{p} \right\rfloor \right\}.$$

Here $v(z) \geq t' + 1$, so the second entry is no larger than the first. Since $v_1N_1(\Delta + \xi) = -t$, the required difference has valuation at least

$$\frac{V}{p} - t + (p-1)\delta + \left\lceil \frac{t}{p} \right\rceil \geq (p-2)t + 2(p-1)\delta + \left\lceil \frac{t}{p} \right\rceil \geq t_2 + 1.$$

For the last inequality, subtract $t + \delta + 1$ and use $p \geq 3$, $t \geq 1$, $\lceil t/p \rceil \geq 1$. Equal characteristic is exact. $\square$

Define $f(X) = X/(1 + X/W)$. For $v(X) = -pt$ its denominator is a unit. The exact norm polynomial gives

$$N_1(\Delta + \xi) = b + \xi - \xi s + R_\xi, \quad v(R_\xi) \geq p((p-1)t_2 - (p-2)t) \geq pt_2.$$

The exact identity

$$f(X+z) - f(X) = \frac{z}{(1 + X/W)(1 + (X+z)/W)}$$

and its first-order expansion show

$$(91) \quad f(N_1\Delta_\xi) \equiv \frac{W(b+\xi)}{W+b+\xi} - \frac{\xi s}{(1+b/W)^2} \pmod{\mathfrak{P}_B^{pt_2}}.$$

Indeed, the quadratic s -error has depth at least $pM + 2S \geq pt_2$; replacing $1 + (b+\xi)/W$ by $1 + b/W$ in that linear error costs at least $pM + S \geq pt_2$. Lemma 9.1 is deeper still.

The power factors, including $p = 3$. The actual trace is

$$W_i = S_2 \left((\Delta/w)^i \frac{b}{1+b/W} \right), \quad 1 \leq i \leq p-1.$$

Each summand's power factor has valuation $i(M-t) \geq 1$; therefore (91) may be substituted at modulus K .

For $p \geq 5$, replacing Δ_ξ by $\Delta + \xi$ in that power factor costs at least

$$V + iM - (2p+i-2)t \geq K.$$

At the weakest values $M = t + 1$, $V = p(p-1)t_2$, the excess over K is $(p^2 - 4p + 2)t + p(p-2)\delta + i - 1 \geq 0$.

Suppose $p = 3$. Hensel's equation at $\Delta + \xi$, $\xi = \pm 1$, gives the sufficient refinement

$$(92) \quad \Delta_\xi = \Delta + (1 + 3\Delta^2)\xi + \eta'_\xi, \quad v(\eta'_\xi) \geq V - t.$$

To see this, the initial residual is $3\xi\Delta^2 + 3\xi^2\Delta$; substituting the first Hensel correction back into the equation adds terms of valuation at least $2V - 4t$, which is at least $V - t$. Thus no coefficient

beyond the displayed $3\xi\Delta^2$ is needed. Deleting η'_ξ in W_i costs at least $V + iM - (i + 3)t \geq K$ for $i = 1, 2$.

Put $\lambda = 1 + 3\Delta^2$ and

$$C_i(\lambda) = w^{-i} \sum_{\xi \in \mathcal{T}} (\Delta + \lambda\xi)^i \frac{W(b + \xi)}{Y + \xi}.$$

Direct summation of the three terms gives the exact identities

$$\begin{aligned} C_1(\lambda) - C_1(1) &= \frac{2(\lambda - 1)W^2}{w(Y^2 - 1)}, \\ C_2(\lambda) - C_2(1) &= \frac{2W(\lambda - 1)}{w^2(Y^2 - 1)} ((\lambda + 1)(bY - 1) + 2W\Delta). \end{aligned}$$

Their valuations are at least $V - 2t + M$ and $V - 5t + 2M$, respectively, hence at least K . In the signed exceptional sum the exact formulas

$$\sum_{\xi = \pm 1} \xi(\Delta + \lambda\xi) = 2\lambda, \quad \sum_{\xi = \pm 1} \xi(\Delta + \lambda\xi)^2 = 4\lambda\Delta$$

give errors of depths $S + M + V - 2t$ and $S + 2M + V - 3t$, also at least K . Thus the unscaled Teichmüller powers may be used for every odd p .

Proposition 9.2 (the actual split). *For $1 \leq i \leq p - 1$,*

$$(93) \quad \begin{aligned} W_i &\equiv w^{-i} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^i \frac{W(b + \xi)}{Y + \xi} \\ &\quad - \frac{s}{w^i(1 + b/W)^2} \sum_{\xi \in \mu_{p-1}} \xi(\Delta + \xi)^i. \end{aligned}$$

In particular, the exceptional sum has no denominator $Y + \xi$.

Proof. Combine (91) with the power estimates just proved. □

Exact interpolation and the four replacements.

Lemma 9.3 (Finite interpolation and power sums). *For $0 \leq i < p$ and $Z \notin -\mathcal{T}$,*

$$\sum_{\xi \in \mathcal{T}} \frac{(\Delta + \xi)^i}{Z + \xi} = \frac{p\Delta^i}{Z} + (p - 1) \frac{(\Delta - Z)^i}{Z^p - Z}.$$

For $1 \leq i < p$, the accompanying power sums are

$$\begin{aligned} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^i &= p\Delta^i + (p - 1)\mathbf{1}_{i=p-1}, \\ \sum_{\xi \in \mu_{p-1}} \xi(\Delta + \xi)^i &= (p - 1)(\mathbf{1}_{i=p-2} + i\Delta\mathbf{1}_{i=p-1}). \end{aligned}$$

All three identities are exact in mixed and equal characteristic.

Proof. Regard the first identity as an equality of rational functions in Z . Both sides have only simple poles at $-\mathcal{T}$ and vanish at infinity. At zero the residue on each side is Δ^i ; at $-\xi$ with $\xi^{p-1} = 1$, it is $(\Delta + \xi)^i$, since the derivative of $Z^p - Z$ there is $p - 1$. Subtraction gives a rational function with no poles and zero value at infinity, hence zero. This works in both characteristics because $p - 1$ is nonzero. The last two identities follow by the binomial theorem and $\sum_{\xi \in \mu_{p-1}} \xi^j = 0$ unless $p - 1$ divides j , when the sum is $p - 1$. □

Apply the lemma with $Z = Y = W + b$. This value has negative valuation, since W dominates b , so the denominators are nonzero. We obtain

$$(94) \quad \begin{aligned} W_i &\equiv \frac{pWb\Delta^i}{w^iY} + (p - 1) \frac{W}{w^{p-1}} \mathbf{1}_{i=p-1} - (p - 1) \frac{W^2(\Delta - Y)^i}{w^iD} \\ &\quad - (p - 1) \frac{s}{(1 + b/W)^2} \left(\frac{\mathbf{1}_{i=p-2}}{w^{p-2}} + \frac{i\Delta\mathbf{1}_{i=p-1}}{w^{p-1}} \right). \end{aligned}$$

The first term has depth at least $V - pt + i(M - t) \geq K$.

The characteristic polynomial of Δ over B_1 gives two useful forms, with all signs fixed by $s = -e_{p-1}$:

$$(95) \quad \begin{aligned} b &= \Delta + a + a\omega, & v(\omega) &\geq pt_2 - t, \\ b &= \Delta + a - \Delta s + \rho, & v(\rho) &\geq p(p-1)t_2 - (p^2 - 2p + 2)t. \end{aligned}$$

Each follows by solving the characteristic equation for its constant term and applying (90) to the remaining coefficients.

Lemma 9.4 (replace the denominator). $W^2(\Delta - Y)^i/(w^i D)$ may be replaced by $W^2(\Delta - Y)^i/(w^i D_0)$.

Proof. Write $q = b/W$, $a_1 = a/W$, $d_1 = \Delta/W$, $h_1 = a\omega/W$. Then $q = a_1 + d_1 + h_1$ and

$$D - D_0 = W^p E, \quad E = (1 + q)^p - 1 - a_1^p - W^{1-p} q.$$

Since $d_1^p = (\Delta + a)/W^p$, $W^{1-p} q = d_1^p + W^{1-p} h_1$. Every term of E is p -divisible, or contains h_1^p , or contains $W^{1-p} h_1$. Consequently

$$\begin{aligned} v(E) &\geq \min\{V + p(M - t), p(pM + pt_2 - (p + 1)t), \\ &\quad p(p - 1)M + pM + pt_2 - (p + 1)t\} \geq K + M. \end{aligned}$$

The remaining factor in the difference has valuation $(p(p - 2) - i(p - 1))M \geq -M$. $\square$

Lemma 9.5 (replace the numerator). Put $A_0 = -a - W$. Modulo the required ideal,

$$\frac{W^2(\Delta - Y)^i}{w^i D_0} \equiv \frac{W^2}{w^i D_0} \begin{cases} A_0^i, & i \leq p - 2, \\ A_0^{p-1} + (p - 1)A_0^{p-2}\Delta s, & i = p - 1. \end{cases}$$

Proof. By (95), $\Delta - Y = A_0 + \Delta s - \rho$. Set $Z = S - t$ and $R = p(p - 1)t_2 - (p^2 - 2p + 2)t$. For $i \leq p - 2$ the first perturbation after the displayed weight has depth at least $2(p - 1)M + Z \geq K$. At $i = p - 1$, deleting ρ costs $(p - 1)M + R \geq K$; every term quadratic or higher in Δs costs $(2p - 1)M + 2Z \geq K$. At $p = 3$, $\delta = 0$, $M = t + 1$ the three excesses are 3, 1, 4, respectively, so none is lost at the boundary. $\square$

Lemma 9.6 (the inverse-square replacement and cancellation). The signed term $s/[w^{p-2}(1 + b/W)^2]$ may use C^{-2} instead, as may its multiple Δ/w . The two remaining linear Δs terms at $i = p - 1$ cancel.

Proof. The difference of the inverse squares is

$$\frac{(a - b)/W [2 + (a + b)/W]}{(1 + b/W)^2 (1 + a/W)^2}.$$

Since $v(a - b) \geq -t$, its weighted first error has depth $S + 2(p - 1)M - t \geq K$; multiplication by Δ/w only increases it. After extracting the common $(p - 1)^2 \Delta s / w^{p-1}$, the two linear terms have bracket

$$-\frac{1}{(1 + a/W)^2} + \frac{(1 + a/W)^{p-2}}{1 - W^{1-p} + (a/W)^p}.$$

The numerator of this difference is $W^{1-p} + \sum_{j=1}^{p-1} \binom{p}{j} (a/W)^j$. The first weighted term has depth $S - t + (p^2 - 1)M \geq K$; the others have depth at least $S - t + (p - 1)M + V + p(M - t) \geq K$. $\square$

Theorem 9.7 (the verified linear packet). For every odd p and $1 \leq i \leq p - 1$,

$$\boxed{\begin{aligned} W_i &\equiv -(p - 1) \frac{s}{w^{p-2} C^2} \mathbf{1}_{i=p-2} + (p - 1) \frac{W}{w^{p-1}} \mathbf{1}_{i=p-1} \\ &\quad - (p - 1) \frac{W^2(-a - W)^i}{w^i (W^p - W + a^p)} \pmod{\mathfrak{P}_B^{1+pt_2}}. \end{aligned}}$$

Proof. Insert Lemmas 9.4–9.6 into (94). $\square$

Remark 9.8. This proves the actual packet needed in the Q_1 logarithm, not only an identity for artificial translates. The delicate $p = 3$ replacement was completed before applying interpolation. The exceptional signed sum in (93) is not changed into a rational sum at any stage.

9.2. The quadratic trace calculation.

The character data and local estimates. Let $A \subset B_1, B_2 \subset B$ be a totally ramified C_p^2 diamond, where p is odd. Put

$$t = t_1, \quad t_2 = t + \delta, \quad t' = t + p\delta, \quad \delta \geq 0, \quad p \nmid t,$$

$$N_i = N_{B/B_i}, \quad n_i = N_{B_i/A}, \quad S_i = \text{Tr}_{B/B_i}, \quad T_i = \text{Tr}_{B_i/A}.$$

Valuations are normalized on the named field; thus $v_B|_{B_i} = pv_i$ and $v_i|_A = pv_A$. Fix

$$\Delta^p - \Delta = a \in B_2, \quad v_B \Delta = -t, \quad b = N_1 \Delta, \quad N_2 \Delta = a, \quad s = -E_{p-1}^{B/B_1}(\Delta).$$

Let $u = \alpha/\gamma \in A$, $m = v_A u$, $M = pm > t$, and choose $w \in B_2^\times$ with $n_2 w = u^{-1}$ and $v_2 w = -m$. The legitimate ordering of these choices is proved in Subsection 8.3. Write

$$W = u^{-1}, \quad C_b = 1 + b/W, \quad C = 1 + a/W, \quad D_0 = W^p - W + a^p,$$

$$q = p - 1, \quad S = p(p - 1)\delta, \quad K = 1 + pt_2, \quad F = 1 + t + t' = 1 + 2t + p\delta.$$

In common-field valuation $v = v_B$ one has

$$v(w) = -M, \quad v(W) = -pM, \quad v(a) = v(b) = -pt, \quad v(C) = v(C_b) = 0, \quad v(D_0) = -p^2 M.$$

In mixed characteristic set $e = v_B(p)$; the cyclic ramification bound gives $e \geq p(p - 1)t_2$. In equal characteristic all p -divisible terms vanish.

Normalize $\Psi_E(x) = e_E(\alpha x)$, with trace-compatible e_E . Its largest trivial ideals on A, B_1, B_2 have exponents $1 + t_2, 1 + t', 1 + t_2$, respectively. The actual minimal first inducing character satisfies

$$(96) \quad \chi(1 - x) = \Psi_{B_1}(bP(x)), \quad x \in \mathfrak{P}_1^{[F/p]}, \quad P(X) = \sum_{j=1}^{p-1} X^j/j.$$

This full formula is the conclusion of Subsection 8.1; it is not replaced by a half-conductor assertion.

We use Lemma 7.4, with all domains visible:

$$(97) \quad v_1(E_j^{B/B_1}(Y\Delta)) \geq (p - 1)t_2 + j(v_2 Y - t), \quad 1 \leq j < p,$$

$$(98) \quad v_1(S_1(Y\Delta^j)) \geq (p - 1)t_2 - jt + v_2 Y, \quad 0 \leq j < p.$$

Here $Y \in B_2$. These are the explicit local estimates of Section 7, not an appeal to Lakkis's Satz 2. We also use Ueda's cyclic estimate

$$(99) \quad v_F(E_j^{E/F} z) \geq \left\lfloor \frac{jev_E z + (p - 1)(b_0 + 1)}{p} \right\rfloor, \quad 1 \leq j < p,$$

for a cyclic edge of break b_0 , and the corresponding trace-ideal formula. The floor in this formula is essential.

For the quadratic packet put

$$(100) \quad T = S_1((\Delta/w)^{p-1}), \quad \mathcal{Q} = S_2\left(\frac{b(\Delta/w)^{p-1}T}{C_b^2}\right).$$

Here $T \in B_1$ and $\mathcal{Q} \in B_2$; the calculation below does not apply a B_2 character to any expression outside B_2 .

The quadratic term: actual conjugates and one finite sum. All congruences in this and the next subsection are in B modulo $\mathfrak{P}_B^K$. Set $f(X) = X/(1 + X/W)^2$ and $\mathcal{T} = \{0\} \cup \mu_{p-1} \subset A$. The symmetric-coefficient estimates give

$$(101) \quad v(T) \geq S + qM, \quad v\left(S_1(\Delta^j/w^q)\right) \geq p(p-1)t_2 - pjt + qM \quad (0 \leq j \leq q).$$

The characteristic polynomial of Δ over B_1 also gives

$$(102) \quad b = \Delta + a + a\omega, \quad v(\omega) \geq pt_2 - t.$$

Indeed the error divided by a has terms of valuation at least $(p-1)(pt_2 - jt)$ for $1 \leq j < p$, by (97); their minimum is at least $pt_2 - t$.

Lemma 9.9 (uniform conjugate replacement for the quadratic term). *In the sum defining $\mathcal{Q}$ one may replace every actual B/B_2 conjugate of Δ by its corresponding $\Delta + \xi$, $\xi \in \mathcal{T}$, modulo $\mathfrak{P}_B^K$. This holds for every odd p , including $p = 3$ in mixed characteristic.*

Proof. In equal characteristic the statement is exact. In mixed characteristic, Hensel's lemma applied at $\Delta + \xi$ gives

$$\Delta_\xi = \Delta + \xi + \epsilon_\xi, \quad v(\epsilon_\xi) \geq e - qt.$$

The derivative is a unit since $e - qt > 0$, and the p approximations have distinct residues relative to one another. They are all the conjugates. The outer power difference, after division by w^q , has valuation at least $e - (2p-3)t + qM$. The inner trace difference has this bound plus qt' , by the trace-ideal formula for B/B_1 . Both resulting product errors have valuation at least

$$(103) \quad e + S + 2qM - 3qt.$$

Here $f(N_1\Delta_\xi)$ has valuation $-pt$, and the unchanged outer power and inner trace have bounds $q(M-t)$ and $S + qM$.

For the norm argument itself put $a_0 = e - (p-2)t$. The quotient $\epsilon_\xi/(\Delta + \xi)$ has valuation at least a_0 . By (99),

$$v\{N_1\Delta_\xi - N_1(\Delta + \xi)\} \geq -pt + \min\{pa_0, a_0 + qt'\} = e - qt + S.$$

The equality for this lower bound uses $a_0 \geq t'$, which follows from $e \geq pqt_2$. The exact identity

$$f(X) - f(Y) = (X - Y) \frac{1 - XY/W^2}{(1 + X/W)^2(1 + Y/W)^2}$$

shows that f does not lower this difference valuation. Multiplying by the outer power and the inner trace gives the further bound $e + 2S + 2qM - 2qt$. After $e \geq pq(t + \delta)$ and $M \geq t + 1$, the excesses of the two bounds over K are at least

$$(p^2 - 3p + 1)t + p(2p - 3)\delta + 2p - 3,$$

$$p(p - 2)t + p(3p - 4)\delta + 2p - 3,$$

respectively. Both are positive already at $p = 3$. Thus the entire Hensel error is controlled, without deleting a factor 3 formally. $\square$

Lemma 9.10 (reduction to a single finite rational sum). *One has*

$$(104) \quad \mathcal{Q} \equiv \frac{T}{w^q} \sum_{\xi \in \mathcal{T}} (\Delta + \xi)^q f(b + \xi).$$

Proof. After Lemma 9.9, expand the translated inner trace as

$$S_1((\Delta + \xi)^q/w^q) = \sum_{j=0}^q \binom{q}{j} \xi^{q-j} \mathcal{U}_j, \quad \mathcal{U}_j = S_1(\Delta^j/w^q).$$

For $j \leq p-3$, the product valuation is at least

$$pqt_2 + 2qM - (2p - 1 + pj)t \geq K.$$

The weakest case $j = p-3$ has excess at least $(p-1)t + p(p-2)\delta + 2p - 3$. For $j = p-2$, its scalar is a multiple of ξ . Subtract the constant product $f(b)\Delta^q/w^q$, whose weighted sum is

zero because $\sum_{\xi \neq 0} \xi = 0$. The outer-power difference has valuation at least $qM - (q-1)t$; also $f(N_1(\Delta + \xi)) - f(b)$ has valuation at least zero. Using (101), the two errors are bounded below by

$$pqt_2 + 2qM - (p^2 - 2)t, \quad pqt_2 + 2qM - (p^2 - p - 1)t.$$

Their excesses are at least $p(p-2)\delta + 2p - 3$ and $(p-1)t + p(p-2)\delta + 2p - 3$. Thus only $\mathcal{U}_q = T$ remains.

The exact norm polynomial now reads

$$N_1(\Delta + \xi) = b + \xi - \xi s + r_\xi, \quad v(r_\xi) \geq pt_2.$$

First delete r_ξ using the exact difference formula for f . Next expand in $-\xi s$. Expansion of the inverse square as a geometric power series shows that its Taylor coefficients of degree $j \geq 2$ have valuation at least $(j-1)pM$; no division by a factorial divisible by p is used. The series converges because $v(s/W) > 0$. Finally replace $f'(b + \xi)$ by $f'(b)$; their difference has valuation at least pM . After multiplication by the power and trace factors, the respective error bounds are

$$S + 2qM - qt + pt_2, \quad 3S + (3p-2)M - qt, \quad 2S + (3p-2)M - qt.$$

At $M = t + 1$, their excesses over K are, respectively,

$$qt + S + 2p - 3, \quad qt + p(3p-4)\delta + 3p - 3, \quad qt + p(2p-3)\delta + 3p - 3,$$

all positive. Increasing M only improves them. The only remaining signed term contains

$$\sum_{\xi \neq 0} \xi(\Delta + \xi)^q = q^2 \Delta.$$

Its valuation is at least $2S + 2qM - t$, whose excess is at least $(p-3)t + p(2p-3)\delta + 2p - 3 \geq 0$. This proves (104). $\square$

Exact differentiation replaces the long quadratic sum. We differentiate the exact rational identity of Lemma 9.3; no error congruence is differentiated.

Theorem 9.11 (quadratic trace at the full modulus). *With the genuine B_2 element $\mathcal{Q}$ of (100),*

$$(105) \quad \mathcal{Q} \equiv (p-1) \frac{aT}{w^{p-1}C^2} \pmod{\mathfrak{P}_B^{1+pt_2}}.$$

The right side is an expression in B ; a B_2 representative is constructed in Subsection 9.6 before its character is evaluated.

Proof. Put $L = W + b$, $A_* = \Delta - L$, and $D_L = L^p - L$. Thus $v(L) = v(A_*) = -pM$ and $v(D_L) = -p^2M$. Apply Lemma 9.3 at $i = q$ and differentiate its *exact rational identity* in the variable Z , holding W, b, Δ constant. Writing

$$H(Z) = p\Delta^q/Z + q(\Delta - Z)^q/(Z^p - Z),$$

we obtain the exact equality

$$\sum_{\xi} (\Delta + \xi)^q f(b + \xi) = W^2 \{H(L) + WH'(L)\}.$$

After collecting terms this is

$$(106) \quad \frac{qW^2 A_*^{p-2}(\Delta - b)}{D_L} + \frac{pW^2 \Delta^q b}{L^2} - \frac{pqW^3 A_*^{p-2}}{D_L} - \frac{qW^3 A_*^q (pL^{p-1} - 1)}{D_L^2}.$$

This calculation differentiates no congruence and divides by no p . The valuations of the last three terms, splitting the last one at $pL^{p-1} - 1$, are bounded below by

$$e - (2p-1)t, \quad e - pM, \quad e - pM, \quad p(p-2)M.$$

The factor T/w^q has valuation at least $S + 2qM$. Consequently their products have the three lower bounds

$$e + S + 2qM - (2p-1)t, \quad e + S + (p-2)M, \quad S + (p^2 - 2)M.$$

At $e = pq(t + \delta)$, $M = t + 1$, their excesses over K are, respectively,

$$\begin{aligned} (p^2 - 2p - 1)t + p(2p - 3)\delta + 2p - 3, \\ (p^2 - p - 2)t + p(2p - 3)\delta + p - 3, \\ (p^2 - p - 2)t + p(p - 2)\delta + p^2 - 3. \end{aligned}$$

All are nonnegative for $p \geq 3$; increasing e or M preserves this. In equal characteristic the p -divisible terms are simply absent.

It remains to treat the first term of (106). By (102), $b - \Delta = a(1 + \omega)$ and $v(a\omega) \geq p\delta - t \geq -t$. Both $A_*/[-(W + a)]$ and $L/(W + a)$ belong to U_B^{pM-t} , and $D_L/L^p = 1 - L^{1-p} \in U_B^{p(p-1)M}$. Since $p - 2$ is odd,

$$\frac{W^2 A_*^{p-2}}{D_L} = -C^{-2}(1 + \eta_0), \quad v(\eta_0) \geq pM - t.$$

The first term is therefore $qaC^{-2}(1 + \omega)(1 + \eta_0)$. After multiplication by T/w^q , the two replacement errors have lower bounds

$$S + 2qM + pt_2 - (p + 1)t, \quad S + (3p - 2)M - (p + 1)t.$$

Their excesses over K , at $M = t + 1$, are $(p - 3)t + pq\delta + 2p - 3$ and $(p - 3)t + p(p - 2)\delta + 3p - 3$. The product error is deeper. Combining this with Lemma 9.10 proves (105), including the full $p = 3$ boundary. $\square$

9.3. The first factor Q_1 .

Actual characters and the norm quotient. Let B/A be a totally ramified C_p^2 diamond, p odd, with $t_1 = t$, $t_2 = t + \delta$, $t' = t + p\delta$, and $p \nmid t$. Write $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$, $S_i = \text{Tr}_{B/B_i}$. Let

$$\Delta^p - \Delta = a \in B_2, \quad v_B \Delta = -t, \quad b = N_1 \Delta, \quad s = -E_{p-1}^{B/B_1}(\Delta).$$

The actual minimal characters χ, φ are compatible and have full coefficients $\alpha b, \alpha a$, with conductors $m_1 = 1 + t + t'$, $m_2 = 1 + t + t_2$. Put $\Psi_L(x) = e_L(\alpha x)$; its largest trivial ideals over A, B_1, B_2 have exponents $1 + t_2, 1 + t', 1 + t_2$, respectively. Let

$$u = \alpha/\gamma, \quad W = 1/u = n_2 w, \quad w = w_2 \in B_2, \quad v_A u = m, \quad pm > t, \quad M = pm.$$

The actual factor is

$$Q_1 = \frac{\chi(\gamma + \alpha b)}{\varphi(\gamma - \alpha w + \alpha a)}.$$

Lemma 9.12 (entry from the actual factors). *For $z = \Delta/w$ and $n = N_1 z = ub$, one has*

$$Q_1^{-1} = \chi\left(\frac{N_1(1 + z)}{1 + n}\right).$$

Proof. The exact Artin-Schreier polynomial gives

$$N_2(1 + z) = 1 - w^{1-p} + a/w^p.$$

The relative difference between this and $1 - uw + ua$ is controlled by

$$(1 - uw + ua) - N_2(1 + z) = (uw^p - 1) \frac{a - w}{w^p}.$$

The cyclic norm-power estimate gives $v_2(uw^p - 1) \geq (p - 1)t_2$. Hence the difference has valuation at least

$$(p - 1)t_2 + pm - \max(t, m) \geq 1 + t + t_2 = m_2.$$

For $m \leq t$, subtracting m_2 leaves $(p - 2)t_2 + pm - 2t - 1 \geq (p - 3)t$. For $m > t$ it leaves $(p - 2)t_2 + (p - 1)m - t - 1 \geq 0$. All compared elements are units, so their φ -values agree. Compatibility gives $\varphi N_2(1 + z) = \chi N_1(1 + z)$. Finally $\chi(\gamma) = \varphi(\gamma)$ by the proved common $A^\times$ restriction, and $N_1 z = ub$ exactly. This proves the formula. $\square$

The full logarithm with its actual signs. Put

$$e_i = E_i^{B/B_1}(z), \quad \mathcal{U}_i = S_1(z^i), \quad T = \mathcal{U}_{p-1}, \quad E = \frac{\sum_{i=1}^{p-1} e_i}{1+n},$$

$$W_i = S_2\left(\frac{bz^i}{1+n}\right), \quad \mathcal{Q} = S_2\left(\frac{bz^{p-1}T}{(1+n)^2}\right).$$

Proposition 9.13 (the corrected logarithmic entry). *One has*

$$(107) \quad Q_1^{-1} = \Psi_2\left(\sum_{i=1}^{p-1} \frac{(-1)^i}{i} W_i + \frac{\mathcal{Q}}{2(p-1)^2}\right).$$

Every discarded term lies in the full character kernel, before any trace to B_2 is taken.

Proof. Lemma 7.4 gives

$$v_1(e_i) \geq (p-1)t_2 + i(m-t).$$

Write $m_0 = \lceil t/p \rceil$; then $m \geq m_0$ because $p \nmid t$. All the indicated bounds are increasing in m , and at $m = m_0 \leq t$ their minimum is $\nu_0 = (p-1)(\delta + m_0)$. Writing $t = pa + b_0$, $1 \leq b_0 \leq p-1$, checks directly that

$$\nu_0 \geq \delta + \lceil (2t+1)/p \rceil = \lceil m_1/p \rceil, \quad 3\nu_0 \geq m_1.$$

Thus the full P -formula for χ applies to $1+E$, and

$$\chi(1+E) = \Psi_1(b(-E + E^2/2)).$$

This truncation is at conductor m_1 , also when $m > t$; it does not use the incorrect lower bound $(p-1)(\delta + m)$ for every e_i in that range.

In a nonlinear Newton term, or in a product $e_i e_j$ other than e_{p-1}^2 , the total symmetric index is at most $2p-3$ and there are at least two factors. After multiplication by b , its B -valuation is at least

$$2p(p-1)t_2 + (i+j)(M-pt) - pt.$$

Its worst bound occurs at $M = t+1$ and $i+j = 2p-3$; the excess over $p(1+t')$ is

$$(p-3)t + p(p-2)\delta + p-3 \geq 0.$$

Terms with additional factors have no smaller bound. Consequently the only square term which must be retained is the top square. Newton's identities, dividing by $1, \dots, p-1$, give

$$\sum_{i=1}^{p-1} e_i \equiv \sum_{i=1}^{p-1} \frac{(-1)^{i-1}}{i} \mathcal{U}_i, \quad e_{p-1}^2 \equiv T^2/(p-1)^2$$

inside this weighted character calculation. In the second replacement the error has at least three positive-depth symmetric factors and is killed by $3\nu_0 \geq m_1$. The remaining exponent is therefore

$$-\frac{b}{1+n} \sum_i \frac{(-1)^{i-1}}{i} \mathcal{U}_i + \frac{bT^2}{2(p-1)^2(1+n)^2}.$$

Pull one trace S_1 out of each summand, then use $\Psi_B = \Psi_2 S_2$. This is precisely (107). $\square$

Consequences for the final Q_1 expression. Use the independently proved actual-conjugate W_i formula and the independently proved quadratic trace identity, all at $K = 1 + pt_2$:

$$W_i \equiv -(p-1) \frac{s \mathbf{1}_{i=p-2}}{w^{p-2}(1+a/W)^2} + (p-1) \frac{W \mathbf{1}_{i=p-1}}{w^{p-1}} - (p-1) \frac{W^2(-a-W)^i}{w^i(W^p - W + a^p)},$$

$$\mathcal{Q} \equiv (p-1) \frac{aT}{w^{p-1}(1+a/W)^2}.$$

Put

$$C = 1 + a/W, \quad F_i = \frac{W}{W^p - W + a^p} \left(\frac{W+a}{w} \right)^i,$$

$$R = W \left(-F_{p-1} + \sum_{i=1}^{p-2} \frac{F_i}{i} + w^{1-p} \right).$$

Let $\mathcal{L}_1 \in B_2$ denote the actual exponent in (107). Substitution gives the following congruence in B , not a character evaluated outside its domain:

$$\mathcal{L}_1 \equiv R + \frac{s}{2w^{p-2}C^2} - \frac{aT}{2w^{p-1}C^2} \pmod{\mathfrak{P}_B^K}.$$

With the literal projections $y_s, y_T \in B_2$ constructed in Lemma 9.25, this implies the well-typed identity

$$Q_1^{-1} = \Psi_2(R + (y_s - y_T)/2).$$

Indeed $(-1)^i$ cancels the $(-1)^i$ in $(-a - W)^i$. For $i < p - 1$ one replaces $-(p - 1)/i$ by $1/i$ only after its factor p is weighted; $i = p - 1$ gives exactly $-F_{p-1}$ and is not changed. The exceptional coefficient $(p - 1)/(p - 2)$ becomes $1/2$ modulo its weighted kernel, and $1/[2(p - 1)]$ becomes $-1/2$ for the quadratic term. For $i \leq p - 2$, direct valuation gives

$$v_2(F_i) = (p - 1)(p - i)m, \quad v_B(pWF_i) \geq v_B(p) + (p - 2)M \geq K.$$

The coefficient of F_{p-1} was never changed. The errors from the two remaining rational-coefficient replacements are weighted by pX_s and pX_T ; their valuations are at least K by the displayed positive source depths in Lemma 9.25. These terms are zero in equal characteristic. Thus the formula is valid for every m with $pm > t$. The B_2 representatives invoked here are constructed in Subsection 9.6; until then these are congruences in B .

9.4. The second factor Q_2 .

An elementary cyclic norm-quotient lemma. Let E/F be totally ramified cyclic of odd prime degree p , with break $t \geq 1$ and different exponent $D = (p-1)(t+1)$. Let λ have conductor $n \geq 2$ and full coefficient γ for a fixed additive character ψ_F :

$$\lambda(1-z) = \psi_F(\gamma Pz), \quad P(Z) = \sum_{j=1}^{p-1} Z^j/j, \quad z \in \mathfrak{p}_F^{[n/p]}.$$

Thus $v_F\gamma = -n_F(\psi_F) - n$ and $\psi_E = \psi_F \text{Tr}_{E/F}$.

Lemma 9.14. *Let $k \in \mathfrak{P}_E^a$ with $a \geq 1$, and put*

$$b = \left\lfloor \frac{a+D}{p} \right\rfloor.$$

If $2b \geq n$, then

$$(108) \quad \lambda\left(\frac{N_{E/F}(1-k)}{1-N_{E/F}k}\right) = \psi_E\left(\frac{\gamma}{1-N_{E/F}k}P(k)\right).$$

No formula for λ on either factor separately is assumed.

Proof. Write $e_i = E_i^{E/F}(k)$ and $q_j = \text{Tr}_{E/F}(k^j)$. The exact norm expansion is

$$\frac{N(1-k)}{1-Nk} = 1 + Q, \quad Q = \frac{\sum_{i=1}^{p-1} (-1)^i e_i}{1-Nk}.$$

The denominator is a unit, since $v_F(Nk) = v_E k \geq a > 0$. Ueda's strong symmetric estimate gives $v_F(e_i) \geq b$ for every $1 \leq i < p$, and hence $v_F Q \geq b$. As $2b \geq n$, one has $b \geq [n/2] \geq [n/p]$, and all terms of degree at least two in $P(-Q)$ are killed at conductor n . Consequently

$$\lambda(1+Q) = \psi_F(-\gamma Q).$$

Newton's identities, dividing only by integers $j < p$, give

$$e_j \equiv (-1)^{j-1} q_j/j \pmod{\mathfrak{p}_F^{2b}}, \quad 1 \leq j < p.$$

Indeed q_i also has valuation at least b , by the trace-ideal formula, and every non-leading term in the Newton recursion contains a product of two such positive-depth factors. Therefore

$$-Q \equiv \frac{\sum_{j=1}^{p-1} q_j/j}{1-Nk} = \frac{\text{Tr}_{E/F} P(k)}{1-Nk} \pmod{\mathfrak{p}_F^n}.$$

Multiplication by γ and trace compatibility prove (108). The proof is valid in both characteristics, including $p = 3$. $\square$

The literal factorization of Q_2 . Use the totally ramified notation $t_1 = t$, $t_2 = t + \delta$, $r = t_2 + m$, $pm > t$, $u = \alpha/\gamma$, $W = 1/u$, $w = w_2$, $n_2 w = 1/u$, $a = N_2 \Delta$, $b_0 = N_1 \Delta$ and $H = n_1(s)$, with $s = -E_{p-1}^{B/B_1}(\Delta)$. Put

$$C = 1+ua, \quad P_* = n_2 C, \quad q_0 = u^{p-1}, \quad Y = P_* - q_0, \quad Z = \frac{n_1(1+ub_0)}{Y}, \quad k = \frac{uw}{C}, \quad F_* = \frac{P_*}{Y}.$$

All of C, P_*, Y, Z are units, and

$$n_2 k = q_0/P_*, \quad v_2 k = (p-1)m, \quad F_* = (1-n_2 k)^{-1}.$$

The actual factor is

$$Q_2 = \lambda\left(\frac{n_1(\gamma + \alpha b_0)}{n_2(\gamma - \alpha w + \alpha a)}\right).$$

Cancellation of the common γ^p , followed by multiplicativity, gives the *exact* identity

$$(109) \quad Q_2^{-1} = \lambda(Z^{-1}) \lambda\left(\frac{n_2(1-k)}{1-n_2 k}\right).$$

In particular this does not require evaluation of λn_2 at $1-k$.

The moment comparison and the exact Z modulus. Set $A_0 = n_2 a$, $D_0 = 1 + u^p A_0$. Write $E_i^{(1)}, E_i^{(2)}$ for the lower elementary symmetric functions of b_0, a , and put $F_i = u^i(E_i^{(1)} - E_i^{(2)})$.

Lemma 9.15 (weighted moments). *For $1 \leq j \leq p-1$,*

$$u^j [T_1(b_0^j) - T_2(a^j)] \equiv (p-1) \mathbf{1}_{j=p-1} u^{p-1} (1-s^p) \pmod{\mathfrak{P}_1^{1+pr}}.$$

Consequently

$$F_i \equiv -\mathbf{1}_{i=p-1} u^{p-1} (1-s^p) \pmod{\mathfrak{P}_1^{1+pr}}, \quad 1 \leq i < p.$$

Proof. For $j = 1$ Theorem 7.5, multiplied by u , gives the required modulus. For $j \geq 2$, apply Theorem 7.8 to $w^{-j} \Delta^{j-1}$, whose B -valuation is $jpm - (j-1)t > 0$. Its error exponent $pt_2 + jpm - (j-1)t$ is at least $1+pr$, since their difference is $(j-1)(pm-t) - 1 \geq 0$. The exact discrepancy from multiplicative homogeneity is

$$T_2((u^j - w^{-pj})a^j).$$

The lower cyclic norm-power estimate gives $v_2(u^j - w^{-pj}) \geq jpm + (p-1)t_2$. After adding the different, the trace is in $\mathfrak{p}_A^{r+1}$, because

$$(j-1)pm + (p-2)t_2 - jt - 1 \geq (p-2)t_2 - t + j - 2 \geq 0.$$

This proves the moment congruence, including its exceptional term. Apply Newton's identities to the two integral tuples ub_0 and ua . Induction over $i < p$ introduces only integral factors and division by i , a unit. At $i = p-1$ its leading sign is negative, so division of $(p-1)u^{p-1}(1-s^p)$ by $-(p-1)$ gives the displayed F_{p-1} . All other F_i vanish at the same modulus. $\square$

Proposition 9.16 (the full Z congruence). *One has*

$$Z \equiv 1 + u^{p-1} \frac{H}{D_0} \pmod{\mathfrak{p}_A^{r+1}}.$$

Proof. The norm endpoints of ub_0 and ua agree by norm transitivity. Summing Lemma 9.15 over the remaining symmetric degrees gives

$$n_1(1 + ub_0) - Y \equiv u^{p-1} s^p \pmod{\mathfrak{P}_1^{1+pr}}.$$

There is no need to transfer $(1-s)^p$ between fields. The ordinary lower cyclic norm-power estimate gives directly

$$v_1(u^{p-1}(s^p - H)) \geq p(p-1)m + p(p-1)\delta + (p-1)t \geq 1+pr,$$

because the excess is $p(p-2)(m+\delta) - t - 1 \geq 0$. Both sides after this substitution belong to A , so intersection with A improves the modulus to $\mathfrak{p}_A^{r+1}$.

It remains to replace the unit denominator Y by D_0 with its actual weight. One has

$$Y - D_0 = \sum_{i=1}^{p-1} E_i^{B_2/A}(ua) - u^{p-1}, \quad v_A E_i(ua) \geq L_0 = \left\lfloor \frac{pm - t + (p-1)(t_2 + 1)}{p} \right\rfloor.$$

The weighted intermediate terms have valuation at least $(p-1)m + (p-1)\delta + L_0 \geq r+1$. For completeness write $t = pa + b$, $1 \leq b \leq p-1$, and use $m \geq a+1$. At $\delta = 0$ the excess is at least

$$(p-3)a + p-2-b + \left\lfloor \frac{(p-2)b + p-1}{p} \right\rfloor \geq 0.$$

For $b \leq p-2$ every indicated term is nonnegative; for $b = p-1$ the final floor is $p-2$, giving again a nonnegative result. Increasing δ or m cannot decrease the excess. The weighted final $-u^{p-1}$ term has excess

$$(2p-3)m + (p-2)\delta - t - 1 \geq pm - t - 1 \geq 0.$$

All denominators are units. These estimates prove $u^{p-1}H(Y^{-1} - D_0^{-1}) \in \mathfrak{p}_A^{r+1}$ and hence the proposition. $\square$

Theorem 9.17 (the actual Q_2 phase). *The full actual factor is*

$$Q_2^{-1} = \Psi_A \left(Wu^{p-1} \frac{H}{D_0} \right) \Psi_2(WF_*P(k)), \quad \Psi_L(x) = e_L(\alpha x).$$

Proof. The conductor of λ is $n = r + 1 = t_2 + m + 1$. For the second factor of (109), Lemma 9.14 has

$$a = (p-1)m, \quad D = (p-1)(t_2 + 1), \quad b = \left\lfloor \frac{(p-1)n}{p} \right\rfloor.$$

For every $n \geq 2$ and odd p , $2\lfloor (p-1)n/p \rfloor \geq n$; this is $n \geq 2\lceil n/p \rceil$, checked at $p = 3$ and then monotone in p . The lemma therefore gives the second displayed phase.

For the first factor, $v_A H \geq (p-1)\delta$. Its small argument has valuation at least $L = (p-1)(m+\delta)$, and

$$2L - n = (2p-3)m + (2p-3)\delta - t - 1 \geq pm - t - 1 \geq 0.$$

Hence $\lambda((1+x)^{-1}) = e_A(\gamma x)$ for this argument. Proposition 9.16, at the full conductor, gives the first phase. These are two different depth estimates: the floor b controls the norm quotient, while L controls Z^{-1} . No ceiling strengthening or shallow individual-factor linearization has been used. $\square$

9.5. The weighted first terminal estimate.

The weight that must be checked. Use a totally ramified diamond $A \subset B_1, B_2 \subset B$ with $\text{Gal}(B/A) = C_p^2$, p odd. Put

$$t = t_1, \quad \delta = t_2 - t \geq 0, \quad p \nmid t, \quad N_i = N_{B/B_i}, \quad n_i = N_{B_i/A}, \quad S_i = \text{Tr}_{B/B_i}.$$

Let $\Delta^p - \Delta = a \in B_2$, $v_B \Delta = -t$, and set

$$s = -E_{p-1}^{B/B_1}(\Delta), \quad Y = w_2^{-1} \in B_2, \quad v_2(Y) = m, \quad n_2 Y = u, \quad M = pm > t.$$

Thus $v_B(Y) = M$, $v_A(u) = m$. Define

$$T = S_1((Y\Delta)^{p-1}), \quad H = n_1(s), \quad J = n_1(T), \quad A_0 = n_2(a), \quad v_A(A_0) = -t.$$

The first bracket in the terminal phase is multiplied by $\alpha u^{p-2} A_0 / (2D^2)$, where D is a unit and α gives ideal conductor $1 + t_2$ on A . The actual assertion needed is therefore

$$(110) \quad L_A := u^{p-1} A_0 (J + u^{p-1} H) \in \mathfrak{p}_A^{1+t_2}.$$

The estimate will be proved with its weight already inserted; an unweighted congruence is not used as a substitute for (110).

A semilinear trace estimate. Write $v = v_B$ in this subsection, and put $z = Y\Delta$. Lemma 7.4 gives

$$(111) \quad v(E_i \Delta) \geq p(p-1)t_2 - pit,$$

$$(112) \quad v(E_i z) \geq p(p-1)t_2 + i(M - pt), \quad 1 \leq i < p.$$

The cyclic norm-power estimate, independently obtained from the strong symmetric bound and the characteristic polynomial, is

$$(113) \quad v_E(N_{E/F} x - x^p) \geq pv_E(x) + (p-1) \text{break}(E/F).$$

It holds also for negative valuations and in both characteristics. Set

$$R = (p-1)M + p(p-1)\delta, \quad B_0 = R + (p-1)t.$$

Lemma 9.18. *The actual signed trace has the approximation*

$$(114) \quad v_B(T + Y^{p-1}s) \geq B_0.$$

Moreover $v_B(T), v_B(Y^{p-1}s) \geq R$.

Proof. The last assertion is the trace estimate in Lemma 7.4 and (111). Subtract Y^p times the characteristic-polynomial identity for Δ from the identity for z . The result is

$$\sum_{i=1}^{p-1} (-1)^i (E_i z - Y^i E_i \Delta) z^{p-i} - N_1 \Delta (n_2 Y - Y^p) = 0.$$

For $i \leq p-2$, (111)–(112) bound the corresponding summand below by

$$pM + p(p-1)t_2 - [p + (p-1)i]t.$$

Equation (113) for B_2/A bounds the last summand below by $pM + p(p-1)t_2 - pt$. Solving for the $i = p-1$ summand and dividing by z , of valuation $M - t > 0$, gives

$$(115) \quad v_B(E_{p-1}z - Y^{p-1}E_{p-1}\Delta) \geq (p-1)M + p(p-1)t_2 - (p-1)^2t = B_0.$$

No division by p occurs.

Newton's identity for degree $p-1$ says

$$S_1(z^{p-1}) = -(p-1)E_{p-1}z + \mathcal{R}(E_1z, \dots, E_{p-2}z),$$

where $\mathcal{R}$ is an integral polynomial whose monomials have weight $p-1$ and at least two elementary-symmetric factors. By (112), every such monomial has valuation at least

$$(p-1)M + p(p-1)(t+2\delta) \geq B_0.$$

Using $E_{p-1}\Delta = -s$ and (115), we obtain $T \equiv (p-1)Y^{p-1}s \pmod{\mathfrak{P}_B^{B_0}}$. In mixed characteristic $v_B(p) \geq p(p-1)t_2$, so

$$v_B(pY^{p-1}s) \geq (p-1)M + p(p-1)(t+2\delta) \geq B_0.$$

In equal characteristic this term is zero. Adding $Y^{p-1}s$ now proves (114), with its plus sign forced by $s = -E_{p-1}$. $\square$

Insert the weight before taking powers.

Lemma 9.19. *In B_1 , put*

$$L_1 = u^{p-1}A_0(T^p + u^{p-1}s^p).$$

Then

$$(116) \quad v_B(L_1) \geq V := 2p(p-1)M - pt + p^2(p-1)\delta.$$

Proof. Let $Z = Y^{p-1}s$ and $E = T + Z$. The identity

$$T^p + Z^p = E^p - \sum_{i=1}^{p-1} \binom{p}{i} T^i Z^{p-i}$$

is exact. Lemma 9.18 gives $v(E) \geq B_0$ and $v(T), v(Z) \geq R$. The weight has valuation

$$v_B(u^{p-1}A_0) = p(p-1)M - p^2t.$$

Its product with E^p therefore has valuation at least V . Every mixed binomial term has valuation at least

$$p(p-1)M - p^2t + v_B(p) + pR \geq V.$$

In equal characteristic these mixed terms vanish.

It remains to replace Z^p by $u^{p-1}s^p$. Equation (113), applied to $Y \in B_2$, gives

$$v_B(u/Y^p - 1) \geq p(p-1)t_2,$$

so

$$v_B(u^{p-1} - Y^{p(p-1)}) \geq p(p-1)M + p(p-1)t_2.$$

After multiplying by $u^{p-1}A_0s^p$, the bound is again at least V by (111). This proves (116). $\square$

Theorem 9.20 (strong weighted bound). *The actual A -valued expression in (110) satisfies*

$$(117) \quad v_A(L_A) \geq 2(p-1)m + (p-1)\delta - \lfloor t/p \rfloor \geq 1 + t_2.$$

Thus the fully weighted first terminal phase is trivial.

Proof. The symmetric-coefficient bounds give $v_1(s) \geq (p-1)\delta$ and $v_1(T) \geq (p-1)(m+\delta)$. Using (113) for B_1/A ,

$$\begin{aligned} v_1(u^{p-1}A_0(T^p - J)) &\geq 2(p-1)M - t + p(p-1)\delta, \\ v_1(u^{2(p-1)}A_0(s^p - H)) &\geq 2(p-1)M - t + p(p-1)\delta. \end{aligned}$$

Both lower bounds are V/p . Lemma 9.19 therefore gives $v_1(L_A) \geq V/p$. Since $L_A \in A$, divide by p again and use integrality of v_A :

$$v_A(L_A) \geq \left\lceil V/p^2 \right\rceil = 2(p-1)m + (p-1)\delta - \lfloor t/p \rfloor.$$

Here $\lceil V/p^2 \rceil$ denotes the ceiling, as usual. Write $t = pa + b$, $a \geq 0$, $1 \leq b \leq p-1$; then $m \geq a+1$. The right side minus $1+t_2$ is at least

$$(p-3)a + 2p - 3 - b + (p-2)\delta \geq p-2 \geq 1.$$

This proves (117). Multiplication by a unit denominator D^{-2} and the rational scalar $1/2$ preserves this ideal bound; then multiplication by α reaches the actual additive conductor. $\square$

9.6. The four-factor identity and transition cancellation.

The stationary data. Use the notation of Subsections 9.2 and 9.3:

$$\begin{aligned} t &= t_1, & t_2 &= t + \delta, & t' &= t + p\delta, & p \nmid t, & \delta \geq 0, & p \text{ odd}, \\ N_i &= N_{B/B_i}, & n_i &= N_{B_i/A}, & S_i &= \text{Tr}_{B/B_i}, & T_i &= \text{Tr}_{B_i/A}, & \Delta^p - \Delta = a \in B_2, \\ b &= N_1\Delta, & N_2\Delta &= a, & v_B\Delta &= -t, & s &= -E_{p-1}^{B/B_1}(\Delta). \end{aligned}$$

The actual minimal characters χ, φ have equal norm pullbacks and conductors $1+t+t'$ and $1+t+t_2$. They are the actual models of Subsection 8.1, not auxiliary characters assumed to have the right stationary coefficients. Put $\theta_1 = \chi(\lambda n_1)$ and $\theta_2 = \varphi(\lambda n_2)$, where $a(\lambda) = 1+r$, $r = t_2 + m$, $M = pm > t$. The exact choices of Subsection 8.3 give

$$\begin{aligned} u &= \alpha/\gamma, & W &= u^{-1}, & v &= \beta/\alpha, & v_A u &= m, & v_A v &= \delta, \\ n_2 w &= \gamma/\alpha = W, & v_2 w &= -m, & n_1 w_1 &= \gamma/\beta, & v_1 w_1 &= -(m + \delta). \end{aligned}$$

Let $\Psi_E(x) = e_E(\alpha x)$, trace compatibly. Its largest trivial ideal on A has exponent $1+t_2$, and those on B_1, B_2 have exponents $1+t', 1+t_2$. The largest trivial ideal for $x \mapsto \Psi_{B_2}(Wx)$ has exponent

$$(118) \quad n_\Gamma = 1 + t_2 + pm.$$

The largest trivial ideal of an additive character is not generally its entire kernel; no converse valuation inference from a single character value will be used.

Set

$$\begin{aligned} C &= 1 + ua, & A_0 &= n_2 a, & D &= 1 + u^p A_0, & P_* &= n_2 C, & H &= n_1 s, \\ T &= S_1((\Delta/w)^{p-1}), & J &= n_1 T, & S &= p(p-1)\delta, & K &= 1 + pt_2. \end{aligned}$$

Thus C, P_*, D are units and $v_A A_0 = -t$. The symmetric-coefficient bounds give

$$(119) \quad v_1 s \geq (p-1)\delta, \quad v_1 T \geq (p-1)(m + \delta).$$

We use their coefficient version as well: for $z \in B_1$,

$$(120) \quad z = \sum_{i=0}^{p-1} c_i \Delta^i, \quad c_i \in B_2 \implies v_2 c_i \geq v_1 z + it.$$

The cyclic symmetric estimate is

$$(121) \quad v_F(E_j^{E/F} x) \geq \left\lfloor \frac{jv_E x + (p-1)(b_0 + 1)}{p} \right\rfloor, \quad 1 \leq j < p,$$

for an edge of break b_0 . It implies

$$(122) \quad v_E(Nx - x^p) \geq pv_E x + (p-1)b_0$$

by Lemma 7.1. All these estimates hold in both characteristics.

The trace-norm theorem in Section 7 gives

$$(123) \quad T_1 b - T_2 a \in \mathfrak{p}_A^{1+t_2}.$$

The actual four factors of Proposition 8.20 are

$$(124) \quad \begin{aligned} Q_1 &= \chi(A_1)/\varphi(C_2), & Q_2 &= \lambda(n_1 A_1/n_2 C_2), \\ Q_3 &= e_{B_1}(-\beta w_1)\theta_1(C_1/A_1), & Q_4 &= \Psi_{B_2}(w), \end{aligned}$$

where $A_1 = \gamma + \alpha b$, $C_1 = A_1 - \beta w_1$ and $C_2 = \gamma - \alpha w + \alpha a$. The full coefficient of θ_1 is C_1 , on the entire P -subgroup of depth $\lceil (1 + t' + pm)/p \rceil$. We use the actual Q_1 logarithm and quadratic evaluation proved in Subsections 9.2 and 9.3, together with the corrected linear W_i formula of Theorem 9.7. All character arguments are in their stated fields.

Weighted denominator bounds.

Lemma 9.21. *One has*

$$(125) \quad \begin{aligned} u^{p-2} H(P_*^{-2} - D^{-2}) &\in \mathfrak{p}_A^{1+t_2}, \\ u^{p-1} A_0 J(P_*^{-2} - D^{-2}) &\in \mathfrak{p}_A^{1+t_2}. \end{aligned}$$

The first assertion also holds with inverse first powers instead of inverse squares.

Proof. The norm expansion and (121) give

$$v_A(P_* - D) \geq \ell := m + \left\lfloor \frac{(p-2)t + (p-1)(\delta+1)}{p} \right\rfloor.$$

Indeed the j th intermediate coefficient has bound $\lfloor [j(pm-t) + (p-1)(t_2+1)]/p \rfloor$, nondecreasing in j . Write $t = pa + b_0$, $1 \leq b_0 \leq p-1$, and $m = a + 1 + c$, $c \geq 0$. Put $f = \lfloor [(p-2)b_0 + (p-1)(\delta+1)]/p \rfloor$. Using (119), the two weighted valuation excesses over $1 + t_2$ are bounded below by

$$(p-3)a + (p-1)c + p-2 + (p-2)\delta - b_0 + f,$$

$$(p-3)a + (2p-1)c + 2p-2 + (p-2)\delta - 2b_0 + f.$$

The first is nonnegative for $b_0 \leq p-2$; if $b_0 = p-1$, use $f \geq p-2$. The second is nonnegative because $b_0 \leq p-1$. All denominators are units; differences of their inverse powers have valuation at least ℓ . This proves every assertion. $\square$

The rational cancellation for the actual second factor. Use the notation of the final formula in Subsections 9.2 and 9.3:

$$F_i = \frac{W}{W^p - W + a^p} \left(\frac{W+a}{w} \right)^i.$$

Put

$$k = \frac{uw}{C}, \quad q_0 = u^{p-1}, \quad F_* = \frac{P_*}{P_* - q_0}, \quad P(X) = \sum_{j=1}^{p-1} \frac{X^j}{j}.$$

These are exactly the variables in Theorem 9.17; in particular $v_2(k) = (p-1)m$. Put $Q = 1 + u^p a^p$ in B_2 , and put

$$\eta = W/w^p, \quad G = \frac{\eta C^p}{Q - q_0}.$$

Both $Q - q_0$ and $P_* - q_0$ are units. Direct multiplication gives

$$(126) \quad F_i = G k^{p-i}, \quad G k^p = \frac{q_0}{Q - q_0}.$$

Lemma 9.22 (the two rational sums cancel to one term). *At the full γ -conductor (118),*

$$(127) \quad -F_{p-1} + \sum_{i=1}^{p-2} \frac{F_i}{i} + F_* P(k) \equiv k(1 - \eta) \pmod{\mathfrak{P}_2^{n_\Gamma}}.$$

Proof. For $2 \leq j \leq p-1$, $1/(p-j) + 1/j = p/[j(p-j)]$. Thus (126) changes the first two terms on the left into $-GP(k)$, with errors divisible by pk^j for $j \geq 2$. In mixed characteristic their valuation is at least $(p-1)t_2 + 2(p-1)m$, whose excess over n_Γ is $(p-2)(t_2+m) - 1 \geq 0$. In equal characteristic the errors vanish. The coefficient of F_{p-1} has not been changed independently.

The norm–power estimate and the binomial theorem give

$$\begin{aligned} v_2(P_* - C^p) &\geq (p-1)t_2, \\ v_2(C^p - Q) &\geq (p-1)t_2 + pm - t, \\ v_2(1 - \eta) &\geq (p-1)t_2. \end{aligned}$$

In particular $P_* - Q$ has depth at least $(p-1)t_2$. The following identity is exact:

$$(128) \quad F_* - G - (1 - \eta) = q_0 \left(\frac{1}{P_* - q_0} - \frac{\eta}{Q - q_0} \right) - \frac{\eta(C^p - Q)}{Q - q_0}.$$

The bracket has depth at least $(p-1)t_2$, as does $F_* - G$. For $j \geq 2$, the valuation of $(F_* - G)k^j$ is at least $(p-1)t_2 + 2(p-1)m \geq n_\Gamma$. For $j = 1$, the first term on the right of (128), multiplied by k , has depth at least

$$(p^2 - 1)m + (p-1)t_2 \geq n_\Gamma.$$

The second has depth at least $(p-1)t_2 + pm - t + (p-1)m$; its excess is

$$(p-2)t_2 + (p-1)m - t - 1 = (p-3)t + (p-2)\delta + (p-1)m - 1 \geq 0.$$

This proves (127) without deleting an unweighted norm–power discrepancy. $\square$

Lemma 9.23 (the remaining elementary rational phase). *There is the exact identity in B_2*

$$(129) \quad W[k(1 - \eta) + w^{-(p-1)}] - w = \frac{a}{w^{p-1}C}(1 - uw^p) \in \mathfrak{P}_2^{1+t_2}.$$

Proof. Use $Wu = 1$, $\eta = 1/(uw^p)$ and $C - 1 = ua$ to obtain the identity by subtraction of the two fractions. The norm–power bound gives $v_2(1 - uw^p) \geq (p-1)t_2$, so the right side has valuation at least $-t + (p-1)m + (p-1)t_2$. Its excess over $1 + t_2$ is $(p-3)t + (p-2)\delta + (p-1)m - 1 \geq 0$. It is its additive phase, not necessarily the element itself, which vanishes. $\square$

Representatives in B_2 and norm transfer. For $z = \sum_{i=0}^{p-1} z_i \Delta^i$ with $z_i \in B_2$, let $\text{pr}_0 z = z_0$. Define

$$(130) \quad X_s = \frac{s}{w^{p-2}C^2}, \quad X_T = \frac{aT}{w^{p-1}C^2}, \quad y_s = \text{pr}_0 X_s, \quad y_T = \text{pr}_0 X_T.$$

The first two expressions are in B ; the last two are actual elements of B_2 . No B_2 character will be applied to X_s or X_T .

Lemma 9.24 (constant projection and its norm). *Let $z = xw_0$ with $x \in B_2$, $w_0 \in B_1$. Suppose*

$$(131) \quad h := v_B z \geq 1 + t', \quad h \geq 1 + t_2 + (p-3)t.$$

Then $y = \text{pr}_0 z \in B_2$ satisfies

$$(132) \quad z \equiv y \pmod{\mathfrak{P}_B^K}, \quad N_1 z \equiv n_2 y \pmod{\mathfrak{P}_1^K}.$$

If $z \neq 0$, then $v_B y = h$.

Proof. The zero case is immediate. Write $z = \sum y_i \Delta^i$ using (120) for w_0 . It gives

$$v_2 y_i \geq h/p + it, \quad v_B(y_i \Delta^i) \geq h + i(p-1)t.$$

Thus all nonconstant terms are strictly deeper than h and the constant term has valuation h . Since $1 + t' + (p-1)t = K$, all nonconstant terms belong to $\mathfrak{P}_B^K$. This proves the first congruence, with $y = y_0$.

The second bound in (131) and the congruence just proved are precisely the hypotheses of Lemma 7.11, with $K_0 = K$. Applying that lemma to $z = xw_0$ and $y = y_0$ proves the norm congruence in (132). $\square$

Lemma 9.25 (all source and target depths). *The elements in (130) satisfy*

$$(133) \quad \begin{aligned} X_s &\equiv y_s \pmod{\mathfrak{P}_B^K}, & X_T &\equiv y_T \pmod{\mathfrak{P}_B^K}, \\ n_2 y_s &\equiv u^{p-2} H / D^2 \pmod{\mathfrak{p}_A^{1+t_2}}, & n_2 y_T &\equiv u^{p-1} A_0 J / D^2 \pmod{\mathfrak{p}_A^{1+t_2}}. \end{aligned}$$

Moreover $y_s, y_T \in \mathfrak{P}_2^{c_2}$, where $c_2 = \lceil (1+t_2)/p \rceil$.

Proof. By (119),

$$h_s := v_B X_s \geq (p-2)M + S, \quad h_T := v_B X_T \geq 2(p-1)M - pt + S.$$

At $M \geq t+1$, their excesses over $1+t'$ are at least

$$(p-3)(t+1) + p(p-2)\delta, \quad (p-3)t + 2p - 3 + p(p-2)\delta,$$

and their excesses over $1+t_2 + (p-3)t$ are at least

$$p-3 + [p(p-1)-1]\delta, \quad 2p-3 + [p(p-1)-1]\delta.$$

All are nonnegative, including $p=3, \delta=0$. Lemma 9.24 therefore applies to both products and proves the first two assertions of (133). It also gives $v_B y_s, v_B y_T \geq 1+t_2$, hence the stated depth c_2 in B_2 . The norm part of that lemma gives

$$n_2 y_s \equiv \frac{u^{p-2} s^p}{P_*^2}, \quad n_2 y_T \equiv \frac{u^{p-1} A_0 T^p}{P_*^2} \pmod{\mathfrak{P}_1^K}.$$

Here N_1 fixes B_1 with norm the p th power, and its restriction to B_2 is n_2 ; these two domains have not been interchanged. By (122) on B_1/A ,

$$\begin{aligned} v_1(u^{p-2}(s^p - H)) &\geq (p-2)M + p(p-1)\delta + (p-1)t, \\ v_1(u^{p-1}A_0(T^p - J)) &\geq 2(p-1)M + p(p-1)\delta - t. \end{aligned}$$

The first excess over K is at least $(p-3)(t+1) + p(p-2)\delta$, and the second at least $(p-3)t + 2p - 3 + p(p-2)\delta$. Thus the two powers may be replaced by H, J . Now both sides lie in A , so $A \cap \mathfrak{P}_1^K = \mathfrak{p}_A^{1+t_2}$ applies. Lemma 9.21 replaces P_*^{-2} by D^{-2} at precisely that weighted modulus, proving the remaining assertions. $\square$

Lemma 9.26 (actual norm character and the rational half). *One has*

$$(134) \quad \begin{aligned} \Psi_{B_2}(y_s/2) &= \Psi_A(-u^{p-2}H/(2D^2)), \\ \Psi_{B_2}(-y_T/2) &= \Psi_A(u^{p-1}A_0J/(2D^2)). \end{aligned}$$

Proof. The actual norm character of B_2/A has full coefficient α . For any $y \in \mathfrak{P}_2^{c_2}$ choose x with $P(x) = y$, using bijectivity of P on that positive ideal. The cyclic coefficient estimate places $n_2(1-x)$ in the full P -domain of that character. Its value on this norm is 1. The norm-logarithm identity of Lemma 6.3, with break t_2 and $h=0$, gives at modulus $\mathfrak{p}_A^{1+t_2}$

$$P(1 - n_2(1-x)) \equiv T_2 P(x) + n_2(P(x)).$$

Consequently $\Psi_{B_2}(y)\Psi_A(n_2y) = 1$; this is the actual norm-character phase of Corollary 6.4, not an auxiliary model equality. For the half, the product $\Psi_{B_2}(y/2)\Psi_A(n_2y/2)$ has square 1 and has p -primary order. Indeed the image of a continuous additive character of a local field of residue characteristic p is p -primary: sufficiently large p -powers of each argument lie in an open trivial ideal. As p is odd, this product is 1. This argument does not move $1/2$ inside the norm, and does not extend phase vanishing under an arbitrary local-unit multiplier. Apply it to y_s, y_T and use Lemma 9.25. The errors after multiplication by $1/2$ still lie in the whole trivial ideal $\mathfrak{p}_A^{1+t_2}$. $\square$

Proposition 9.27 (the actual $Q_1Q_2Q_4$ product). *The genuine character values satisfy*

$$(135) \quad (Q_1Q_2Q_4)^{-1} = \Psi_A \left(W u^{p-1} \frac{H}{D} - \frac{u^{p-2}H}{2D^2} + \frac{u^{p-1}A_0J}{2D^2} \right).$$

Proof. The end of Subsections 9.2 and 9.3 expresses the actual Q_1 exponent, a member of B_2 , modulo $\mathfrak{P}_B^K$ as

$$-WF_{p-1} + W \sum_{i=1}^{p-2} F_i/i + W/w^{p-1} + \frac{p-1}{p-2} X_s + \frac{1}{2(p-1)} X_T.$$

Replace X_s, X_T by y_s, y_T using Lemma 9.25. The resulting expression belongs to B_2 , so its difference from the genuine exponent lies in $B_2 \cap \mathfrak{P}_B^K = \mathfrak{P}_B^{1+t_2}$ and has trivial Ψ_{B_2} phase. This supplies the field-valued representatives required by the calculation, rather than silently applying Ψ_{B_2} to B .

The two coefficient replacements are

$$\frac{p-1}{p-2} - \frac{1}{2} = \frac{p}{2(p-2)}, \quad \frac{1}{2(p-1)} + \frac{1}{2} = \frac{p}{2(p-1)}.$$

Their errors have common-field valuations at least $v_B(p) + h_s$ and $v_B(p) + h_T$, respectively, and hence at least K . They vanish identically in equal characteristic. Thus the surviving exceptional phase is $\Psi_{B_2}((y_s - y_T)/2)$. Theorem 9.17 and Lemma 9.22 combine the two rational sums. Their error is at the full γ modulus, not only the α modulus. Including W/w^{p-1} and $Q_4^{-1} = \Psi_{B_2}(-w)$, Lemma 9.23 kills the remaining elementary phase. The product is therefore exactly

$$\Psi_A(Wu^{p-1}H/D) \Psi_{B_2}((y_s - y_T)/2).$$

Finally apply (134). This proves (135) with every scalar outside the appropriate norm. $\square$

The third factor Q_3 . Put $h_0 = m + \delta$ and $z = \beta w_1/A_1$. Then

$$v_1 z = (p-1)h_0, \quad n_1^{\text{char}} = 1 + t' + pm = 1 + t + ph_0$$

is the conductor of θ_1 .

Lemma 9.28 (direct Q_3 evaluation). *One has*

$$(136) \quad Q_3^{-1} = \Psi_A \left(-\frac{W(vu)^{p-1}}{2D} \right).$$

Proof. The inequalities $p(p-1)h_0 \geq n_1^{\text{char}}$ and $3(p-1)h_0 \geq n_1^{\text{char}}$ follow from $pm \geq t+1$: for the latter the excess is $(2p-3)h_0 - t - 1 \geq 0$, and the former is no smaller. Thus the full P -formula of θ_1 , whose coefficient is $C_1 = A_1(1-z)$, applies at z with all terms of degree at least three in the following expression beyond its conductor:

$$Q_3 = e_{B_1}(A_1[(1-z)P(z) - z]) = e_{B_1}(-A_1 z^2/2).$$

The identity used here is exact before deleting the higher terms; for $p=3$ the cubic term is $-z^3/2$ and the same displayed inequality includes it. Consequently

$$Q_3^{-1} = e_{B_1} \left(\frac{\beta^2 w_1^2}{2A_1} \right).$$

Set

$$y = \frac{(\beta/\gamma)w_1^2}{1+ub}.$$

Its valuation is $(p-2)h_0 \geq \rho := \lceil t/p \rceil$. The actual norm character of B_1/A , with coefficient β , has the norm phase on all of this ideal. The proof is exactly the argument of Lemma 9.26 on that edge; here $\rho = \lceil (t+1)/p \rceil$ because $p \nmid t$. It gives

$$e_{B_1}(\beta y/2) = e_A(-\beta n_1 y/2).$$

The exact norm choice $n_1 w_1 = \gamma/\beta$ now implies

$$(137) \quad Q_3^{-1} = e_A \left(-\frac{\gamma}{2} \frac{(\beta/\gamma)^{p-1}}{n_1(1+ub)} \right).$$

No norm is replaced by a p th power in this calculation.

The endpoint of the norm expansion in the denominator is $n_1(ub) = u^p A_0$, so (121) on the lower edge of break t gives

$$v_A(n_1(1 + ub) - D) \geq m + \left\lfloor \frac{(p-2)t + p-1}{p} \right\rfloor.$$

The numerator weight $(\beta/\gamma)^{p-1}$ has valuation $(p-1)(m + \delta)$. After subtracting the γ -conductor $r + 1$, the weighted denominator error has excess at least

$$(p-1)m + (p-2)\delta + \left\lfloor \frac{(p-2)t + p-1}{p} \right\rfloor - t - 1 \geq 0.$$

To verify the last inequality write $t = pa + b_0$, $1 \leq b_0 \leq p-1$, and $m = a + 1 + c$. The expression becomes

$$(p-3)a + (p-1)c + p-2 - b_0 + \left\lfloor \frac{(p-2)b_0 + p-1}{p} \right\rfloor + (p-2)\delta.$$

It is nonnegative for $b_0 \leq p-2$; for $b_0 = p-1$ the floor is $p-2$. This includes the equality boundary $p=3, b_0=2$. Both denominators are units. Hence (137) permits replacement by D at the actual additive modulus, yielding (136) because $\gamma = \alpha W$ and $\beta/\gamma = vu$. $\square$

The terminal phase and its cancellation.

Theorem 9.29 (actual transition phase). *The actual four character factors satisfy*

$$(138) \quad (Q_1 Q_2 Q_3 Q_4)^{-1} = \Psi_A(\phi),$$

where the expression on the right has now been derived from those factors and is

$$(139) \quad \phi = -\frac{W(vu)^{p-1}}{2} \frac{H}{D} + W u^{p-1} \frac{H}{D} - \frac{u^{p-2} H}{2D^2} + \frac{u^{p-1} A_0 J}{2D^2}.$$

Proof. Combine Proposition 9.27 and Lemma 9.28. In particular, (138) is not inferred from vanishing of a candidate terminal expression or from the desired equality of local constants. $\square$

Corollary 9.30 (transition cancellation). *For the actual model supplied by Subsections 8.1 and 8.3 and the local inputs specified above, $Q_1 Q_2 Q_3 Q_4 = 1$ in both characteristics.*

Proof. Since $Wu = 1$ and $D = 1 + u^p A_0$, direct expansion gives the exact algebraic identity

$$(140) \quad \phi = \frac{L_A}{2D^2} + \frac{u^{p-2}}{2D} (H - v^{p-1}), \quad L_A = u^{p-1} A_0 (J + u^{p-1} H).$$

The independently proved, fully weighted estimate of Theorem 9.20 is

$$v_A L_A \geq 2(p-1)m + (p-1)\delta - [t/p] \geq 1 + t_2.$$

The actual-character R_2 of Theorem 8.14 is

$$H - v^{p-1} \in \mathfrak{p}_A^{L+(p-1)\delta}, \quad L = 1 + t - [t/p].$$

The second term of (140) has depth at least $1 + t_2$ because, with $\rho = [t/p]$ and $m \geq \rho$,

$$(p-2)m + L + (p-1)\delta - (1 + t_2) = (p-2)(m + \delta) - \rho \geq 0.$$

The unit denominator and rational half preserve these whole ideal bounds. Both terms therefore lie in the largest trivial ideal of Ψ_A , and Theorem 9.29 proves the assertion. Neither an unweighted substitute for the first estimate nor an exact auxiliary norm for β/α is used. $\square$

Corollary 9.31 (actual local factors in this transition row). *Under the same explicit inputs, the two actual normalized local factors agree in the totally ramified transition row $pm > t$.*

Proof. Subsection 8.3 derives the full Ueda-normalized Lamprecht formula for these actual characters, with its complete odd-conductor term:

$$\frac{\Delta_{B_2}(\theta_2, e_{B_2})}{\Delta_{B_1}(\theta_1, e_{B_1})} = \frac{g_2}{g_1} Q_1 Q_2 Q_3 Q_4 \Psi_A(T_1 b - T_2 a), \quad g_i \in \mu_4.$$

The g_i are the full evaluated residue factors, not merely their polar forms. Equation (123) and Corollary 9.30 put this actual quotient in μ_4 . Independently, Ueda's First Main Lemma and

the odd cyclic lambda products give its p th power equal to 1, as proved for the actual pair in Corollary 4.3. Since p is odd, $\mu_p \cap \mu_4 = \{1\}$, proving equality. No equality of local constants was used to establish any field congruence or character phase above. $\square$

9.7. Conductor reduction and completion of the odd-prime case.

Local notation. Let B/A be totally ramified with Galois group C_p^2 , p odd. Let B_1, B_2 be the distinguished lower fields, with breaks

$$t = t_1 \leq t_2 = t + \delta, \quad t' = t + p\delta, \quad p \nmid t.$$

The upper breaks are $u_1 = t'$ and $u_2 = t$. Set $N_i = N_{B/B_i}$, $n_i = N_{B_i/A}$ and $T_i = \text{Tr}_{B_i/A}$ and $e_E = e_A \text{Tr}_{E/A}$. Suppose $\psi_1 N_1 = \psi_2 N_2 = \psi_B$ is non-descending from A . It is invariant under all of $\text{Gal}(B/A)$, because it is invariant under each of the two distinct upper cyclic subgroups.

The exact conductor reduction. Let $a(\theta)$ denote a multiplicative conductor exponent. For a generator τ_i of $\text{Gal}(B_i/A)$ put $\mu_i = \psi_i^{\tau_i} / \psi_i$. By Lemma 4.2, μ_i generates the upper norm-character group, the restrictions $\psi_1|_A, \psi_2|_A$ agree, and Corollary 4.3 supplies the common p th power.

Lemma 9.32 (primitive lower bound). *For $i = 1, 2$,*

$$a(\psi_i) \geq 1 + t_i + u_i.$$

Proof. The nontrivial μ_i of Lemma 4.2 has conductor $u_i + 1$. Choose $y \in U_i^{u_i}$ with $\mu_i(y) \neq 1$. Write $y = 1 + z$, with $v_i z \geq u_i$. The ramification expansion gives $v_i(\tau_i^{-1} z - z) \geq t_i + v_i z$, so $\tau_i^{-1} y / y \in U_i^{t_i + u_i}$. Its value under ψ_i is $\mu_i(y) \neq 1$. Thus $a(\psi_i) > t_i + u_i$, as asserted. No equality in the ramification estimate is needed. $\square$

Lemma 9.33 (deep norm-kernel descent). *Let E/A be a ramified cyclic edge with break s , and let M/E be the other upper edge, of break u with $p \nmid u$. If θ is an endpoint of a primitive compatible pair, then*

$$\theta(x) = 1 \quad \text{for } x \in U_E^{1+s+u}, \quad N_{E/A} x = 1.$$

Proof. Hilbert 90 gives $x = \sigma(y)/y$. If $x = 1$ there is nothing to prove. Multiply y by an element of $A^\times$ so that its valuation is in $\{0, \dots, p-1\}$. If this valuation is nonzero, the first ramification term of $\sigma(y)/y$ has depth exactly s : its uniformizer contribution is nonzero modulo p , while the unit contribution has depth at least $s+1$. This contradicts the depth of x . Thus y is a unit, and a residue-field lift from A makes $y \in U_E^1$.

Choose the representative of $yA^\times$ with maximal possible unit depth q . Such a finite maximum exists: otherwise representatives from the closed subfield A approximate y arbitrarily closely, forcing $y \in A^\times$ and $x = 1$. One has $p \nmid q$. In fact

$$U_E^{pk} = U_A^k U_E^{pk+1} \quad (k \geq 1),$$

because the image of $\mathfrak{p}_A^k$ in $\mathfrak{P}_E^{pk} / \mathfrak{P}_E^{pk+1}$ is the entire common residue field. If $q = pk$, this would improve the representative's depth, a contradiction. The first ramification term for a depth- q unit is consequently nonzero and has depth $q + s$. Since $x \in U_E^{1+s+u}$, it follows that $q \geq 1 + u$. The commutator θ^σ / θ has conductor $u + 1$. It is trivial on this y , so $\theta(\sigma(y)/y) = 1$, with either consistent convention for the generator. This proves the lemma. $\square$

Proposition 9.34. *Every primitive compatible pair can be written*

$$\psi_1 = \chi(\lambda n_1), \quad \psi_2 = \varphi(\lambda n_2),$$

where $\chi N_1 = \varphi N_2$ is primitive and

$$a(\chi) = m_1 = 1 + t + t', \quad a(\varphi) = m_2 = 1 + t + t_2.$$

If $a(\psi_2) > m_2$, then $a(\lambda) = 1 + r$ with $r > t_2$, and $p(r - t_2) > t$. Thus this is precisely the transition range of Subsection 8.3, without a missing intermediate conductor row.

Proof. By Lemma 9.33, ψ_1 on $U_1^{m_1}$ factors through $n_1(U_1^{m_1})$. The factored character is continuous, because the unit groups are compact and the norm map is a quotient onto its image. Extend it to a continuous character λ of $A^\times$; the image is open, and divisibility of $\mathbf{C}^\times$ extends a character across the resulting discrete quotient. Remove λn_i from both endpoints. The pair remains compatible and primitive, and its first conductor is at most m_1 . Lemma 9.32 makes it exactly m_1 .

Both endpoint conductors are greater than their upper breaks plus one, by Lemma 9.32. Therefore the high cyclic conductor formula for their common norm pullback gives

$$p a(\chi) - (p-1)(t'+1) = p a(\varphi) - (p-1)(t+1).$$

It follows that $a(\varphi) = m_2$. If $a(\psi_2) > m_2$, then $a(\lambda n_2) = a(\psi_2) > m_2$. The lower conductor formula forces $a(\lambda) = 1 + r > 1 + t_2$ and

$$a(\psi_2) = 1 + t_2 + p(r - t_2) > 1 + t + t_2,$$

which is equivalent to the asserted range. $\square$

The minimal-conductor row.

Theorem 9.35. *If the primitive compatible pair has conductors m_1, m_2 , its two complete Second-Main-Lemma factors are equal.*

Proof. Choose the actual Δ furnished by Theorem 8.8 and write $b = N_1\Delta$, $a = N_2\Delta$. Its full coefficients are $C_1 = \alpha b$, $C_2 = \alpha a$, where $e_A(\alpha \cdot)$ has ideal conductor $1 + t_2$. The full Lamprecht formula of Subsection 8.3, including its evaluated odd residual factor, gives

$$\frac{\Delta_{B_2}(\varphi)}{\Delta_{B_1}(\chi)} = \frac{g_2}{g_1} \frac{\chi(\alpha b)}{\varphi(\alpha a)} e_A(\alpha[T_1 b - T_2 a]), \quad g_i \in \mu_4.$$

Lemma 4.2 cancels the two values at α . Compatibility, evaluated on $\Delta \in B^\times$, cancels $\chi(b)/\varphi(a)$. The trace-norm bound of Theorem 7.5 $T_1 b - T_2 a \in \mathfrak{p}_A^{1+t_2}$ kills the last factor. Thus the ratio lies in μ_4 . Corollary 4.3 gives its p th power equal to 1; since p is odd, the ratio is 1. That lemma also identifies each lower cyclic norm-character product with 1, so these are the complete target factors, not just a quotient missing lambda constants. $\square$

Theorem 9.36 (The totally ramified branch). *For a totally ramified C_p^2 diamond over a local field of odd residue characteristic, every compatible non-descending pair gives equal complete Second Main Lemma factors. This holds in mixed and equal characteristic.*

Proof. First compare the distinguished smallest-break field with an arbitrary other lower field, using the notation fixed above. If the second conductor is m_2 , the conductor equality in Proposition 9.34 gives m_1 on the first side, and Theorem 9.35 applies. Otherwise Proposition 9.34 writes the pair as a common base twist of an actual minimal pair, with $p(r - t_2) > t$. The simultaneous choices of Subsection 8.3 realize that pair without changing either local constant. The actual four-factor calculation of Subsection 9.6, using Subsections 9.3 and 9.4, puts the quotient in μ_4 . Corollary 4.3 puts it in μ_p , so it is 1. The lower cyclic products are also 1 by that corollary. For a pair not containing the distinguished field, extend its invariant common pullback to that field by Hilbert 90 and character extension; the two distinguished comparisons then give the desired equality by transitivity. The preceding proofs include both characteristics and $p = 3$. $\square$

10. DYADIC EXTENSIONS WITH AN UNRAMIFIED INTERMEDIATE FIELD

10.1. The local configuration and notation. Let F be a nonarchimedean local field of residue characteristic 2, in mixed or equal characteristic. Let K/F be biquadratic. For a quadratic intermediate field L , let ω_L be the nontrivial character of $F^\times / N_{L/F} L^\times$. A primitive compatible pair means characters θ_i of $L_i^\times$ with the same pullback Θ to $K^\times$, where Θ does not descend from F through $N_{K/F}$. Fix ψ_F and use $\psi_L = \psi_F \text{Tr}_{L/F}$. The target is equality of

$$(141) \quad A_L(\theta_L) = \Delta_L(\theta_L, \psi_L) \Delta_F(\omega_L, \psi_F).$$

The omitted trivial-character factor is 1. Write $a_L(\theta)$ for the multiplicative conductor, n_L for the additive conductor in the convention that $\mathfrak{p}_L^{-n_L}$ is the largest trivial ideal, and $v_L(\mathfrak{p}_L) = 1$. On a ramified quadratic edge L/F of break t , put $T = t + 1$. Its different exponent is T , and

$$(142) \quad n_L = 2n_F + T, \quad \mathrm{Tr}_{L/F}(\mathfrak{p}_L^j) = \mathfrak{p}_F^{\lfloor (j+T)/2 \rfloor}, \quad v_F(\mathrm{N}_{L/F} x) = v_L(x).$$

For an unramified edge, conductors and normalized valuations restrict without a ramification multiplier.

We use the one-dimensional results recalled in Sections 2–4, and the local norm-character identifications of Appendix D.

Put $N_i = \mathrm{N}_{K/L_i}$, $n_i = \mathrm{N}_{L_i/F}$, and let ν_i denote the upper norm character. Lemma 4.2 gives $\theta_i^{\sigma_i} = \theta_i \nu_i$ and the common restriction $D = \theta_i|_F \omega_i$. Proposition 4.4 leaves a possible sign; the residual calculations below determine that sign.

10.2. The quadratic norm-character phase. Let E/F be ramified quadratic of break $t \geq 1$, put $T = t + 1$ and $q = \lceil T/2 \rceil$, and let τ be its actual norm character. By additive duality on U_F^q/U_F^T , choose α such that

$$(143) \quad \tau(1 - z) = \Psi_F(z), \quad z \in \mathfrak{p}_F^q, \quad \Psi_L(w) := \psi_L(\alpha w), \quad v_F \alpha = -n_F - T.$$

Both Ψ_F and Ψ_E have largest trivial ideal with exponent T . The linear chart is valid because $2q \geq T$; there is no coefficient $1/2$. In mixed characteristic put $e = v_F(2)$; in equal characteristic put $e = +\infty$. The trace of 1 in (142) gives

$$(144) \quad e \geq \lfloor T/2 \rfloor, \quad e + q \geq T.$$

Lemma 10.1 (Actual dyadic norm phase). *For every $z \in \mathfrak{p}_E^q$,*

$$(145) \quad \Psi_E(z) = \Psi_F(\mathrm{N}_{E/F} z).$$

Proof. The exact quadratic identity is

$$\mathrm{N}(1 - z) = 1 - \mathrm{Tr} z + \mathrm{N} z.$$

Both $\mathrm{Tr} z$ and $\mathrm{N} z$ have valuation at least q , by (142). Evaluate (143) at $\mathrm{Tr} z - \mathrm{N} z$ and use $\tau(\mathrm{N}(1 - z)) = 1$. This gives $\Psi_F(\mathrm{Tr} z - \mathrm{N} z) = 1$, which is (145). $\square$

Remark 10.2. The sign in (145) is positive. The odd-prime expression with a negative norm term on the right must not simply be imported at $p = 2$. This proof is exact on its entire stated ideal; it does not assert (145) on shallower ideals.

Lemma 10.3 (The odd quadratic conductor). *If T is odd, then $\mathrm{char} F = 0$ and $T = 2e + 1$. In that case, with k the residue field,*

$$(146) \quad \Psi_F(4z) = (-1)^{\mathrm{Tr}_{k/\mathbb{F}_2}(\bar{z})}, \quad z \in \mathcal{O}_F.$$

Proof. Write $T = 2r + 1$, so $r \geq 1$, and $e \geq r$ by (144). If $e > r$, including equal characteristic, the square $(1 + \pi^r z)^2$ is $1 + \pi^{2r} z^2$ modulo $\mathfrak{p}_F^T$. Since τ has order 2, this would make its nontrivial last-layer additive character trivial on every residue square. Frobenius is onto k , a contradiction. Hence $e = r$. Now $v_F 4 = 2e = T - 1$, so $z \mapsto \Psi_F(4z)$ is a nontrivial residue additive character. The equality $(1 + 2z)^2 = 1 + 4(z + z^2)$ and the order of τ show that it kills $\bar{z} + \bar{z}^2$. The image of this Artin–Schreier map is exactly the absolute-trace-zero hyperplane: its kernel has two elements and its image has zero trace. The unique nontrivial character of the quotient is the right side of (146). $\square$

10.3. Minimal characters and common twists. Assume from now through Section 10.6 that U/F is unramified quadratic, E/F ramified quadratic of break t , and $K = EU$. Write

$$N_U = \mathrm{N}_{K/U}, \quad N_E = \mathrm{N}_{K/E}, \quad n_U = \mathrm{N}_{U/F}, \quad n_E = \mathrm{N}_{E/F}.$$

Let $\eta = \omega_U$ and $\tau = \omega_E$. The norm character on K/U is $\tau_U = \tau n_U$. All four $\Psi_L = \psi_L(\alpha \cdot)$ have largest trivial ideal $\mathfrak{p}_L^T$. Let $k \subset \kappa$ be the residue fields of $F \subset U$, and put $Q = |k|$. Choose $c \in \mathcal{O}_F$ whose residue has absolute trace 1, and choose

$$(147) \quad d \in \mathcal{O}_U, \quad d^2 - d = c.$$

This polynomial gives the unramified quadratic extension. For its nontrivial automorphism σ ,

$$\sigma(d) = 1 - d, \quad \text{Tr}_{U/F} d = 1, \quad n_U(d) = -c.$$

In particular c, d are units and $\bar{d} \notin k$.

Theorem 10.4 (Compatible minimal characters on the full multiplicative groups). *There exist characters χ_U^0, χ_E^0 , both of conductor T , with a primitive common norm pullback, such that*

$$(148) \quad (\chi_U^0)^\sigma / \chi_U^0 = \tau_U,$$

$$(149) \quad \chi_U^0(1 - z) = \Psi_U(d^2 z) \quad (z \in \mathfrak{p}_U^q),$$

$$(150) \quad \chi_E^0(1 - z) = \Psi_E(cz) \quad (z \in \mathfrak{p}_E^q).$$

These are identities on the full specified subgroups, with actual characters on the entire multiplicative groups.

Proof. The right side of (149) is a character on $H = U_U^q$, since $2q \geq T$. Its exact conductor is T . Unramified norm expansion shows

$$\tau_U(1 - z) = \Psi_U(z) \quad (z \in \mathfrak{p}_U^q),$$

because $v_F n_U z \geq 2q \geq T$. Further, $\sigma(d^2) - d^2 = 1 - 2d$; the $2d$ error is killed on this whole ideal by (144). Thus the prescribed character on H has the required conjugate quotient on H .

Here is the extension to all units, with the commutator retained. Put $V = U_U^1$ and $I = \{\sigma(v)/v : v \in V\}$. Prescribe $\kappa_I(\sigma(v)/v) = \tau_U(v)$. This is well defined because $\tau_U(f) = \tau(f^2) = 1$ on U_F^1 . If $\sigma(v)/v \in H$, then v modulo $\mathfrak{p}_U^q$ is fixed, hence is represented by a unit of F ; write $v = fh$ with $f \in U_F^1$, $h \in H$. (For unramified extensions this follows coefficient by coefficient in the uniformizer expansion.) The characters prescribed on I and H consequently agree on the intersection. Extend their product character from IH/U_U^T across the finite abelian group V/U_U^T . Make it trivial on the odd-order Teichmüller group and on a uniformizer of F . The character τ_U is trivial on both these factors. The resulting χ_U^0 has (148) on all of $U^\times$ and has exact conductor T .

Its pullback to K is invariant under both quadratic automorphisms, since τ_U kills $N_U K^\times$. Hilbert 90 makes this pullback trivial on $\ker N_E$; it therefore descends and extends to χ_E^0 on $E^\times$. The unramified norm is onto units, so only an unramified value is left to choose in this extension. For $z \in \mathfrak{p}_K^q$, the exact ramified quadratic norm gives

$$\chi_U^0(N_U(1 - z)) = \Psi_K(d^2 z) \Psi_U(-d^2 N_U z) = \Psi_K((d^2 - d)z) = \Psi_K(cz).$$

The middle equality uses Lemma 10.1 on dz and $N_U(dz) = d^2 N_U z$, an exact equality. The unramified norm onto U_E^q and its expansion modulo $\mathfrak{p}_E^T$ now give (150). Since c is a unit, its conductor is exactly T . Nontriviality of (148) excludes descent of the common pullback from F . $\square$

Proposition 10.5 (Realizing any given pair). *For every primitive compatible pair (θ_U, θ_E) there is a character λ of $F^\times$, after an unramified adjustment of χ_E^0 if necessary, such that*

$$(151) \quad \theta_U = \chi_U^0(\lambda n_U), \quad \theta_E = \chi_E^0(\lambda n_E).$$

There is an integer $h \geq 0$ with

$$(152) \quad a_U(\theta_U) = T + h, \quad a_E(\theta_E) = T + 2h.$$

For $h > 0$, $a_F(\lambda) = T + h$; for $h = 0$, $a_F(\lambda) \leq T$.

Proof. Lemma 4.2 identifies the actual conjugate quotient over U with τ_U , with no choice of power in a group of order 2. Thus θ_U / χ_U^0 is invariant. Hilbert 90 and character extension make it λn_U . The other quotient is a norm character for the unramified edge K/E , so it is absorbed by the unramified adjustment. The conductor of θ_U is at least T , the conductor of its commutator. If it is greater, the common twist has this conductor, since χ_U^0 has conductor T and unramified pullback preserves conductor. At the endpoint it has conductor at most T . Both twists of θ_U by τ_U are conjugate and have the same conductor. Ueda's norm-pullback conductor formula,

including its no-drop criterion at T , gives upstairs conductor $T + 2h$. The unramified edge K/E preserves it. $\square$

10.4. Common norm representatives. For $0 \leq h \leq t$ put

$$s_U = \lceil (T + h)/2 \rceil, \quad s_E = \lceil T/2 \rceil + h, \quad d_U = \lfloor (T + h)/2 \rfloor.$$

Choose a coefficient A_* of the base twist on its half-conductor layer:

$$(153) \quad \lambda(1 - z) = \Psi_F(A_* z) \quad (z \in \mathfrak{p}_F^{s_U}).$$

For $h > 0$, $v_F A_* = -h$; at $h = 0$ the coefficient can be chosen integral. Its ambiguity is $\mathfrak{p}_F^{T-s_U}$.

Lemma 10.6. *There is $x \in E$, with $A = n_E x$, such that A can replace A_* in (153). If $h > 0$, $v_E x = -h$; at $h = 0$, x is integral. The zero coefficient class may be represented by $x = 0$.*

Proof. For $h > 0$ the required relative precision is $T - s_U + h = d_U \leq t$, since $T + h \leq 2T - 1$. Thus the subcritical norm representative theorem applies. For an integral coefficient of valuation $a < T - s_U$ use relative precision $T - s_U - a \leq t$. If the integral class is zero, choose zero. No approximation beyond the break is requested. $\square$

Put

$$(154) \quad B_U = A + d^2, \quad B_E = A - x + c, \quad C = x + d, \quad Z_U = N_U C, \quad Z_E = N_E C, \quad b = \text{Tr}_{E/F} x.$$

Lemma 10.7 (The actual coefficients). *For the given characters in (151),*

$$(155) \quad \theta_U(1 - z) = \Psi_U(B_U z) \quad (z \in \mathfrak{p}_U^{s_U}), \quad \theta_E(1 - z) = \Psi_E(B_E z) \quad (z \in \mathfrak{p}_E^{s_E}).$$

Their coefficient valuations are $-h$ and $-2h$, respectively.

Proof. For the unramified pullback of λ , the extra norm term has phase valuation at least $-h + 2s_U \geq T$; use (149). On the ramified side, $u = \text{Tr } z - n_E z$ belongs to $\mathfrak{p}_F^{s_U}$ for $z \in \mathfrak{p}_E^{s_E}$, because

$$s_E \geq s_U, \quad \lfloor (s_E + T)/2 \rfloor \geq s_U.$$

Then (153) and Lemma 10.1, applied to xz of valuation at least $\lceil T/2 \rceil$, give

$$\lambda(n_E(1 - z)) = \Psi_F(A \text{Tr } z - A n_E z) = \Psi_E((A - x)z).$$

Together with (150) this proves the formulas. For $h > 0$, A is the unique leading term in each coefficient. At $h = 0$, write $\xi = \bar{x} \in k$. The residues are $\xi^2 + \bar{d}^2$ and $\xi^2 - \xi + \bar{c}$. The first is nonzero since $\bar{d} \notin k$, and the second has absolute trace 1. $\square$

The norms of the *same* element C have the exact expressions

$$(156) \quad Z_U = A + bd + d^2,$$

$$(157) \quad Z_E = x^2 + x - c,$$

$$(158) \quad S := \text{Tr}_{U/F} Z_U - \text{Tr}_{E/F} Z_E = 1 + 4c - \mathcal{D}_x, \quad \mathcal{D}_x = b^2 - 4A = (2x - b)^2.$$

Indeed $\text{Tr } d = 1$, $\text{Tr } d^2 = 1 + 2c$, and $\text{Tr } x^2 = b^2 - 2A$. These identities do not use translated roots in characteristic zero.

The normalized factors and residual functions are those of Lemma 4.1, with $J = T$. We denote them by H_L, g_L .

10.5. The minimal-conductor comparison.

Theorem 10.8. *Every primitive compatible pair with $a_U(\theta_U) = a_E(\theta_E) = T$ satisfies*

$$(159) \quad \Delta_U(\theta_U) \Delta_F(\eta) = \Delta_E(\theta_E) \Delta_F(\tau).$$

This includes the odd-conductor maximal-break case in mixed characteristic and all equal-characteristic minimal pairs.

Even T . Here $T = 2r$ and x in Lemma 10.6 is integral. The differences $Z_U - B_U = bd$ and

$$Z_E - B_E = bx - 2A + 2x - 2c$$

have respective valuations at least r in U and $2r$ in E , by (142) and $v_F 2 \geq r$. Thus Z_U, Z_E are legitimate stationary coefficients. All three conductors T are even, so there are no residual sums. Put $c_0 = -\alpha^{-1}$. Compatibility gives $\theta_U(Z_U) = \theta_E(Z_E)$; Lemma 4.2 gives $\theta_U(c_0)/\theta_E(c_0) = \eta(c_0)\tau(c_0)$. Using (7) and $\Delta_F(\tau) = \tau(c_0)\Psi_F(-1)$ therefore gives

$$(160) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^T \Psi_F(1 - S).$$

Here $\Delta_F(\eta) = (-1)^{n_F}$ and $\eta(c_0) = (-1)^{n_F+T}$; this explains the sign instead of discarding it. Now $2x - b$ is trace zero and has E -valuation at least $2r$. Lemma 10.1 gives $\Psi_F(\mathcal{D}_x) = 1$ since $n_E(2x - b) = -\mathcal{D}_x$. Also $v_F(4c) \geq 2r = T$. Equation (158) proves that (160) is 1.

Odd T : the elementary sign. By Lemma 10.3, $T = 2e + 1$, $\text{char } F = 0$ and $e \geq 1$. We still have integral x , $b \in \mathfrak{p}_F^e$ and legitimate stationary coefficients Z_U, Z_E : their errors from (155) have depths at least e and $2e$, which exceed their ambiguity depth e as required (with equality allowed in U). In addition put

$$(161) \quad P_C = \text{Tr}_{K/E} C = 2x + 1, \quad W_E = n_E(P_C), \quad Q_C = \text{Tr}_{K/U} C = b + 2d, \quad W_U = n_U(Q_C).$$

The element P_C is a unit and $W_E = 1 + 2b + 4A \equiv 1 \pmod{\mathfrak{p}_F^{2e}}$. Thus W_E is a legitimate stationary coefficient for τ in the normalized Ψ_F convention, and $\tau(W_E) = 1$ because it is an actual norm.

Lemma 4.5, in the present notation, gives

$$(162) \quad \text{Tr}_{U/F} N_U Y + n_U(\text{Tr}_{K/U} Y) = \text{Tr}_{E/F} N_E Y + n_E(\text{Tr}_{K/E} Y).$$

Use the three stationary coefficients Z_U, Z_E, W_E in (7). Compatibility, (9), and (162) give

$$(163) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^T \Psi_F(W_U) \frac{g_U}{g_E g_\tau}.$$

The prefactor on the right is 1. Indeed $a = b/2 \in \mathcal{O}_F$ and

$$W_U = 4n_U(d + a), \quad \overline{n_U(d + a)} = \bar{a}^2 + \bar{a} - \bar{c}.$$

Its absolute trace is 1. Lemma 10.3 gives $\Psi_F(W_U) = -1$, which cancels $(-1)^T = -1$.

Odd T : an identity between the three actual residual functions. Choose a uniformizer Π of E and put $\pi = n_E \Pi$; this is a uniformizer of F and U . In (8) use critical elements π^e for U, F and Π^e for E . For $z \in \mathcal{O}_K$ set

$$Y_z = C(1 + \Pi^e z).$$

All traces whose norms are used below are nonzero. The trace over E stays a unit. The trace over U stays at valuation e , as will follow below. The three normalized critical residues are

$$(164) \quad \begin{aligned} \frac{N_U Y_z / Z_U - 1}{\pi^e} &\equiv \bar{z}^2 && \text{in } \kappa, \\ \frac{N_E Y_z / Z_E - 1}{\Pi^e} &\equiv \text{Tr}_{\kappa/k} \bar{z} && \text{in } k, \\ \frac{n_E(\text{Tr}_{K/E} Y_z) / W_E - 1}{\pi^e} &\equiv (\text{Tr}_{\kappa/k}(\bar{C}\bar{z}))^2 && \text{in } k. \end{aligned}$$

To check the first row, the trace of $\Pi^e z$ on the ramified edge has U -valuation at least $\lfloor (e + T)/2 \rfloor \geq e + 1$; its norm has leading term $\pi^e \bar{z}^2$. For the second row the unramified norm's trace term has leading coefficient $\text{Tr } \bar{z}$, and its quadratic term has depth $2e \geq e + 1$. For the third, the relative trace change has leading E -coefficient $\text{Tr}(\bar{C}\bar{z})$ at depth e , since $\overline{P_C} = 1$; apply the first ramified calculation to its norm. These verifications include the smallest case $e = 1$.

Moreover

$$(165) \quad n_U(\text{Tr}_{K/U} Y_z) - W_U \in \mathfrak{p}_F^T.$$

In fact $v_U Q_C = e$, because $Q_C = 2(d + b/2)$ and $\bar{d} \notin k$. The trace change $\text{Tr}_{K/U}(C\Pi^e z)$ has valuation at least $\lfloor (e + T)/2 \rfloor \geq e + 1$. Expanding its unramified quadratic norm proves (165).

Apply (162) to Y_z and to C , and subtract. The common multiplicative character values cancel by compatibility. The two τ -values of the norms of the E -traces are both 1. Equation (165) removes precisely the remaining U -trace-norm phase. Lemma 4.1 and (164) give the *pointwise identity of actual residual functions*

$$(166) \quad H_U(z^2) = H_E(\text{Tr}_{\kappa/k} z) H_\tau((\text{Tr}_{\kappa/k}(\bar{C}z))^2), \quad z \in \kappa.$$

No identification of polar forms alone has been used.

The map

$$z \mapsto (\text{Tr } z, (\text{Tr}(\bar{C}z))^2) : \kappa \longrightarrow k \times k$$

is bijective. Write $z = u + v\bar{d}$ and $\bar{x} = \xi \in k$; its coordinates are v and $(u + (\xi + 1)v)^2$. Frobenius on k is bijective. Since Frobenius on κ is bijective too, summing (166) gives

$$\sum_{\kappa} H_U = \left(\sum_k H_E \right) \left(\sum_k H_\tau \right), \quad g_U = g_E g_\tau.$$

Together with (163), this proves Theorem 10.8 in the odd row as well. Notice that the local norm maps themselves produced the affine phases and their coordinates, before any finite sum was evaluated.

10.6. Positive critical conductors. The naive pair of norms in (156)–(157) need not both be stationary near the critical boundary. We replace the second norm by a conjugate multiple. This changes its character value by an explicitly known sign, not by an uncontrolled character error.

Throughout this subsection let

$$1 \leq h \leq t = T - 1, \quad m_U = T + h, \quad m_E = T + 2h.$$

Use the actual model and base twist of Proposition 10.5, and choose $A = n_E x$ in the stationary coefficient class by Lemma 10.6. In particular $v_E x = -h$, so $x \neq 0$. Let x' be its nontrivial E/F conjugate, and put

$$(167) \quad \rho = \frac{x'}{x}, \quad \hat{Z}_U = A + bd + d^2, \quad \hat{Z}_E = \rho(x^2 + x - c), \quad b = \text{Tr}_{E/F} x.$$

The trace parity lemma.

Lemma 10.9 (An exact last-layer sign). *Put $\epsilon = (T - h) \bmod 2$ and $\mathbf{u} = b^2/A$. If $\epsilon = 0$, then $v_F \mathbf{u} \geq T$, with $v_F 0 = +\infty$. If $\epsilon = 1$, then*

$$v_F \mathbf{u} = T - 1, \quad \Psi_F(\mathbf{u}z) = (-1)^{\text{Tr}_{k/\mathbb{F}_2}(\bar{z})} \quad (z \in \mathcal{O}_F).$$

Consequently, for the fixed c in (147),

$$(168) \quad (-1)^{T+h} \Psi_F(-c\mathbf{u}) = 1.$$

This holds in both mixed and equal characteristic.

Proof. The trace-ideal formula gives

$$v_F b \geq B := \lfloor (T - h)/2 \rfloor.$$

If $T - h$ is even, $2B + h = T$, proving the first assertion. If $T - h$ is odd, the induced map

$$\mathfrak{p}_E^{-h} / \mathfrak{p}_E^{1-h} \longrightarrow \mathfrak{p}_F^B / \mathfrak{p}_F^{B+1}$$

is a nonzero k -linear map between one-dimensional k -spaces. Indeed the trace images of the two consecutive source ideals are exactly $\mathfrak{p}_F^B$ and $\mathfrak{p}_F^{B+1}$. Thus an element of exact valuation $-h$ has trace of exact valuation B . Hence $b \neq 0$ and $v_F \mathbf{u} = 2B + h = T - 1$.

For $z \in \mathcal{O}_F$, quadratic norm expansion gives the exact identity

$$(169) \quad n_E \left(1 + \frac{b}{A} xz \right) = 1 + \mathbf{u}(z + z^2).$$

Here $v_E((b/A)x) = 2B + h = T - 1 > 0$, so the argument is a unit. Since $T - 1 \geq q$, the actual norm-character formula (143) applies to the right side. It follows that the residue additive character $z \mapsto \Psi_F(uz)$ annihilates $z + z^2$. It is nontrivial, because Ψ_F has largest trivial ideal $\mathfrak{p}_F^T$ and $v_F \mathfrak{u} = T - 1$. The image of $z \mapsto z + z^2$ in k is the absolute-trace-zero hyperplane. The character is therefore exactly $(-1)^{\text{Tr}_{k/\mathbb{F}_2} z}$. Since $\bar{c}$ has absolute trace one, $\Psi_F(-cu) = -1$ in this parity and is 1 in the other. Finally $T - h$ and $T + h$ have the same parity, proving (168). $\square$

Stationarity and the exact elementary factor.

Lemma 10.10 (Corrected common representatives). *The two elements $\hat{Z}_U, \hat{Z}_E$ in (167) are actual ordinary stationary coefficients for θ_U, θ_E in the normalization of (7), for every $1 \leq h < T$. Moreover*

$$(170) \quad \hat{Z}_U - B_U = bd,$$

$$(171) \quad \hat{Z}_E - B_E = b(1 - c/x),$$

$$(172) \quad \hat{S} := \text{Tr}_{U/F} \hat{Z}_U - \text{Tr}_{E/F} \hat{Z}_E = 1 + cu,$$

$$(173) \quad \frac{\theta_E(\hat{Z}_E)}{\theta_U(\hat{Z}_U)} = (-1)^h.$$

Proof. The first error is immediate. Since $xx' = A$ and $x + x' = b$,

$$\rho(x^2 + x - c) = A + x' - c\rho,$$

whose difference from $A - x + c$ is $b - c(x'/x + 1) = b(1 - c/x)$. As $v_E x = -h < 0$, the latter parenthesis is a unit. With $B = \lfloor (T - h)/2 \rfloor$, the error valuations are at least B over U and $2B$ over E . The required ambiguity depths are

$$T - s_U = B, \quad T - s_E = \lfloor T/2 \rfloor - h.$$

For $\epsilon = (T - h) \bmod 2$,

$$2B - (T - s_E) = \lceil T/2 \rceil - \epsilon \geq 0.$$

Both errors therefore lie in the correct coefficient ambiguity ideals. The actual coefficients have valuations $-h, -2h$; the corrected ones do also, either by these estimates or directly by their unique leading terms.

The traces of ρ and of the corrected coefficients are

$$\text{Tr}_{E/F} \rho = \frac{b^2 - 2A}{A} = \mathfrak{u} - 2,$$

$$\text{Tr}_{E/F} \hat{Z}_E = 2A + b + 2c - cu, \quad \text{Tr}_{U/F} \hat{Z}_U = 2A + b + 1 + 2c.$$

This proves (172). The uncorrected Z_U, Z_E are the two norms of $C = x + d$, so compatibility gives $\theta_U(Z_U) = \theta_E(Z_E)$. Lemma 4.2 on E gives

$$\theta_E(\rho) = \frac{\theta_E(x')}{\theta_E(x)} = \eta(n_E x) = (-1)^{v_F A} = (-1)^h,$$

proving (173). No equation $n_E x = A_*$ was assumed: the exact norm $A = n_E x$ was chosen inside its justified coefficient class. $\square$

Remark 10.11 (A literal common element remains available). The correction can also be recorded using a common element and a base scalar. Put $Y = x'(x + d) \in K$ and $\lambda_0 = A^{-1} \in F$. Then $\lambda_0 N_U Y = \hat{Z}_U$ and $\lambda_0 N_E Y = \hat{Z}_E$. The symbol λ_0 is a field element, not the twisting character λ . We use (173) directly rather than suppressing the character value introduced by this scalar.

Proposition 10.12 (Reduction to actual residual factors). *Use the corrected coefficients in the two Lamprecht formulas, and use coefficient 1 for τ . Let g_U, g_E, g_τ be their full residual factors, with $g = 1$ in even conductor. Then*

$$(174) \quad \frac{\Delta_U(\theta_U)\Delta_F(\eta)}{\Delta_E(\theta_E)\Delta_F(\tau)} = (-1)^{T+h}\Psi_F(-cu)\frac{g_U}{g_E g_\tau} = \frac{g_U}{g_E g_\tau}.$$

Proof. Let $c_0 = -\alpha^{-1}$. Lemma 4.2 gives $\theta_U(c_0)/\theta_E(c_0) = \eta(c_0)\tau(c_0)$. The product $\eta(c_0)\Delta_F(\eta)$ is $(-1)^T$, since $v_F c_0 = n_F + T$ and $\Delta_F(\eta) = (-1)^{n_F}$. The quotient of the multiplicative stationary values is (173). The additive phase from (7) and the norm-character factor is $\Psi_F(1 - \hat{S}) = \Psi_F(-cu)$. These exact identities give the first equality in (174). Lemma 10.9 gives the second. In particular the elementary sign has been evaluated, not inferred from the square relation. $\square$

The unramified residual sum.

Lemma 10.13. *If $T + h$ is odd and $h > 0$, the actual residual function for θ_U with coefficient $\hat{Z}_U$ is 1 on the residue subfield $k \subset \kappa$. Its normalized sum is therefore $g_U = 1$.*

Proof. Set $a = (T + h - 1)/2$ and choose the critical element $\delta_U = \pi^a \in F$, for any uniformizer π of F . For $z \in \mathcal{O}_F$ put $v = \delta_U z$. One has

$$2a - s_E = \lfloor T/2 \rfloor - 1 \geq 0.$$

Also $a \geq q$: for even T this follows from odd $h \geq 1$; for odd T the present parity requires even $h \geq 2$. Thus both $\theta_E(1 + v)$ and $\tau(1 + v)$ are evaluated by their legitimate stationary formulas. The determinant identity and triviality of η on units give

$$\theta_U(1 + v) = \theta_E(1 + v)\tau(1 + v).$$

Substitution into the actual residual function yields

$$H_U(\bar{z}) = \Psi_F((1 - \hat{S})v) = \Psi_F(-cu \delta_U z) = 1.$$

Indeed $v_F u \geq T - 1$ and $a \geq 1$.

For completeness, the complete Lamprecht function is a nondegenerate quadratic phase on the additive group of κ . Its restriction to k is 1, so k is isotropic for its polar pairing. Since $|k|^2 = |\kappa|$, nondegeneracy gives $k^\perp = k$. Summation on a coset of k is zero unless the coset is k itself, and the sum on k is $|k|$. Thus $|\kappa|^{-1/2} \sum_\kappa H_U = 1$. This computes the full refinement, including any linear term. $\square$

Reciprocal residual functions at odd T . If T is even, the two remaining factors g_E, g_τ are already 1. For odd T write $T = 2e + 1$, where $e = v_F(2) \geq 1$ by Lemma 10.3. We prove their exact cancellation, without dividing a characteristic-two phase by two or guessing a square root.

Lemma 10.14 (Critical quadratic norm reciprocity). *For $w \in \mathfrak{p}_E^e$ one has*

$$(175) \quad \tau(1 + n_E w) = \Psi_E(w).$$

In particular, writing the actual norm-character residual function as

$$\mathcal{H}_\tau(z) = \Psi_F(-z)\tau(1 + z)^{-1}, \quad z \in \mathfrak{p}_F^e,$$

one has

$$(176) \quad \mathcal{H}_\tau(n_E w)^{-1} = \Psi_E(w)\Psi_F(n_E w).$$

Proof. The trace of w has valuation at least $\lfloor (e + T)/2 \rfloor \geq e + 1$, and $v_F n_E w \geq e$. The exact identity

$$n_E(1 + w) = (1 + n_E w) \left(1 + \frac{\text{Tr}_{E/F} w}{1 + n_E w} \right)$$

can therefore be evaluated by τ , using its linear formula on the second factor only. Since the left side is a norm,

$$\tau(1 + n_E w) = \Psi_F \left(\frac{\text{Tr} w}{1 + n_E w} \right) = \Psi_E \left(\frac{w}{1 + n_E w} \right).$$

The difference between $w/(1 + n_E w)$ and w has E -valuation at least $3e \geq 2e + 1 = T$. This proves (175), including $e = 1$. Equation (176) follows from the definition of the actual residual function. It is a statement at the critical depth e ; it was not obtained by extending Lemma 10.1 below its domain. $\square$

Lemma 10.15 (The ramified residual factor is the reciprocal). *For every $1 \leq h < T$ when T is odd,*

$$g_E = g_\tau^{-1}.$$

Proof. The critical E -ideal is $\mathfrak{p}_E^{e+h}$. For v in this ideal, both the core formula for χ_E^0 and the base-twist formula apply. The first is valid since $e + h \geq e + 1 = q$. For the second, the two terms in $n_E(1 + v) - 1 = \text{Tr } v + n_E v$ have F -valuations at least

$$\lfloor (3e + h + 1)/2 \rfloor \quad \text{and} \quad e + h,$$

respectively, both at least $s_U = e + \lceil (h + 1)/2 \rceil$. It follows that

$$(177) \quad \theta_E(1 + v) = \Psi_E(-(A + c)v) \Psi_F(-An_E v).$$

The difference between $\widehat{Z}_E$ and $A - x + c$ is $b(1 - c/x)$. Its product with v has valuation at least

$$2 \lfloor (T - h)/2 \rfloor + e + h = 3e + (h \bmod 2) \geq T.$$

Consequently the actual residual function at v is

$$\begin{aligned} \mathcal{H}_E(v) &= \Psi_E(-\widehat{Z}_E v) \theta_E(1 + v)^{-1} \\ &= \Psi_E(xv) \Psi_F(An_E v) \\ &= \mathcal{H}_\tau(n_E(xv))^{-1}, \end{aligned}$$

where the last equality is Lemma 10.14, since $v_E(xv) \geq e$ and $n_E(xv) = An_E v$ exactly.

Choose a uniformizer Π of E and put $\pi = n_E \Pi$. Write $x = \Pi^{-h} u$ with $u \in \mathcal{O}_E^\times$, and use Π^{e+h} and π^e as the two critical elements. On the two residue fields k the map $v \mapsto n_E(xv)$ induces

$$z \mapsto \bar{u}^2 z^2.$$

This is a bijection. Terminal representative invariance therefore gives

$$g_E = |k|^{-1/2} \sum_{z \in k} H_\tau(z)^{-1} = \overline{g_\tau} = g_\tau^{-1}.$$

The final equality uses the unit magnitude of the full normalized Lamprecht sum. No individual value of either Gauss factor was assumed. $\square$

Theorem 10.16 (Every positive critical unramified–ramified row). *Every primitive compatible pair with $1 \leq h < T$ satisfies (159), in mixed and equal characteristic.*

Proof. Proposition 10.12 leaves $g_U/(g_E g_\tau)$. If $T + h$ is even, $g_U = 1$ by the even-conductor formula; if it is odd, Lemma 10.13 gives $g_U = 1$. If T is even, $g_E = g_\tau = 1$. If T is odd, Lemma 10.15 gives $g_E g_\tau = 1$. Thus the actual full ratio is 1 in every case. $\square$

10.7. Sufficiently ramified common twists.

Lemma 10.17 (Comparison of high stationary covectors). *Let L/F be ramified quadratic with different exponent δ , and let λ have conductor $n > \delta$. Let b_F be a stationary covector for λ , so $\lambda(1 + z) = \psi_F(b_F z)$ for $z \in \mathfrak{p}_F^{\lceil n/2 \rceil}$. A stationary covector b_L for $\lambda \circ N_{L/F}$ satisfies*

$$(178) \quad b_L/b_F \in U_L^{n-\delta}.$$

For an unramified quadratic edge b_F itself is a stationary covector for the pullback.

Proof. On the ramified edge the pullback conductor is $M = 2n - \delta$, and $n_L(\psi_L) = 2n_F + \delta$, so $v_L b_L = v_L b_F = -2(n + n_F)$. For $z \in \mathfrak{p}_L^n$, its trace belongs to $\mathfrak{p}_F^{\lceil n/2 \rceil}$ because $\lfloor (n + \delta)/2 \rfloor \geq \lceil n/2 \rceil$, and its norm belongs to $\mathfrak{p}_F^n$. Exact norm expansion and the downstairs linear formula give $(\lambda \circ N_{L/F})(1 + z) = \psi_L(b_F z)$ on this whole ideal. The stationary formula for b_L also applies there. Perfect additive duality gives

$$b_L - b_F \in \mathfrak{p}_L^{-2n_F - \delta - n}.$$

Divide by b_F to obtain (178). On an unramified edge at depth $s = \lceil n/2 \rceil$, the norm endpoint has depth $2s \geq n$ and the trace has depth at least s , so the same expansion gives the final assertion at the full stationary depth. $\square$

Theorem 10.18 (Exact stable comparison). *Let (χ_1, χ_2) be any primitive compatible pair in a biquadratic diagram, and put $a_i = a_{L_i}(\chi_i)$ and $\delta_i = a_F(\omega_{L_i})$. Let λ have conductor $n \geq 2$. Assume for every ramified L_i/F that*

$$n > \delta_i, \quad n \geq a_i + \delta_i, \quad n \geq 2\delta_i,$$

and for every unramified L_i/F that $n \geq 2a_i$. For the actual twisted characters $\theta_i = \chi_i(\lambda n_i)$,

$$(179) \quad A_{L_i}(\theta_i) = D(c_F)\Delta_F(\lambda)^2, \quad c_F = b_F^{-1}, \quad D = \chi_i|_F \omega_{L_i}.$$

In particular these full expressions are independent of i .

Proof. The inequalities ensure that χ_i is within the exact stable-twist range of λn_i . On a ramified edge its stationary depth is $n - \lceil \delta_i/2 \rceil \geq a_i$; the unramified assertion follows from $n \geq 2a_i$. Ueda's exact stable twist gives

$$\Delta_{L_i}(\chi_i(\lambda n_i)) = \chi_i(c_i)\Delta_{L_i}(\lambda n_i), \quad c_i = b_{L_i}^{-1}.$$

Lemma 10.17 and $n - \delta_i \geq a_i$ show $\chi_i(c_i) = \chi_i(c_F)$. On the unramified edge $c_i = c_F$ may be used. The First Main Lemma on L_i/F is

$$\Delta_{L_i}(\lambda n_i)\Delta_F(\omega_{L_i}) = \Delta_F(\lambda)\Delta_F(\lambda\omega_{L_i}).$$

A downstairs stable twist is valid since $n \geq 2\delta_i$ (and $\delta_i = 0$ on the unramified edge). Its right side is $\omega_{L_i}(c_F)\Delta_F(\lambda)^2$. Multiply the two formulas and use Lemma 4.2. This argument retains all quadratic norm-character factors and all residual factors; stable twisting cancels the latter exactly. $\square$

Corollary 10.19. *In the unramified–ramified diagram, every primitive pair with $h \geq T$ in (152) satisfies the Second Main comparison.*

Proof. Use the actual minimal pair from Theorem 10.4 and Proposition 10.5. Its two conductors are T , the lower different exponents are $0, T$, and $a_F(\lambda) = T + h \geq 2T$. All hypotheses of Theorem 10.18 hold. $\square$

10.8. Completion of the unramified–ramified case.

Theorem 10.20 (Complete dyadic unramified–ramified comparison). *Let K/F be biquadratic over a local field of residue characteristic two, and assume it has an unramified quadratic intermediate field U . For every other quadratic intermediate field E and every primitive compatible pair (θ_U, θ_E) , one has*

$$\Delta_U(\theta_U, \psi_U)\Delta_F(\omega_U, \psi_F) = \Delta_E(\theta_E, \psi_E)\Delta_F(\omega_E, \psi_F).$$

The complete induction expression is therefore independent of all three quadratic intermediate fields. The assertion holds in mixed and equal characteristic, for every conductor.

Proof. The other quadratic fields are ramified, since an unramified local Galois group is cyclic. For a comparison with either such E , Proposition 10.5 gives conductors $T + h, T + 2h$ with $h \geq 0$. The case $h = 0$ is Theorem 10.8; the cases $1 \leq h < T$ are Theorem 10.16; and every $h \geq T$ is Corollary 10.19. This exhausts the conductors. Comparing each ramified field with U gives independence of all three by transitivity. The possible extensions of a fixed upstairs character differ by the upper norm character and are conjugate, by Lemma 4.2; their local constants agree by an automorphism change of variables. Thus the conclusion does not depend on those choices either. $\square$

The ordinary common norm $N_E(x + d)$ was not used as stationary outside its proved range. Its replacement is $\rho N_E(x + d)$, with $\rho = x'/x$, and the nontrivial character value of ρ was retained. The sign from the last trace row was derived from an actual norm identity, and the two odd- T residual factors were compared pointwise, not merely through their squares or polar forms.

11. TOTALLY RAMIFIED EXTENSIONS IN CHARACTERISTIC TWO

11.1. Statement and normalization. Let $F = k((\varpi))$, where k is a finite field of characteristic 2. Let K/F be totally ramified and Galois with group C_2^2 . Write its three nontrivial automorphisms as g_1, g_2, g_3 , with $g_1 g_2 = g_3$, and put

$$L_i = K^{\langle g_i \rangle}, \quad N_i = N_{K/L_i}, \quad S_i = \text{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}, \quad T_i^{\text{tr}} = \text{Tr}_{L_i/F}.$$

We reserve T_i without a superscript for a different/conductor exponent. Let ω_i be the nontrivial character of $F^\times / n_i L_i^\times$.

Theorem 11.1. *Suppose Θ is $\text{Gal}(K/F)$ -invariant and is not a norm pullback from F . Choose characters θ_i of $L_i^\times$ with $\theta_i N_i = \Theta$. For any nontrivial additive character ψ_F , put $\psi_{L_i} = \psi_F T_i^{\text{tr}}$. Then*

$$(180) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

is independent of i . Continuous quasi-characters are allowed.

The local constants and their admissible denominators have Ueda's normalization. In particular, the omitted trivial-character factor is 1. Only restrictions to compact unit groups enter the finite calculations.

We use the First Main Lemma, Lamprecht's formula, exact stable twisting, additive scaling and duality, norm filtration, and the trace and conductor formulas from [8], recalled in Sections 2–4. The subcritical norm representatives are those of [8, Lemma 6.6]. On a ramified quadratic edge E/D of break t , these give

$$(181) \quad \text{Tr}_{E/D}(\mathfrak{p}_E^a) = \mathfrak{p}_D^{\lfloor (a+t+1)/2 \rfloor}, \quad v_D(Nx) = v_E(x), \quad a_D(\omega_{E/D}) = t + 1.$$

If $a_D(\lambda) > t + 1$, its norm pullback has conductor $2a_D(\lambda) - (t + 1)$. For $0 \leq j \leq t$, an arbitrary $A \in D^\times$ has a norm approximation with relative error in U_D^j and the correct valuation. No approximation modulo U_D^{t+1} is assumed. We also use Hilbert 90, extension of abelian characters, elementary Artin–Schreier theory, and the local trace–residue identity for separable Laurent-series fields. The residue formula for the norm characters needed here is proved next. There is no higher-dimensional induction theorem among the inputs.

Throughout this section K/F is totally ramified. Fix a uniformizer π of K and take

$$(182) \quad \Pi_i = N_i \pi, \quad \varpi = N_{K/F} \pi = n_i \Pi_i.$$

These are compatible uniformizers; the coefficient field k is contained in all four fields. Write $\text{tr} = \text{Tr}_{k/\mathbb{F}_2}$. We first prove the theorem for

$$(183) \quad \psi_F(x) = (-1)^{\text{tr Res}_F(x d\varpi)}.$$

Its largest trivial ideal is $\mathcal{O}_F$. The passage to an arbitrary additive character is given at the end.

11.2. Local Artin–Schreier characters and a residue identity. For a Laurent differential in a uniformizer v , define

$$\mathcal{C} \left(\sum_n a_n v^n dv \right) = \sum_m a_{2m+1}^{1/2} v^m dv.$$

Coefficient square roots are unique because the residue field is perfect. The identities used below follow from this formula:

$$(184) \quad \mathcal{C}(f^2 \eta) = f \mathcal{C}(\eta), \quad \mathcal{C}(dv/v) = dv/v, \quad \text{tr Res } \mathcal{C}(\eta) = \text{tr Res } \eta.$$

Also $\mathcal{C}(du/u) = du/u$ for every nonzero Laurent series u . One direct verification writes a principal unit, successively in its coefficients, as a convergent product of factors $1 - cv^m$. Factors with even m have zero logarithmic derivative. For odd m ,

$$d \log(1 - cv^m) = \sum_{j \geq 1} c^j v^{mj-1} dv,$$

and the displayed formula for $\mathcal{C}$ retains $j = 2j'$ and returns the same series. The valuation and constant factors of u give the other parts of the assertion. Every residue calculation uses only finitely many terms of these convergent formal series.

We use the local trace–residue identity proved in Theorem B.4

$$(185) \quad \mathrm{Tr}_{k_E/k_D} \mathrm{Res}_E(\eta) = \mathrm{Res}_D(\mathrm{Tr}_{E/D} \eta)$$

for a finite separable extension of equal-characteristic local fields. Here the trace of $x d\varpi_D$ is $\mathrm{Tr}_{E/D}(x) d\varpi_D$. Appendix B proves this formal identity by finite free Laurent-series expansions and coefficient specialization; wild ramification is included. The logarithmic norm identity needed with it is Lemma B.5. Thus both ingredients have proofs here, independently of a residue theorem on a global curve.

Lemma 11.2 (The actual norm character). *Let $z^2 + z = f \in F$ define a nontrivial quadratic extension E/F . Then its actual norm character is*

$$(186) \quad \omega_f(u) = (-1)^{\mathrm{tr} \mathrm{Res}_F(f du/u)}.$$

If $f = f_0 + \sum_{j>0, j \text{ odd}} f_j \varpi^{-j}$ is reduced and its largest pole is $t > 0$, this character has conductor $t + 1$.

Proof. The formula is multiplicative in u . Adding $w^2 + w$ to f does not change it: by (184) and the logarithmic identity, the residues of $w^2 du/u$ and $w du/u$ have the same absolute trace. Every Artin–Schreier class has a reduced representative of the stated form. Remove a negative even power by adding $w^2 + w$ and repeat; remove the positive-depth part using the bijectivity of $w \mapsto w^2 + w$ on the maximal ideal. Constants are taken modulo the same map on k .

For $u = N_{E/F} v$, (185) makes its exponent equal to the absolute trace of $\mathrm{Res}_E((z^2 + z) dv/v)$, which is zero by (184). Thus the formula is trivial on norms. For a reduced representative with pole t , it is trivial on U_F^{t+1} , where $f du/u$ has nonnegative valuation. At $u = 1 + c\varpi^t$ its last-layer exponent is $\mathrm{tr}(f_t c)$, which is nontrivial as c varies. If instead the class is an unramified nonzero constant, evaluation at a uniformizer is nontrivial. Consequently the formula is nontrivial for every nonzero Artin–Schreier class. The quadratic norm quotient has order two, so this nontrivial character trivial on norms is precisely its norm character. The conductor assertion follows as well. $\square$

Lemma 11.3 (A local Cartier calculation). *For $f, w \in F$, the denominator $1 + \varpi w^2$ is nonzero and*

$$(187) \quad \mathrm{tr} \mathrm{Res}_F \left(\frac{\varpi f^2}{1 + \varpi w^2} d\varpi \right) = \mathrm{tr} \mathrm{Res}_F \left(\frac{f}{1 + \varpi w^2} d\varpi \right).$$

Proof. Nonvanishing follows from the parity of valuations. Put $H = 1 + \varpi w^2$ and write

$$\frac{\varpi f^2}{H} = \varpi(f/H)^2 + \varpi^2(fw/H)^2.$$

Applying (184) gives $(f/H) d\varpi$, and taking absolute residue traces proves the assertion. $\square$

11.3. A simultaneous origin and its ramification data. Choose the labeling so that L_1/F has the smallest break, and take reduced representatives

$$z_i^2 + z_i = f_i \quad (i = 1, 2), \quad z_3 = z_1 + z_2, \quad f_3 = f_1 + f_2.$$

Because K/F is totally ramified, all three nonzero classes have a pole. Write their breaks as $t_1 \leq t_2 = t_3$, all odd, and put

$$(188) \quad T_1 = t_1 + 1 = 2r_1, \quad T_2 = t_2 + 1 = 2r_2, \quad \delta = r_2 - r_1 \geq 0.$$

In the one-break case the two leading pole coefficients are distinct; otherwise their sum would have smaller conductor and would be the first field. In the two-break case $t_2 - t_1 = 2\delta$.

Write $f_i = f_{i,0} + f_i^-$, where f_i^- is the negative-power part. Define

$$(189) \quad \beta_i = \varpi^{t_2} f_i^- = d_i^2 \in \mathcal{O}_F, \quad d_3 = d_1 + d_2.$$

The square roots exist in F : every exponent $t_2 - j$ is even, and k is perfect. Thus $v_F d_1 = \delta$ and d_2, d_3 are units. Set

$$(190) \quad Y = d_1 z_2 + d_2 z_1, \quad B = \beta_1 f_2 + \beta_2 f_1 = \beta_1 f_{2,0} + \beta_2 f_{1,0} \in \mathcal{O}_F, \quad \kappa_i = d_j d_k \quad (\{i, j, k\} = \{1, 2, 3\}).$$

The automorphisms are labeled so that g_1 changes z_2 by 1 and fixes z_1 , while g_2 changes z_1 by 1 and fixes z_2 . Exact identities in the fields are

$$(191) \quad g_i Y - Y = d_i, \quad S_i Y = d_i, \quad a_i := N_i Y = \kappa_i z_i + B.$$

For example $Y^2 = \beta_1 z_2 + \beta_2 z_1 + B$, which proves every formula. In particular,

$$(192) \quad v_K Y = -t_1, \quad v_{L_i} a_i = -t_1, \quad \mathcal{N} := N_{K/F} Y = n_i a_i, \quad v_F \mathcal{N} = -t_1.$$

For $i = 2, 3$, $v_{L_i} \kappa_i = 2\delta = t_2 - t_1$; for $i = 1$ its valuation is zero. These observations prove the second assertion of (192), and the first follows by the norm valuation formula.

The upper breaks are $u_1 = 2t_2 - t_1$ and $u_2 = u_3 = t_1$. For completeness, a ramification automorphism g changes an element of odd valuation v by an element of exact valuation $v + u_g$: expand in a uniformizer and use the first nonzero ramification term. Apply this to Y in (191). The valuation of d_1 in K is 4δ , and those of d_2, d_3 are zero, proving these upper breaks. Thus the upper different exponents are

$$(193) \quad D'_1 = 2T_2 - T_1, \quad D'_2 = D'_3 = T_1.$$

Put

$$(194) \quad \alpha = \varpi^{-T_2}, \quad \Psi_F(x) = \psi_F(\alpha x), \quad \Psi_{L_i}(x) = \Psi_F(T_i^{\text{tr}} x).$$

The largest trivial ideals for these characters have exponents

$$(195) \quad J_F = T_2, \quad J_1 = 2T_2 - T_1, \quad J_2 = J_3 = T_2.$$

Do not confuse $J_2 = T_2$ with the upper different $D'_2 = T_1$. Since T_2 is even, $\Psi_F(x^2) = 1$ for every $x \in F$. Lemma 11.2, integration by parts for a formal residue, and $df_i = \alpha \beta_i d\varpi$ give

$$(196) \quad \omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{r_i}), \quad r_3 = r_2.$$

Indeed replacing $(1 - z)^{-1}$ by 1 in $f_i dz/(1 - z)$ costs no residue at depth r_i : the error has valuation at least $-t_i + r_i + (r_i - 1) = 0$. Notice the exact simultaneous norm property

$$(197) \quad \beta_i = n_i(S_i Y) = n_i(d_i), \quad \omega_i(\beta_i) = 1.$$

These are full stationary coefficients, not just their leading residues. Finally exact quadratic norm expansion yields the phase conversion

$$(198) \quad \Psi_{L_i}(\beta_i z) = \Psi_F(\beta_i n_i z) \quad (z \in \mathfrak{p}_{L_i}^{r_i}).$$

Both trace and norm of z lie in $\mathfrak{p}_F^{r_i}$; evaluate (196) on $n_i(1 - z) = 1 + T_i^{\text{tr}} z + n_i z$ and use triviality on norms. This establishes the conversion on its whole specified ideal.

11.4. Construction of compatible characters. Define the minimal conductors and stationary depths

$$(199) \quad \begin{aligned} m_1 &= 2T_2 - 1 = 2t_2 + 1, & s_1 &= T_2, \\ m_2 &= m_3 = T_1 + T_2 - 1 = t_1 + t_2 + 1, & s_2 &= s_3 = r_1 + r_2. \end{aligned}$$

Thus $m_i = J_i + t_1$, $m_i = 2s_i - 1$, and the formulas

$$(200) \quad R_i(1 - z) = \Psi_{L_i}(a_i z), \quad z \in \mathfrak{p}_{L_i}^{s_i}$$

define characters of $U_{L_i}^{s_i}$ with exact conductor m_i . Multiplicativity follows since $2s_i - t_1 \geq J_i$.

Lemma 11.4 (The required full commutator compatibility). *Let σ be the nontrivial automorphism of L_1/F and $\nu_1 = \omega_2 n_1$, the actual norm character for K/L_1 . For every $v \in L_1^\times$ with $\sigma(v)/v \in U_{L_1}^{s_1}$,*

$$(201) \quad R_1(\sigma(v)/v) = \nu_1(v).$$

Proof. The description of ν_1 follows from Lemmas D.5 and D.6, whose characteristic-two proof uses only (186) and (185). Multiplication of v by an element of $F^\times$ does not affect either side. Reduce its valuation to 0 or 1. In valuation 1 its conjugate quotient has exact depth $t_1 < s_1$, by the first ramification term. Thus only units occur. Write $v = a + bz_1$, with $a, b \in F$. The two summands have valuations of opposite parity, since t_1 is odd. Consequently a is a unit and $v_F b \geq r_1$. The conjugate quotient is $1 + b/v$; the assumed depth $s_1 = 2r_2$ forces $v_F b \geq r_2$. Put $w = b/a$, so $v_F w \geq r_2$. The case $b = 0$ is immediate. Exact quadratic trace calculation gives

$$(202) \quad T_1^{\text{tr}} \left(a \frac{b}{v} \right) = \frac{\kappa_1 ab + Bb^2}{n_1 v} = \frac{\kappa_1 w + Bw^2}{1 + w + w^2 f_1}.$$

Here is an independent evaluation of the other side. Put $R = w^2 f_1$, $H = 1 + R$ and $V = H + w$. These are units, since $v_F R \geq 2r_2 - t_1 \geq 1$. Factor $V = H(1 + w/H)$. The residue formula evaluates $\omega_2(H)$ exactly, and the full linear chart (196) applies to the second factor. They give

$$\omega_2(V) = \Psi_F \left(\frac{\beta_1 f_2 w^2 + \beta_2 w}{H} \right) = \Psi_F \left(\frac{\beta_2 R + \beta_2 w}{H} \right).$$

In the last equality Bw^2/H was discarded at depth at least T_2 , using $\beta_1 f_2 + \beta_2 f_1 = B \in \mathcal{O}_F$. Write $R_0 = \varpi W^2$, where $W = \varpi^{-r_2} d_1 w \in \mathcal{O}_F$. Then $R - R_0 = f_{1,0} w^2 \in \mathfrak{p}_F^{T_2}$, so it can be discarded in all these unit denominators. Lemma 11.3, with $f = \varpi^{-r_2} d_2 W$, proves

$$\Psi_F \left(\frac{\beta_2 R_0}{1 + R_0} \right) = \Psi_F \left(\frac{d_1 d_2 w}{1 + R_0} \right).$$

It follows that $\omega_2(V) = \Psi_F(\kappa_1 w/H)$ because $\beta_2 + d_1 d_2 = d_2 d_3 = \kappa_1$. Replacing H by V here costs only $\kappa_1 w^2/(HV)$, of depth at least T_2 . The same is true of the Bw^2/V term in (202). Since $n_1 v = a^2 V$ and $\omega_2(a^2) = 1$, (202) proves (201). Every discarded expression was evaluated at its actual F -character modulus $\mathfrak{p}_F^{T_2}$. $\square$

Lemma 11.5 (Compatibility on the full common unit subgroup). *For $z \in \mathfrak{p}_K^{T_2}$ all $N_i(1 - z)$ lie in the domains of R_i , and $R_i(N_i(1 - z))$ is independent of i .*

Proof. The norm term has L_i -valuation at least T_2 . The upper trace terms have respective depths at least $3r_2 - r_1$ and $r_1 + r_2$; these are at least s_1 and s_2 . The identity of Lemma 4.5 is

$$(203) \quad E_2(C) = T_i^{\text{tr}} N_i C + n_i S_i C.$$

Apply it to $Y(1 - z)$ and Y . Put $h_i = S_i(Yz)/d_i$. The upper trace formula and (192) give

$$v_{L_1} S_1(Yz) \geq r_1 + 3\delta, \quad v_{L_2} S_2(Yz), v_{L_3} S_3(Yz) \geq r_2.$$

After division by d_1 , of L_1 -valuation 2δ , the first bound is also r_2 . Thus $v_{L_i} h_i \geq r_i$ on every side. Since $S_i(Y(1 - z)) = d_i(1 - h_i)$, its lower norm divided by β_i is an actual norm. Formula (196) applied to that norm quotient shows that the additive phase of $n_i S_i(Y(1 - z)) - \beta_i$ is 1. The remaining term in (203) is $T_i^{\text{tr}}(a_i(S_i z + N_i z))$, exactly the exponent of $R_i(N_i(1 - z))$. Therefore this value is $\Psi_F(E_2(Y(1 - z)) - E_2(Y))$, independent of i . $\square$

Theorem 11.6 (Existence of actual minimal models). *There are characters χ_i on $L_i^\times$, with exact conductors m_i , primitive common norm pullback, and*

$$(204) \quad \chi_i(1 - z) = \Psi_{L_i}(a_i z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}).$$

Proof. Let $I = \{\sigma(v)/v : v \in L_1^\times\}$ and prescribe $\kappa_I(\sigma(v)/v) = \nu_1(v)$. This is well defined because ν_1 is trivial on $F^\times$: it is $\omega_2(n_1 x) = \omega_2(x^2)$ there. Lemma 11.4 proves agreement with R_1 on $I \cap U_{L_1}^{s_1}$. Their product character on $IU_{L_1}^{s_1}$ is trivial on $U_{L_1}^{m_1}$. Extend it across the abelian group $L_1^\times/U_{L_1}^{m_1}$. One can first extend on its finite unit subgroup and then choose the uniformizer value. The resulting χ_1 is continuous, has exact conductor m_1 , and satisfies $\chi_1^\sigma/\chi_1 = \nu_1$ globally.

The character $\chi_1 N_1$ is invariant under G : its other conjugate quotient is $\nu_1 N_1 = \omega_2 N_{K/F} = 1$. Hilbert 90 therefore makes it trivial on $\ker N_i$ for $i = 2, 3$; extend its descent from the norm image to χ_i on $L_i^\times$. Above the upper break t_1 , norm filtration gives

$$N_i(U_K^{T_2+1}) = U_{L_i}^{s_i} \quad (i = 2, 3),$$

since $2s_i - t_1 = T_2 + 1$. Lemma 11.5 then proves (204) on those entire groups, not just on a last layer. Their conductors are at least m_i . The common pullback has conductor $2m_1 - D'_1 = t_1 + 2t_2 + 1$; since $m_i > D'_i$, the high pullback-conductor formula forces their conductors to equal m_i . Finally, descent of the common pullback from F would make χ_1 a base pullback times its upper norm character. Both are invariant under σ , contradicting the nontrivial commutator ν_1 . $\square$

11.5. Common-twist realization and conductor ranges. Write ν_i for the nontrivial norm character of K/L_i . The common restriction $D_\theta = \theta_i|_{F \times \omega_i}$ is the one in Lemma 4.2.

Proposition 11.7 (Actual common-twist realization). *For every primitive compatible family there is a core family satisfying (204), and a character λ of $F^\times$, such that*

$$(205) \quad \theta_i = \chi_i(\lambda n_i).$$

Put $n = a_F(\lambda)$ and $h = n - T_2$. If $n \leq T_2 + r_1 - 1$, the conductors are exactly m_i . If $n \geq T_2 + r_1$, they are $M_i = 2n - T_i$, all even. These alternatives exhaust all primitive conductors.

Proof. Start with Theorem 11.6. By Lemma 4.2, θ_1/χ_1 is invariant under $\text{Gal}(L_1/F)$. Hilbert 90 and abelian character extension make it λn_1 . For $i = 2, 3$, the remaining quotient is a power of ν_i . Absorb it into χ_i . This does not change its common pullback or (204), because $a_{L_i}(\nu_i) = T_1 \leq s_i$.

Primitivity forces $a_{L_i}(\theta_i) \geq t_i + u_i + 1 = m_i$: on the last unit row of the nontrivial commutator, the first ramification term in $\sigma(v)/v$ has exact depth $t_i + u_i$. Its coefficient is nonzero since u_i is odd; a character of smaller conductor would make the commutator trivial on that row. For $n > T_i$, the pullback λn_i has conductor $2n - T_i$; for $n \leq T_i$ it has conductor at most T_i . The number $2n - T_i$ is even, whereas m_i is odd, so cancellation at equal conductors is impossible. The same threshold $n = T_2 + r_1$ is obtained for all three fields. This proves the assertions. $\square$

11.6. Moving the common origin by a norm from the third field. We compare fields 1 and 2 and use L_3 for the adjustment. The same construction compares 1 and 3 with their roles exchanged. For $X \in L_3$, put

$$(206) \quad A = n_3 X, \quad a = T_3^{\text{tr}} X, \quad C = Y + X, \quad Q_i = aY + d_i X \in L_i \quad (i = 1, 2).$$

The following are exact algebraic identities:

$$(207) \quad N_i C = a_i + A + Q_i, \quad S_i C = d_i + a,$$

$$(208) \quad n_i Q_i = \beta_i A + E, \quad E = a(aa_3 + \kappa_3 X) \in F.$$

For example, write $X = p + qz_3$. Then $a = q$, $Q_1 = d_1 p + qd_3 z_1$ and $Q_2 = d_2 p + qd_3 z_2$; taking the two quadratic norms proves (208), with $E = q^2 B + q\kappa_3 p$.

If $v_{L_3} X = -h$, trace ideals and (192) give

$$(209) \quad v_F a \geq r_2 - \lceil h/2 \rceil,$$

$$(210) \quad v_{L_1} Q_1 \geq 2\delta - h, \quad v_{L_2} Q_2 \geq -h,$$

$$(211) \quad v_F E \geq r_2 + \delta - h.$$

These statements include negative h and vanishing traces. For the last one, the invariant element $aa_3 + \kappa_3 X$ has L_3 -valuation at least $2\delta - h$, hence F -valuation at least $\delta - \lceil h/2 \rceil$; combine this with (209). For the Q_i bounds one can use directly the upper trace of $Y g_i X$, with valuation $-t_1 - 2h$, and the upper differentials (193).

Suppose $\lambda(1 - z) = \Psi_F(Az)$ on a downstairs ideal containing both trace and norm of a variable $z \in L_i$. Whenever $(Q_i/\beta_i)z \in \mathfrak{p}_{L_i}^{r_i}$, equations (198) and (208) give

$$(212) \quad \Psi_{L_i}(Q_i z) = \Psi_F((A + E/\beta_i)n_i z).$$

Thus, if $(E/\beta_i)n_i z$ is in $\mathfrak{p}_F^{T_2}$, the full pulled-back character has coefficient $A + Q_i$. No exact equation $n_i(Q_i/\beta_i) = A/\beta_i$ is asserted.

11.7. Stationary origins at minimal conductors. Let the family be realized as in (205), with $n \leq T_2 + r_1 - 1$. Put $q_0 = r_1 + r_2$. For $z \in \mathfrak{p}_{L_i}^{s_i}$, both $T_i^{\text{tr}} z$ and $n_i z$ lie in $\mathfrak{p}_F^{q_0}$. For $i = 1$ the trace bound is exactly q_0 ; for $i = 2, 3$ its bound is at least q_0 . If $n \leq q_0$, take $X = 0$ and $C = Y$ below. Otherwise put $h = n - T_2$, so

$$(213) \quad 1 - \delta \leq h \leq r_1 - 1.$$

Since $2q_0 \geq n$, additive duality supplies

$$\lambda(1 - z) = \Psi_F(A_* z) \quad (z \in \mathfrak{p}_F^{q_0}), \quad v_F A_* = -h,$$

with ambiguity $\mathfrak{p}_F^{T_2 - q_0} = \mathfrak{p}_F^\delta$. Choose $X \in L_3$ by norm approximation so that

$$(214) \quad A = n_3 X \equiv A_* \pmod{\mathfrak{p}_F^\delta}, \quad v_{L_3} X = -h.$$

The relative precision requested is $\delta + h \leq r_2 - 1 \leq t_2$, so this choice is legitimate. The exact norm A may replace A_* on the whole indicated ideal.

Lemma 11.8 (The full stationary properties). *The element C just constructed has valuation $-t_1$ in K . For $i = 1, 2$, its actual upper norms and lower trace norms satisfy*

$$(215) \quad \theta_i(1 - z) = \Psi_{L_i}(N_i C z) \quad (z \in \mathfrak{p}_{L_i}^{s_i}),$$

$$(216) \quad \omega_i(1 - z) = \Psi_F(n_i S_i C z) \quad (z \in \mathfrak{p}_F^{r_i}).$$

If $\delta = 0$, the same single C satisfies these identities for all three indices.

Proof. The case $X = 0$ follows from the core formulas. Otherwise $-2h > -t_1$ by (213), so C has the valuation of Y . For $i = 1, 2$, (210) gives

$$v_{L_1}((Q_1/\beta_1)z) \geq T_2 - 2\delta - h = 2r_1 - h \geq r_1, \quad v_{L_2}((Q_2/\beta_2)z) \geq r_1 + r_2 - h \geq r_2.$$

The error in (212), at these depths, has valuations at least $T_2 + r_1 - h$ and $3r_2 - h$, respectively; both are at least T_2 . Therefore exact norm expansion evaluates the actual pullback λn_i with coefficient $A + Q_i$. Combining it with the core coefficient gives $N_i C$ by (207). Moreover $n_i S_i C = (d_i + a)^2 = \beta_i + a^2$. The bound

$$2v_F a \geq 2r_2 - 2 \lceil h/2 \rceil \geq 2r_2 - r_1 = T_2 - r_1$$

proves (216) on both full lower stationary ideals. It also shows $v_F a > \delta$, so the traces $d_i + a$ are nonzero with the original valuations.

When $\delta = 0$, $r_1 = r_2 = r$ and $h \leq r - 1$. On the third side the exact norm-phase conversion gives the pullback coefficient $A + d_3 X$, because $n_3(X/d_3) = A/\beta_3$. Its argument has depth at least $2r - h \geq r$. But

$$N_3 C = a_3 + X^2 + d_3 X = a_3 + A + d_3 X + aX,$$

and $v_{L_3}(aX) \geq 2r - 2h - (h \bmod 2) \geq 0$. The last term is invisible on $\mathfrak{p}_{L_3}^{2r}$. Finally $S_3 C = d_3$, so its lower norm coefficient is unchanged. This proves the assertion for all three fields in the one-break case. $\square$

11.8. The minimal sign and the critical norm image. Use Lemma 4.1 with stationary coefficient A_L and critical element $\Pi_L^{[m/2]}$, writing its residual function and sum as H_L, g_L . In characteristic two the field minus signs in that lemma disappear; complex phases are unchanged.

For the C of Lemma 11.8, use $N_i C$ for the upper coefficient and $n_i S_i C$ for the lower one. The latter is an actual norm and the lower conductor T_i is even. Compatibility and (203) therefore give

$$(217) \quad \mathcal{A}_i(\theta_i) = D_\theta(\alpha^{-1})\Theta(C)^{-1}\Psi_F(E_2(C))g_i.$$

All character values and elementary phases in this identity have been retained.

For $z \in \mathcal{O}_K$ vary C to $C_z = C(1 + \pi^{t_2} z)$. The relative upper norms are on the terminal stationary rows, of depths

$$\ell_1 = t_2, \quad \ell_2 = \ell_3 = (t_1 + t_2)/2.$$

The changes $S_i(C\pi^{t_2}z)$ have valuations at least $r_1 + 3\delta$ and r_2 on sides 1 and 2, respectively. After division by S_iC , both relative changes have valuation at least $r_2 \geq r_i$. The actual lower norm quotient can thus be evaluated by (216); its additive phase is 1. Subtracting (203) for C_z, C , and using compatibility and terminal-coset invariance, proves

$$(218) \quad H_i(p_i(z)) = Q(z),$$

where p_i is the actual graded upper norm map at this row and Q is independent of the compared index. These are identities between full functions, not just between polar forms.

Proposition 11.9 (The two-break minimal row). *If $t_1 < t_2$, every minimal family satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. On side 1, t_2 is below the upper break $2t_2 - t_1$. The trace term is at least one level deeper than the norm term, so $p_1(z) = z^2$, with compatible uniformizers (182). On side 2, the trace induces an isomorphism of the one-dimensional residue lines at source depth t_2 and target depth $(t_1 + t_2)/2$:

$$\lfloor (t_2 + T_1)/2 \rfloor = (t_1 + t_2)/2, \quad \lfloor (t_2 + 1 + T_1)/2 \rfloor = (t_1 + t_2)/2 + 1.$$

The norm endpoint is strictly deeper. Hence p_2 is a nonzero k -linear map. Both p_i are bijections of k . Summing (218) gives $g_1 = g_2$, and (217) proves equality. The same comparison with L_3 proves independence of every field in this configuration. $\square$

Proposition 11.10 (The one-break minimal row). *If $t_1 = t_2 = t$, every minimal family satisfies $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$.*

Proof. Lemma 11.8 supplies one C on all three sides. Put $u = \overline{C\pi^t} \in k^\times$ and $c_i = \overline{g_iC - C} \in k^\times$. The three c_i are distinct, with $c_1 + c_2 = c_3$. Indeed $g_iY - Y = d_i$ and the possible adjustment changes the first two by $a \in \mathfrak{p}_F$, leaving their residues unchanged. The first ramification coefficient on π is c_i/u : expand the leading term $u\pi^{-t}$ of C and use oddness of t . Thus the upper critical norm maps are

$$(219) \quad p_i(z) = z^2 + (c_i/u)z.$$

Write $P = c_1c_2c_3$ and $k_i = c_jc_k$. For the lower edge, the difference of the two conjugates of N_iC has residue k_i , while its leading term is $u^2\Pi_i^{-t}$. Consequently its lower critical norm polynomial has coefficient $\eta_i = k_i/u^2$, namely $z^2 + \eta_iz$.

The coefficient of the actual lower norm character in (216) has residue c_i^2 . Triviality on its critical norms gives

$$\mathrm{tr}(c_i^2(z^2 + \eta_iz)) = 0 \quad (z \in k).$$

Nondegeneracy of the residue trace pairing forces $c_i + c_i^2\eta_i = 0$, hence

$$(220) \quad \eta_i = c_i^{-1}, \quad u^2 = P.$$

This determines the unit coefficients; it is not a valuation-only claim. The polar pairing of the actual upper residual function is therefore

$$(221) \quad H_i(x + z)/(H_i(x)H_i(z)) = (-1)^{\mathrm{tr}(k_izx)}.$$

Here tracing Π_i^t gives the coefficient η_i and the leading coefficient of N_iC contributes u^2 .

The image V_i of (219) is $\{x : \mathrm{tr}(u^2x/c_i^2) = 0\}$, an index-two hyperplane. For $j \neq i$, put $w_i = k_i/u^2 = p_i(c_j/u)$. The common actual function in (218) gives $H_i(w_i) = H_j(0) = 1$ and $w_i \in V_i \setminus \{0\}$. By (220), its polar character in (221) is $x \mapsto (-1)^{\mathrm{tr}(u^2x/c_i^2)}$. Lemma 4.8, applied with ambient group $(k, +)$ and image V_i , gives $\sum_k H_i = \frac{1}{2} \sum_k Q$ for every i . The normalization $|k|^{-1/2}$ is common, so (217) proves the result. $\square$

11.9. Intermediate common twists. Suppose now $n \geq T_2 + r_1$, and write $h = n - T_2 \geq r_1$. The actual conductors and stationary depths are

$$(222) \quad M_i = 2n - T_i, \quad b_i = M_i/2 = n - r_i; \quad b_1 = T_2 + h - r_1, \quad b_2 = r_2 + h.$$

For this subsection assume

$$(223) \quad r_1 \leq h \leq r_1 + 2r_2 - 2.$$

Put

$$q_F = r_2 + \lfloor (h + r_1)/2 \rfloor.$$

The trace and norm of each variable $z \in \mathfrak{p}_{L_i}^{b_i}$, $i = 1, 2$, belong to $\mathfrak{p}_F^{q_F}$. Moreover $q_F \geq \lceil n/2 \rceil$ and $b_i \geq s_i$. Choose the actual coefficient A_* of λ on this whole base ideal. It has valuation $-h$ and ambiguity $\mathfrak{p}_F^{T_2 - q_F}$. A norm from L_3 can represent this class, because its requested relative precision is

$$(224) \quad h + T_2 - q_F = r_2 + \lceil (h - r_1)/2 \rceil \leq 2r_2 - 1 = t_2.$$

Thus choose $X \in L_3$ with $v_{L_3}X = -h$ and $A = n_3X$ in that class, and use (206).

Theorem 11.11 (Exact intermediate comparison). *Every family in (223) satisfies $\mathcal{A}_1(\theta_1) = \mathcal{A}_2(\theta_2)$.*

Proof. The elements $(Q_i/\beta_i)z$ in (212) have depths at least r_1 and r_2 , respectively, because

$$-h - 2\delta + b_1 = r_1, \quad -h + b_2 = r_2.$$

The error terms there have depths at least

$$(r_2 + \delta - h) - 2\delta + b_1 = T_2, \quad (r_2 + \delta - h) + b_2 = T_2 + \delta.$$

Thus the actual pulled-back coefficient is $A + Q_i$ on its full stationary ideal. Combining it with (204) proves that $N_iC = a_i + A + Q_i$ is the stationary coefficient for θ_i . Here $v_KX = -2h < -t_1 = v_KY$, so $C \neq 0$ and $v_{L_i}N_iC = -2h$, exactly the required $J_i - M_i$.

Both upper and lower conductors are even. Use N_iC for the upper coefficient, but retain the original actual coefficient β_i for ω_i . It is unnecessary, and generally unjustified, to claim that n_iS_iC is still stationary for ω_i . The full Lamprecht formula instead gives

$$\mathcal{A}_i(\theta_i) = D_\theta(\alpha^{-1})\Theta(C)^{-1}\Psi_F(T_i^{\text{tr}}N_iC + \beta_i).$$

By (203) and (207), its last exponent is

$$E_2(C) + (d_i + a)^2 + d_i^2 = E_2(C) + a^2,$$

the same for $i = 1, 2$. This proves exact equality without a residual sign argument. In fact $\Psi_F(a^2) = 1$, but this extra simplification is not needed. Relabeling the third field proves the other comparison. $\square$

11.10. Higher twists and their common correction. The remaining range is

$$(225) \quad h \geq h_0 := r_1 + 2r_2 - 1, \quad n = T_2 + h.$$

Let $A \in F^\times$ be an actual stationary coefficient of λ on $\mathfrak{p}_F^{\lceil n/2 \rceil}$, in the normalized Ψ_F convention. Thus $v_FA = -h$, and put $c_F = (\alpha A)^{-1}$. For each $i = 1, 2$, choose only an approximate norm

$$(226) \quad n_i x_i = (A/\beta_i)\varepsilon_i, \quad \varepsilon_i \in U_F^{t_i}, \quad v_{L_i}x_i = -h - (T_2 - T_i).$$

Such x_i exist by subcritical norm approximation at its stated precision t_i .

Lemma 11.12 (The high stationary coefficient and core value). *A stationary coefficient of λn_i is $A + \beta_i x_i$. If $c_i = (\alpha(A + \beta_i x_i))^{-1}$, then*

$$(227) \quad \chi_i(c_i) = \chi_i(c_F)\Psi_F(\mathcal{N}/A).$$

The correction is identical for both fields.

Proof. For $z \in \mathfrak{p}_{L_i}^{b_i}$, $b_i = n - r_i$, both its trace and norm lie in the base stationary ideal. Also $x_i z$ has valuation at least r_i , so (198) applies. The error from (226), after multiplication by $n_i z$, has normalized valuation at least

$$-h + t_i + b_i = T_2 + r_i - 1 \geq T_2.$$

Exact quadratic norm expansion proves the first assertion on the whole stationary ideal. The leading term is A , since

$$v_{L_i}(\beta_i x_i / A) = h + T_2 - T_i > 0.$$

Set $v_i = \beta_i x_i / A$. The lower bound (225) places v_i in the core domain $\mathfrak{p}_{L_i}^{s_i}$, hence

$$\chi_i(c_i / c_F) = \chi_i((1 + v_i)^{-1}) = \Psi_{L_i}(a_i v_i).$$

The last equality keeps the actual core character; its displayed value has order at most two. To convert it, the variable $a_i x_i / A$ has valuation

$$h - t_1 - (T_2 - T_i) \geq r_i.$$

For $i = 1$ this inequality is precisely $h \geq h_0$; for $i = 2$ it is weaker. Apply (198), using $n_i a_i = \mathcal{N}$, to get

$$\Psi_{L_i}(\beta_i a_i x_i / A) = \Psi_F(\mathcal{N} \varepsilon_i / A).$$

The error on deleting ε_i has valuation at least $h - t_1 + t_i$, which is at least T_2 on both sides by (225). This proves (227). $\square$

Theorem 11.13 (Exact high comparison). *For every family in (225),*

$$(228) \quad \mathcal{A}_i(\theta_i) = D_\chi(c_F) \Psi_F(\mathcal{N} / A) \Delta_F(\lambda, \psi_F)^2 \quad (i = 1, 2).$$

In particular the full induction expressions are equal.

Proof. The core conductor lies within the upstairs stable range:

$$b_1 = T_2 + h - r_1 \geq 2T_2 - 1 = m_1, \quad b_2 = r_2 + h \geq T_1 + T_2 - 1 = m_2.$$

Ueda's exact stable twist gives $\Delta_{L_i}(\theta_i) = \chi_i(c_i) \Delta_{L_i}(\lambda n_i)$. The First Main Lemma on L_i / F gives

$$\Delta_{L_i}(\lambda n_i) \Delta_F(\omega_i) = \Delta_F(\lambda) \Delta_F(\lambda \omega_i).$$

Since $n \geq 2T_2 \geq 2T_i$, the downstairs stable twist is valid and the right side is $\omega_i(c_F) \Delta_F(\lambda)^2$. Combine this with Lemma 11.12 and the common determinant restriction of the core family. No arbitrary exact norm or comparison of the desired local constants has been used to obtain the correction. $\square$

11.11. Completion in characteristic two.

Theorem 11.14. *Theorem 11.1 holds when K/F is totally ramified.*

Proof. Proposition 11.7 realizes the actual family by the stated core and a common base twist. If $n \leq T_2 + r_1 - 1$, use Propositions 11.9 and 11.10. Otherwise $h = n - T_2 \geq r_1$. The interval $r_1 \leq h \leq r_1 + 2r_2 - 2$ is Theorem 11.11, and the remaining integers are Theorem 11.13. Their endpoints are consecutive, including $r_1 = r_2 = 1$. In the two-break case repeat the comparison of the distinguished first field with the third field. This proves independence of all three. Changing an inducing extension by an upper norm character conjugates it and preserves its local constant by an automorphism change of variables in the defining local integral. The passage to an arbitrary additive character is made once for all branches at the end. $\square$

Proof of Theorem 11.1. For the canonical additive character this is Theorem 11.14. Every nontrivial additive character of $F = k((\varpi))$ is a scaling $x \mapsto \psi_F(cx)$ of (183), with $c \in F^\times$. Ueda's additive-scaling formula multiplies the complete induction expression by $\theta_i(c) \omega_i(c) = D_\theta(c)$, independent of i by Lemma 4.2. This proves the stated arbitrary-character result. $\square$

12. TOTALLY RAMIFIED MIXED CHARACTERISTIC: THE MAXIMAL BREAK

12.1. Statement and ramification data. Let $F/\mathbf{Q}_2$ be finite, with residue field k , normalized valuation v_F , and $e = v_F(2)$. Suppose K/F is totally ramified, $\text{Gal}(K/F) = C_2^2$. Label its three quadratic fields so that

$$(229) \quad t_1 = 2a - 1, \quad t_2 = t_3 = 2e, \quad 1 \leq a \leq e.$$

Put $T_1 = 2a$, $T_2 = T_3 = T = 2e + 1$. Write g_i for the nontrivial upper automorphism fixing L_i , and

$$N_i = N_{K/L_i}, \quad S_i = \text{Tr}_{K/L_i}, \quad n_i = N_{L_i/F}, \quad \text{tr}_i = \text{Tr}_{L_i/F}.$$

Let ω_i be the actual quadratic norm character for L_i/F . Fix a nontrivial additive ψ_F and compose with trace on all fields.

Theorem 12.1. *For every primitive compatible family $\theta_i N_i = \Theta$, where Θ is not a pullback from F , the full expressions*

$$(230) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

are equal. Continuous quasi-characters are allowed. The conclusion holds for every conductor under (229).

We prove the comparison of L_1, L_2 using L_3 as auxiliary field. Relabeling the two maximal fields then proves the other comparison.

The one-dimensional inputs are the results of [8]: the full Lamprecht formula, stationary additive duality, norm filtration and norm-character conductor, norm-pullback conductor, trace/different formulas, and the trivial-character local constant. Hilbert 90 and character extension are proved in Appendix C; the crossed norm characters and ramification relations are proved in Appendix D. We give the additional quadratic-symbol calculation explicitly. No two-inducing-field comparison or global functional equation is used.

For a ramified quadratic edge E/F of different exponent D ,

$$(231) \quad \text{Tr}_{E/F}(\mathfrak{p}_E^j) = \mathfrak{p}_F^{[(j+D)/2]}, \quad v_F(Nx) = v_E x, \quad n_E(\psi_E) = 2n_F(\psi_F) + D.$$

The upper different exponents in our diagram are

$$(232) \quad D'_1 = 2T - T_1 = 4e - 2a + 2, \quad D'_2 = D'_3 = 2a.$$

Indeed the upper breaks are $2t_2 - t_1$ and t_1 , by the usual break calculation of Lemma D.3. All valuations below name their field; restriction of v_K to F is $4v_F$.

12.2. A local symbol identity and the aligned Kummer origin. For $A, B \in F^\times$ let (A, B) be the quadratic norm symbol: it is 1 exactly when B is a norm from $F(\sqrt{A})$, with the split case interpreted as 1. It is a character in B . Symmetry follows from the symmetric conic equation $z^2 = Ax^2 + By^2$, and hence it is bilinear in both square classes. For $A \neq 1$, one also has $(A, 1 - A) = 1$, since $1 - A$ is the nonzero norm of $1 + \sqrt{A}$. If $A + B \neq 0$, the same conic, under $u = Ax + By$, $v = x - y$, gives

$$(233) \quad (A, B) = (A + B, -AB), \quad (A + B)z^2 = u^2 + ABv^2.$$

The linear change in (x, y) is invertible. Thus this equality is a local quadratic-norm identity, not a local-constant identity.

Lemma 12.2 (A sufficient unit depth). *If $v_F u = m \geq 1$, $v_F v = n \geq 1$, and $m + n > 2e$, then $(1 + u, 1 + v) = 1$.*

Proof. If $1 + u$ is a square, the assertion is immediate; this includes $m > 2e$ by Hensel's lemma. Otherwise put $r = \lfloor m/2 \rfloor \leq e$. The integral element $(\sqrt{1 + u} - 1)/\varpi^r$ generates an order of discriminant valuation $2e - 2r$. The field discriminant divides the order discriminant, so the quadratic norm character has conductor at most $2e - 2r$; the unramified case also satisfies this bound. If m is even, $n \geq 2e - 2r + 1$; if m is odd, $n \geq 2e - 2r$. In both cases the character kills $1 + v$. $\square$

Put $b = 2e - 2a + 1$, an odd positive integer, and $q = \varpi^b$. There are Kummer generators s, R_0 with

$$s^2 = 1 + qu_0, \quad R_0^2 = qw_0, \quad u_0, w_0 \in \mathcal{O}_F^\times.$$

Here is a verification at the stated breaks. For a uniformizer Π_1 of L_1 , its quadratic minimal polynomial and different imply $v_F \operatorname{Tr} \Pi_1 = a$: the term $2\Pi_1$ has L_1 -valuation $2e + 1 > 2a$. Thus $s = 1 - 2\Pi_1 / \operatorname{Tr} \Pi_1$ has the required square defect of valuation b . For a maximal quadratic field, a uniformizer Π has $v_F \operatorname{Tr} \Pi \geq e + 1$, and $\Pi - \operatorname{Tr} \Pi / 2$ is a trace-zero uniformizer. Multiplying it by ϖ^{e-a} gives R_0 .

Lemma 12.3 (Alignment beyond the leading residue). *The generators can be replaced by $S = s(1 + \delta)$ and $R = R_0v$, with v a base unit, so that, writing $\epsilon = S^2 - 1$,*

$$(234) \quad v_F \epsilon = v_F R^2 = b, \quad \epsilon / R^2 \in U_F^a.$$

Proof. Take $\delta = \varpi^{e-a+1}w$, $w \in \mathcal{O}_F$. Modulo $\mathfrak{p}_F^a$,

$$\frac{S^2 - 1 - R^2}{q} \equiv u_0 - w_0v^2 + \varpi(1 + qu_0)w^2.$$

The omitted term $2\delta/q$ has valuation at least a . Because $a \leq e$, $\mathcal{O}_F/\mathfrak{p}_F^a$ has characteristic two and is $k[\varpi]/(\varpi^a)$ after choosing its coefficient field. Such a coefficient field is obtained by the unique lifts of the roots of $X^{|k|} - X$, whose derivative is a unit. In this truncated ring solve

$$w_0v^2 + \varpi(1 + qu_0)w^2 = u_0.$$

Successively in the degree of ϖ , an even degree determines the next coefficient of v^2 , with unit leading coefficient $\bar{w}_0$; an odd degree determines the next coefficient of w^2 , with leading coefficient 1. Contributions of earlier coefficients are already known. Frobenius on k supplies the unique required square roots. The constant coefficient of v is nonzero. Lift v, w to F . The displayed congruence proves (234), and the positive-depth correction to u_0 preserves its unit leading coefficient. $\square$

Set $L_1 = F(S)$, $L_2 = F(R)$, $L_3 = F(SR)$, and define

$$(235) \quad Y = (S + R - 1)/2, \quad a_i = N_i Y, \quad h_i = S_i Y, \quad \beta_i = n_i h_i.$$

For $B_0 = (\epsilon - R^2)/4$ and $C_0 = (\epsilon + R^2)/4$ one has

$$(236) \quad \begin{aligned} a_1 &= (1 - S)/2 + B_0, & h_1 &= S - 1, & \beta_1 &= -\epsilon, \\ a_2 &= -R/2 - B_0, & h_2 &= R - 1, & \beta_2 &= 1 - R^2, \\ a_3 &= -SR/2 - C_0, & h_3 &= -1, & \beta_3 &= 1. \end{aligned}$$

The alignment gives $v_F B_0, v_F C_0 \geq 1 - a$; for the plus sign use also $v_F(2R^2) \geq b + a$. Consequently

$$(237) \quad v_K Y = -t_1, \quad v_{L_i} a_i = -t_1, \quad v_{L_1} h_1 = b, \quad v_{L_2} h_2 = v_{L_3} h_3 = 0, \quad \operatorname{Tr}_{K/F} Y = -2.$$

For instance a_2 has the unique leading term $-R/2$ of valuation $1 - 2a$, while B_0 has L_2 -valuation at least $2 - 2a$. The value of Y follows by norm valuation; the other entries follow from (236) and $(S - 1)(S + 1) = \epsilon$.

12.3. Simultaneous coefficients for the norm characters. Choose the normalization from the *third* maximal character:

$$(238) \quad \omega_3(1 - z) = \Psi_F(z) \quad (z \in \mathfrak{p}_F^{e+1}), \quad \Psi_L(x) = \psi_L(\alpha x), \quad v_F \alpha = -n_F(\psi_F) - T.$$

The largest trivial ideals of $\Psi_F, \Psi_{L_1}, \Psi_{L_2}, \Psi_{L_3}$ have respective exponents

$$(239) \quad T, \quad J_1 = 2T - 2a, \quad J_2 = J_3 = T.$$

The existence and precision of (238) are the full linear chart from additive duality; $2(e + 1) \geq T$.

Lemma 12.4 (Full lower character formulas). *With $q_1 = a$, $q_2 = q_3 = e + 1$, one has*

$$(240) \quad \omega_i(1 - z) = \Psi_F(\beta_i z) \quad (z \in \mathfrak{p}_F^{q_i}).$$

Each β_i is a genuine norm from L_i , and on the whole stated source ideal the resulting phase conversion is

$$(241) \quad \Psi_{L_i}(\beta_i x) = \Psi_F(\beta_i n_i x) \quad (x \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. For $z \in \mathfrak{p}_F^a$ put $C = 1 + \epsilon z$. The Steinberg identity and (233) give

$$(1 + \epsilon, 1 - z) = (1 + \epsilon, -\epsilon(1 - z)) = (C, \epsilon(1 - z)(1 + \epsilon)).$$

Here $v_F(C - 1) \geq b + a$, and $(b + a) + a = 2e + 1$. Lemma 12.2 removes $1 - z$ from the second argument and replaces ϵ by R^2 , since $\epsilon/R^2 \in U_F^a$. By symmetry the result is $(R^2(1 + \epsilon), C) = \omega_3(C)$. Also $b + a = 2e - a + 1 \geq e + 1$, so (238) gives $\omega_1(1 - z) = \Psi_F(-\epsilon z)$ on the entire ideal. On U_F^{e+1} the product $\omega_2 = \omega_1\omega_3$ therefore has coefficient $1 - \epsilon$. Replacing it by $1 - R^2$ costs valuation at least $b + a + e + 1 = 3e - a + 2 \geq T$. This proves (240); the third formula is the chosen normalization. The norm assertions are (236). Finally for $x \in \mathfrak{p}_{L_i}^{q_i}$ both its trace and norm lie in $\mathfrak{p}_F^{q_i}$ by (231). Evaluate (240) at $n_i(1 - x) = 1 - \text{tr}_i x + n_i x$ and use triviality on norms. This proves (241), including its positive norm sign. $\square$

12.4. Construction and realization of the minimal characters. Put

$$(242) \quad m_1 = 4e + 1, \quad m_2 = m_3 = 2e + 2a, \quad s_1 = 2e + 1, \quad s_2 = s_3 = e + a.$$

Thus $m_i = J_i + t_1$ and $s_i = \lceil m_i/2 \rceil$. The formulas

$$(243) \quad R_i(1 - z) = \Psi_{L_i}(a_i z), \quad z \in \mathfrak{p}_{L_i}^{s_i},$$

define characters of their indicated unit groups with exact conductor m_i : $2s_i - t_1 \geq J_i$, and the last-layer coefficient is nonzero.

Lemma 12.5 (Compatibility before extension). *For $z \in \mathfrak{p}_K^{2e+1}$, all $N_i(1 - z)$ lie in the domains of R_i , and $R_i(N_i(1 - z))$ is independent of i .*

Proof. The norm endpoints have valuation at least $2e + 1$ on the target field. The upper traces have depths at least $3e - a + 1$ and $e + a$, hence are in the required ideals. Use the exact identity

$$(244) \quad E_2(W) = \text{tr}_i N_i W + n_i S_i W$$

for the six pair-products of the four conjugates of W . Apply it to $Y(1 - z)$ and Y . The quantities $w_i = S_i(Yz)/h_i$ have valuation at least $e + 1$ on all three fields: the first upper trace bound is $3e - 2a + 2$, from which one subtracts b , and the other two are $e + 1$. Thus $n_i S_i(Y(1 - z))/\beta_i = n_i(1 - w_i)$ is an actual norm in the domain of (240). Its additive phase implies

$$\Psi_F(n_i S_i(Y(1 - z)) - \beta_i) = 1.$$

The remaining difference in (244) identifies $R_i(N_i(1 - z)) = \Psi_F(-E_2(Y(1 - z)) + E_2(Y))$, independent of i . $\square$

Theorem 12.6 (Actual global core models). *There are primitive compatible characters χ_i on $L_i^\times$ of conductors m_i , whose full stationary formulas are (243).*

Proof. First extend R_2 across the abelian group $L_2^\times/U_{L_2}^{m_2}$; extend on the finite unit subgroup and choose a uniformizer value. Call the character χ_2 . Its lower conjugate quotient is determined by R_2 on every element of $L_2^\times$, because every ratio $\sigma(v)/v$ has depth at least $t_2 = 2e \geq s_2$. For $v = N_2 Z$, this ratio is $N_2(g_1 Z/Z)$. The element $g_1 Z/Z$ has depth at least $4e - 2a + 1 \geq 2e + 1$ and its N_1 -norm is 1. Lemma 12.5 therefore proves that the conjugate quotient of χ_2 kills $N_2 K^\times$. It is nontrivial: on $U_{L_2}^{t_1}$ the first nonzero lower ramification term maps onto the last row $U_{L_2}^{t_1+t_2}/U_{L_2}^{t_1+t_2+1}$, since t_1 is odd. Formula (243) is nontrivial on this row. Thus the quotient is the actual upper norm character ν_2 .

Consequently $\chi_2 N_2$ is invariant under all of G ; Hilbert 90 makes it trivial on $\ker N_1$ and $\ker N_3$. Extend its descents to $\tilde{\chi}_1, \chi_3$. On side 3, norm filtration gives $N_3(U_K^{2e+1}) = U_{L_3}^{e+a}$, since

$2(e + a) - t_1 = 2e + 1$. Lemma 12.5 gives the full R_3 formula. On side 1, every preimage of $U_{L_1}^{s_1}$ under N_1 lies in $U_K^{s_1}$: the norm has nonzero graded map at every smaller positive depth, all below the upper break $4e - 2a + 1$, and its residue map is Frobenius. Norm valuation excludes nonunits. Hence $\tilde{\chi}_1$ agrees with R_1 on $U_{L_1}^{s_1} \cap N_1 K^\times$. Their quotient extends across the order-two norm quotient; divide by this norm character to obtain χ_1 with the full R_1 formula and the same pullback. The extensions are trivial at depth m_i and nontrivial at $m_i - 1$, by their prescribed charts and norm filtration, so their conductors are exactly m_i . If the common pullback descended from F , χ_2 would be a base pullback times an upper norm character, both invariant under its lower automorphism. This contradicts its nontrivial commutator. $\square$

Lemma 12.7 (Conjugacy, determinant, and realization). *For every primitive compatible family,*

$$\theta_i^\sigma / \theta_i = \nu_i = \omega_j n_i \quad (j \neq i), \quad D_\theta = \theta_i|_{F^\times} \omega_i$$

is independent of i . For comparison of 1 and 2 we may, without changing either local constant or the common pullback, write

$$(245) \quad \theta_i = \chi_i(\lambda n_i) \quad (i = 1, 2)$$

with cores as in Theorem 12.6. If $n = a_F(\lambda)$, the conductors are m_i for $n \leq 2e + a$; for $n \geq 2e + a + 1$ they are

$$(246) \quad M_1 = 2n - 2a, \quad M_2 = 2n - T.$$

Proof. Lemma 4.2 gives the stated conjugate quotient and corrected common restriction, for the given family and the core family. The quotient θ_2/χ_2 is invariant and is therefore λn_2 . The discrepancy on side 1 is either 1 or ν_1 . Conjugate θ_1 in the latter case. Its local constant is unchanged by a field-automorphism change of variables in the defining integral; its norm pullback is unchanged because $\nu_1 N_1 = 1$. This proves (245); it is not an application of SML. Finally the conductor formula for norm pullback gives the alternatives: below the displayed threshold both twists have smaller conductor than their cores; above it they have larger conductor. On side 1 the core is odd and the higher pullback even; on side 2 the core is even and the higher pullback odd. No equal-conductor cancellation is omitted. $\square$

12.5. Normalization and the exact norm choice.

Lemma 12.8 (Preservation and an exact third-field norm). *Replacing α by αu , with $u \in U_F^{2e}$, leaves (240) and the full actual core formulas (243) unchanged. For any nonzero coefficient A_* of a base character relative to Ψ_F , such a replacement can be made so that $A = A_*/u$ is an exact norm from L_3/F . It remains the coefficient of the same base character relative to the new Ψ_F .*

Proof. For the lower formula the change has valuation at least $2e + v_F \beta_i + q_i \geq T$. For the upper core formula it has L_i -valuation at least $4e - t_1 + s_i \geq J_i$. Thus these formulas for the already fixed characters remain valid on whole ideals, not merely on their final residue rows. The actual ω_3 has conductor $2e + 1$ and is nontrivial on U_F^{2e} . Apply Lemma 4.9 to this subgroup. It gives the claimed norm without changing the raw coefficient αA_* . Norm valuation fixes $v_{L_3} X = v_F A$. $\square$

We henceforth use the new normalization when needed.

For comparison of 1, 2 take $X \in L_3$, and put

$$A = n_3 X, \quad p_X = \text{tr}_3 X, \quad C = Y - X, \quad Q_i = S_i(Y g_i X) \quad (i = 1, 2).$$

Then, exactly,

$$(247) \quad \begin{aligned} N_i C &= a_i + A - Q_i, & S_i C &= h_i - p_X, \\ n_i Q_i &= \beta_i A + \mathcal{E}, & \mathcal{E} &= (X' - X)(a_3 X' - a'_3 X) \in F, \end{aligned}$$

$$(248) \quad n_i S_i C = \beta_i + \Delta, \quad \Delta = p_X^2 + 2p_X.$$

Primes in the middle line denote the automorphism of L_3/F . These identities follow by multiplying the four conjugates, with X fixed by g_3 ; the last uses $\text{tr}_i h_i = \text{Tr}_K Y = -2$.

Lemma 12.9 (Weighted common error). *If $v_{L_3}X = -h$, for any integer h , then*

$$(249) \quad \begin{aligned} v_F p_X &\geq e - \lfloor h/2 \rfloor, & v_F \mathcal{E} &\geq T - a - h, \\ v_{L_1} Q_1 &\geq b - h, & v_{L_2} Q_2 &\geq -h. \end{aligned}$$

Proof. Let $r_3 = SR/\varpi^{e-a}$, a trace-zero uniformizer of L_3 . Write $X = P + Qr_3$; parity of the two valuations gives $v_F P \geq -\lfloor h/2 \rfloor$ and $v_F Q \geq -\lceil h/2 \rceil$. The trace is $2P$, proving its bound. Write $a_3 = A_3 + B_3 r_3$, with $v_F A_3 \geq 1 - a$ and $v_F B_3 = -a$ by (236). Now $X' - X = -2Qr_3$ and $a_3 X' - a'_3 X = 2(B_3 P - A_3 Q)r_3$. Their product has F -valuation at least $2e + 1 - a - h = T - a - h$. The upper trace formula applied to $Y_{g_i} X$, of K -valuation $-t_1 - 2h$, gives respectively $b - h$ and $-h$ for Q_i . $\square$

The conversion (241) and (247) imply, whenever $(Q_i/\beta_i)z \in \mathfrak{p}_{L_i}^{q_i}$,

$$(250) \quad \Psi_{L_i}(Q_i z) = \Psi_F((A + \mathcal{E}/\beta_i)n_i z).$$

This retains the actual norm error with its weight.

12.6. Minimal conductors: upper and lower stationary origins. Suppose $n \leq 2e + a$. Both trace and norm of a variable in $\mathfrak{p}_{L_i}^{s_i}$ lie in $\mathfrak{p}_F^{e+a}$, for $i = 1, 2$. If $n \leq e + a$, the twist is trivial there, and use $X = 0$, $C = Y$. Otherwise put $h = n - T$, so

$$(251) \quad a - e \leq h \leq a - 1.$$

Additive duality on U_F^{e+a} supplies a nonzero base coefficient A_* of valuation $-h$, since $2(e+a) \geq n$. Use Lemma 12.8 to make its new value $A = n_3 X$ an exact norm.

Lemma 12.10 (Full stationary formulas for the given characters). *The resulting C has $v_K C = -t_1$. For $i = 1, 2$,*

$$(252) \quad \begin{aligned} \theta_i(1 - z) &= \Psi_{L_i}(N_i C z) & (z \in \mathfrak{p}_{L_i}^{s_i}), \\ \omega_i(1 - z) &= \Psi_F(n_i S_i C z) & (z \in \mathfrak{p}_F^{q_i}). \end{aligned}$$

Both $S_i C$ are nonzero and have the same valuations as h_i .

Proof. For $X \neq 0$, $-2h > -t_1$, so Y dominates in C . The two arguments of (250) have depths at least $2a - h \geq a + 1$ and $e + a - h \geq e + 1$. The error terms $(\mathcal{E}/\beta_i)n_i z$ have depths at least $T + a - h$ and $T + e - h$, both at least T . Exact quadratic norm expansion evaluates the actual λn_i on its full stationary ideal as $\Psi_{L_i}((A - Q_i)z)$, giving the first formula. For the second, (249) and $h \leq a - 1$ give

$$v_F \Delta \geq \min(2e - 2\lfloor h/2 \rfloor, 2e - \lfloor h/2 \rfloor).$$

Adding a to either bound gives at least T ; adding $e + 1$ does also. Thus (248) preserves both full lower formulas. Moreover $v_{L_1} p_X \geq 2e - 2\lfloor h/2 \rfloor > b$, while p_X has positive valuation on L_2 . The traces retain their nonzero leading terms. $\square$

Use the full stationary formula and terminal invariance of Lemma 4.1, with $c_0 = -\alpha^{-1}$ and residual factors denoted g_θ .

Theorem 12.11 (Exact minimal comparison). *Every family with $n \leq 2e + a$ satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. Use the coefficients of Lemma 12.10 in (7). Their norm-character values are 1, and compatibility cancels the two upper multiplicative values. Formula (244) gives

$$(253) \quad \mathcal{A}_i = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C))g_{\theta_i}g_{\omega_i}.$$

Only θ_1 and ω_2 have odd conductors, with critical depths $2e$ and e respectively. Choose a top uniformizer π and put $C_z = C(1 + \pi^{2e}z)$ for integral $z \in K$. On side 1 the relative upper norm has leading term its norm endpoint at depth $2e$; its trace has depth at least $3e - a + 1 \geq 2e + 1$. Its residue map is a nonzero scalar times Frobenius, hence bijective. On side 2 the relative upper norm is in the even stationary ideal: its trace has depth $e + a$, and its norm endpoint depth $2e \geq e + a$.

The trace change $S_i(C\pi^{2e}z)$ has first-side depth at least $3e - 2a + 1$; after division by $S_1 C$ its depth is at least $e \geq a$. Its lower norm quotient is therefore on the even stationary ideal of ω_1 .

On side 2, the upper trace of an element of K -valuation $b = 2e - 2a + 1$ induces an isomorphism from its residue line to the L_2 row of depth e : the consecutive trace-ideal exponents are e and $e + 1$. Division by the unit S_2C preserves this isomorphism. The lower maximal norm then has leading endpoint at depth e and trace at depth at least $\lfloor (3e + 1)/2 \rfloor \geq e + 1$. Thus its residue map is again bijective.

Subtract (244) for C_z, C . The multiplicative character values cancel by the actual common pullback, and both lower arguments are ratios of actual norms. The complete residual expressions therefore give

$$H_{\theta_1}(p(z)) = H_{\omega_2}(q(z)) \quad (z \in k),$$

where both $p, q : k \rightarrow k$ are bijective maps just constructed. All even expressions are 1; all arguments of multiplicative characters are nonzero, since the relative changes have positive depth. Summing gives $g_{\theta_1} = g_{\omega_2}$. Equation (253) proves the exact sign. $\square$

12.7. Higher conductors: two stationary origins. Now $n \geq 2e + a + 1$, and put $h = n - T \geq a$. The upper stationary depths are

$$(254) \quad b_1 = T + h - a, \quad b_2 = e + 1 + h,$$

with $M_1 = 2b_1$ even and $M_2 = 2b_2 - 1$ odd. Choose an actual coefficient A_* of λ on its ordinary base stationary ideal. Both trace and norm of $z \in \mathfrak{p}_{L_i}^{b_i}$ lie in that base ideal, by (231). Its valuation is $-h$. Lemma 12.8 supplies, after its harmless normalization change, $A = n_3X$ exactly, with $v_{L_3}X = -h$. There is no upper bound on h .

Lemma 12.12. *For the actual given characters, N_iC is a full stationary coefficient on $\mathfrak{p}_{L_i}^{b_i}$, $i = 1, 2$. Its valuation is $-2h$.*

Proof. The arguments $(Q_i/\beta_i)z$ of (250) have respective depths at least a and $e + 1$. The weighted errors there have depths

$$(T - a - h) - b + b_1 = T, \quad (T - a - h) + b_2 = 3e + 2 - a \geq T.$$

Consequently exact quadratic norm expansion yields the pulled-back coefficient $A - Q_i$ on the whole indicated ideals. These ideals are contained in the core domains, so adding a_i gives N_iC . The element X dominates Y because $-2h < -t_1$; hence $v_KC = -2h$ and $v_{L_i}N_iC = -2h$, as required by the actual conductors. $\square$

Retain $\beta_i = n_i h_i$, rather than $n_i S_i C$, for the lower characters. No assertion that the latter remain stationary in this range is needed. Equations (244), (248), and (7) give

$$(255) \quad \mathcal{A}_i = D_{\theta}(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C) + \Delta)g_{\theta_i}g_{\omega_i}.$$

The prefactor is already the same for $i = 1, 2$, even if Δ is shallow. Both residual factors on side 1 are 1; both on side 2 are odd.

Theorem 12.13 (Exact comparison at every higher conductor). *For every $h \geq a$, one has $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. Put

$$V = Y\pi^{2e}z, \quad C_z = C + V, \quad Y_z = Y + V \quad (z \in \mathcal{O}_K).$$

The upper character rows use C_z/C , but the lower norm-character rows use $S_i Y_z/h_i$. These are *two* origins, a distinction necessary when $n_i S_i C$ is not stationary. The element V has K -valuation at least b and V/C has depth $b + 2h$. On side 1, its upper trace depth is at least $3e - 2a + 1 + h \geq b_1$, and its norm endpoint depth $b + 2h \geq b_1$. Thus the upper residual expression is 1 on its even stationary ideal. On side 2 the consecutive upper trace-ideal exponents at source depths $b + 2h, b + 2h + 1$ are $e + h, e + h + 1$. The norm endpoint is strictly deeper, since $b + 2h - (e + h) = e - 2a + 1 + h \geq 1$. It follows that the actual upper odd residue map $p : k \rightarrow k$ is bijective.

For the lower rows, $S_1 V/h_1$ has depth at least $e \geq a$, so its norm quotient is in the even stationary ideal of ω_1 . The map $z \mapsto S_2 V/h_2$ has a nonzero graded trace at depth e , by the consecutive trace exponents at $b, b + 1$. Its lower norm has leading endpoint at depth e and trace

of depth at least $e + 1$. Hence the actual lower odd residue map $q : k \rightarrow k$ is bijective as well. All traces in these lower character arguments stay nonzero.

Here the elementary phase equality is exact, not an error estimate. Since $S_i C = h_i - p_X$, quadratic norm expansion gives

$$[n_i S_i C_z - n_i S_i C] - [n_i S_i Y_z - \beta_i] = -p_X \operatorname{Tr}_{K/F} V.$$

Using (244), we obtain

$$(256) \quad \operatorname{tr}_i(N_i C_z - N_i C) + (n_i S_i Y_z - \beta_i) = E_2(C_z) - E_2(C) + p_X \operatorname{Tr}_{K/F} V,$$

independent of i . Upper multiplicative values are the common $\Theta(C_z/C)$ and lower ones are 1, because their arguments are norms. Thus (256) and the complete residual definition give

$$H_{\theta_2}(p(z))H_{\omega_2}(q(z)) = 1 \quad (z \in k).$$

Both residue maps are additive bijections. Therefore $\sum H_{\theta_2} = \overline{\sum H_{\omega_2}}$, after reindexing. Each normalized sum has absolute value 1 by Ueda's nondegenerate finite quadratic-sum formula. Consequently $g_{\theta_2}g_{\omega_2} = 1$. Equation (255) proves the assertion, for arbitrarily large h , without a separate stability or norm-approximation limit. $\square$

12.8. Completion of the maximal-break case.

Proof of Theorem 12.1. The explicit aligned origin and Lemma 12.4 give full actual lower coefficients. Theorem 12.6 constructs actual cores, and Lemma 12.7 realizes the given comparison, using at most a conjugation that preserves the individual local constant and pullback. Every base conductor is either $n \leq 2e + a$, covered by Theorem 12.11, or $n \geq 2e + a + 1$, covered by Theorem 12.13. Their endpoints are consecutive. Repeat the construction with the two maximal fields interchanged to compare the distinguished L_1 with L_3 . This proves independence of all three expressions for the original given family. $\square$

13. TOTALLY RAMIFIED MIXED CHARACTERISTIC: NONMAXIMAL BREAKS

13.1. Statement and ramification data. Let $F/\mathbf{Q}_2$ be finite, with residue field k , uniformizer ϖ , and $e = v_F(2)$. Normalize all valuations on their named fields. Let K/F be totally ramified with group C_2^2 . Label its lower quadratic fields $L_i = K^{\langle g_i \rangle}$ so that

$$(257) \quad t_1 = 2a - 1, \quad t_2 = t_3 = 2r - 1, \quad 1 \leq a \leq r \leq e.$$

Put $\delta = r - a$, $T = 2r$, and $T_1 = 2a$, $T_2 = T_3 = T$. These T_i are the lower different exponents and the conductors of the actual quadratic norm characters ω_i . Write

$$N_i = \mathbf{N}_{K/L_i}, \quad S_i = \operatorname{Tr}_{K/L_i}, \quad n_i = \mathbf{N}_{L_i/F}, \quad \operatorname{tr}_i = \operatorname{Tr}_{L_i/F}.$$

For an arbitrary nontrivial additive character ψ_F , use $\psi_{L_i} = \psi_F \operatorname{tr}_i$. Set $n_F = n_F(\psi_F)$, so $\mathbf{p}_F^{-n_F}$ is its largest trivial ideal.

Theorem 13.1. *Suppose $\theta_i N_i = \Theta$ is a primitive compatible family: Θ is invariant under $\operatorname{Gal}(K/F)$ and does not descend by $\mathbf{N}_{K/F}$. Then the three expressions*

$$(258) \quad \mathcal{A}_i(\theta_i) = \Delta_{L_i}(\theta_i, \psi_{L_i}) \Delta_F(\omega_i, \psi_F)$$

are equal. This holds for every continuous quasi-character conductor under (257).

A prescribed compatible primitive pair extends to such a family by Hilbert 90 and character extension from the third norm image; its two prescribed members are not changed. The trivial-character factor omitted from (258) is one.

The one-dimensional inputs are the results of [8]: full Lamprecht ([8, Theorem 4.5]), stationary additive duality, quadratic norm filtration and norm-character conductor, norm-pullback conductor ([8, Proposition 3.10]), trace/different formulas, and nondegeneracy and magnitude of the residual quadratic sum. For a ramified quadratic edge E/D of different exponent d these give

$$(259) \quad \operatorname{Tr}_{E/D}(\mathbf{p}_E^j) = \mathbf{p}_D^{\lfloor (j+d)/2 \rfloor}, \quad v_D(\mathbf{N}x) = v_E x, \quad n_E(\psi_E) = 2n_D(\psi_D) + d.$$

A base character of conductor $n > d$ pulls back with conductor $2n - d$; for $n \leq d$ its pullback has conductor at most d . The descent and order-discriminant facts are proved in Appendix C, and the required crossed norm-character and ramification facts in Appendix D. In particular Lemma D.3 gives the upper quadratic different exponents

$$(260) \quad D'_1 = 4r - 2a, \quad D'_2 = D'_3 = 2a.$$

Every additional two-field calculation is proved below.

13.2. Alignment of the quadratic generators. Choose actual generators $x \in L_1, y \in L_2$ with

$$(261) \quad x^2 + x = f, \quad y^2 + y = g, \quad v_F f = 1 - 2a, \quad v_F g = 1 - 2r.$$

Here is a construction. If a uniformizer of a quadratic field has minimal polynomial $X^2 - cX + b$, its different is the valuation of $2\Pi - c$. The two possible valuations $2e + 1$ and $2v_F c$ have different parities. Different $2a < 2e + 1$ therefore forces $v_F c = a$. Then $x = -\Pi/c$ satisfies (261); use the same argument for y . All conjugations are $x \mapsto -1 - x, y \mapsto -1 - y$.

Lemma 13.2. *The generator x may be changed, without changing its field or the valuations in (261), and $d \in F$ may be chosen so that*

$$(262) \quad v_F d = \delta, \quad E := f + d^2 g \in \mathfrak{p}_F^{1-a}.$$

Moreover $k_0 := 1 + d$ is a unit, including when $a = r$.

Proof. Start with x_0, f_0 and put $S_0 = 1 + 2x_0, U_0 = S_0^2 = 1 + 4f_0$. Let $b = 2e - 2a + 1$ and $q = \varpi^b$. Replace

$$S_0 \text{ by } S = S_0(1 + \varpi^{e-a+1}w), \quad x = (S - 1)/2, \quad d = \varpi^{r-a}v.$$

The two base variables v, w will be integral, with v a unit. Modulo $\mathfrak{p}_F^a$,

$$\frac{S^2 - 1 + 4d^2g}{q} \equiv u_0 + \varpi U_0 w^2 + v_0 v^2, \quad u_0 = 4f_0/q, \quad v_0 = 4\varpi^{2r-2a}g/q.$$

The omitted linear term has valuation at least a ; u_0, v_0, U_0 are units. Since $a \leq e$, the ring $\mathcal{O}_F/\mathfrak{p}_F^a$ has characteristic two and is $k[\varpi]/(\varpi^a)$. Its coefficient field can be constructed as the unique lifts of the roots of $Z^{|k|} - Z$, using its unit derivative. Solve $v_0 v^2 + \varpi U_0 w^2 = u_0$ successively in powers of ϖ . At an even degree the next coefficient of v^2 is determined, and at an odd degree the next coefficient of w^2 is determined; earlier contributions are known and the leading coefficients are units. Frobenius on k supplies the required residue square roots. The constant coefficient of v is nonzero. Lifting v, w to F proves $v_F E \geq b + a - 2e = 1 - a$. The perturbation of $4f_0$ is strictly deeper than b , so $v_F f = 1 - 2a$ is unchanged. This did not request a square root of a fixed arbitrary unit modulo $\mathfrak{p}_F^a$.

If $r > a$, then $d \in \mathfrak{p}_F$. Suppose $r = a$ and $\bar{d} = 1$. Then $f + g$ has valuation at least $2 - 2a$, and so does $f + g + 4fg$, since $e \geq a$. The generator $z = x + y + 2xy \in L_3$ satisfies $z^2 + z = f + g + 4fg$. The integral element $\varpi^{a-1}z$ generates an order whose polynomial discriminant has valuation $2a - 2$. The field different is at most this value, contradicting $T_3 = 2a$. Thus $1 + d$ is a unit. $\square$

Put

$$(263) \quad z = x + y + 2xy \in L_3, \quad Y = x + dy, \quad a_i = N_i Y, \quad h_i = S_i Y, \quad \beta_i = n_i h_i.$$

The following are exact identities, not characteristic-two approximations:

$$(264) \quad z^2 + z = f + g + 4fg,$$

$$(265) \quad h_1 = 2x - d, \quad h_2 = 2dy - 1, \quad h_3 = -k_0,$$

$$(266) \quad \beta_1 = d^2 + 2d - 4f, \quad \beta_2 = 1 + 2d - 4d^2g, \quad \beta_3 = k_0^2,$$

$$(267) \quad a_1 = 2f - E - k_0x, \quad a_2 = E - 2f - dk_0y, \quad a_3 = -E - dz.$$

The right side of (264) has valuation $1 - 2r$. If $r > a$, g is its unique leading term; if $r = a$, Lemma 13.2 excludes cancellation of the leading terms of $f + g$. The term $4fg$ is strictly deeper

in both cases. Thus $v_{L_3}z = 1 - 2r$. The constant terms in (267) have base valuation at least $1 - a$, whereas its nonconstant terms have L_i -valuation $1 - 2a$. Hence

$$(268) \quad v_K Y = v_{L_i} a_i = -t_1, \quad v_{L_1} h_1 = 2\delta, \quad v_{L_2} h_2 = v_{L_3} h_3 = 0, \quad \text{Tr}_{K/F} Y = -2k_0.$$

For h_1 , the depth of $2x$ is $2e - 2a + 1 > 2\delta$; the other traces are units. In particular all trace norms in (263) are nonzero, $v_F \beta_1 = 2\delta$, and β_2, β_3 are units.

13.3. The lower stationary coefficients. Use the quadratic norm symbol, the conic identity (233), and the unit-depth estimate of Lemma 12.2. Their hypotheses depend only on F and $e = v_F(2)$, not on maximality of the breaks.

Choose full ordinary stationary covectors b_1, b_2 for the actual lower characters:

$$\omega_1(1 - u) = \psi_F(b_1 u) \quad (u \in \mathfrak{p}_F^a), \quad \omega_2(1 - u) = \psi_F(b_2 u) \quad (u \in \mathfrak{p}_F^r).$$

They exist by additive duality, with valuations $-n_F - 2a$ and $-n_F - 2r$. The product character ω_3 has coefficient $b_1 + b_2$ on $\mathfrak{p}_F^r$. Define

$$(269) \quad \alpha = b_2/\beta_2, \quad \Psi_L(u) = \psi_L(\alpha u), \quad v_F \alpha = -n_F - T.$$

The largest trivial ideals for $\Psi_F, \Psi_{L_1}, \Psi_{L_2}, \Psi_{L_3}$ have exponents

$$(270) \quad T, \quad J_1 = 4r - 2a, \quad J_2 = J_3 = T.$$

Lemma 13.3 (All three full lower formulas). *With $q_1 = a, q_2 = q_3 = r$, one has*

$$(271) \quad \omega_i(1 - u) = \Psi_F(\beta_i u) \quad (u \in \mathfrak{p}_F^{q_i}).$$

Moreover $\omega_i(\beta_i) = 1$, and

$$(272) \quad \Psi_{L_i}(\beta_i v) = \Psi_F(\beta_i n_i v) \quad (v \in \mathfrak{p}_{L_i}^{q_i}).$$

Proof. Write $U = 1 + 4f, V = 1 + 4g$. For $u \in \mathfrak{p}_F^a$, put $C = 1 + 4fu$ and $w = (f/g)u$. Then

$$v_F(C - 1) \geq 2e - a + 1, \quad v_F w \geq 2r - a \geq r.$$

Multiplying the second symbol argument by the norm $-4f$ and using (233) gives

$$\omega_1(1 - u) = (C, f(1 - u)U).$$

The ratio $f/(-d^2g)$ lies in U_F^a , by (262). Lemma 12.2 removes this ratio and $1 - u$ from the second argument. The result is $(C, -gU)$. Applying the same transformation with g, w gives $\omega_2(1 - w) = (C, g(1 - w)V) = (C, gV)$. The factor -1 in the product is also killed by Lemma 12.2, since $v_F(-2) = e$. Consequently the exact identity on this whole ideal is

$$(273) \quad \omega_1(1 - u) = \omega_2(1 - (f/g)u) \omega_3(1 + 4fu).$$

All uses of the lemma have depth sum greater than $2e$; in particular $(2e - a + 1) + a = 2e + 1$. The last argument has depth at least $2e - a + 1 \geq r$, so all three full linear formulas apply. Equation (273) gives, by full-ideal duality,

$$(274) \quad b_1 - b_2 R \in \mathfrak{p}_F^{-n_F - a}, \quad R = \frac{f(1 - 4g)}{g(1 + 4f)}.$$

Here multiplication of the variable u by the unit $1 + 4f$ preserves its whole ideal; it is not an inference from one trivial phase.

We check that the explicit norm coefficients have exactly this ratio. Direct expansion gives

$$(275) \quad f(1 - 4g)\beta_2 - g(1 + 4f)\beta_1 = E(1 + 2d + 16fg) - 8fgd(d + 2) - 2gd(d^2 + d + 1).$$

Every term has valuation at least $1 - a$. For the middle term, $v_F(d + 2) = \delta < e$, so its valuation is at least $3e + 2 - 4a \geq 1 - a$; for the last it is at least $e - r - a + 1 \geq 1 - a$. The first follows from (262) and $v_F(16fg) \geq 2$. Division by $g(1 + 4f)\beta_2$, of valuation $1 - 2r$, gives $R - \beta_1/\beta_2 \in \mathfrak{p}_F^{2r - a}$. Together with (274) this proves the first lower formula. The second is (269). Finally

$$\beta_3 - \beta_1 - \beta_2 = -2d + 4E \in \mathfrak{p}_F^r,$$

which proves the third formula on $\mathfrak{p}_F^r$. The β_i are the actual norms in (263), so their character values are one. For $v \in \mathfrak{p}_{L_i}^{q_i}$ both its trace and norm lie in $\mathfrak{p}_F^{q_i}$. Evaluate (271) at the exact

norm $n_i(1 - v) = 1 - \text{tr}_i v + n_i v$. Its character is one, which proves (272) with the positive sign shown. $\square$

13.4. Construction of the minimal characters. Put

$$(276) \quad m_1 = 4r - 1, \quad m_2 = m_3 = 2a + 2r - 1, \quad s_1 = 2r, \quad s_2 = s_3 = a + r.$$

Then $m_i = J_i + t_1 = 2s_i - 1$. The expressions

$$(277) \quad R_i(1 - v) = \Psi_{L_i}(a_i v) \quad (v \in \mathfrak{p}_{L_i}^{s_i})$$

define characters with exact conductor m_i on the indicated groups: $2s_i - t_1 \geq J_i$, and the last-layer coefficient is nonzero.

Lemma 13.4 (The shallow commutator is evaluated, not assumed). *Let σ be the lower involution on L_1 and $\nu_1 = \omega_2 n_1$, the actual norm character for K/L_1 . For every $v \in L_1^\times$ with $\sigma(v)/v \in U_{L_1}^{2r}$,*

$$(278) \quad R_1(\sigma(v)/v) = \nu_1(v).$$

Proof. Multiplication by a base element changes neither side. Odd valuation would give conjugate quotient of exact depth $t_1 < 2r$, by the first ramification term, so it is excluded. Otherwise scale to a unit and write $v = P + Qx$. Parity of the two summand valuations makes P a unit and $v_F Q \geq a$. The element $1 + 2x$ is a unit of L_1 . Thus the assumed conjugate-quotient depth forces $w = Q/P \in \mathfrak{p}_F^r$. Put

$$V_0 = 1 - w - fw^2 = n_1(1 + wx), \quad Q_0 = 1 - dw - d^2gw^2 = n_2(1 + dwy), \quad H_0 = 1 + d^2gw^2.$$

All three are units. Write $c = f - d^2g = E - 2d^2g$, so $v_F c \geq 1 - a$. The difference $Q_0 - V_0 = (1 - d)w + cw^2$ lies in $\mathfrak{p}_F^r$. Since Q_0 is a norm from L_2 , the actual lower formula gives

$$(279) \quad \nu_1(v) = \omega_2(V_0) = \Psi_F(\beta_2(Q_0 - V_0)/Q_0).$$

On the other hand $\sigma(v)/v = 1 - w(1 + 2x)/(1 + wx)$. Using $a_1 = f - d^2g - k_0x$, an exact quadratic trace calculation yields

$$(280) \quad R_1(\sigma(v)/v) = \Psi_F(B_0/V_0), \quad B_0 = -(1 + 4f)(cw^2 + k_0w).$$

Here is the full modulus check comparing these expressions. Both numerators in (279)–(280) lie in $\mathfrak{p}_F^r$. Also

$$V_0 - H_0 = -w - Ew^2 \in \mathfrak{p}_F^r, \quad Q_0 - H_0 = -dw - 2d^2gw^2 \in \mathfrak{p}_F^r.$$

Thus replacing either unit denominator by H_0 costs an element of $\mathfrak{p}_F^{2r}$, exactly the trivial ideal of Ψ_F . With $U = 1 + 4f$, the remaining numerator difference is

$$B_0 - \beta_2(Q_0 - V_0) = -(U + \beta_2)cw^2 - (Uk_0 + \beta_2(1 - d))w.$$

The identities

$$U + \beta_2 = 2k_0 + 4c, \quad Uk_0 + \beta_2(1 - d) = 2(1 + d - d^2) + 4Ek_0 - 8d^2g$$

show that the first coefficient has valuation at least e , and the second at least $e \geq r$. The first summand therefore has valuation at least $2r + e + 1 - a \geq 2r$ and the second at least $2r$. Equations (279)–(280) now prove the assertion. In particular no character formula on an unjustified shallow ideal was used to evaluate $\nu_1(v)$. $\square$

For any $Z \in K$ we will repeatedly use the exact identity

$$(281) \quad E_2(Z) = \text{tr}_i N_i Z + n_i S_i Z.$$

Both sides are the sum of the six pair-products of its four conjugates.

Lemma 13.5. *For $v \in \mathfrak{p}_K^{2r}$ all $N_i(1 - v)$ belong to the domains in (277), and $R_i(N_i(1 - v))$ is independent of i .*

Proof. The upper traces have depths at least $3r - a$ and $r + a$, and the norm endpoints have depth $2r$. These meet all three domain requirements. Apply (281) to $Y(1 - v)$ and Y . The upper trace changes $S_i(Yv)$ have depths at least $3r - 2a$ and r , respectively. After division by h_i , each relative change has depth at least r , which is at least q_i . Its lower norm is an actual norm, and Lemma 13.3 makes the phase of $n_i S_i(Y(1 - v)) - \beta_i$ trivial. The other part of (281) is $\text{tr}_i(a_i(-S_i v + N_i v))$. It follows that

$$R_i(N_i(1 - v)) = \Psi_F(-E_2(Y(1 - v)) + E_2(Y)),$$

independent of i . $\square$

Theorem 13.6 (Full global core models). *There exist characters χ_i of the full multiplicative groups $L_i^\times$, with primitive common norm pullback, conductors m_i , and the full formulas (277).*

Proof. Let $I = \{\sigma(v)/v : v \in L_1^\times\}$ and prescribe $\kappa_I(\sigma(v)/v) = \nu_1(v)$. This is well defined since ν_1 is trivial on $F^\times$. Lemma 13.4 proves agreement with R_1 on $I \cap U_{L_1}^{s_1}$. Their product character factors through $IU_{L_1}^{s_1}/U_{L_1}^{m_1}$; extend it across $L_1^\times/U_{L_1}^{m_1}$, first on its finite unit subgroup and then on a uniformizer. The resulting χ_1 has exact conductor m_1 and satisfies $\chi_1^\sigma/\chi_1 = \nu_1$ globally.

Its norm pullback to K is invariant under both involutions, since $\nu_1 N_1 = 1$. Hilbert 90 makes it trivial on each other norm kernel; extend its descents to χ_2, χ_3 . Norm filtration gives

$$N_i(U_K^{2r+1}) = U_{L_i}^{a+r} \quad (i = 2, 3),$$

since these upper edges have break $2a - 1$ and $2(a + r) - (2a - 1) = 2r + 1$. Lemma 13.5 gives the full R_i formulas on these groups. Their conductors are at least $m_i > D'_i$. The common upstairs conductor, computed from χ_1 , is $2m_1 - D'_1 = 4r + 2a - 2$. The high norm-pullback conductor formula then forces the other conductors to be exactly m_i . If the common pullback descended from F , χ_1 would be a base pullback times its upper norm character, both invariant under σ . This contradicts its nontrivial commutator. Thus the family is primitive. $\square$

13.5. Common-twist realization and norm choices.

Lemma 13.7. *For every primitive compatible family,*

$$\theta_i^\sigma/\theta_i = \nu_i = \omega_j n_i \quad (j \neq i), \quad D_\theta = \theta_i|_{F^\times} \omega_i$$

is independent of i . There are cores with the formulas of Theorem 13.6 and a base character λ such that

$$(282) \quad \theta_i = \chi_i(\lambda n_i).$$

If $n = a_F(\lambda)$, their conductors are m_i for $n \leq 2r + a - 1$; for $n \geq 2r + a$ they are $M_i = 2n - T_i$, all even.

Proof. Lemma 4.2 gives the stated conjugate quotient and corrected common restriction, for the given family and the core family. The quotient θ_1/χ_1 is invariant, so Hilbert 90 and character extension give λn_1 . The remaining discrepancies on sides 2 and 3 are powers of ν_i . Absorb them into the cores: their conductors $2a$ are at most $s_i = a + r$, so the full core charts do not change. This proves (282) for the given family. The pullback of a base character with $n > T_i$ has conductor $2n - T_i$; otherwise it has conductor at most $T_i < m_i$. Since m_i is odd and $2n - T_i$ is even, equal-conductor cancellation does not occur. The common threshold is $n = 2r + a$, giving precisely the alternatives. $\square$

Lemma 13.8 (Normalization freedom after realization). *Replacing α by αu with $u \in U_F^{T-1}$ preserves all full core and lower formulas for the already fixed characters. For any nonzero coefficient A_* of λ relative to Ψ_F , such a replacement makes $A = A_*/u$ an exact norm from L_3/F . The raw coefficient αA_* and the base character itself are unchanged.*

Proof. On each lower ideal the normalized error has depth at least $T - 1 + v_F \beta_i + q_i \geq T$. On the upper core ideals it has depth at least $2(T - 1) - t_1 + s_i \geq J_i$. These inequalities include $a = r = 1$. Thus the full formulas are preserved. The actual ω_3 is onto $\{1, -1\}$ on U_F^{T-1} , so Lemma 4.9 applies with this subgroup and gives the exact norm while keeping αA_* fixed. $\square$

13.6. Adjustment through the third quadratic field. For comparison of fields 1 and 2, take $X \in L_3$ and put

$$(283) \quad A = n_3 X, \quad p_X = \text{tr}_3 X, \quad C = Y - X, \quad Q_i = S_i(Y g_i X) \quad (i = 1, 2).$$

For $X = 0$ all expressions below have their evident zero values. With primes denoting conjugation in L_3/F , direct multiplication gives

$$(284) \quad N_i C = a_i + A - Q_i, \quad S_i C = h_i - p_X,$$

$$(285) \quad n_i Q_i = \beta_i A + \mathcal{E}, \quad \mathcal{E} = (X' - X)(a_3 X' - a'_3 X) \in F,$$

$$(286) \quad n_i S_i C = \beta_i + \Delta, \quad \Delta = p_X^2 + 2k_0 p_X.$$

The last identity uses $\text{tr}_i h_i = -2k_0$, the same for both fields. It is valid even when a trace of C vanishes; no character is evaluated at such a zero.

Lemma 13.9. *If $v_{L_3} X = -h$ for any integer h , then*

$$(287) \quad \begin{aligned} v_F p_X &\geq r - \lceil h/2 \rceil, & v_F \mathcal{E} &\geq T - a - h, \\ v_{L_1} Q_1 &\geq 2\delta - h, & v_{L_2} Q_2 &\geq -h. \end{aligned}$$

Proof. Write $X = P + Qz$, with z as in (264). Parity of valuations gives $v_F P \geq -\lceil h/2 \rceil$ and $v_F Q \geq r - \lceil h/2 \rceil$. The trace $2P - Q$ has the asserted bound because $e \geq r$. Using $a_3 = -E - dz$, the exact common error is

$$\mathcal{E} = Q(dP - EQ)(1 + 4(f + g + 4fg)).$$

Its last factor is a unit. The first product term has depth at least $r + \delta - h = T - a - h$; the second at least $1 - a + 2r - 2\lceil h/2 \rceil \geq T - a - h$. Finally $Y g_i X$ has K -valuation $-t_1 - 2h$. The upper trace formula (259), with (260), gives the two Q_i bounds exactly as stated. $\square$

Whenever $(Q_i/\beta_i)v \in \mathfrak{p}_{L_i}^{q_i}$, the actual norm phase gives

$$(288) \quad \Psi_{L_i}(Q_i v) = \Psi_F((A + \mathcal{E}/\beta_i)n_i v).$$

If the error term is in $\mathfrak{p}_F^T$, exact quadratic norm expansion therefore evaluates λn_i with coefficient $A - Q_i$ on its whole stationary ideal. Adding the core coefficient gives $N_i C$. The shared error, not an unjustified individual exact norm equation, is what matters in this step.

13.7. Common stationary origins at minimal conductors. Assume $n \leq 2r + a - 1$ in (282). Both trace and norm of a variable in the core stationary ideal $\mathfrak{p}_{L_i}^{s_i}$ lie in $\mathfrak{p}_F^{q_F}$, where $q_F = a + r$. If $n \leq q_F$, take $X = 0$. Otherwise $h = n - T$ satisfies

$$(289) \quad 1 - \delta \leq h \leq a - 1.$$

Since $2q_F \geq n$, the base character has a linear formula $\lambda(1 - v) = \Psi_F(A_* v)$ on $\mathfrak{p}_F^{q_F}$, with $v_F A_* = -h$. Apply Lemma 13.8 and choose $A = n_3 X$ exactly, with $v_{L_3} X = -h$. Use the adjusted α henceforth.

Lemma 13.10. *The element C so constructed has $v_K C = -t_1$. For $i = 1, 2$,*

$$(290) \quad \theta_i(1 - v) = \Psi_{L_i}(N_i C v) \quad (v \in \mathfrak{p}_{L_i}^{s_i}),$$

$$(291) \quad \omega_i(1 - v) = \Psi_F(n_i S_i C v) \quad (v \in \mathfrak{p}_F^{q_i}).$$

The traces $S_i C$ are nonzero with the valuations of h_i . If $a = r$, this same C satisfies both formulas for all three indices.

Proof. For $X \neq 0$, the inequality $h \leq a - 1$ gives $-2h > -t_1$, so Y dominates in C . On the two upper ideals the arguments of (288) have respective depths at least $2a - h \geq a$ and $a + r - h \geq r$. Its weighted errors have depths at least $T + a - h$ and $3r - h$, respectively, both at least T . The trace and norm variables lie in the full base ideal already specified. Thus exact norm expansion and (284) prove (290) for the actual characters.

By Lemma 13.9,

$$v_F \Delta \geq \min\{2r - 2\lceil h/2 \rceil, e + r - \lceil h/2 \rceil\}.$$

Adding a to both bounds gives at least T , since $h \leq a - 1$; adding r does also. Equation (286) therefore proves (291). Moreover $2r - 2 \lceil h/2 \rceil > 2\delta$ and $r - \lceil h/2 \rceil > 0$, so the trace perturbations cannot cancel the leading terms of h_1, h_2 .

When $a = r$, one has $s_3 = T$, $q_3 = r$, and $h \leq r - 1$. On side 3 the exact norm phase applied to $(X/k_0)v$ gives the pullback coefficient $A - k_0X$, since $\beta_3 = k_0^2$. But

$$N_3C = a_3 + X^2 + k_0X, \quad (X^2 + k_0X) - (A - k_0X) = p_XX - 2A + 2k_0X.$$

The three terms on the right have L_3 -valuations at least $2r - 2h - (h \bmod 2)$, $2e - 2h$, and $2e - h$, all nonnegative. They are invisible on $\mathfrak{p}_{L_3}^T$, proving the third upper formula. Also $S_3C = -k_0 - 2X$ is a unit and

$$n_3S_3C - \beta_3 = 2k_0p_X + 4A.$$

Both terms have base valuation at least r , giving the third lower formula. The case $X = 0$ is already contained in the core and lower formulas. $\square$

13.8. The exact minimal sign. Write $c_0 = -\alpha^{-1}$ and use the full factors and terminal invariance of Lemma 4.1. Their residual functions and normalized sums will be denoted by H_i, g_i .

Use N_iC for the upper coefficient and n_iS_iC for the lower one in Lemma 13.10. The latter is an actual norm and all T_i are even. By compatibility, the determinant restriction, and (281), the full expression is

$$(292) \quad \mathcal{A}_i = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C))g_i.$$

Here g_i is the full upper residual factor.

Choose a top uniformizer π and compatible lower uniformizers $\Pi_i = N_i\pi$, $\varpi = n_i\Pi_i$. These choices need not be the ones used to construct the generators earlier. Vary C to $C_v = C(1 + \pi^{t_2}v)$ for integral $v \in K$. The upper critical depths are t_2 on side 1 and $a + r - 1$ on sides 2,3. The changes $S_i(C\pi^{t_2}v)$ have depths at least $3r - 2a$ and r . After division by S_iC they have depth at least $r \geq q_i$. Their lower norm quotients are actual norms on the lower stationary ideals. Subtracting (281) at C_v, C , using the full lower formulas and canceling the common multiplicative value, gives

$$(293) \quad H_i(p_i(\bar{v})) = Q(\bar{v}),$$

where p_i is the actual upper graded norm map and Q is independent of the compared index. All these assertions concern full functions, not merely their polar forms. Every trace used as a multiplicative argument stays nonzero, since its relative change has positive depth.

Proposition 13.11 (Two unequal breaks). *If $a < r$, the minimal family satisfies $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. The source depth $t_2 = 2r - 1$ is below the first upper break $4r - 2a - 1$. Thus $p_1(v) = v^2$, with its trace term deeper. On the second edge the consecutive trace ideals at source depths $2r - 1, 2r$ have depths $a + r - 1, a + r$. Hence the trace induces a nonzero k -linear map of residue lines, and the norm endpoint is strictly deeper because $r > a$. Thus p_2 is also bijective. Summing (293) proves $g_1 = g_2$. Equation (292) gives the assertion. The relabeled construction compares the first with the third field. $\square$

Proposition 13.12 (One equal break). *If $a = r$, every minimal family satisfies $\mathcal{A}_1 = \mathcal{A}_2 = \mathcal{A}_3$.*

Proof. Now $t = t_1 = t_2 = 2r - 1$ and Lemma 13.10 provides the same C on all three sides. Put

$$u = \overline{C\pi^t} \in k^\times, \quad c_i = \overline{g_iC - C} \in k^\times, \quad P = c_1c_2c_3, \quad k_i = c_jc_k \quad (\{i, j, k\} = \{1, 2, 3\}).$$

The residues are $\bar{d}, 1, 1 + \bar{d}$, respectively. Indeed the origin formulas give these residues for Y , and the adjustment X changes none: its conjugate differences have positive valuation by $h \leq r - 1$ and the ramification bound. Thus the c_i are distinct and $c_1 + c_2 = c_3$.

Expansion of the odd-valuation leading term $u\pi^{-t}$ shows that the first ramification coefficient on π is c_i/u . The exact quadratic critical norm polynomial consequently is

$$(294) \quad p_i(v) = v^2 + (c_i/u)v.$$

One can fix the trace coefficient without a convention choice: the norm endpoint has leading coefficient one, and $g_i\pi/\pi$ is the nontrivial critical kernel element with residue c_i/u . The lower conjugate difference of N_iC has residue k_i . For Y this follows immediately from (267); the change to C has conjugate difference of positive valuation. Its leading term is $u^2\Pi_i^{-t}$. Therefore the lower critical norm polynomial is $v^2 + \eta_i v$ with $\eta_i = k_i/u^2$.

Write the actual last-layer additive character as

$$\Psi_F(\varpi^t v) = (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2}(\gamma \bar{v})}, \quad \gamma \in k^\times.$$

The lower stationary norm coefficient $n_i S_i C$ has residue c_i^2 : $2C$ is of positive K -valuation. Triviality of ω_i on its actual critical norms and full residue duality give

$$\mathrm{Tr}_{k/\mathbf{F}_2}(\gamma c_i^2(v^2 + \eta_i v)) = 0 \quad (v \in k), \quad \eta_i = (\sqrt{\gamma} c_i)^{-1}, \quad u^2 = \sqrt{\gamma} P.$$

Every unit scalar is thus fixed. Tracing the leading term of $N_i C \Pi_i^{2t}$ in the Lamprecht polar formula gives the polar pairing of the actual H_i :

$$(295) \quad \frac{H_i(x+y)}{H_i(x)H_i(y)} = (-1)^{\mathrm{Tr}_{k/\mathbf{F}_2}(\gamma k_i xy)}.$$

The omitted next term in that trace lies in $\mathfrak{p}_F^{t+1}$ by (259). No normalized coefficient $\gamma = 1$ has been presumed.

The image of (294) is the index-two hyperplane

$$V_i = \{x : \mathrm{Tr}_{k/\mathbf{F}_2}(u^2 x/c_i^2) = 0\}.$$

For $j \neq i$, put $w_i = p_i(c_j/u) = k_i/u^2 \neq 0$. The common *actual* function (293) gives $H_i(w_i) = H_j(0) = 1$. The polar character of w_i in (295) is the character defining V_i , since $u^4 = \gamma P^2$. All hypotheses of Lemma 4.8 now hold on $(k, +)$, with image V_i . It gives $\sum_k H_i = \frac{1}{2} \sum_k Q$, independently of i . The common normalization and (292) therefore identify the complete factors. $\square$

13.9. All higher conductors. Now $n = T + h \geq 2r + a$, so $h \geq a$. The upper conductors and stationary depths are

$$(296) \quad M_i = 2n - T_i, \quad b_1 = 2r + h - a, \quad b_2 = r + h, \quad M_i = 2b_i.$$

For a variable in either $\mathfrak{p}_{L_i}^{b_i}$, $i = 1, 2$, its trace and norm belong to the ordinary base stationary ideal $\mathfrak{p}_F^{[n/2]}$. For the first trace this follows from $r + \lfloor (h+a)/2 \rfloor \geq r + \lceil h/2 \rceil$; for the second use $r \geq 1$. Choose the actual coefficient A_* of λ there, of valuation $-h$. Apply Lemma 13.8 to make $A = n_3 X$ exact, and use (283). There is no upper bound on h .

Theorem 13.13 (Exact comparison for every higher family). *For every $h \geq a$, one has $\mathcal{A}_1 = \mathcal{A}_2$.*

Proof. On the full stationary ideals the arguments $(Q_i/\beta_i)v$ of (288) have depths at least a and r , respectively:

$$(2\delta - h) - 4\delta + b_1 = a, \quad -h + b_2 = r.$$

The weighted errors have base depths at least

$$(T - a - h) - 2\delta + b_1 = T, \quad (T - a - h) + b_2 = 3r - a \geq T.$$

Thus the actual pullbacks λn_i have full coefficient $A - Q_i$, and the core charts, valid because $b_i \geq s_i$, give coefficient $N_i C$ for the actual θ_i . Moreover X dominates Y , as $-2h < -t_1$. Hence $C \neq 0$ and $v_{L_i} N_i C = -2h = J_i - M_i$.

Retain the original lower coefficients β_i , which are actual norms. It is unnecessary, and not asserted, that $n_i S_i C$ are still lower stationary coefficients in this range. All upper and lower conductors are even, so every residual factor is one. The complete formula gives

$$\mathcal{A}_i = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-\mathrm{tr}_i N_i C - \beta_i) = D_\theta(c_0)\Theta(C)^{-1}\Psi_F(-E_2(C) + \Delta).$$

The last identity is (281) and (286). Every factor is the same for $i = 1, 2$. In particular a possibly shallow Δ is retained as a common phase rather than discarded. Relabel the third field for the other comparison. $\square$

13.10. Completion of the nonmaximal case.

Proof of Theorem 13.1. Lemmas 13.2 and 13.3 construct an actual common origin and full norm-character coefficients. Lemmas 13.4 and 13.5 prove the extension and compatibility conditions on their whole domains, giving the actual cores of Theorem 13.6. Lemma 13.7 realizes the prescribed family by a common base twist. For $n \leq 2r + a - 1$ use Lemma 13.10 and Propositions 13.11–13.12. For $n \geq 2r + a$ use Theorem 13.13. These two conductor intervals are consecutive. The one-break and two-break ramification possibilities in (257) are both included. No step uses an equality of the desired local constants to deduce a field congruence or a norm equation. $\square$

14. COMPLETION OF THE SECOND MAIN LEMMA

14.1. Ramification possibilities.

Lemma 14.1 (Selection of fields in wild odd degree). *Suppose p is odd and K/F is totally ramified with group C_p^2 . One can choose a degree- p lower field B_1 so that, for every other lower field B_2 , their lower breaks are $t \leq t_2 = t + \delta$, and the breaks of $K/B_2, K/B_1$ are $t, t' = t + p\delta$, respectively. Here $\delta \geq 0$ is an integer and $p \nmid t$.*

Proof. Lemma D.3 gives the distinguished field B_1 , the lower breaks t, t_2 , and the upper-edge breaks $t' = t + p(t_2 - t), t$. Its proof uses only the subgroup definition and transitivity of the different, not a quotient ramification theorem. In particular $\delta = t_2 - t$ is a nonnegative integer. In mixed characteristic, the trace of 1 on B_2/F and (3) imply $v_K(p) \geq p(p-1)t_2$. The proof of Proposition 7.2, before using any assumption $p \nmid t$, constructs an exact Artin–Schreier coordinate by additive Hilbert 90 and best approximation and proves precisely that divisibility assertion. Its hypotheses are the edge breaks and the displayed p -valuation bound, so it applies here without circularity. The equal-characteristic argument of that proposition proves the same assertion. There is consequently no extra maximal-break case missing from the odd total-ramification calculation. $\square$

Lemma 14.2 (Quadratic conductors in mixed characteristic). *Let $F/\mathbf{Q}_2$ be finite and $e = v_F(2)$. A ramified quadratic norm character has conductor $2a$, $1 \leq a \leq e$, or $2e + 1$. Any two such characters of conductor $2e + 1$ have the same restriction to U_F^{2e} , so their product has conductor at most $2e$.*

Proof. The trace of 1 gives $e \geq \lfloor D/2 \rfloor$ for the quadratic different exponent D , so $2 \leq D \leq 2e + 1$. The even possibilities are as stated. Lemma 10.3 proves that an odd D is $2e + 1$ and fixes the last-layer value in coordinates $1 + 4z$ as $(-1)^{\text{Tr}_{k_F/\mathbf{F}_2} \bar{z}}$. This value is independent of the maximal character; hence the product of two is trivial on U_F^{2e} . $\square$

Lemma 14.3 (Exhaustive local classification). *Every C_ℓ^2 local diamond belongs to one of the proof branches in Sections 5–13.*

Proof. By Lemma D.2 the finite residue Galois group is cyclic; a C_ℓ^2 extension cannot be unramified. Its inertia has order ℓ or ℓ^2 . If $\ell \neq p$, wild inertia is trivial and tame inertia is cyclic, so inertia has order ℓ . Abelianness and the residue-Frobenius conjugation action imply $\ell \mid |k_F| - 1$. There is a unique unramified lower field and every other lower field is tamely ramified. These are exactly the hypotheses of Section 5.

Suppose $\ell = p$. Inertia of order p gives the unramified–ramified configuration, treated in Sections 6–9 for odd p and Section 10 for $p = 2$. If inertia has order p^2 , the extension is totally ramified. For odd p , Lemma 14.1 provides all the hypotheses of the totally ramified proof in Sections 7–9.

It remains to classify totally ramified biquadratic extensions. In characteristic exactly two, the reduced Artin–Schreier representatives of Section 11 have positive odd breaks and the two largest equal, including the one-break case. Theorem 11.1 treats all of them. In mixed characteristic the three nontrivial lower norm characters obey $\omega_3 = \omega_1\omega_2$ by Lemma D.6, and all are ramified. The product of two characters of unequal conductors has the larger conductor, so

the maximum of their three conductors occurs at least twice. If it is $2e + 1$, Lemma 14.2 makes the third smaller, and the lower breaks are $(2a - 1, 2e, 2e)$ with $1 \leq a \leq e$, precisely Section 12. Otherwise their conductors are $(2a, 2r, 2r)$ after relabeling, with $1 \leq a \leq r \leq e$, giving the breaks $(2a - 1, 2r - 1, 2r - 1)$ of Section 13. The upper different formulas in those parts follow from Lemma D.3. No further parity or divisibility case is assumed away. $\square$

14.2. Proof of the main theorem.

Proof of Theorem 1.1. For $\ell \neq p$, Theorem 5.3 compares the unique unramified lower field with every other lower field. This includes $\ell = 2$ and its exact tame sign when p is odd. For $\ell = p$ odd and inertia of order p , use Theorem 6.1; its conductor dispatch includes the final cubic affine correction. If inertia has order p^2 , choose the distinguished lower field as in Lemma 14.1 and apply Theorem 9.36 to its comparison with every other field.

For $p = \ell = 2$ and inertia of order two, use Theorem 10.20. For total ramification in characteristic two, use Theorem 11.1. For total ramification in mixed characteristic, Theorem 12.1 or Theorem 13.1 applies according to Lemma 14.3. These branches exhaust every character conductor: each has an actual common-twist realization and consecutive minimal, intermediate, and high conductor ranges, or a direct all-high-conductor comparison. All quadratic correction factors and full residual functions are retained.

In every case the given compatible characters have precisely the common norm pullback required by the branch theorem, and primitivity is the non-descent hypothesis in those theorems. Where realization conjugates an inducing character, it is an upper norm-character twist. An automorphism change of variables in the defining local integral preserves its local constant and its common pullback. This uses no two-field comparison. Comparisons with a distinguished field give independence of all the intermediate fields by transitivity of equality. The canonical additive character in the equal-characteristic calculation was removed by its proved scaling argument and the common determinant restriction. All other arguments already use the given trace-compatible additive characters. The result is therefore the full normalized expression of Theorem 1.1, for the given quasi-characters in either characteristic. $\square$

APPENDIX A. THE LEADING GAUSS CONGRUENCE, WITH PROOF

This appendix proves the precise leading congruence used in Section 5. Only finite-field character orthogonality, the binomial theorem, and the elementary local lifting of roots of unity are needed. In particular, neither a factorization theorem for Gauss sums nor a formula involving the p -adic gamma function is an input.

A.1. The coefficient field and notation. Let $k = \mathbf{F}_q$, $q = p^f$, and choose an unramified extension $K_0/\mathbf{Q}_p$ with residue field k . For clarity, this elementary lift can be constructed as follows. Lift a monic irreducible polynomial of degree f over $\mathbf{F}_p$ to a monic polynomial over $\mathbf{Z}_p$, and take its quotient ring R . It is finite free and p -adically complete over $\mathbf{Z}_p$, with $R/pR = k$. Every element with nonzero residue is a unit: lift an inverse modulo p and invert the remaining element of $1 + pR$ by its convergent geometric series. Every nonzero element is p^n times such a unit, since R is finite free and separated over $\mathbf{Z}_p$. Thus R is a discrete valuation ring with uniformizer p , and its fraction field is the required K_0 .

Hensel lifting of $X^{q-1} - 1$ gives a unique multiplicative lift $x \mapsto [x]$ from $k^\times$ into $R^\times$: its derivative is a unit at every nonzero residue root. The lifts multiply because their products are again roots with the prescribed residue. Choose a primitive p th root ζ and put

$$\varpi = \zeta - 1, \quad K_1 = K_0(\zeta), \quad \mathcal{O}_1 = \mathcal{O}_{K_1}.$$

For $p > 2$, the polynomial

$$\frac{(1 + X)^p - 1}{X} = p + \binom{p}{2}X + \cdots + X^{p-1}$$

is Eisenstein; hence ϖ is a uniformizer of K_1 and its residue field is k . For $p = 2$, $K_1 = K_0$ and $\varpi = -2$ is already a uniformizer. All root-of-unity values below are viewed in K_1 ; the identities are algebraic identities among these values.

Let $\text{tr}_k = \text{Tr}_{k/\mathbb{F}_p}$ and $\psi(x) = \zeta^{\text{tr}_k(x)}$. For $0 < a < q - 1$, write

$$a = \sum_{i=0}^{f-1} a_i p^i, \quad s_p(a) = \sum_i a_i, \quad a!_p = \prod_i a_i!.$$

Define the plus-sign Gauss sum

$$G_a = \sum_{x \in k^\times} [x]^{-a} \psi(x).$$

The convention of Section 5 is $\tau_k(\omega^a) = -G_a$, where $\omega(x) = [x]$.

A.2. A first-order calculation and a Jacobi sum.

Lemma A.1. *If $q > 2$, then $G_1 \equiv -\varpi \pmod{\varpi^2 \mathcal{O}_1}$. Moreover $G_{p^j} = G_1$ whenever $0 \leq j < f$ and $p^j < q - 1$.*

Proof. For each x , let $t_x \in \{0, \dots, p-1\}$ represent $\text{tr}_k(x)$. Then $\psi(x) = (1 + \varpi)^{t_x} \equiv 1 + t_x \varpi \pmod{\varpi^2}$. The exact identity $\sum_{x \neq 0} [x]^{-1} = 0$ follows by summing a nontrivial character of the cyclic group $k^\times$. Therefore

$$G_1 \equiv \varpi \sum_{x \neq 0} [x]^{-1} t_x \pmod{\varpi^2}.$$

The residue of the last sum is

$$\sum_{x \neq 0} x^{-1} \text{tr}_k(x) = \sum_{i=0}^{f-1} \sum_{x \neq 0} x^{p^i-1} = -1 \quad \text{in } k.$$

Indeed the $i = 0$ inner sum is $q - 1 = -1$, while every other exponent is strictly between 0 and $q - 1$ and its power sum is zero. A power sum vanishes by multiplying its variable by an element on which the corresponding character is nontrivial. Finally, Frobenius permutes $k^\times$ and preserves its absolute trace. Reindexing by the inverse Frobenius shows that $G_{pa} = G_a$, with indices reduced modulo $q - 1$. Iteration gives the last assertion. $\square$

Lemma A.2 (Jacobi sums when the digits add without carries). *Suppose $a, b > 0$, $a + b < q - 1$, and $a_i + b_i < p$ for every digit. Put*

$$J(a, b) = \sum_{x \in k \setminus \{0, 1\}} [x]^{-a} [1 - x]^{-b}.$$

Then

$$(297) \quad G_a G_b = J(a, b) G_{a+b},$$

and $J(a, b)$ is a unit with

$$(298) \quad J(a, b) \equiv - \prod_{i=0}^{f-1} \binom{a_i + b_i}{a_i} \pmod{\varpi \mathcal{O}_1}.$$

Proof. Expand the product of the Gauss sums and separate $t = x + y$. When $t = 0$ the contribution is zero because the character $[x]^{-(a+b)}$ is nontrivial. When $t \neq 0$, write $x = tu$, $y = t(1 - u)$. The two sums separate, giving (297).

For the reduction of the Jacobi sum, both exponents below are positive, so the terms at 0 and 1 may be included. In k the reduction is

$$\sum_{x \in k} x^{q-1-a} (1-x)^{q-1-b} = \sum_{j=0}^{q-1-b} (-1)^j \binom{q-1-b}{j} \sum_{x \in k} x^{q-1-a+j}.$$

The exponent is positive and smaller than $2(q-1)$. Its only possible positive multiple of $q-1$ occurs at $j = a$. This index occurs because $a + b < q - 1$. The reduction is consequently $-(-1)^a \binom{q-1-b}{a}$.

To evaluate this binomial coefficient, use in $\mathbf{F}_p[X]$ the identity

$$(1 + X)^{q-1-b} = \prod_{i=0}^{f-1} (1 + X^{p^i})^{p-1-b_i}.$$

Comparison of coefficients gives $\binom{q-1-b}{a} = \prod_i \binom{p-1-b_i}{a_i}$ modulo p ; each exponent on the right has a unique digit expansion. Since $a_i + b_i < p$, ordinary factorials give

$$\binom{p-1-b_i}{a_i} \equiv (-1)^{a_i} \binom{a_i + b_i}{a_i} \pmod{p}.$$

For odd p , $a \equiv \sum_i a_i \pmod{2}$, so the signs cancel. For $p = 2$ the same sign equality holds in the residue field. This proves (298). None of its factorials is zero modulo p , proving that $J(a, b)$ is a unit. $\square$

A.3. The complete leading unit.

Theorem A.3 (Leading Gauss congruence). *For every prime p , every $q = p^f$, and $0 < a < q - 1$,*

$$(299) \quad G_a \equiv -\frac{\varpi^{s_p(a)}}{a!_p} \pmod{\varpi^{s_p(a)+1}\mathcal{O}_1}.$$

In particular G_a has ϖ -valuation exactly $s_p(a)$, and

$$(300) \quad \tau_k(\omega^a) \equiv_{\times} \frac{\varpi^{s_p(a)}}{\prod_i a_i!}.$$

Here $X \equiv_{\times} Y$ means $X/Y \in 1 + \varpi\mathcal{O}_1$.

Proof. If $q = 2$ there is no index in the specified interval. Otherwise use induction on $s_p(a)$. For digit sum one, $a = p^j$, and Lemma A.1 proves the assertion. For larger digit sum choose j with $a_j > 0$ and put $c = a - p^j$. Then $c > 0$, and $c + p^j = a < q - 1$ is an addition without carries. Lemma A.2 gives $J(c, p^j) \equiv -a_j \pmod{\varpi}$. It is a unit, so the exact product formula and the induction hypothesis give

$$G_a = \frac{G_c G_{p^j}}{J(c, p^j)} \equiv -\frac{\varpi^{s_p(a)}}{a_j(a_j - 1)! \prod_{i \neq j} a_i!} \pmod{\varpi^{s_p(a)+1}}.$$

Division by the unit does not lower the error precision. This proves (299), including its nonzero leading coefficient. Changing from G_a to $-G_a$ proves (300) with the positive sign required in (11). The argument applies unchanged when $p = 2$ and $\varpi = -2$. $\square$

Thus the leading unit used to resolve the root of unity in the tame comparison is proved here, not imported as an additional Gauss-sum theorem. The base- q grouping and factorial manipulations following (11) are consequences of this theorem and the proofs already given in Section 5.

APPENDIX B. FORMAL LOCAL RESIDUES AND THEIR TRACE COMPATIBILITY

This appendix proves the local identity used in the characteristic-two argument. A differential on $k((t))$ means a *formal differential* $h(t) dt$, with termwise differentiation of Laurent series, and

$$\text{Res}_t \left(\sum_n h_n t^n dt \right) = h_{-1}.$$

No global curve or residue theorem is used. We first prove substitution and trace compatibility algebraically, including in wild ramification.

B.1. Substitution and a finite free expansion.

Lemma B.1 (Formal substitution). *Let k be a field and let $u(t) \in t^e k[[t]]$ have exact order $e \geq 1$. For every $H(s) \in k((s))$,*

$$(301) \quad \text{Res}_t(H(u(t))u'(t) dt) = e \text{Res}_s(H(s) ds).$$

In particular the residue is unchanged by a change of uniformizer.

Proof. It suffices to treat $H(s) = s^n$: only finitely many of its Laurent monomials can contribute to the residue after substitution. Write $u = t^e v$ with $v \in k[[t]]^\times$. If $n = -1$, then $u'/u = e/t + v'/v$ and $v'/v \in k[[t]]$, proving the assertion. For $n \neq -1$ one must not divide by $n+1$ in characteristic dividing $n+1$. Instead, perform the calculation first over the torsion-free ring

$$R = \mathbf{Z}[c_0, c_0^{-1}, c_1, c_2, \dots], \quad v = c_0 + c_1 t + c_2 t^2 + \dots.$$

The coefficient giving the residue uses only finitely many c_i and lies in this ring, also for negative n . The residue of a derivative is zero, so differentiating u^{n+1} gives $(n+1) \text{Res}_t(u^n u' dt) = 0$. Torsion-freeness gives residue zero in R . Specializing the coefficients to any field proves the desired identity without division in that field. Linearity now proves (301); for a change of uniformizer, $e = 1$. $\square$

Lemma B.2 (A basis compatible with coefficients). *Let R be a commutative ring and $u(t) = t^e(c_0 + c_1 t + \dots)$, where $c_0 \in R^\times$. Under $s \mapsto u(t)$, the module $R[[t]]$ is free over $R[[s]]$ with basis $1, t, \dots, t^{e-1}$. After inverting s it is $R((t))$, free over $R((s))$ with the same basis. These expansions and the resulting multiplication matrices commute with specialization of the coefficients.*

Proof. For $n = ej + i$, $0 \leq i < e$, the series $s^j t^i$ has leading term $c_0^j t^n$. Given a power series, cancel its lowest coefficient using the corresponding multiple of $s^j t^i$ and continue successively. This constructs an expansion $\sum_{i=0}^{e-1} H_i(s) t^i$, with $H_i \in R[[s]]$. It is unique: the first nonzero terms of the nonzero $H_i(s) t^i$ have different degrees modulo e , and hence cannot cancel. The algorithm uses only ring operations and c_0^{-1} . Every output coefficient is determined after finitely many operations, which proves compatibility with specialization. Since $s = t^e v(t)$ with $v(t)$ a unit, inverting s is equivalent to inverting t . This proves the assertions for Laurent series as well. In particular, the algebraic trace of multiplication by an element of $R((t))$ is an element of $R((s))$, obtained by the trace of its finite matrix, and it commutes with specialization. $\square$

B.2. Trace compatibility.

Lemma B.3 (Totally ramified substitution). *For a field k and $s = u(t)$ of positive order e ,*

$$(302) \quad \text{Res}_s \left(\text{Tr}_{k((t))/k((s))}(h(t)) ds \right) = \text{Res}_t (h(t) u'(t) dt) \quad (h \in k((t))).$$

The trace here is the algebraic trace on the finite free extension. The identity remains valid even when $u' = 0$.

Proof. First suppose that k has characteristic zero. After a finite extension of the coefficient field, we can take an e th root of the leading coefficient of u and include all e th roots of unity. This scalar extension preserves the trace matrix in Lemma B.2, so an identity proved there descends. The binomial series then gives a formal parameter $z = t(c_0 + c_1 t + \dots)^{1/e}$ with $s = z^e$. Write $h = \sum_n b_n z^n$. The extension $k((z))/k((s))$ is now Galois with automorphisms $z \mapsto \xi z$, $\xi^e = 1$. Thus

$$\text{Tr}(h) = e \sum_{j \in \mathbf{Z}} b_{ej} s^j, \quad h ds = e \sum_n b_n z^{n+e-1} dz.$$

Both residues equal eb_{-e} . The change from t to z preserves residue by Lemma B.1. The trace is continuous for the Laurent-series topology, either from the displayed automorphisms or from its finite multiplication matrix, so these calculations apply to infinite Laurent series, not just Laurent polynomials.

Now take a field of arbitrary characteristic. Lift every coefficient of $u(t)$ and $h(t)$ to an independent indeterminate in a polynomial ring over $\mathbf{Z}$, inverting the nonzero leading coefficient of u . This ring is a characteristic-zero integral domain. The two residues in (302) belong to

this ring: Lemma B.2 gives this for the trace, and the right side is an ordinary coefficient of a product and derivative. Each residue uses only finitely many of the indeterminates. Over the fraction field of this ring the characteristic-zero proof gives equality. It is therefore an equality in the ring itself. Specialize the coefficients to their values in k , using Lemma B.2 to specialize the trace. This proves (302) in every characteristic. This coefficient argument does not invert the possibly zero integer e in k and does not assume a tame extension. $\square$

Theorem B.4 (Local trace–residue identity). *Let E/F be a finite separable extension of equal-characteristic local fields with finite residue fields. For every formal differential η on E ,*

$$(303) \quad \mathrm{Tr}_{k_E/k_F} \mathrm{Res}_E(\eta) = \mathrm{Res}_F(\mathrm{Tr}_{E/F} \eta).$$

Here if s is a uniformizer of F and $\eta = x ds$, the trace of the differential is $\mathrm{Tr}_{E/F}(x) ds$.

Proof. We explain the Laurent-series presentations used in the proof. In an equal-characteristic complete discrete valuation field with finite residue field k of size q , each residue has a unique lift solving $X^q - X = 0$, by Hensel's lemma with derivative -1 . In characteristic p these lifts form a field: sums and products are again roots with the expected residues. Choosing a uniformizer t and successively subtracting residue coefficients gives a unique convergent Laurent expansion. Thus the field is $k((t))$. In a finite extension, these coefficient fields are compatible, because the lifts of elements of the smaller residue field are the same unique roots.

Write $F = k_F((s))$ and $E = k_E((t))$. The intermediate field $F' = k_E((s))$ is unramified over F ; its trace is the coefficientwise finite-field trace. In E , $s = u(t)$ has positive order e , and Lemma B.2 gives $[E : F'] = e$. Separability implies $u'(t) \neq 0$: otherwise $u \in k_E((t^p))$, and the intermediate extension $k_E((t))/k_E((t^p))$ would be purely inseparable of degree p , contradicting separability of E/F' . Therefore $ds \neq 0$ and every formal differential on E can be written as $x ds$.

Apply Lemma B.3 to E/F' . It gives $\mathrm{Res}_{F'}(\mathrm{Tr}_{E/F'}(x) ds) = \mathrm{Res}_E(x ds)$ in k_E . Apply Tr_{k_E/k_F} and commute it with extraction of the s^{-1} coefficient. Transitivity of matrix traces proves (303). Residues do not depend on the chosen uniformizers by Lemma B.1. $\square$

Lemma B.5 (Logarithmic derivative of a norm). *For $v \in E^\times$ in the same separable extension,*

$$(304) \quad \frac{dN_{E/F}(v)}{N_{E/F}(v)} = \mathrm{Tr}_{E/F} \left(\frac{dv}{v} \right).$$

Proof. A derivation extends uniquely across a finite separable algebraic extension: for a root z of a separable polynomial P , differentiating $P(z) = 0$ gives $dz = -(dP)(z)/P'(z)$. This formula also shows that differentiation commutes with every F -embedding into a normal closure. Differentiate the finite product $N_{E/F}(v) = \prod_{\sigma} \sigma(v)$ and divide by that nonzero product. The result is $\sum_{\sigma} \sigma(dv/v)$, which is the trace of the differential. Thus (304) is elementary algebra, independent of (303). $\square$

Theorem B.4 and Lemma B.5 supply all local residue statements formerly cited in the Artin–Schreier norm-character calculation. The Cartier identities used there are proved there by their coefficient formulas; no statement from Tate's paper is required to complete that argument.

APPENDIX C. ELEMENTARY DESCENT AND RAMIFICATION ARGUMENTS

We include the short algebraic arguments used in character descent, so Hilbert 90 need not be imported from a reference. The additional prime-square norm and ramification facts needed for the diamond are proved in Appendix D.

Lemma C.1 (Cyclic Hilbert 90, in both forms). *Let E/F be cyclic of degree n , with generator σ . Then*

$$\ker N_{E/F} = \{y/\sigma(y) : y \in E^\times\}, \quad \ker \mathrm{Tr}_{E/F} = \{\sigma(y) - y : y \in E\}.$$

Proof. Distinct field embeddings are linearly independent as maps. Here is the short argument: take a nonzero relation of shortest length. Evaluating at tx and subtracting one embedding's value at t times the relation at x removes that embedding. Choose t on which another embedding differs from it. The resulting shorter nonzero relation is a contradiction.

If $a \in E^\times$ has norm one, put $b_0 = 1$ and $b_i = \prod_{j=0}^{i-1} \sigma^j(a)$ for $1 \leq i \leq n$. By the independence just proved the map $x \mapsto \sum_{i=0}^{n-1} b_i \sigma^i(x)$ is not identically zero. Choose x for which its value y is nonzero. Since $b_n = 1$ and $a\sigma(b_i) = b_{i+1}$, one has $a\sigma(y) = y$. This gives $a = y/\sigma(y)$. The reverse inclusion telescopes under the norm.

For the additive statement, the trace map is nonzero by the same independence, and hence is a surjective F -linear map to F . Its kernel has dimension $n - 1$. The map $\sigma - 1$ has kernel F , so its image also has dimension $n - 1$. Its image lies in the trace kernel by telescoping, proving equality. This proof applies when the characteristic divides n . $\square$

Lemma C.2 (Continuous extension of the characters used here). *Let H be a subgroup of $E^\times$ containing U_E^m for some $m \geq 1$, and let $\chi : H \rightarrow \mathbf{C}^\times$ be trivial on U_E^m . Then χ extends to a continuous character of $E^\times$. The same assertion holds if χ is first prescribed on several subgroups, agrees on their intersections, and defines a character on the subgroup they generate containing such a U_E^m .*

Proof. The group $E^\times/U_E^m$ is a product of an infinite cyclic group and the finite group $\mathcal{O}_E^\times/U_E^m$. Extend a character of a subgroup one generator at a time. If the first relation of a new generator g with the existing subgroup is $g^n = h$, choose a nonzero complex n th root of the prescribed value of h . If there is no such relation, choose any nonzero complex value. This defines an extension at each step, and finitely many steps suffice. Pullback to $E^\times$ is continuous because U_E^m is open and remains in the kernel. Agreement and multiplicativity on the generated subgroup are explicit hypotheses in the last assertion, not conclusions of the extension argument. $\square$

Lemma C.3 (The first wild ramification term). *Let σ be an inertia automorphism of a finite Galois extension of local fields, let π be an upstairs uniformizer, and suppose*

$$\sigma(\pi) = \pi(1 + c\pi^u), \quad v(c) = 0, \quad u \geq 1.$$

For $x \neq 0$ with normalized upstairs valuation m , one has $v(\sigma x - x) \geq m + u$. If m is prime to the residue characteristic, then equality holds.

Proof. Expand $x = \sum_{j \geq m} a_j \pi^j$ using the multiplicative lifts of residue representatives. Inertia fixes those lifts: they are the unique lifts of their residues among roots of unity of order prime to the residue characteristic (and zero). The series converges in the local field, with only finitely many negative powers. For any integer j , the binomial expansion at the principal unit gives

$$(1 + c\pi^u)^j - 1 \equiv jc\pi^u \pmod{\mathfrak{p}^{u+1}}.$$

This is also valid for negative j , by the convergent inverse series. Consequently the j th summand changes by an element of valuation at least $j + u$. When m is prime to the residue characteristic, the first summand changes with exact valuation $m + u$, and all following summands have strictly larger valuation. This proves the assertions. The hypothesis $u \geq 1$ matters: in a tame row the first coefficient is instead the residue of $(\sigma\pi/\pi)^m - 1$ and may vanish. $\square$

Lemma C.4 (Transitivity of the different). *For a tower of finite separable local extensions $L/E/F$,*

$$\mathfrak{D}_{L/F} = \mathfrak{D}_{L/E} \mathfrak{D}_{E/F} \mathcal{O}_L.$$

Proof. Write $A = \mathcal{O}_F$, $B = \mathcal{O}_E$ and $C = \mathcal{O}_L$. These are finite free modules along the tower. Via the nondegenerate trace pairings their dual modules identify with the inverse differentials. The elementary adjunction

$$\mathrm{Hom}_A(C, A) \simeq \mathrm{Hom}_B(C, \mathrm{Hom}_A(B, A))$$

sends a functional ℓ to $c \mapsto (b \mapsto \ell(bc))$; its inverse evaluates at $b = 1$. Write the principal fractional B -ideal $\mathfrak{D}_{E/F}^{-1} = dB$. Under the trace identifications, the right side is $d\mathrm{Hom}_B(C, B) = d\mathfrak{D}_{L/E}^{-1}$. Transitivity of field trace shows that these identifications send $x \in L$ to the same functional $c \mapsto \mathrm{Tr}_{L/F}(xc)$, with no scalar discrepancy. Thus $\mathfrak{D}_{L/F}^{-1} = d\mathfrak{D}_{L/E}^{-1}$ as fractional C -ideals. Inversion gives the assertion. $\square$

Lemma C.5 (The discriminant of an integral order). *Let E/F be finite separable and let $B \subset \mathcal{O}_E$ be an $\mathcal{O}_F$ -order of full rank. Then its discriminant ideal is the field discriminant ideal multiplied by the square of its index ideal. In particular the field discriminant divides the order discriminant. For a totally ramified extension, the normalized different exponent is at most the F -valuation of the order discriminant.*

Proof. Choose $\mathcal{O}_F$ -bases of B and $\mathcal{O}_E$, and let M be the inclusion matrix. The two trace Gram matrices satisfy $G_B = M^\top G_{\mathcal{O}_E} M$, so $\det G_B = (\det M)^2 \det G_{\mathcal{O}_E}$. This proves the first assertion and divisibility. The valuation of the field discriminant is the length over $\mathcal{O}_F$ of the trace-dual lattice modulo $\mathcal{O}_E$: this follows by diagonalizing the trace matrix over the discrete valuation ring. The trace dual is $\mathfrak{D}_{E/F}^{-1}$ by the definition of the inverse different. Its cyclic-prime specialization underlies the trace-ideal formula of [8, Theorem 3.7]. If the different exponent is d , this quotient has $\mathcal{O}_F$ -length $[k_E : k_F]d$, since each successive $\mathcal{O}_E$ -lattice quotient is k_E . In the totally ramified case $[k_E : k_F] = 1$, proving the stated bound. $\square$

APPENDIX D. THE DIAMOND-LEVEL LOCAL FACTS, WITHOUT LOCAL RECIPROCITY

The general local reciprocity theorem and the general quotient theorem for ramification groups are unnecessary here. We prove their particular consequences used in this paper. Serre [5, Chapters III–V, XIII–XIV] remains a reference for the classical background and its more general formulations. The inputs below are the cyclic-prime results of [8, Theorem 3.7, Corollary 3.9, and Proposition 3.3], elementary finite-field and Galois theory, Hensel lifting, and the usual monogenic different identity. The latter says $\mathfrak{D}_{E/F} = (f'(\pi))$ when $\mathcal{O}_E = \mathcal{O}_F[\pi]$ and f is the minimal polynomial of π . It is the specialization of the trace-dual description of the different to a power basis. Its general algebraic form, simple-root Hensel lifting, and finite-field cyclicity are already available in Mathlib [4]. For completeness we give the short quantitative Newton argument below; this avoids confusing a theorem for $\mathbf{Z}_p$ with one for every local field. The declaration and hypothesis matches are recorded separately, not inserted into the mathematical argument.

We also use the elementary degree formula $[E : F] = e(E/F)f(E/F)$, which is part of the local-field infrastructure of the FML development. For a totally ramified extension, any uniformizer π generates the integer ring: $F(\pi) = E$ follows from the degree formula, and in the basis $1, \pi, \dots, \pi^{n-1}$, the valuations $nv_F(a_i) + i$ of the terms $a_i\pi^i$ are distinct modulo n . Thus an integral sum has every a_i integral. This supplies the monogenic hypothesis whenever the displayed different identity is used.

The cyclic coordinates used below need no further classification input. In characteristic p , additive Hilbert 90 applied to 1 in a cyclic degree- p extension gives $\sigma z - z = 1$. Then $z^p - z \in F$, and z generates the extension. In degree two and characteristic other than two, a nonzero $w = u - \sigma u$ has $\sigma w = -w$; therefore $w^2 \in F$ and $E = F(w)$. The reductions of these coordinates at a specified valuation are proved in the corresponding case calculations.

Lemma D.1 (Quantitative lifting over the local fields used here). *Let R be a complete discrete valuation ring with valuation v , let $f \in R[X]$, and let $a \in R$. Suppose $s = v(f'(a)) < \infty$ and $v(f(a)) > 2s$. Then there is a unique root b in the ball $v(b - a) > s$. Moreover $v(b - a) \geq v(f(a)) - s$.*

Proof. The case $f(a) = 0$ is immediate, including uniqueness by the argument below. Otherwise iterate $a_{n+1} = a_n - f(a_n)/f'(a_n)$. Taylor's polynomial identity with integral coefficients shows inductively that $v(f'(a_n)) = s$, and, writing $r_n = v(f(a_n)) - 2s > 0$, that

$$v(a_{n+1} - a_n) = s + r_n, \quad r_{n+1} \geq 2r_n.$$

If an iterate is a root, stop. The derivative differences have valuation at least $s + r_n > s$, so they cannot change s ; the linear Taylor term cancels and the remaining terms have valuation at least $2(s + r_n)$. Thus the iterates are integral and converge to a root with the asserted error bound. For b, c in the stated ball, divided differences give $f(b) - f(c) = (b - c)(f'(a) + \varepsilon)$ with $v(\varepsilon) > s$. The second factor is nonzero; hence two roots there are equal. The proof uses only completeness,

polynomial identities, and the nonarchimedean valuation inequalities. Rescaling the variable and the polynomial gives the fractional-coordinate applications wherever used. $\square$

Lemma D.2 (Inertia, tame action, and subgroup numbering). *For a finite Galois extension K/F of local fields, let $G = \text{Gal}(K/F)$, let I be the kernel of its residue action, and put $q = |k_F|$. The residue action identifies G/I with the cyclic group $\text{Gal}(k_K/k_F)$. Its fixed field K^I/F is unramified. The quotient I/G_1 is cyclic of order prime to the residue characteristic p , and G_1 is a p -group. A lift of residue Frobenius conjugates I/G_1 by the q th power map. For $H \subset G$, the lower ramification groups on K/K^H are $H_j = H \cap G_j$ for every integer $j \geq 0$.*

Proof. Choose a generator $\bar{a}$ of k_K/k_F , lift its monic irreducible polynomial to $\mathcal{O}_F[X]$, and lift $\bar{a}$ to a root $a \in \mathcal{O}_K$ by Hensel's lemma. The field $F(a)$ has degree at most $f = [k_K : k_F]$ and residue degree at least f , hence degree exactly f and ramification index one. All residue conjugates of $\bar{a}$ lift uniquely inside $F(a)$, so $F(a)/F$ is Galois and reduction identifies its group with $\text{Gal}(k_K/k_F)$. Normality of K/F extends its automorphisms to K . An element of G acts trivially on the residue field precisely when it fixes a , by uniqueness of the simple-root lifts. This proves the assertions about I and G/I .

Choose a uniformizer π of K . Inertia fixes the multiplicative lifts of every residue element. Expanding integers in powers of π therefore tests lower ramification on π alone. In particular,

$$I \longrightarrow k_K^\times, \quad g \longmapsto \overline{g\pi/\pi}$$

is a homomorphism with kernel G_1 . For $j \geq 1$ the map

$$G_j \longrightarrow k_K, \quad g \longmapsto \overline{(g\pi/\pi - 1)/\pi^j}$$

is additive with kernel G_{j+1} . These statements follow by multiplying the two uniformizer ratios and reducing the first nonzero coefficient. The finite subgroup of $k_K^\times$ is cyclic and has order prime to p ; each additive image is a p -group. The filtration eventually equals 1, since every nonidentity inertia automorphism moves π . This proves the assertion about G_1 .

The tame residue is unchanged on replacing π by $u\pi$ for a unit u , since inertia fixes its residue. For a Frobenius lift ϕ , compute using $\pi' = \phi^{-1}\pi$:

$$\overline{(\phi g \phi^{-1})(\pi)/\pi} = \overline{\phi(g(\pi')/\pi')} = \overline{g\pi/\pi}^q.$$

Finally, membership in a lower ramification group is the condition $v_K(gx - x) \geq j + 1$ for all $x \in \mathcal{O}_K$. For an element of H this condition is identical over F and over K^H : both use the same normalized valuation on K . This proves the subgroup assertion. $\square$

Lemma D.3 (The breaks of a totally ramified prime-square diamond). *Let K/F be totally ramified with $G = C_p^2$, where p is the residue characteristic. There are integers $1 \leq t \leq b$ such that the filtration is G through t , possibly one line H_0 of order p through b , and then 1. Choose $L_0 = K^{H_0}$ if $b > t$, and choose any degree- p field if $b = t$. The break of L_0/F is t ; the break of K/L_0 is b . Every other lower field L/F has break*

$$s = t + (b - t)/p,$$

and K/L has break t . In particular s is an integer, and $b = t + p(s - t)$. Inertia of order p , rather than p^2 , gives instead one unramified lower field and ramified lower fields whose breaks are unchanged by the unramified base extension.

Proof. Lemma D.2 gives $G_1 = G$. A decreasing filtration of C_p^2 can contain at most one proper nontrivial subgroup. The subgroup assertion gives the breaks on the two upper edges. For a top uniformizer π , the monogenic different formula gives

$$d_{K/F} = v_K \prod_{g \neq 1} (\pi - g\pi) = \sum_{j \geq 0} (|G_j| - 1) = (p^2 - 1)(t + 1) + (p - 1)(b - t).$$

Here each nonidentity g is counted exactly $v_K(\pi - g\pi)$ times. Transitivity of the different and the cyclic-prime formula give

$$d_{K/F} = (p - 1)(b + 1) + p d_{L_0/F} = (p - 1)(t + 1) + p d_{L/F}.$$

Solving these equalities proves the stated lower-edge breaks, including integrality of s , without a quotient-numbering theorem.

If inertia has order p , its fixed field U is the unique unramified lower field, and the other lower fields E are totally ramified. In $K = EU$, a uniformizer of E is still a uniformizer of K . The generator on K/U restricts to the generator on E/F , and the normalized valuation restricts without a multiplier on E . Its uniformizer difference thus has the same valuation on both edges. $\square$

Proposition D.4 (Separation of the norm kernels needed for SML). *Let K/F have group C_ℓ^2 . The norm kernels $N_{L_0/F}L_0^\times$ and $N_{L/F}L^\times$ are distinct if L_0 is the unramified lower field, or, in total ramification, L_0 is the distinguished field of Lemma D.3 and $L \neq L_0$. In the totally ramified one-break case all lower norm kernels are pairwise distinct.*

Proof. Every lower norm subgroup has index ℓ by the cyclic-prime results of FML. With an unramified lower field its nontrivial norm characters have conductor zero, whereas those of every ramified lower field have positive conductor. Equal norm kernels would give equal character groups, contrary to those conductors. In the totally ramified two-break case, Lemma D.3 gives the distinct lower conductors $t+1 < s+1$, proving the assertion in the same way.

It remains to distinguish the kernels when every edge has break t . Here $\ell = p$. Fix $\pi \in K$ of valuation one and put

$$c_g = \overline{(g\pi/\pi - 1)/\pi^t} \in k, \quad k = k_F = k_K.$$

Lemma D.2 makes $g \mapsto c_g$ an injective additive map, with two-dimensional $\mathbf{F}_p$ -image V . For a line $H = \langle \sigma \rangle$, set $L_H = K^H$, $c = c_\sigma$, $\Pi_H = N_{K/L_H}\pi$, and $\varpi = N_{K/F}\pi$. For $\tau \notin H$ write $e = c_\tau$. FML's critical norm polynomial ([8, Proposition 3.3]) gives

$$P_H(X) = X^p - c^{p-1}X, \quad \overline{(\tau\Pi_H/\Pi_H - 1)/\Pi_H^t} = e^p - c^{p-1}e =: d_H \neq 0.$$

The second equality is the first applied to $N_{K/L_H}(\tau\pi/\pi) = \tau\Pi_H/\Pi_H$. Apply the same critical norm theorem now to L_H/F with the compatible uniformizers Π_H, ϖ . Its critical norm image is

$$d_H^p \ker \text{Tr}_{k/\mathbf{F}_p},$$

whose annihilator is the line $d_H^{-p}\mathbf{F}_p$ in the residue trace pairing. This line is precisely the restriction of the norm-character group to the last unit layer, by the norm-filtration theorem and the exact norm-character conductor.

Choose a basis c_1, c_2 of V and put $D = c_1c_2^p - c_1^pc_2 \neq 0$. For any ordered basis c, e of V ,

$$ce^p - c^pe = aD \quad \text{for some } a \in \mathbf{F}_p^\times, \quad d_H = aD/c.$$

Thus the annihilator line is $(c^p/D^p)\mathbf{F}_p$. Distinct lines H give distinct lines $c\mathbf{F}_p$, and Frobenius is bijective, so these annihilators are distinct. The norm-character groups, and hence their norm kernels in $F^\times$, are distinct. This argument identifies whole last-layer characters, not just one trivial character value. $\square$

Lemma D.5 (Crossed norm characters from distinct norm kernels). *Let L_i, L_j be distinct lower fields in a C_ℓ^2 diamond, and suppose their lower norm kernels are distinct. Then*

$$S(L_j/F) \longrightarrow S(K/L_i), \quad \omega \longmapsto \omega \circ N_{L_i/F}$$

is an isomorphism. In particular this applies to every distinguished comparison in Proposition D.4.

Proof. Norm transitivity shows that every displayed pullback kills $N_{K/L_i}K^\times$. A nontrivial ω has kernel $N_{L_j/F}L_j^\times$, since that quotient has prime order ℓ . Its pullback cannot be trivial: otherwise one index- ℓ lower norm subgroup would be contained in the other and hence equal to it. Both character groups have order ℓ by FML, so the displayed homomorphism is an isomorphism. This is the required base-change identity proved directly, not an appeal to local reciprocity. $\square$

Lemma D.6 (All three quadratic norm characters). *In a biquadratic local extension, the three nontrivial lower norm characters are pairwise distinct and, after labeling, satisfy $\omega_3 = \omega_1\omega_2$. Consequently Lemma D.5 applies to every pair of its lower fields.*

Proof. If $\text{char } F \neq 2$, associate to a square class a the norm character of $F(\sqrt{a})/F$, and associate the trivial character to the square class of 1. Denote its value at b by (a, b) . The existence of a nonzero solution of

$$X^2 = aY^2 + bZ^2$$

is equivalent to $(a, b) = 1$. If a is nonsquare, any such solution has $Z \neq 0$ and gives $b = N(X/Z + (Y/Z)\sqrt{a})$; the split case is immediate. The conic is symmetric in a, b , so $(a, b) = (b, a)$. Multiplicativity in the second variable is the norm-character property; symmetry gives it in the first variable. A nonsquare class has nontrivial character by FML's index-two norm theorem. Thus the map from square classes to quadratic characters is an injective homomorphism. The three fields have classes a, b, ab , proving the claim.

If $\text{char } F = 2$, use the explicitly proved formula of Lemma 11.2:

$$\omega_f(u) = (-1)^{\text{Tr}_{k/\mathbb{F}_2} \text{Res}_F(f du/u)}.$$

Its proof uses only the single quadratic norm index, the Cartier calculation, and Appendix B; in particular it uses no crossed norm-character assertion or SML comparison. The formula is additive in the Artin–Schreier class f , and the same proof establishes nontriviality for every nonzero class, ramified or unramified. The classes of the three fields are $f, g, f + g$, giving pairwise distinctness and the product identity. $\square$

Remark D.7 (The scope of the replacements). For odd ℓ the proof of SML compares a distinguished lower field with every other one and then uses transitivity of equality. It does not require norm separation for an arbitrary pair of larger-break lower fields, or the general norm correspondence for all abelian extensions. For $\ell = 2$, Lemma D.6 supplies every pair as well as the product of the three lower characters. All the ramification numbers used in the final case classification follow from Lemmas D.2–D.3. Thus neither full local class field theory nor the general Herbrand quotient theorem is an unstated prerequisite. The proofs above give the additional local arguments; the elementary matrix, field, and finite-group bookkeeping need not be reproduced in greater generality here.

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RESEARCH INSTITUTE FOR MATHEMATICAL SCIENCES, KYOTO UNIVERSITY, KYOTO 606–8502, JAPAN
 Email address: `fueda@kurims.kyoto-u.ac.jp`